Mastering Microsoft Fabric

SAASification of Analytics

Debananda Ghosh

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Mastering Microsoft Fabric: SAASification of Analytics

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Table of Contents

About the Author ix

About the Technical Reviewers xi

Chapter 1: The Evolution of Analytics 1

Cloud Analytics Evolution 2

Introduction to Microsoft Fabric 5

Analytics SaaSification 6

Anatomy of Microsoft Fabric 9

Synapse Data Engineering 10

Synapse Data Science 11

Synapse Data Warehouse 11

Data Factory 12

Synapse Real-Time Analytics 12

Data Activator 13

Power BI 13

Summary 13

Further Reading 14

Chapter 2: Microsoft Fabric: The SaaSification of Analytics 17

Microsoft Fabric Tenants, Capacities, and Workspace 18

Microsoft Fabric Trial 25

Power BI Premium per Capacity License (P SKU) 26

Microsoft Fabric Capacity: Azure Portal (F SKU) 27

How to Provision Microsoft Fabric 28
Provision Fabric Using Power BI Premium per Capacity 29
Provisioning Fabric Using the Azure Platform 34
Differentiators of SaaS Analytics 39
OneLake: OneDrive for the Organization 39
Multicloud Virtualization (AWS S3 Support) 42
Built-in Data Mesh 45
Summary 47
Further Reading 48

Chapter 3: OneLake and Lakehouses for Data Engineers 49

The Lakehouse Concept 51
Fabric Lakehouse: An Optimized Delta Lake 52
Creating a Fabric Lakehouse 54
Lakehouse Data Engineering 60
Data Ingestion and Pipeline 61
Lakehouse Explorer 70
Data Preparation and Transformation Using a Notebook 72
Defining a Spark Job 78
Monitoring a Spark Job 80
SQL Endpoint of a Lakehouse 87
Data Visualization 90
Summary 92
Further Reading 93

Chapter 4: Microsoft Fabric for Data Scientists 95

Fabric Data Science Overview 98
Exploring, Ingesting, and Preparing Data 100

Ingesting Data 100
Exploring the Data 103
Preparing the Data Using a Data Wrangling Tool 104
Exploratory Data Analysis (EDA) and Data Visualization 108
Preparing the Data with Notebook Code 110
Data Science with VS Code 112
Developing the Model 116
Setting Up Experiment Tracking and Registering the Model 118
Large Language Modeling and Azure OpenAI Integration 124
Summary 127
Further Reading 128

Chapter 5: Data Warehousing in Microsoft Fabric 131

Fabric Data Warehouse Introduction 133
Provisioning a Data Warehouse 134
Ingesting Data into a Data Warehouse 136
Data Warehouse Development 141
Visual Interface Transformation 142
Cross-Database and Virtual Datawarehouse Queries 146
SQL Query 147
SQL Server Management Studio Connectivity 148
Advanced DW Capabilities 152
Workload Management 154
Automated Multitiered Coaching and Query Optimization 154
Fabric DW Transaction Support 156
Data Models with Fabric DWs 157
Integration with Power BI and Microsoft Office Tools 159

Monitoring a Fabric DW Pipeline 162
Summary 164
Further Reading 165

Chapter 6: Data Integration for Office Users 167

Data Flow Gen2 170
Fabric Copilot Experience 189
Data Pipeline 192
Data Factory Mount 204
Summary 207
Further Reading 208

Chapter 7: Real-Time Analytics with Microsoft Fabric 209

Fabric Real-Time Analytics 211
Create a Kusto Database 213
Ingesting Data into the Kusto Database 216
KQL Database Query 219
Kusto Query Development 223
Python Plugin 228
Kusto API and SDK 230
Data Retention, Caching, and OneLake Integration 231
Using an Event Stream 232
Summary 239
Further Reading 240

Chapter 8: Microsoft Fabric Activator Real-Time Monitoring and Alerts 243

Data Activator Anatomy 246
Connect with Data Sources 250
Power BI as the Data Source 252

Event Stream as a Source....255
Define and Detect Actionable Patterns....260
Trigger an Automated Workflow: Custom Action....262
Summary....263
Further Reading....264

Chapter 9: Power BI in the Microsoft Fabric Workspace....265

Power BI Fundamentals....267
Power BI Desktop and Power BI Report Builder....268
Power BI Service....283
Power BI Mobile Apps....290
Power BI Key New Features (Fabric)....294
Report Auto-Create....294
Quick Insights....297
Lineage....298
Paginated Report....299
Getting Insights....302
Power BI Direct Lake Mode....302
Report Sharing....306
Datamarts....309
Fabric Power BI Copilot....313
Summary....316
Further Reading....317

Chapter 10: Microsoft Fabric: Inside and Out....319

Fabric Security and Governance....320
Fabric Tenant Security....323
Fabric Workspace Security....325

Fabric Item Security....327
Fabric Computation Security....330
Storage Encryption....331
Fabric Conditional Access....331
Fabric and Purview: Data Governance....335
Fabric Pricing and Licensing....340
Summary....348
Further Reading....350

Index....351

About the Author



Debananda Ghosh is a data and AI specialist based in Singapore and has been working in the data and AI field for past two decades. His expertise includes data warehouses, database administration, data engineering, big data and AI, data architecture, and related cloud analytics. Deb has worked with customers in multiple industries such as finance, manufacturing, utilities, telecom, retail, e-commerce, and aviation.

He currently works with Microsoft cloud analytics global team and helps enteprise customers achieve their digital transformation journey using advanced analytics and AI. Prior to Microsoft, he worked in the Rolls-Royce Singapore data lab developing aviation analytics products in the cloud and working with big data and AI. He has a degree from Jadavpur University, Kolkata, and a post-graduate degree in data science and business analytics from the McCombs School of Business at the University of Texas, Austin, also currently pursuing Chief Data, Analytics and AI Officer Programme by NUS School of Computing. He is prominent author, speaker and blogger in Cloud analytics and AI field. He plays pivotal role in Microsoft Fabric ASIA community and contributor of Microsoft cloud analytics documentation.

About the Technical Reviewers



Kasam Shaikh is a prominent figure in India's artificial intelligence landscape, holding the distinction of being one of India's first four Microsoft Most Valuable Professionals (MVPs) in AI. Currently serving as a senior architect at Capgemini, Kasam boasts an impressive track record as an author, having written five best-selling books dedicated to Azure and AI technologies. Beyond his writing endeavors, Kasam is recognized as a Microsoft Certified Trainer (MCT) and influential tech YouTuber (@mekasamshaikh). He also leads the largest online Azure AI community, known as DearAzure | Azure INDIA, and is a globally renowned AI speaker. His commitment to knowledge sharing extends to Microsoft Learn, where he plays a pivotal role.

Within the realm of AI, Kasam is a respected subject-matter expert in generative AI for the cloud. He actively promotes the adoption of no-code and Azure OpenAI solutions and possesses a strong foundation in hybrid and cross-cloud practices. Kasam's versatility and expertise make him an invaluable asset in the rapidly evolving landscape of technology, contributing significantly to the advancement of Azure and AI.

Kasam was recently recognized as a LinkedIn Top Voice in AI, making him the sole Indian professional to be acknowledged by both Microsoft and LinkedIn for his contributions to the world of artificial intelligence.

In summary, Kasam Shaikh is a multifaceted professional.



Samarendra, a seasoned data analytics and AI engineer at Microsoft, has dedicated his career to leveraging the potential of the Microsoft data platform. With a wealth of experience, he excels in crafting innovative solutions that harness the power of data. He is committed to exploring and implementing the latest advancements in data technologies and contributing to the ongoing transformation of the industry.

CHAPTER 1

The Evolution of Analytics

At the May 2023 Microsoft Build event, Microsoft introduced the public preview of Fabric, an end-to-end unified analytics system based on a software-as-a-service (SaaS) architecture. Microsoft Fabric provides high-performing business intelligence, machine learning, big data, and artificial intelligence capabilities at scale. Microsoft Fabric compiles data engineering, data science, business intelligence, data integration, real-time analytics, data exploration, event-based actions, and alerts into one unified and simplified SaaS foundation.

This book covers the end-to-end technical capabilities of Microsoft Fabrics in depth. By the end of this book, we expect you to have a decent understanding of Microsoft Fabric. You will be proficient enough to start or manage a production analytics workload using Microsoft Fabric.

In this first chapter, we focus on the high-level concepts of Microsoft Fabric. To make it more interesting, you will learn how the analytics field has evolved over the last few decades and where Microsoft Fabric fits in. Specifically, we will cover the following topics in this chapter:

- The evolution of cloud analytics
- What Microsoft Fabric is
- The SaaSification of analytics
- The anatomy of Microsoft Fabric

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By the end of this chapter, you will understand the basics of Microsoft Fabric. Then we will move on to in-depth coverage of each Fabric capability in subsequent chapters.

Cloud Analytics Evolution

Today's world is exploding with data. Most organizations today are on a journey of data-first modernization. Organizations want their data to be more accessible to both business users and technical users. Every large enterprise wants to democratize its data and AI capabilities for its business users.

On the consumer side, there are billions of users who interact with data every day. When we watch Netflix at home, wear a smartwatch, drive the latest car model, book a taxi, get food online, travel on a flight, or even use social media, we generate tons of data through smart applications. As of 2023, about 2.5 quintillion (that's 18 zeros!) bytes of data are generated every day, primarily through social media, gaming, and video.

This data unlocks a lot of business value. Any organization harnessing the data and gaining business insights has a higher potential for profits. However, each organization needs to unify its data and simplify the data silos and data integrations, which is a huge challenge. The tasks of managing the data and analyzing it require advanced data management capabilities.

Data management capabilities have evolved in several ways. In the late 1980s, data warehouses performed cost-effective transformations for decision-making purposes. For example, on-premises data warehouse appliances had massively parallel processing (MPP) architectures to analyze data.

When Internet usage surged in the late 1990s, the data volume, velocity, variety, and veracity also changed. To solve such big data problems, organizations needed more advanced and scalable capabilities

for cost-effective data management processes. On-premises Hadoop distributors such as Hortonworks, Cloudera, and MAPR offered big data and analytics frameworks and tools to solve such problems.

As data continued to explode, analytics frameworks needed even higher performance and scalability. At the same time, the industry needed to be agile enough to adopt new business use cases during their digital transformation journey. Thus, the industry started to move to the cloud platform for its scalability, pay-per-usage model, low IT overhead, and lower total cost of ownership. Adopting cloud analytics went hand in hand with migrating to a cloud infrastructure. Initially cloud analytics were based on an infrastructure-as-a-service (IaaS) foundation.

Figure 1-1 provides a quick overview of the main types of cloud infrastructure—IaaS, platform as a service (PaaS), and software as a service (SaaS)—as well as on-premises capabilities. The highlighted tasks in Figure 1-1 are managed by customers, and the rest are managed by cloud providers. For example, for PaaS, the operating system (OS) is managed by the cloud provider, and the application is managed by the organization.

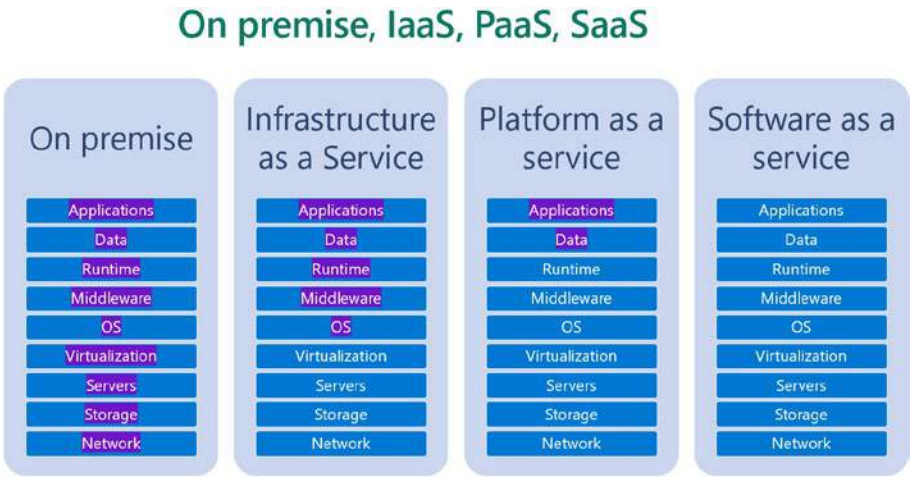


Figure 1-1. *On-premises, IaaS, PaaS, and SaaS differences*

The cloud analytics IaaS journey starts by installing open-source tools such as Spark, Python, etc., for users on multiple scalable virtual machines on the cloud. To simplify this experience, preconfigured cloud data science virtual machines (for example, the Azure Data Science Virtual Machine) became available. Analytics partners such as Hortonworks (Cloudbreak) and Cloudera Data Platform (CDP) also started offering preconfigured cloud analytics marketplace solutions in 2014. These solutions were based on multiple scalable virtual machines.

During a similar timeframe (2010 to 2020), cloud analytics capabilities evolved through PaaS foundations. Services like the Azure SQL Data Warehouse (gen1/gen2), Azure HDinsight EMR, and Google BigQuery are examples of such PaaS services. The advantages of these services include no patching of analytics virtual machines, no upgrade of the analytics engine, and no maintenance overhead. These capabilities further evolved into the unified analytics platforms (for example, Azure Synapse analytics in 2020), which enhance developer productivity and involve various options, including low-code, no-code, and code-centric experiences approaches.

SaaS based Microsoft Fabric release (released in 2023) is the latest tool in the cloud analytics space during the course of the book writing. The Microsoft Fabric workspace further simplifies the unified analytics experience and empowers users to achieve more in the data and AI field. Figure 1-2 depicts the cloud analytics evolution and related timeline.

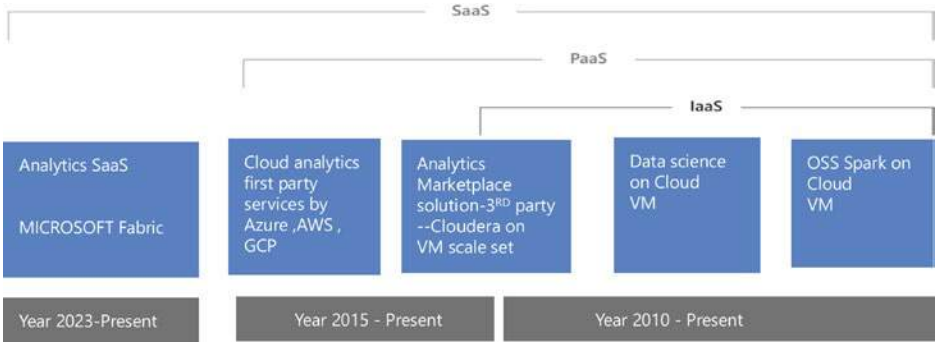


Figure 1-2. *Cloud analytics evolution from IaaS to SaaS*

Let's focus now on Microsoft Fabric in the following section. You will then learn more about the SaaSification benefits of Microsoft Fabric cloud analytics in subsequent sections.

Introduction to Microsoft Fabric

Microsoft Fabric evolved from Microsoft Azure Synapse analytics, Azure Data Factory, and Power BI services to SaaS architecture. Microsoft Fabric empowers every user to innovate and act faster on data insights. The Fabric workspace offers a secure process to discover, analyze, and manage the data from diverse sources, from centralized ones to governed sources. Collaboration capabilities inside the workspace provide a "single pane of glass" data view for enterprises.

In addition, conversational language integration with ChatGPT, Copilot integration (an upcoming feature), the ability to manage powerful AI models, one-click visualizations from datasets, code-free data wrangling, and many more capabilities are available in this SaaS foundation. Fabric also helps organizations to optimize their expense management and provides fast deployment capabilities for greenfield scenarios.

Organizations that are already using Azure Synapse analytics will continue to enjoy the existing capabilities. Additionally, they will be able to adopt the newest data and AI capabilities using the Fabric workspace. For Power BI users, Fabric brings additional new data analytics capabilities to the same Power BI interface. As part of its SaaS foundation, Microsoft Fabric offers a simplified and unified lakehouse concept known as OneLake. (We will discuss the OneLake capability more in Chapter 2.) Figure 1-3 depicts the Azure Data Factory, Synapse analytics, and Power BI components of the unified Microsoft Fabric platform.



Figure 1-3. *Microsoft Fabric foundation*

In the next section, we will discuss the advantages of being a cloud analytics SaaS service compared to a cloud analytics PaaS offering.

Analytics SaaSification

The PaaS cloud analytics capabilities are simpler to adopt for the majority of enterprises. Products like Azure Synapse analytics have brought the following capabilities to developers and data analysts:

- A simplified user experience
- Unification of data engineering, data science, data pipeline, data exploration, and real-time analytics in a single workspace
- Increased developer productivity via code-free mechanisms
- Code-free data pipeline design and big data transformation capabilities
- Scalability using auto-scale, scale-up, and scale-out features
- Easier collaboration

- A DataOps mechanism through CI/CD
- Integration with Power BI
- Integration with a data governance tool

As next step journey towards cloud analytics SaaS product, Microsoft Fabric provides the following capabilities:

- **Intuitive experience:** Microsoft Fabric brings an Office 365–like user experience to its workspace, simplifying the development experience for its users.
- **OneLake-One drive for Organization:** OneLake capability brings a Windows Explorer/OneDrive–like experience to file exploration. The OneLake shortcut capability creates multicloud virtualization with one click. This experience is quite similar to the Windows desktop's Create Shortcut command.
- **Built-in data mesh:** Microsoft Fabric allows developers to associate and deploy its components (artifacts) based on a customized domain. Thus, it natively helps developers create a data mesh framework.
- **Visual query editor:** The Fabric lakehouse comes with a visual editor that helps to develop queries in a completely code-free manner.
- **Dataflow Gen2:** Fabric incorporates a Dataflow Gen2 capability for extraction, transformation, and loading (ETL) development purposes, which creates an easy development experience like with Office tools. It inherits a Power Query–based data orchestration experience. (Learn more about Power Query in the "Further Reading" section.)

- **Code-free data wrangling:** The Fabric notebook provides data wrangling capabilities through a visual editor. It enhances productivity for data scientists, who can then focus on core experimentation work.
- **World-class lakehouse:** Users can create a unified Fabric lakehouse with one click. The Fabric lakehouse introduces Spark auto-tuning, fast Spark pool activation, and higher concurrency support as part of its advanced Spark offering.
- **Auto-visualization:** With one click users can generate insights from data inside a Microsoft Fabrics dataset.
- **Simplified licensing:** Purchasing Fabric is easier through the Azure and Power BI portals and comes with simplified line items like storage cost and computation cost. A built-in capacity metrics application provides simpler visibility for cost management.
- **Code-free real-time alert configuration:** The workspace has a Data Activator interface that helps users configure alerts based on any event in a user-friendly way.
- **Machine learning user interface:** A user-friendly interface for machine learning model tracking and experimentation capability is available in the Microsoft Fabric Data Science interface.
- **Virtual data warehouse:** The Fabric workspace provides the ability to query across virtual data warehouses.

- **Fabric data insights inside the workspace:** The workspace provides a dedicated hub with a Microsoft Purview tab for Fabric data demographics and insights.
- **Built-in data governance:** Microsoft Fabric provides built-in data estate insights and integration with a Microsoft Purview–based data governance capability. Thus, users can natively leverage data asset insights across the organization within the Fabric portal.

We will revisit these capabilities in detail in Chapter 2. Let's now focus on the key pillars of a Microsoft Fabric workspace.

Anatomy of Microsoft Fabric

Fabric covers the entire spectrum of analytics capabilities including business intelligence, data engineering, and data science within the same workspace. Using this workspace, we do not need to stitch together different data when generating analytics. Figure 1-4 shows the Microsoft Fabric landing page, which consists of seven verticals.

To sign up and evaluate Microsoft Fabric, you can refer to https://msit.fabric.microsoft.com/home.

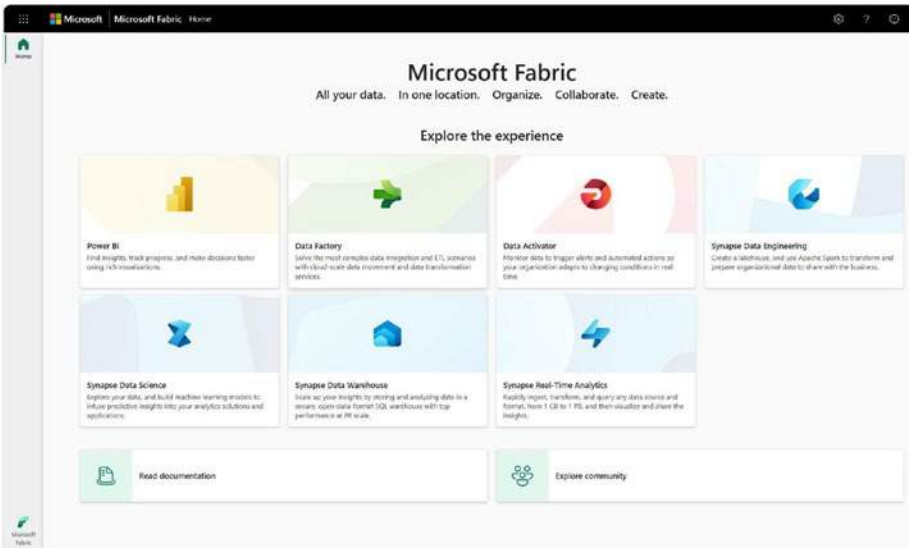


Figure 1-4. *Microsoft Fabric workspace landing page*

The seven sections of the Fabric workspace are as follows:

- Synapse Data Engineering
- Synapse Data Science
- Synapse Data Warehouse
- Data Factory
- Synapse Real-Time Analytics
- Data Activator
- Power BI

Synapse Data Engineering

Microsoft Fabric provides premium data engineering capabilities including world-class Spark, an advanced lakehouse, intuitive notebooks for writing Spark and SQL code, a visual query editor, and a code-free data pipeline.

The lakehouse helps organizations easily manage and analyze huge data volumes in a seamless way. The lakehouse provides capabilities such as easy ingestion, automatic table discovery, the ability to auto-register a fully managed file into a table experience, and a SQL endpoint for users on such delta files. The lakehouse is based on serverless computation capabilities that allow SQL, the Kusto Query Language (KQL), and Spark to run on such computations.

The core of the lakehouse table format is Delta Lake, which is an optimized storage layer for storing data. Delta Lake is open-source and provides atomicity/consistency/isolation/durability (ACID) support that is compatible with the Spark API. In Chapter 3, you will learn more about the data engineering capabilities in the Fabric workspace.

Synapse Data Science

Microsoft Fabric provides end-to-end data science capabilities for data scientists. The workspace comes with an enhanced UI experience. Users can now use both a notebook and a user interface–driven mechanism for machine learning mode development and tracking capabilities. The notebook also has UI-based data wrangling capabilities, which can cleanse data, generate code, and increase a data scientist's productivity. The workspace provides multilanguage support for Python, Spark, and the R language. Users can natively integrate Azure Open AI cognitive services inside the notebook. In Chapter 4, you will learn more about the data science experience in the Fabric workspace.

Synapse Data Warehouse

Microsoft Fabric offers a lake-centric data warehouse capability for enterprise-scale data management using T-SQL. The Microsoft Fabric data warehouse capability comes with cross-database querying and thus a virtual warehouse. The Fabric data warehouse totally decouples storage and computation.

Note that Fabric provides two distinct SQL flavors: a SQL endpoint with the lakehouse and a warehouse endpoint. While both warehouses are based on T-SQL, the lakehouse SQL endpoint is read-only, and the warehouse endpoint is read-write. In Chapter 5, we will walk through the data warehouse capabilities in detail.

Data Factory

Azure Data Factory is a native Azure PaaS solution that provides big data orchestration mechanisms for developers. Azure Data Factory provides ingestion mechanisms using a user interface; code-free Spark-based transformation tools; and scheduling, pipeline monitoring, alert, and event-based trigger capabilities.

Data Flow (part of the Power platform) is another no-code tool for users to do data ingestion and transformation. Fabric brings the best of Azure Data Factory and Power Query to users. In Chapter 6, we will discuss both the Power Query and Azure Data Factory capabilities.

Synapse Real-Time Analytics

Microsoft Fabric brings Azure Data Explorer capabilities to its workspace for big data streaming workloads. Azure Data Explorer provides full and native integration with the Fabric workspace and brings a SaaS-like experience to users. Users can run a KQL script for streaming workloads. Additionally, an event stream capability introduced in the Fabric workspace provides a no-code experience for adding streaming sources and destinations. Enhanced capabilities such as confidential computing (in memory computation) are available in the Fabric real-time analytics features. In Chapter 7, you will learn about Azure Data Explorer in more detail.

Data Activator

Converting data insights into executable tasks is an important way to make data valuable. Previously, data analysts needed to tweak different configurations in real-time analytics services to generate low-latency alerts. Data Activator helps users monitor data services and generate alerts based on any event. This is a no-code feature within the Microsoft Fabric workspace to connect with data services, detect actions, and trigger actions. In Chapter 8, you will learn about Data Activator in depth.

Power BI

Power BI is a data visualization tool that provides end-to-end business intelligence to end users. This tool helps both business users and technical users aggregate, analyze, and visualize data. Power BI was initially released by Microsoft in 2011, and since then there are various licensing options available based on feature and collaboration needs.

The purpose of this book is not to go through the existing Power BI capabilities but rather to focus on the new capabilities of Fabric that you get in the Power BI workspace. Power BI premium users will continue to enjoy Power BI capabilities as well as the new Fabric capabilities. In Chapter 9, you will learn about existing Power BI capabilities as well as the new features. You can learn even more about Power BI in the "Further Reading" section.

Summary

Let's do a quick recap of this introductory chapter. We discussed how the cloud analytics field has evolved. We introduced Microsoft Fabric and explained how it fits into the cloud analytics journey. You learned about

the main parts of the Microsoft Fabric workspace. In subsequent chapters, we will deep dive into each segment of the Microsoft Fabric workspace and learn about its features in detail.

Further Reading

Learn more about the topics in this chapter at the following locations:

- **Fabric documentation:** https://learn.microsoft.com/en-us/fabric/
- **Fabric cloud SaaSification benefits:** https://www.linkedin.com/pulse/microsoft-fabric-saasification-cloud-analytics-debananda-ghosh/
- **Signing up with Microsoft Fabric:** https://msit.fabric.microsoft.com/home
- **Microsoft Fabric data engineering:** https://learn.microsoft.com/en-us/fabric/data-engineering/data-engineering-overview
- **Power BI documentation:** https://powerbi.microsoft.com/en-au/
- **Power BI Gartner reports:** https://powerbi.microsoft.com/en-us/blog/microsoft-named-a-leader-in-the-2023-gartner-magic-quadrant-for-analytics-and-bi-platforms/
- **Power BI Power query:** https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview

- **Data Activator:** https://blog.fabric.microsoft.com/en-us/blog/driving-actions-from-your-data-with-data-activator/
- **Delta lake table:** https://learn.microsoft.com/en-us/azure/databricks/delta/
- **Big data 2023 fun fact:** https://explodingtopics.com/blog/big-data-stats#top-big-data-stats

CHAPTER 2

Microsoft Fabric: The SaaSification of Analytics

In the previous chapter, you learned about the high-level concepts and features of Microsoft Fabric. You also looked at the SaaS features that differentiate it when compared to PaaS cloud analytics offerings.

In this chapter, we will dive deeply into the SaaS cloud analytics features of Microsoft Fabric, and we will explain why you should use Microsoft Fabric in enterprise-scale analytics ecosystems. You will also learn about the different licensing needs as a prerequisite to provision Microsoft Fabric. You will understand how to deploy Fabric capacities with all the license types inside a Microsoft tenant. You will also discover details about Fabric deployments, Fabric tenants, and the workspace. Specifically, we will cover the following topics:

- Microsoft Fabric tenants, capacities, and workspace
- How to provision Microsoft Fabric
- Differentiators for SaaS analytics

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Microsoft Fabric Tenants, Capacities, and Workspace

In this section, we will cover the Microsoft Fabric tenant, workspace, capacities, and other provisioning-related concepts. In Microsoft cloud offerings, an organization sits at the top of the hierarchy. An organization consists of multiple subscriptions such as Microsoft 365 subscriptions, Dynamics 365 subscriptions, and Azure subscriptions. Cloud offerings come with a license-per-user model or consumption model. Figure 2-1 illustrates this subscription/organization/license concept. The Microsoft Power platform provides code-free and low-code tooling. The Power platform also has user-based and capacity-based licenses.

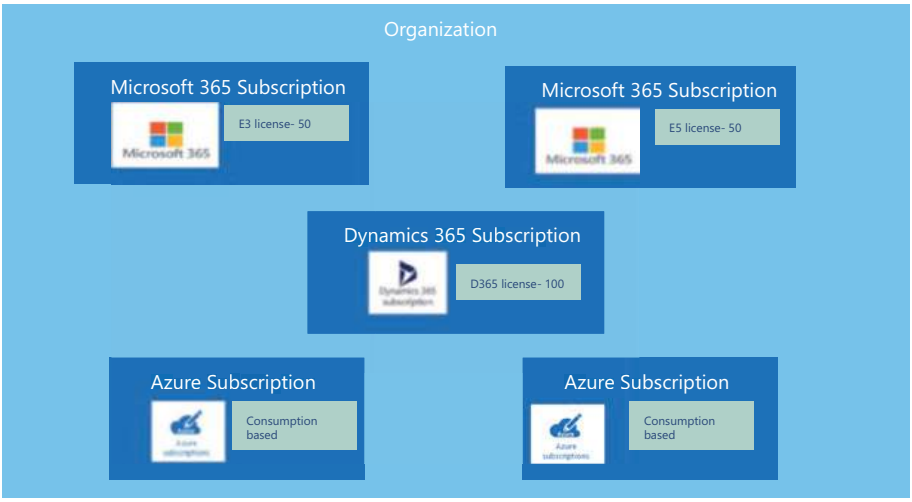


Figure 2-1. *Microsoft organization/subscription/license concept*

A Microsoft tenant is tied to a domain name. It is a centralized consolidation of your assets that includes the Microsoft subscription, license, users, and domains, and it represents a logical mapping of your organization.

For SaaS offerings like the Power platform, Dynamics 365, and Microsoft 365, tenants house the servers in a regional location. Platform as a service (PaaS) and infrastructure as a service (IaaS) like Azure do not have region and tenancy hard linking; the services can be deployed in any Azure data center in the world. An organization can have one tenant or multiple tenants. Multiple tenants are mostly based on business and technical situations like conglomerates and M&A boundaries. Multitenancy needs a strategy prior to deployment to find the right architecture.

Tenants are also the foundation of Microsoft Fabric. You can enable Microsoft Fabric under your tenant using the Microsoft Power platform admin settings, as shown in Figure 2-2.

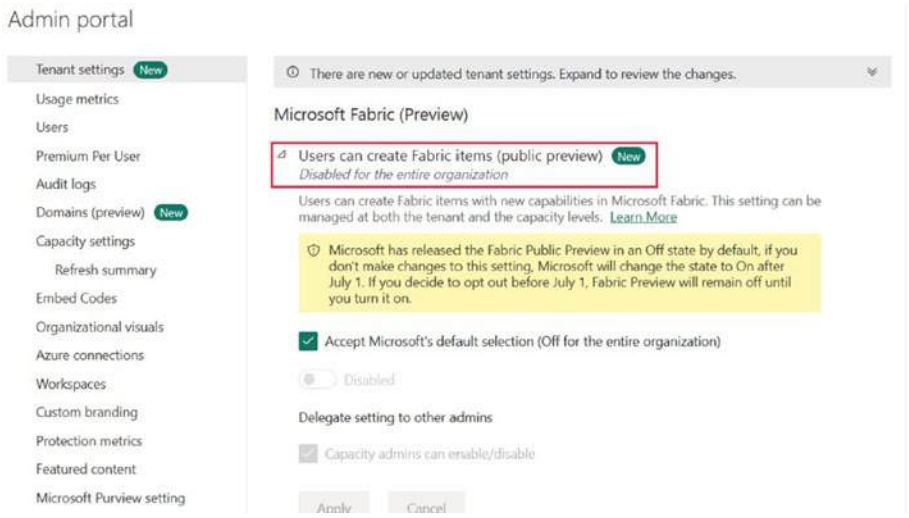


Figure 2-2. *Enabling Microsoft Fabric*

One tenant can host multiple Fabric capacities. A Fabric *capacity* sits under the tenants and holds dedicated resources and computation units for Fabric. A single capacity contains multiple workspaces. The workspace concept is more of a logical grouping of Fabric items. The Fabric hierarchy is as follows:

- Tenant
- Fabric capacity
- Fabric workspace
- Fabric items

Figure 2-3 shows this hierarchy.

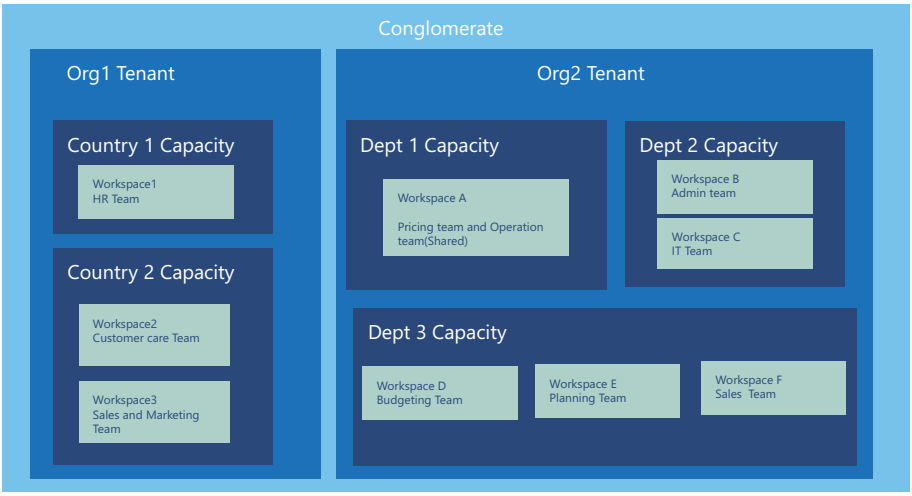


Figure 2-3. *Tenant/capacity/Fabric workspace concept*

The Fabric workspace is designed for collaboration. To create a new workspace in Fabric, take the following steps:

1. Go `https://msit.fabric.microsoft.com/home`.
2. Click Power BI on the left.

3. Click Workspaces on the left.
4. Click New Workspace at the bottom.
5. Fill in the details of the "Create a workspace" form, as shown in Figure 2-4.

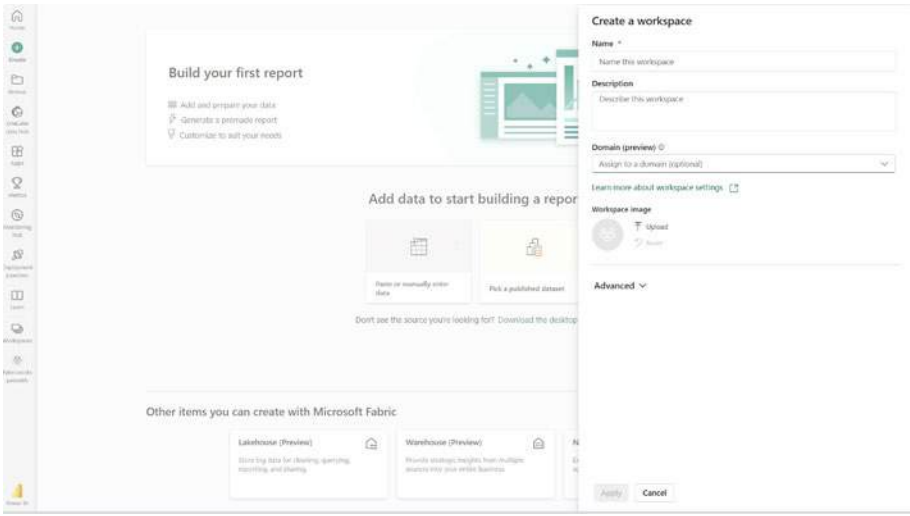


Figure 2-4. *Creating a workspace*

Each workspace consists of multiple Fabric items. Fabric items are components of the workspace that provide all the capabilities such as the data warehouse, data lakehouse, data pipelines, real-time data analytics, data visualization tools, and more. You can search for pre-created Fabric items in a Fabric workspace using the Filter drop-down on the home page, as shown in Figure 2-5. This helps you quickly find specific Fabric items. The user interface (UI) also provides recently accessed items. You can also mark Fabric items as Favorites so you can find them easily.

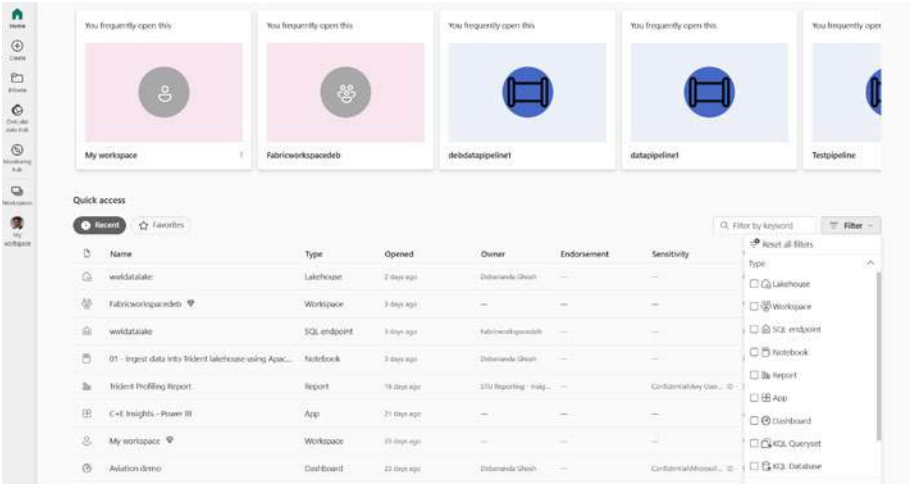


Figure 2-5. *Finding Fabric items in the workspace*

The following are all the Fabric items (non-Power BI) at the time of this writing:

- Data Pipeline
- Data Flow Gen2
- Data Factory Mount
- Environment
- KQL Database
- KQL Queryset
- Lakehouse
- Spark job definition
- Notebook
- Reflex

- Report
- Streaming semantic model
- Streaming data flow
- Datamart
- Data flow
- Dashboard
- Scorecard
- Paginated report
- ML Model
- ExperimentNotebook
- Report
- App
- Dashboard
- KQL Query set
- KQL database
- EventStream
- Event house
- Real-time dashboard
- Datawarehouse
- Mirroring Snowflake
- Mirroring Azure SQL DB
- Mirroring Cosmos DB

Figure 2-6 shows all the icons for these Fabric items.



Figure 2-6. *Fabric item icons*

You have now understood the Fabric workspace and it's hierarchy. You will now learn about the licenses and dedicated capacity and platform you need to leverage to provision Microsoft Fabric capacities. The following are the ways that the Microsoft technology landscape provides Fabric capacities:

- Microsoft Fabric trial
- Power BI Premium per capacity (P SKU)
- Microsoft Fabric capacity through the Azure portal (F SKU)

Microsoft Fabric comes with a dedicated set of resources known as *capacity units* (CUs). A CU can be provisioned by both the Microsoft Azure platform and a Power BI license. To enable Microsoft Fabric via Power BI licensing, you need a Microsoft Power BI Premium capacity license

(P series SKU). Alternately to enable Microsoft Fabric (F series SKU) on the Azure platform, you can use the Azure portal. Developers can both create and share Fabric content using both of these SKUs.

Note that the following Power BI licenses do *not* provide Fabric capacity at this time. However, these licenses do provide non-Fabric / Power BI items.

- Power BI free sign-up (but not on Fabric trial)
- Power BI Premium per user
- Power BI Pro per user license/Office 365 E5 license
- Azure Power BI Embedded capacity (A/EM SKU)

Now we will explain three ways to provision Fabric capacity in the following three sections.

Microsoft Fabric Trial

The Microsoft Fabric trial is available for free for 60 days for all users. For users who have Power BI Premium per user, Power BI Pro, or even Power BI, the free trial can leverage this temporary trial license. Go to `https://app.fabric.microsoft.com` to sign up for a free Power BI license. Once you access the Power BI portal, you need to click the Start Trial button. Your portal will be upgraded to the free Microsoft Fabric trial. Note that the Fabric trial license will be available for 60 days and is appropriate only when you are test-driving the product. Figure 2-7 shows a sample trial license.

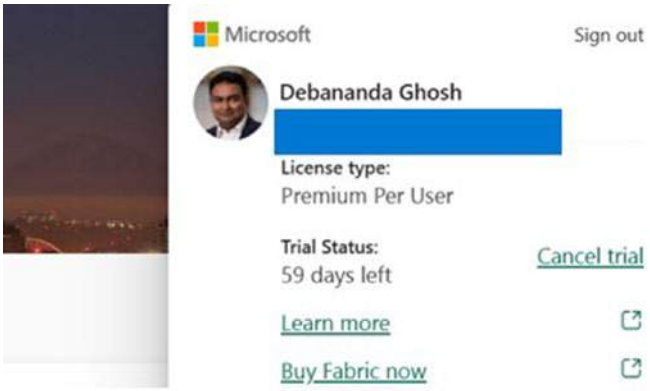


Figure 2-7. *Activating a Fabric trial license*

A trial license is good for a test-drive of the Fabric product. For proof of concept, development and production workloads, you need to consider the licenses described in the following sections.

Power BI Premium per Capacity License (P SKU)

The Power BI Premium series provides extensive and comprehensive features within the Power BI portfolio. Microsoft Fabric can be provisioned using a Microsoft Power BI Premium per capacity license. The Power BI Premium per capacity license is available via the stock keeping units (SKUs) known as the P series. The Power BI Premium P series SKUs provide the ability for users to create and connect Fabric items. The Power BI Premium capacity can be enabled via the Microsoft 365 admin center, as shown in Figure 2-8.

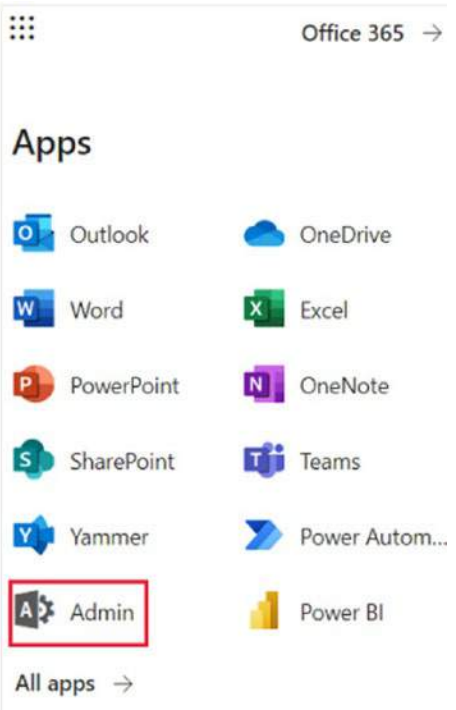


Figure 2-8. *Enabling Power BI premium capacity*

You can learn about the Power BI Premium pricing at `https://powerbi.microsoft.com/en-sg/pricing/`. In the next section, we will focus on the capacity that is available via the Power BI Premium capacity on the Azure platform.

Microsoft Fabric Capacity: Azure Portal (F SKU)

Microsoft Fabric can also be provisioned using the Power BI Premium capacity via the Azure platform. You need to go to Power BI Premium capacity's Azure portal at `https://ms.portal.azure.com/#home` to provision this capacity. You will go through the steps of provisioning in the

"Provision Fabric Using the Azure platform." This provisioning comes with the F series SKU. Table 2-1 shows the Fabric P and F SKU details and how they translate to the Fabric CUs.

Table 2-1. *Fabric F SKU and PSKU Offerings*

Power BI Premium per Capacity (P Series)	Azure SKU (F Series)	Fabric CU
	F2	2
	F4	4
	F18	8
	F16	16
	F32	32
P1	F64	64
P2	F128	128
P3	F256	256
P4	F512	512
P5	F1024	1024
	F2048	2048

You will learn more about the F SKU and P SKU pricing in Chapter 10. In the next section, we will discuss how to provision Microsoft Fabric using the previously mentioned licenses/platforms.

How to Provision Microsoft Fabric

In this section, you will learn how to enable or deploy Microsoft Fabric within your organization tenant. We will walk you through both Power BI Premium per capacity enablement and Azure Microsoft Fabric SKU provisioning.

Provision Fabric Using Power BI Premium per Capacity

Organizations that want to provision Microsoft Fabric via the Power BI Premium capacity need to have the Power BI Premium per capacity license. You can refer to https://powerbi.microsoft.com/en-us/power-bi-premium/ for more information. Those who already have a Power BI Premium capacity (P1 or above license) do not need any new allow-listing or procurement mechanism.

As a configuration prerequisite, you need to ensure the Power BI or Fabric workspace is assigned to the Premium-dedicated capacity. The workspace can be allocated to Power BI Premium from the Power BI admin portal. Once that workspace is within the Premium capacity and the user has access to the workspace, the user can continue to leverage the Fabric capabilities.

In the following section, you will learn how to assign a workspace within the Premium-dedicated capacity. This can be done both with the Fabric admin portal and with the workspace settings.

Enabling Power BI Premium Capacity via the Admin Portal

In this section, you will learn how to do enable Fabric using the Power BI Premium capacity via the admin portal. Fabric admins need to go to Settings and then to the admin portal, as shown in Figure 2-9.

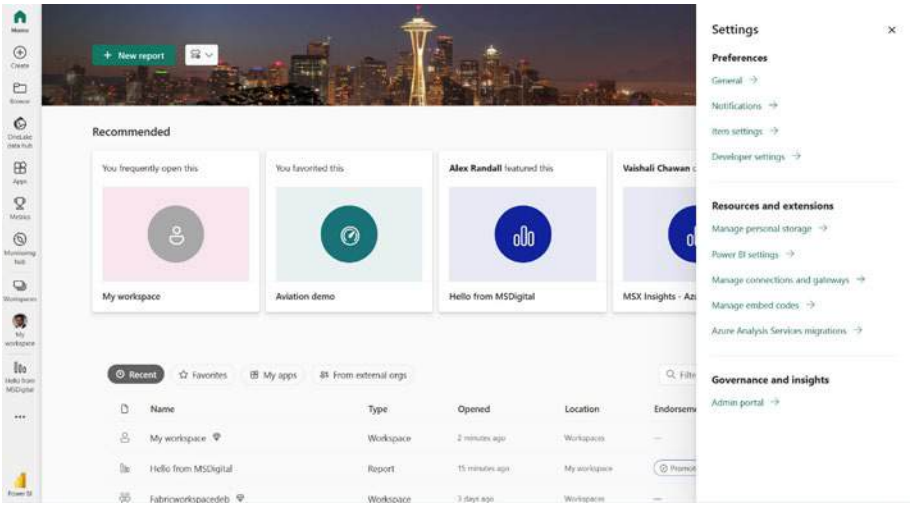


Figure 2-9. *Power BI portal settings*

Within the "Capacity management" section, you can assign workspaces and apply the assignment for the Premium-dedicated capacity. Choose "Specific workspaces" and then click "Assign workspaces" to assign capacity to the specific workspace. Figure 2-10 shows the workspace assignment feature inside the Fabric portal.



Figure 2-10. *Power BI workspace assignment to the Premium capacity*

Enabling Power BI Premium Capacity via the Workspace

The Power BI Premium capacity can be enabled via the workspace directly. Follow these steps to enable Fabric using the Power BI Premium capacity via the workspace settings:

1. Go to the workspace and click … (the three dots).
2. Select the "Workspace settings" feature, as shown in Figure 2-11.

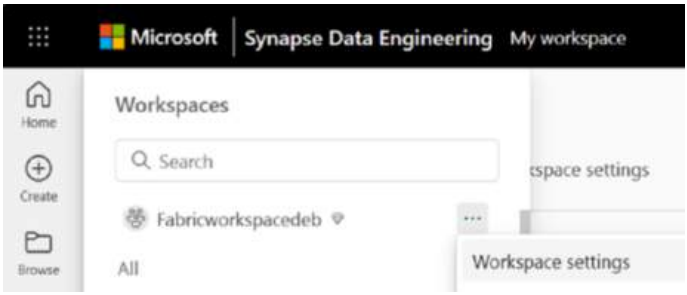


Figure 2-11. *Power BI workspace assignment to Premium capacity*

Once you open the workspace settings, you can assign the Premium capacity under Premium tab. In next Figure 2-12, you can see the workspace is assigned under the trial (Fabric Trial) license. This is applicable when you enable Fabric via the Fabric trial license. In this case, you need to choose "Premium capacity" from the "License mode" categories for the Premium capacity license (P series). Note that for the Azure platform SKU (F series), you need to select "Fabric capacity" from this list. Note that Pro, Premium per user, Embedded does not provide capabilities to create Fabric items at this time.

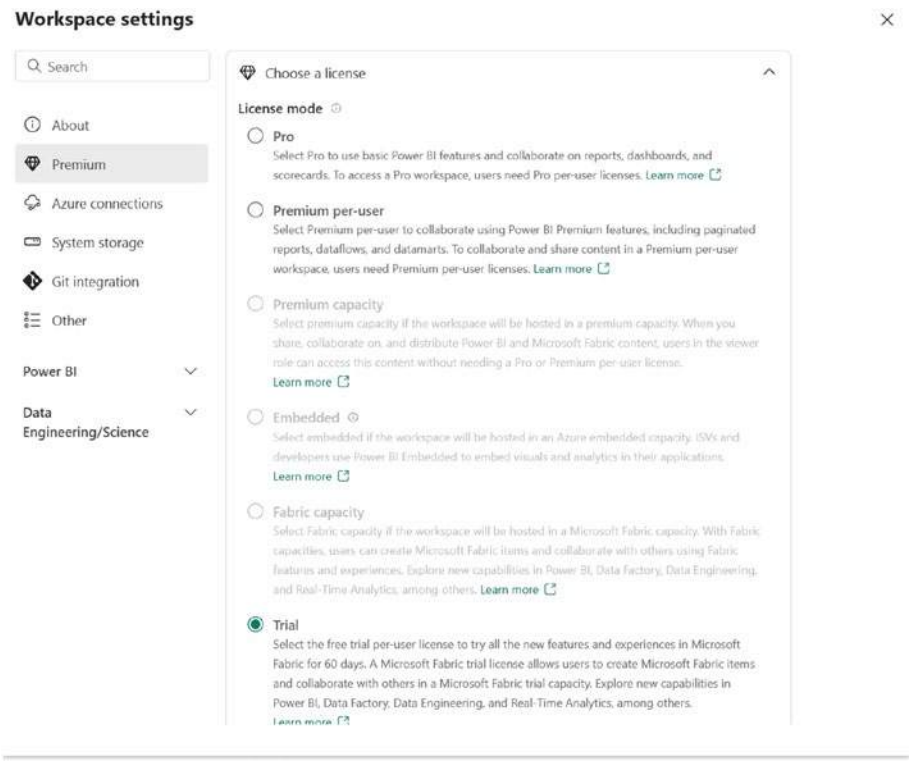


Figure 2-12. *Choose "Premium capacity."*

Subsequent users who have access to the workspace can reuse their existing Power BI URL (`https://powerbi.microsoft.com/en-us/`) to access the Fabric capabilities. On the left of the landing page, when you click the Power BI logo, you can see the Fabric capabilities, as shown in Figure 2-13.

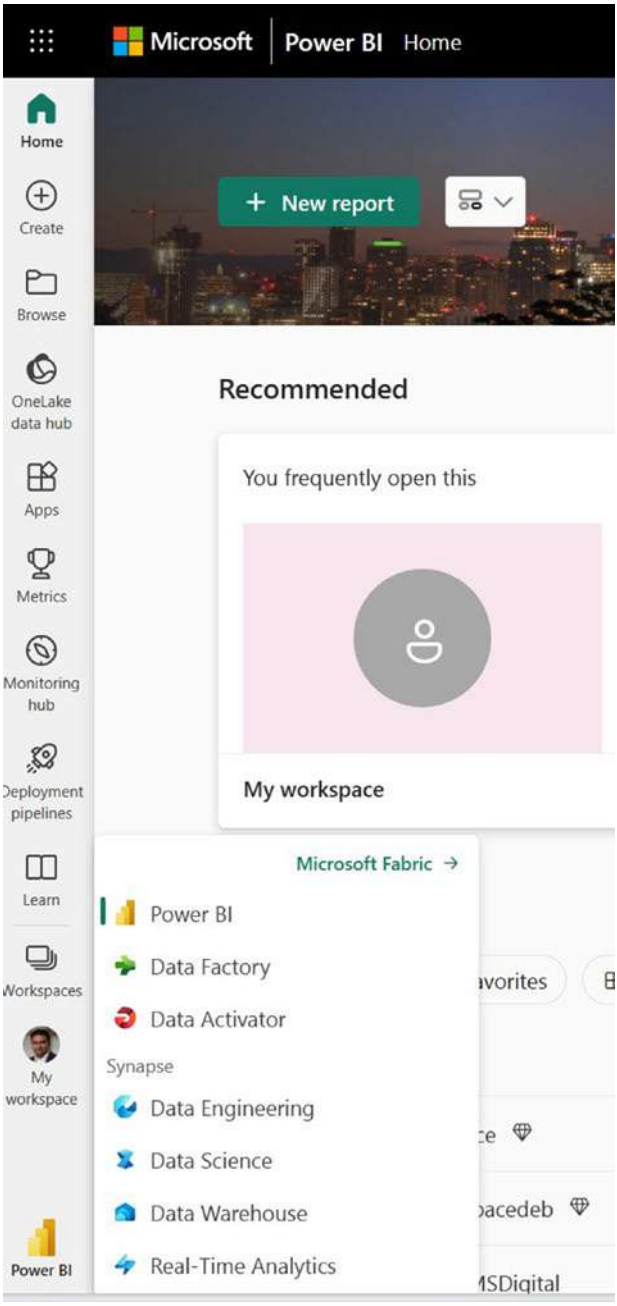


Figure 2-13. *Power BI landing page*

Users can also access this page directly at https://msit.fabric.microsoft.com/home, as shown in Figure 2-14.

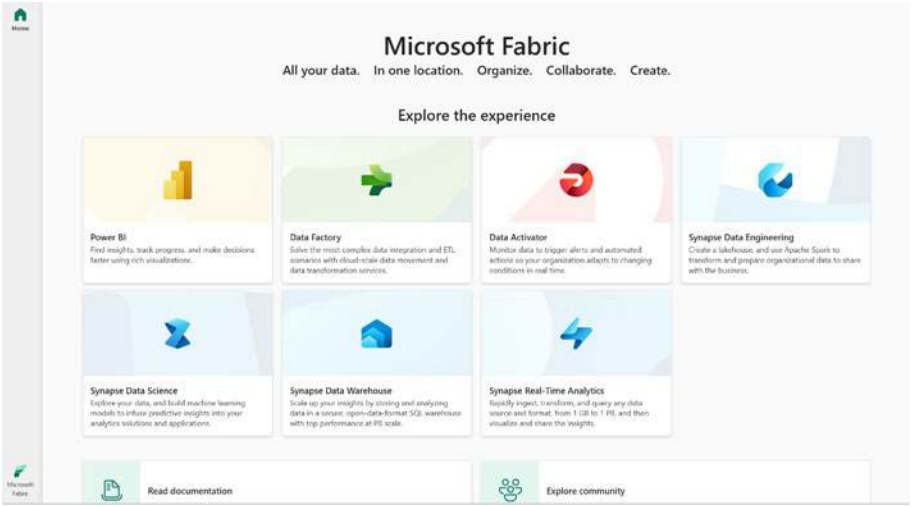


Figure 2-14. *Fabric workspace landing page*

You will now learn how to provision Microsoft Fabric in the Azure platform (F series SKU).

Provisioning Fabric Using the Azure Platform

To provision Fabric via the Fabric capacity (F series SKU), you need to log in to the Azure portal at https://ms.portal.azure.com/#home. The Azure portal is a single place that manages the Azure platform resources. To provision Fabric, you need to sign up in the Azure portal using your domain ID. In the search bar of the Azure portal, search for *Fabric* and then select Microsoft Fabric from the drop-down, as shown in Figure 2-15.

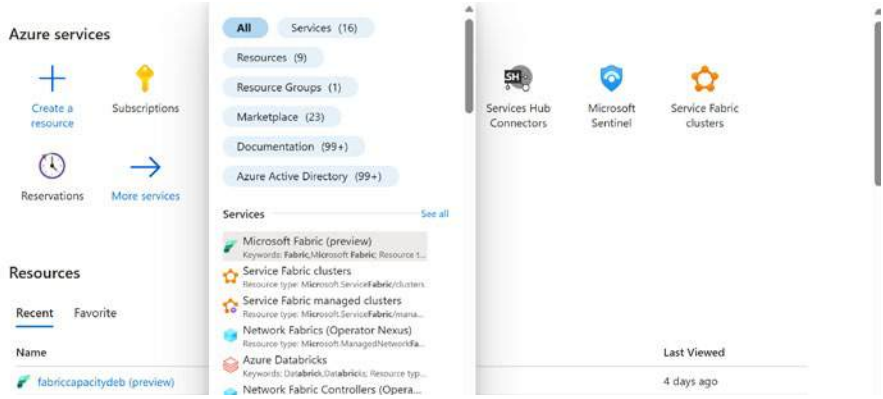


Figure 2-15. *Fabric service in the Azure portal*

Now perform the following steps:

1. Select the Fabric service and click the Create button. The portal provides a simple UI to create a Fabric capacity, as shown in Figure 2-16.

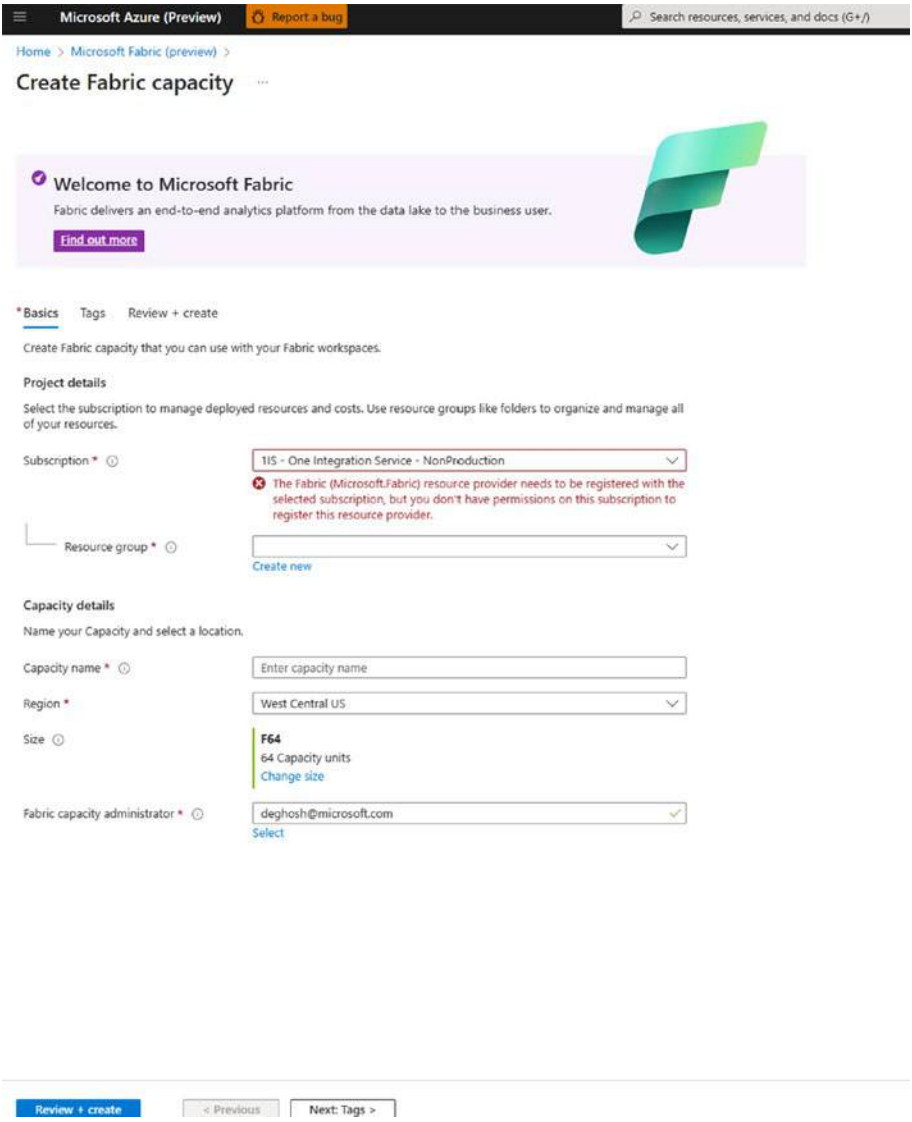


Figure 2-16. *Creating a capacity*

2. Select the desired subscription from the drop-down and provide a capacity. Then choose the region and size, as shown in Figure 2-16.
3. Assign a Fabric capacity administrator for admin purposes.
4. Once you fill up relevant information, you can go to Review + Create and create a new Fabric capacity.

You have now created a demo of a Fabric capacity, as shown in Figure 2-17.

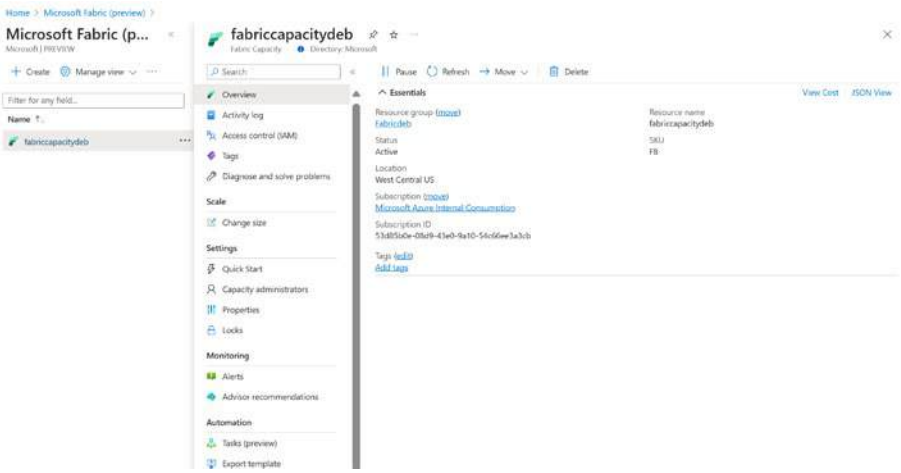


Figure 2-17. *Complete Fabric capacity*

Follow these steps for workspace allocation:

1. Go to `https://msit.fabric.microsoft.com/home`.
2. On the left side, click Workspace.
3. Select the workspace from the workspace list.
4. In the selected workspace, click ... (three dots).

5. Within the workspace settings, select Premium, as shown in Figure 2-18.
6. Select "License mode" as the Fabric capacity, as shown in Figure 2-18.
7. Within the license capacity, at the bottom select the right license capacity from the drop-down. In this case, it's fabriccapacitydeb, as shown in Figure 2-18.
8. For a larger dataset (more than 10 GB), you can choose the "Large dataset" option.

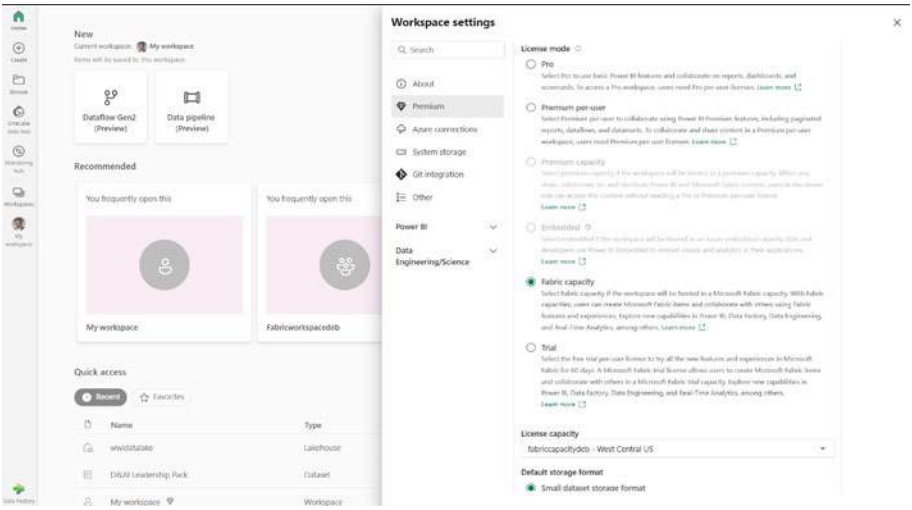


Figure 2-18. *Workspace allocation to Fabric capacity (F series SKU)*

Note that users who have workspace access can now start using this workspace. For workspace access management, you can leverage the workspace access feature (you need to click three dots, …), as shown in Figure 2-19.

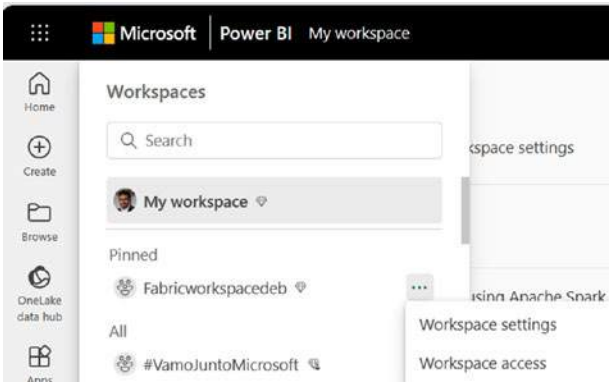


Figure 2-19. *Workspace access management*

To recap, you have learned about the different licensing needs as prerequisites to provision Fabric. Also, you have learned how to provision Fabric through various licenses and platforms. In the next section, we will explain why you should use Microsoft Fabric for an enterprise-scale analytics ecosystem.

Differentiators of SaaS Analytics

In this section, we will do a deep dive into the SaaSification of analytics. We will go into more detail about a few SaaS analytics capabilities. These concepts will be used going forward in the subsequent chapters as well for further discussion of each of the Fabric workspace pillars.

OneLake: OneDrive for the Organization

A lakehouse is the foundation of the Microsoft Fabric workspace. The Microsoft Fabric lakehouse is also known as OneLake. Previously, organizations built multiple lake houses for their business needs, and managing multiple resources required extra effort. Fabric OneLake is a

unified storage option for an organization. This feature is built on top of the Azure Data Lake Gen2 (ADLS Gen2) cloud service. The Fabric data stays inside the boundary of tenants as part of the cloud SaaS model. Figure 2-20 depicts decoupled computation, storage, and the surrounding capabilities available inside Fabric.



Figure 2-20. *OneLake, the security and serverless computation concept from Microsoft*

Fabric provides a Microsoft OneDrive–like, intuitive user experience for lakehouse developers and business users via the OneLake explorer. Lakehouse files automatically sync when a user's Windows device is online. You can download the OneLake explorer thick client from `https://www.microsoft.com/en-us/download/details.aspx?id=105222`. Figure 2-21 shows how you can explore lakehouse files using the Windows Explorer–like experience.

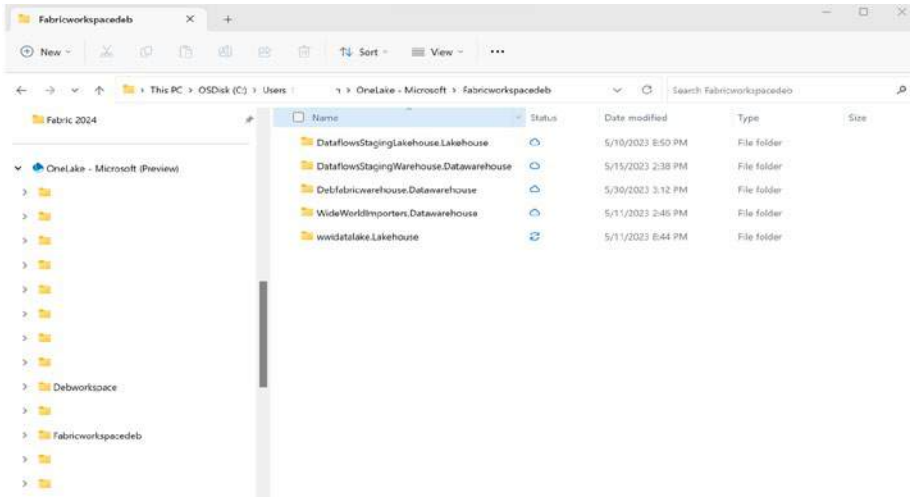


Figure 2-21. *OneLake explorer experience*

While this experience is desktop-based and offers dataset exploration capabilities, the OneLake data hub inside the Fabric workspace provides a filter mechanism to search all Fabric items. To browse the data hub, click the OneLake data hub inside the Fabric workspace, as shown in Figure 2-22.

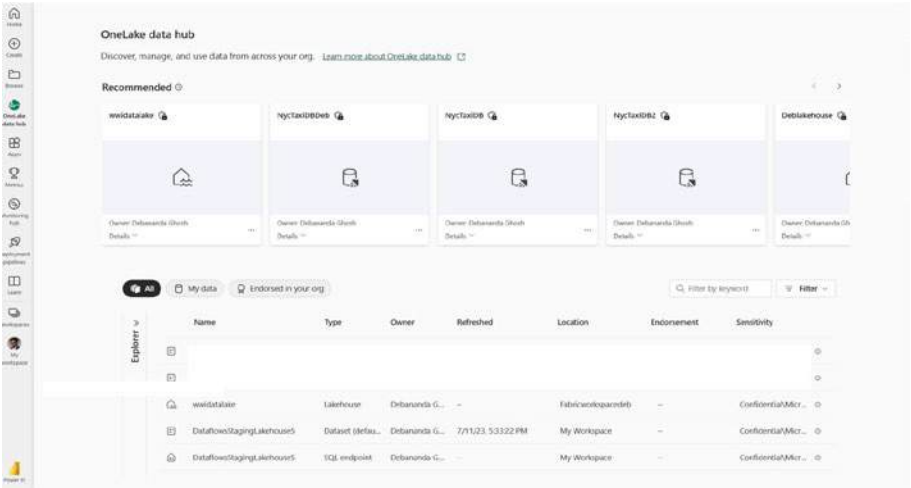


Figure 2-22. *Accessing the data hub*

In the next chapter, you will see that all the serverless computations (SQL, KQL, Spark, Analysis Service) are being built using OneLake as the foundational data lake. Since data and computation are completely decoupled, other services that have native integration with OneLake/Azure Data Lake Gen2 can access seamlessly the files created via serverless computations. Multiple analytics engines within the Fabric workspace and outside Fabric can leverage OneLake. Thus, it provides the openness shown in Figure 2-23.

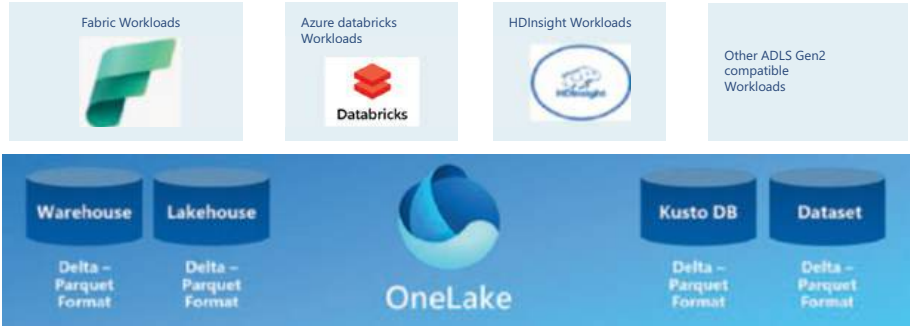


Figure 2-23. *OneLake openness*

Multicloud Virtualization (AWS S3 Support)

Most organizations want to create a multicloud data strategy. Hence, we will continue to see multiple lakehouses/fragmented analytics ecosystems across multiple clouds. To unify such multicloud analytic ecosystems, either you need one more virtualization layer or you need another data pipeline to bring data from one cloud to another cloud storage. For example, if data is in AWS S3 and you need to bring it into ADLS Gen2, it needs a lot of physical data movement between these two cloud storage accounts. Microsoft Fabric OneLake provides shortcut features that create a logical link with AWS S3. Data consumers in the Fabric workspace can browse the AWS S3 data within the workspace like an internal Fabric dataset. This avoids any data movement between the two cloud storage accounts.

To leverage Microsoft Fabric's new shortcut features, perform the following steps:

1. Go to https://msit.fabric.microsoft.com/home.
2. On the left side, click Workspace.
3. Select the workspace from the workspace list, in this case workspacedeb.
4. On the right side you can use Filter and select Lakehouse, as shown in Figure 2-24.

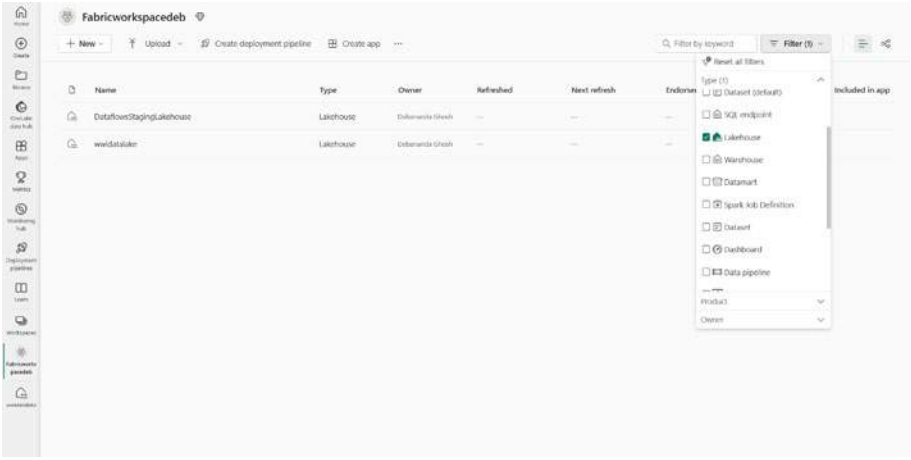


Figure 2-24. *OneLake shortcut feature*

5. Select the lakehouse you need to explore, in this case wwidatalake.
6. Click the three dots (…) beside Files.

7. Click the "New shortcut" feature, as shown in Figure 2-25.

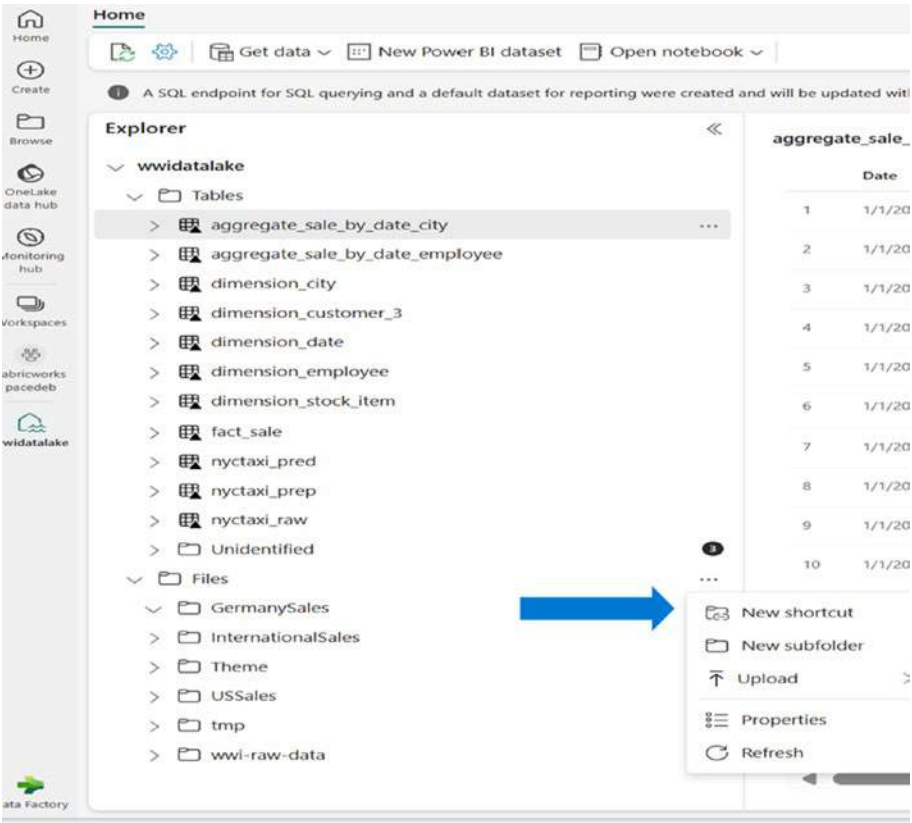


Figure 2-25. *Selecting the "New shortcut" feature*

8. You can now see the supported source system (refer to Figure 2-26).
9. Click AWS S3 and then provide the necessary configuration details to link with AWS S3.

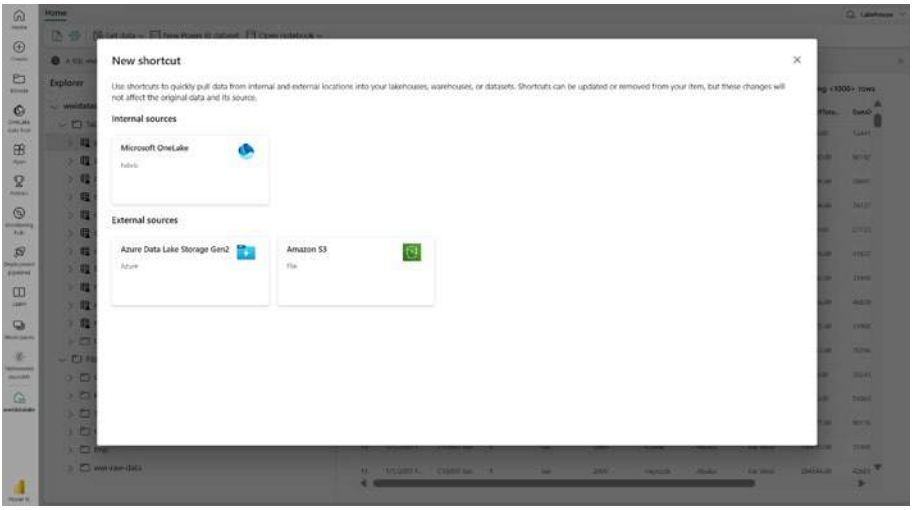


Figure 2-26. *Providing the necessary configuration details*

These features have provided a multiload storage virtualization layer with a few clicks. This is another unique feature that is part of the cloud analytics SaaS foundation.

Built-in Data Mesh

Microsoft Fabric provides data mesh capabilities natively as part of its SaaS offerings. We will explore in built data mesh capabilities in this sub section. However let's take one step back and learn the data mesh framework concept at a high level.

For certain enteprise organizations, their centralized data strategy becomes a bottleneck. This happens because of regulations, data residency, data compliance, data ownership, and few more reasons. Also, having quick answers to data questions is sometimes key for business decisions. Therefore, a centralized data team can also becoming a bottleneck. In such scenarios, shifting the ownership and responsibility

from the centralized data team to a domain team is helpful. Thus, the decentralization of analytics becomes another key foundation of the data mesh capability.

A data mesh framework is based on these four principles:

- Domain ownership
- Data as a product
- Self-service data infrastructure platform
- Federated governance

Microsoft Fabric provides domain assignment capabilities natively. Fabric items can be assigned, organized, and tagged under a specific data domain. Figure 2-27 depicts decentralization analytics features using the Microsoft Fabric platform.

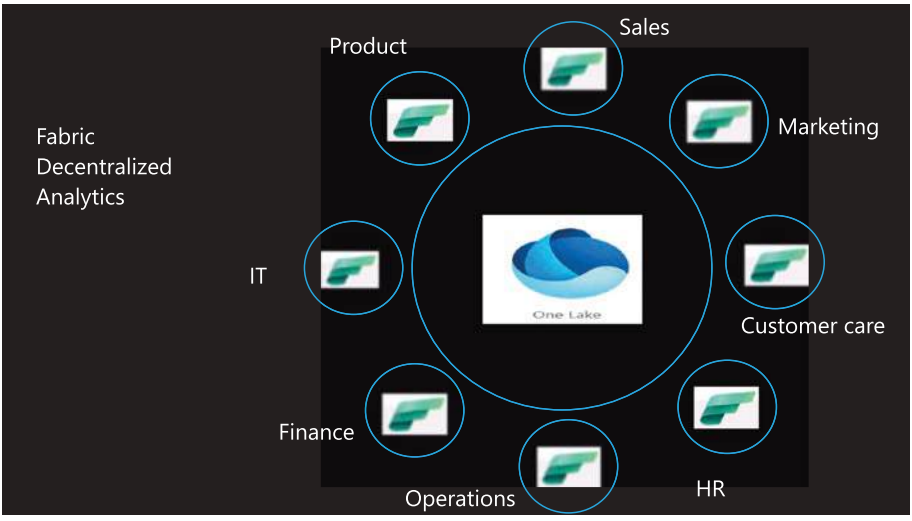


Figure 2-27. *Data mesh features*

To assign and group the Fabric items in a domain, you need to have the Fabric admin use the admin portal. Once the domains are created, the workspaces can be assigned to a domain when they are created.

These are the steps to assign a Fabric workspace with a domain:

1. On the left side of the Fabric workspace, click "Workspaces".
2. Then click "New workspaces." In the "Create a workspace" UI, you will be able to see the Domain feature, as shown in Figure 2-28. Pre-created domains can be selected from the drop-down and assigned a workspace.

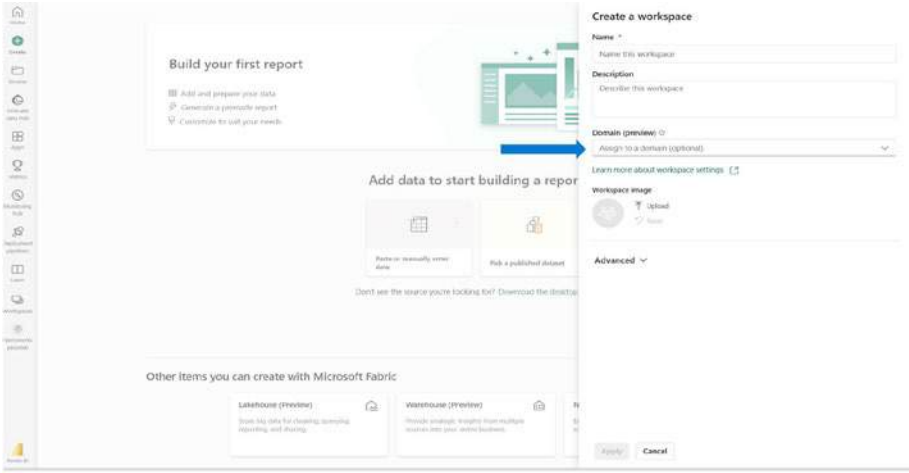


Figure 2-28. *Data mesh with the Fabric lakehouse platform*

Summary

In this chapter, we covered the Fabric infrastructure-related details that are managed by Microsoft. You learned about Fabric tenants and capacities. We also walked you through the Microsoft Fabric workspace and capacity assignment capabilities. You learned a few key features of Microsoft Fabric including OneLake. In the following chapters, you will hear more details about Fabric OneLake and lakehouses.

Further Reading

Learn more about the topics in this chapter at these locations:

- **Microsoft organization hierarchy:** https://learn.microsoft.com/en-us/microsoft-365/enterprise/subscriptions-licenses-accounts-and-tenants-for-microsoft-cloud-offerings?view=o365-worldwide
- **Microsoft tenant strategy:** https://learn.microsoft.com/en-us/dynamics365/guidance/implementation-guide/environment-strategy-tenant-strategy
- **Explorer Power BI plans:** https://powerbi.microsoft.com/en-sg/pricing/
- **Azure portal:** https://azure.microsoft.com/en-us/get-started/azure-portal
- **Data meshes:** https://martinfowler.com/articles/data-mesh-principles.html
- **Microsoft Fabric:** https://learn.microsoft.com/en-us/fabric/get-started/microsoft-fabric-overview

CHAPTER 3

OneLake and Lakehouses for Data Engineers

In this chapter, you will learn about the Fabric data engineering features in depth. You will use the Synapse data engineering features inside the Fabric workspace to explore some sample datasets. You will also learn how to explore some datasets using Spark notebooks in Fabric and Fabric SQL endpoints. We will go through the steps to visualize and monitor query results with a few clicks. Figure 3-1 depicts the Synapse Data Engineering canvas of the Fabric workspace.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_3

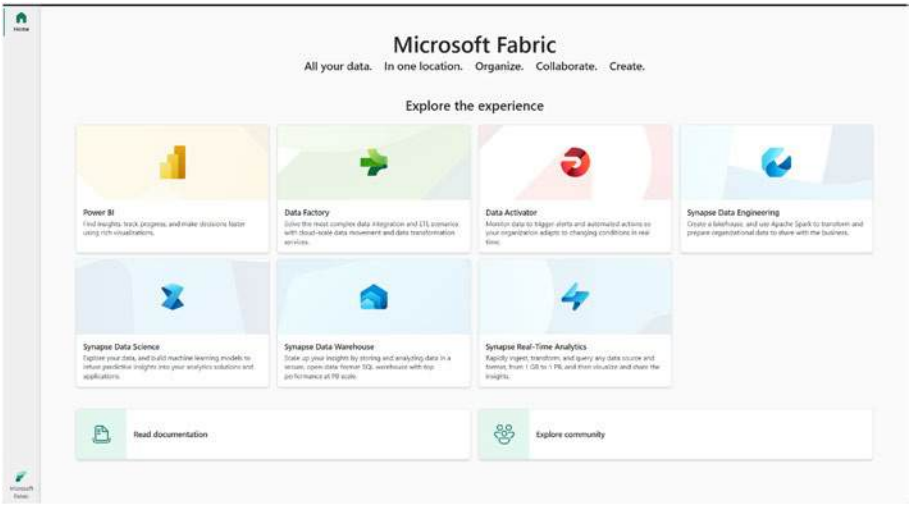


Figure 3-1. *Synapse Data Engineering in the Fabric workspace*

Specifically, you will learn about the following topics:

- Lakehouse concepts
- Lakehouse data engineering development
- SQL endpoints
- Lakehouse security

Note that this book does not aim to go through the Python, Spark, or SQL programming code in detail; rather, it focuses on how to use such languages inside the Fabric workspace. By the end of this chapter, you will have good understanding of the Synapse Data Engineering capability of Fabric; for more information, see `https://msit.fabric.microsoft.com/home`.

Let's first start by introducing lakehouses and looking at their benefits.

The Lakehouse Concept

In Chapter 1, we briefly touched upon the evolution from on-premises to cloud analytics. On-premise data warehouses have a long history of being used for business intelligence workloads. However, data warehouses have performance limitations for high-volume data. For example, they do not support all file formats.

Data lakes evolved as a good alternate for high-volume data processing engines and provided decent cost for the performance for high-volume data workloads. However, data lakes come with drawbacks such as consistency, atomicity, isolation, and durability related issues. A lakehouse is a combination of a data warehouse and a data lake. The lakehouse is constructed on the foundation of a delta table, thereby incorporating the functionalities of ACID and consistency. It brings advantages to both data lakes and data warehouses and eliminates disadvantages. Table 3-1 compares datawarehouses, datalakes, and lakehouses. Microsoft Fabric embraces Lakehouse foundation and we will learn why its an Optimized Lakehouse in next subsection.

Table 3-1. *Datawarehouse, Datalake, Lakehouse Comparison*

	Datawarehouse	Datalake	Lakehouse
File formats	Supports structured data; does not support semi-structured and unstructured data	Supports structured, unstructured, and semi-structured data	Supports structured, unstructured, and semi-structured data
Atomicity, consistency, isolation, durability (ACID) features	ACID compliant using database management system foundation	Not ACID compliant	ACID compliant using its transaction log features
Cost versus performance	Not optimal for large-scale data	Optimal for large-scale data	Optimal for large-scale data
Consumers	Business intelligence	Artificial intelligence	Both business intelligence and artificial intelligence

Fabric Lakehouse: An Optimized Delta Lake

Microsoft Fabric has chosen delta file as its lakehouse format. The following capabilities make it a highly optimized lakehouse:

- **Auto-discovery:** Fabric lakehouse provides the key capability of OneLake storage and One Lake shortcut-based tables discovery mechanism. Thus we do not need to separately define schema when storage is mapped with Fabric workspace.
- **V-ordered Parquet:** In Microsoft Fabric, all the compute engines use a V-ordered Parquet file for fast reads. A V-order file is a writing technique that helps to achieve a significant increase in file readability.

For example, here is a PySpark V-ordered Parquet file:

```
%%pyspark
spark.conf.get('spark.sql.parquet.vorder.enabled')
```

- **Low-shuffle merge optimization:** The delta Spark feature of Microsoft Fabric provides low-shuffle merge optimization, which excludes unmodified rows during shuffling. Thus, the `MERGE` command of the Spark delta table is highly optimized.

  For example, here is how to enable Spark low-shuffle enablement in Spark SQL:

  ```
  %%sql
  SET 'spark.microsoft.delta.merge.lowShuffle.
  enabled' = true
  ```

- **Optimize, VACUUM, Z-order:** The delta Spark feature of Microsoft Fabric leverages delta Spark utilities. Hence, Z-order (colocated information for efficient data skipping) can be used with V-order techniques. Spark performs well when file sizes are large. Spark's optimized write capability in Microsoft Fabric provides an optimal file size and reduces the number of files written, thus providing Spark efficiency. Merging all the changes to a bigger consolidated Parquet file is known as *bin compaction* and is achieved with `OPTIMIZE`. To clear any file that is no longer referenced by the delta table, the `VACUUM` utility is used.

  For example, here is how to do bin compaction:

  ```
  %%sql
  OPTIMIZE bing_covid_19_data VORDER;
  ```

- **Auto-tune:** Fabric Spark automatically tunes the workload and reduces execution time.

  The Spark auto-tune feature optimizes the following Spark configuration based on the baseline machine learning model:

  - `spark.sql.shuffle.partitions`
  - `spark.sql.autoBroadcastJoinThreshold`
  - `spark.sql.files.maxPartitionBytes`

  For example, here is a code snippet in Spark SQL to enable auto-tune:

```
%%sql
SET spark.ms.autotune.queryTuning.enabled=TRUE
```

Overall, such features make Microsoft Fabric a high-performing lakehouse. Since Fabric is based on the SaaS model, deploying a lakehouse is completely code-free. In the following sections, you will learn how to create a lakehouse and load it with sample datasets.

Creating a Fabric Lakehouse

In the following steps, you will first create a workspace and allocate it under the Fabric capacity. Then you will spin up a Fabric lakehouse inside the precreated Microsoft Fabric workspace.

1. Click Synapse Data Engineering.
2. Click the Workspaces tab on the left side to create a new workspace, as shown in Figure 3-2.

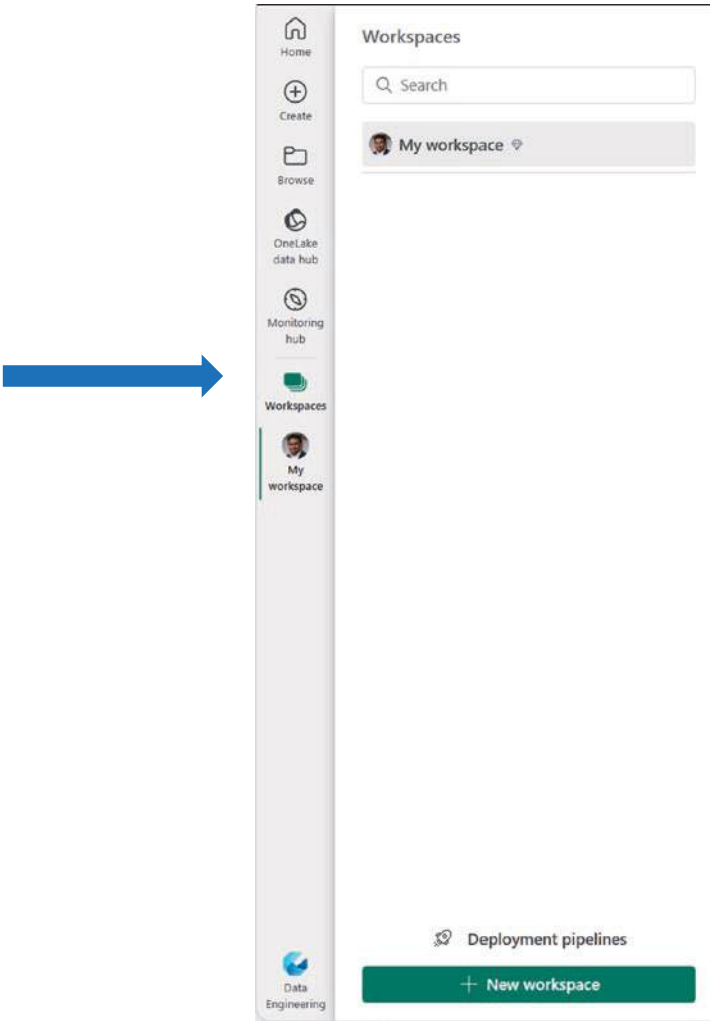


Figure 3-2. *Creating a new workspace inside Fabric*

3. Provide the name of the workspace. In this case, we used Fabricworkspacebook as the workspace name, as shown in Figure 3-3.

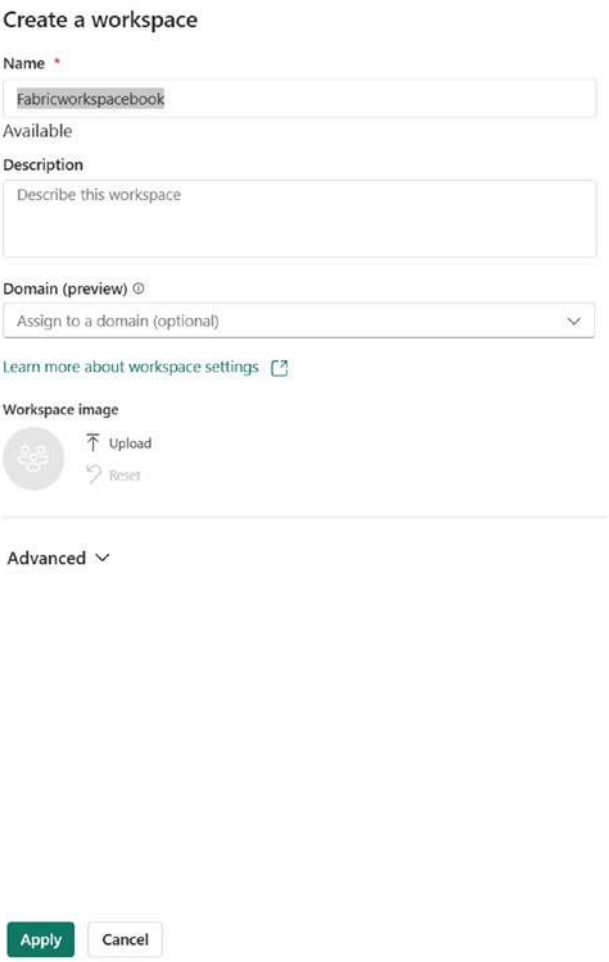


Figure 3-3. *Naming the workspace*

4. Going forward, you need to always select the workspace Fabricworkspacebook to leverage the Fabric capabilities in these steps. To make the workspace selection easier, click the Workspaces tab again on the left side. Then search for the workspace name (Fabricworkspacebook in this case) and click

"Pin to top." Now this workspace is pinned in the Workspaces canvas, and you don't need to search the workspace every time you need it. Figure 3-4 shows the "Pin to top" feature.

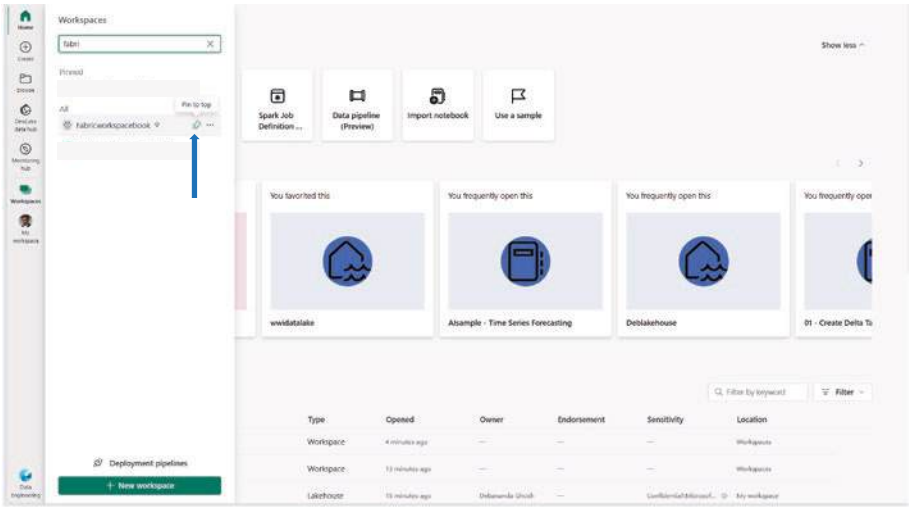


Figure 3-4. *Workspace pinning*

5. Click the + New button and then select the Lakehouse artifact from the drop-down, as shown in Figure 3-5.

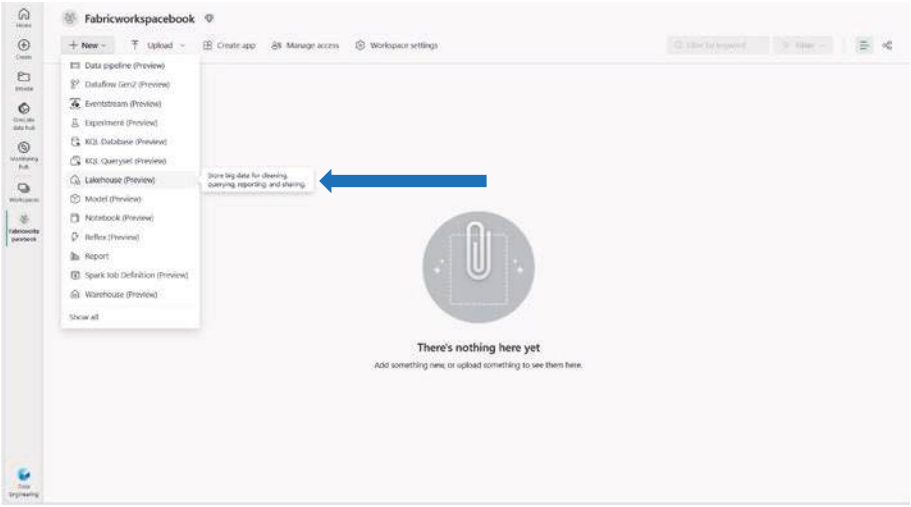


Figure 3-5. *Creating a lakehouse*

6. Provide the lakehouse name (in this case Fabricworkspacebook). In this example, you are creating a Fabric lakehouse called Fabricworkspacebook under the precreated workspace Fabricworkspacebook, as shown in Figure 3-6.

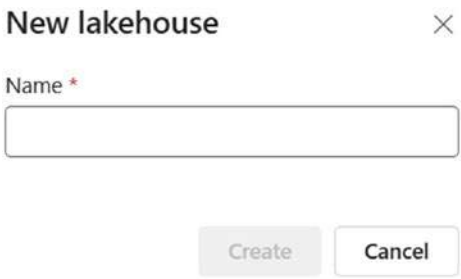


Figure 3-6. *Naming the lakehouse*

7. After you click Create, the new lakehouse will be created, as shown in Figure 3-7.

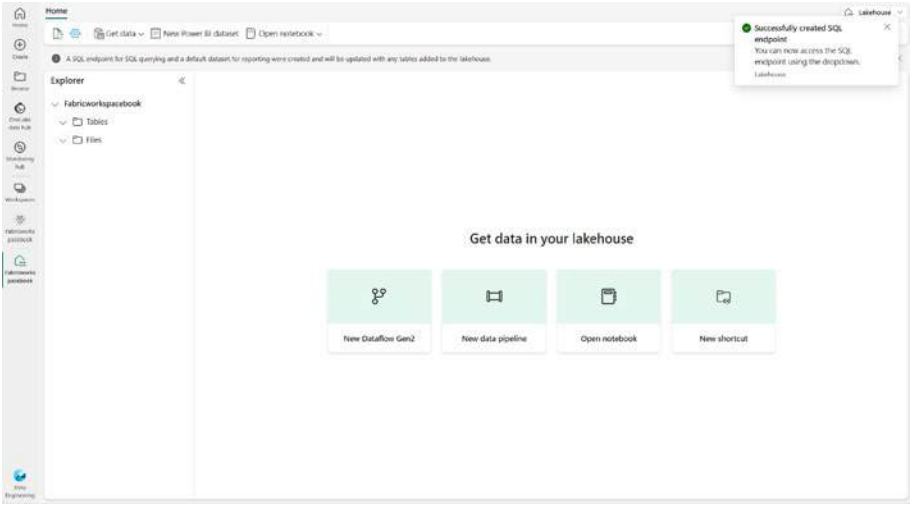


Figure 3-7. *Fabric lakehouse creation finishes*

In the previous steps, you created a lakehouse through one specific canvas. Alternately we can create Lakehouse using '+Create' tab in the left side of the interface, as shown in Figure 3-8.

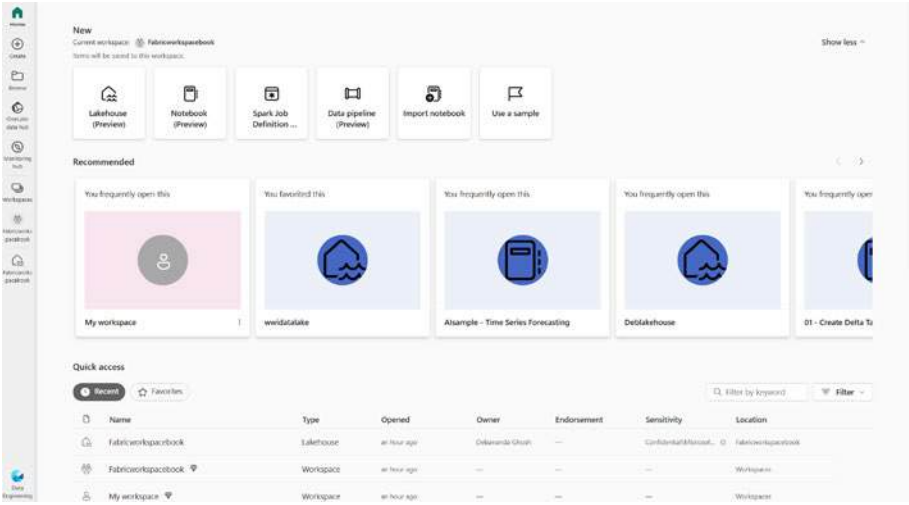


Figure 3-8. *Lakehouse canvas*

To recap, you just learned how to create a Fabric workspace and deploy lakehouse in the same workspace. In the next section, we will focus on the lakehouse data engineering features.

Lakehouse Data Engineering

Let's shift our focus from the infrastructure, configuration, settings, and deployment to the data architecture key components. Figure 3-9 shows Microsoft Fabric-based end-to-end data architecture diagram. We will get into each components of this architecture diagram throughout the book. In this chapter, we will discuss the specific data ingestion, data preparation, and data visualization features available in the Fabric SaaS tool. To start, you will ingest a sample dataset provided by Fabric to explore its data engineering features.

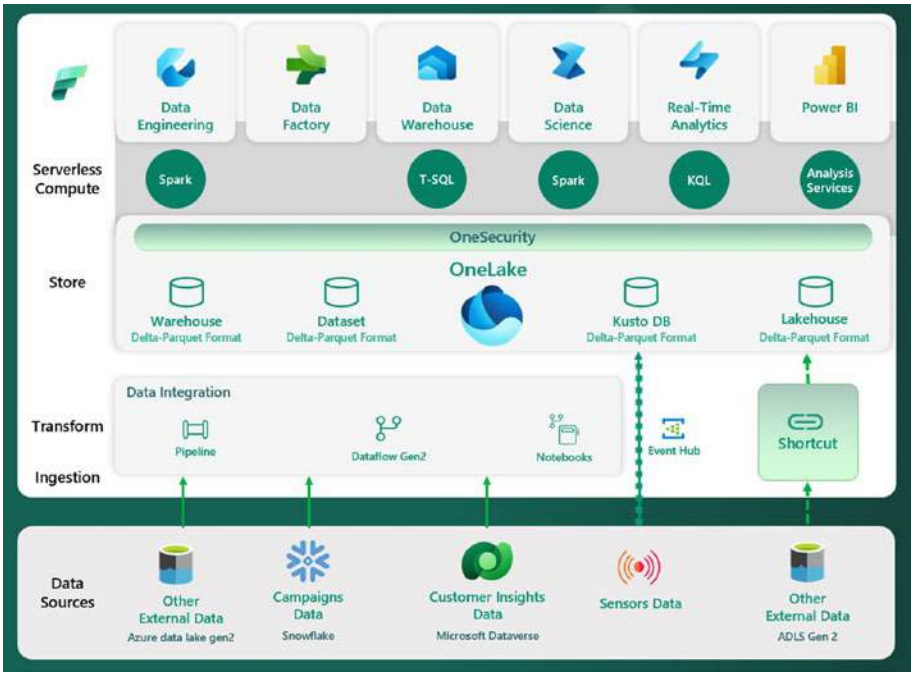


Figure 3-9. *Fabric data engineering architecture*

We will now focus on data ingestion and pipelines in the following section and load sample data inside Fabric newly created Lakehouse.

Data Ingestion and Pipeline

There are different ways to create a data pipeline and perform data ingestion inside a lakehouse. In this section, you will leverage the "Copy data" feature. The Fabric Copy data tool allows users to connect with multiple sources via its canvas. This tool allows users to configure destinations and eventually create a copy task that does data ingestion in a seamless manner from source to destination. And there are other ways to do data ingestion inside the Fabric canvas such as using Data Flow Gen2.

1. Use Power Query/Data Flow Gen2 and upload the file using a dataflow.
2. Open the "Copy data" tool.
3. Upload the files directly from the local file system via the user interface.
4. Use a Fabric notebook to directly access the file path and load the data.
5. Leverage the shortcut mechanism to connect the data remotely.

In the following steps, you will load some COVID-19 data from an example dataset provided by the Fabric workspace via the "Copy data" tool. We will cover a few other ingestion techniques in Chapter 6. Note that the "Copy data" tool steps are similar to the Azure Data Factory native copy tool steps in case you are familiar with Azure Data Factory.

1. Go to the Fabric workspace, click Synapse Data Engineering, and then click "Data pipeline (preview)," as shown in Figure 3-10.

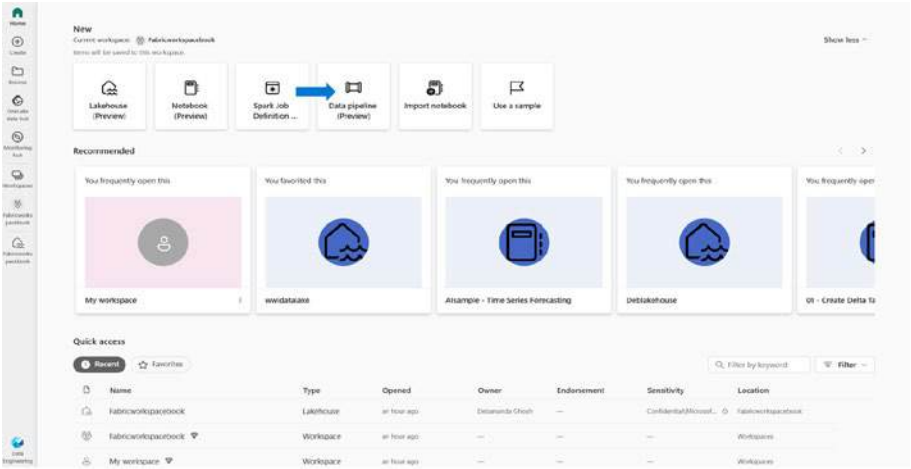


Figure 3-10. *Creating a Fabric data pipeline*

2. Enter a new pipeline name (FabricBookPipeline in this scenario), as shown in Figure 3-11.

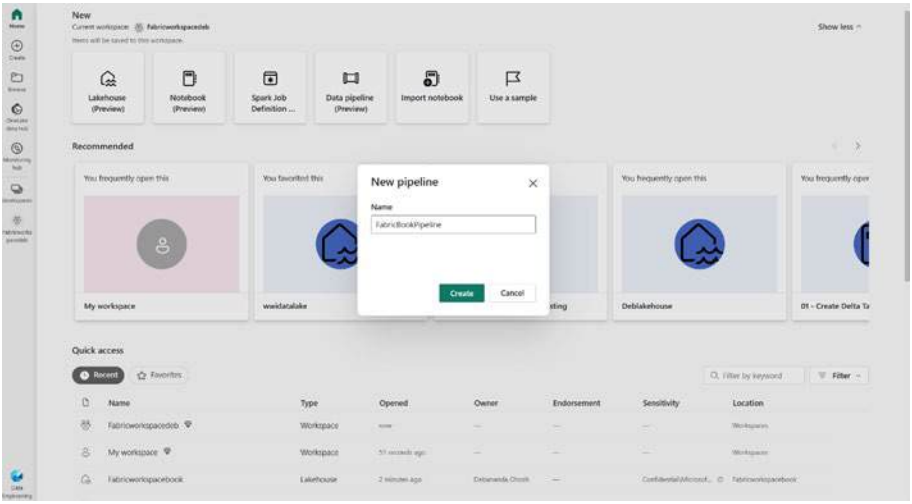


Figure 3-11. *Naming the pipeline*

3. Click Create, as shown in Figure 3-11. Now you will be able to see the "Data pipeline" canvas, as shown in Figure 3-12. You need to choose "Copy data" to create a data pipeline.

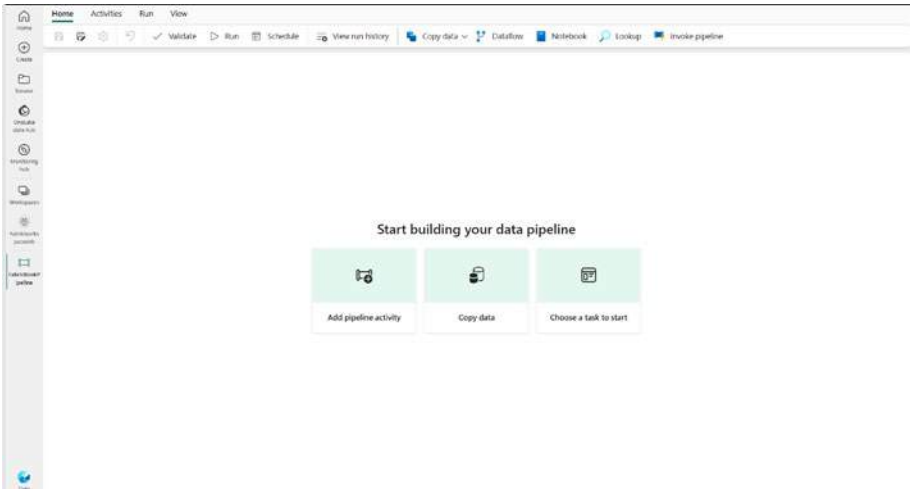


Figure 3-12. *"Data pipeline" canvas*

4. We will use some sample data provided by Fabric for academic purposes. The COVID-19 Data Lake dataset will be loaded in the new lakehouse named Fabricworkspacebook. Figure 3-13 illustrates the sample datasets available in Fabric.

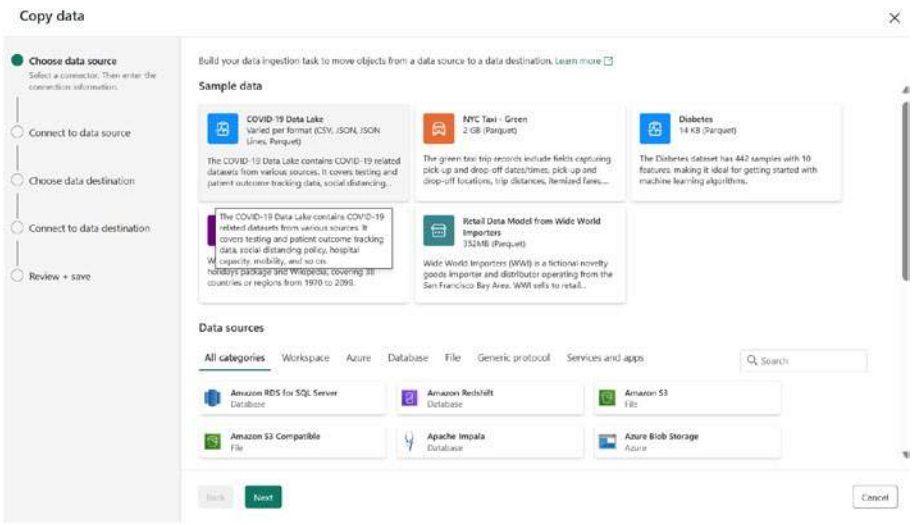


Figure 3-13. *Fabric sample datasets*

5. After you click Next, choose the Bing COVID-19 dataset as the input dataset for the subsequent copy steps as shown in Figure 3-14.

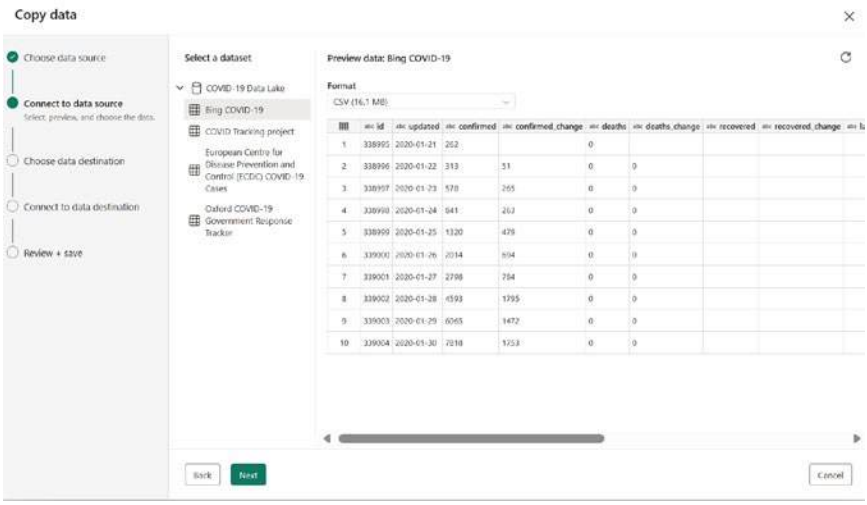


Figure 3-14. *"Choose to data source data destination" canvas*

6. Now you need to select "Choose data destination" to write the COVID-19 data to the Fabric lakehouse storage. Figure 3-15 shows the "Choose data destination" canvas in the "Copy data" tool.

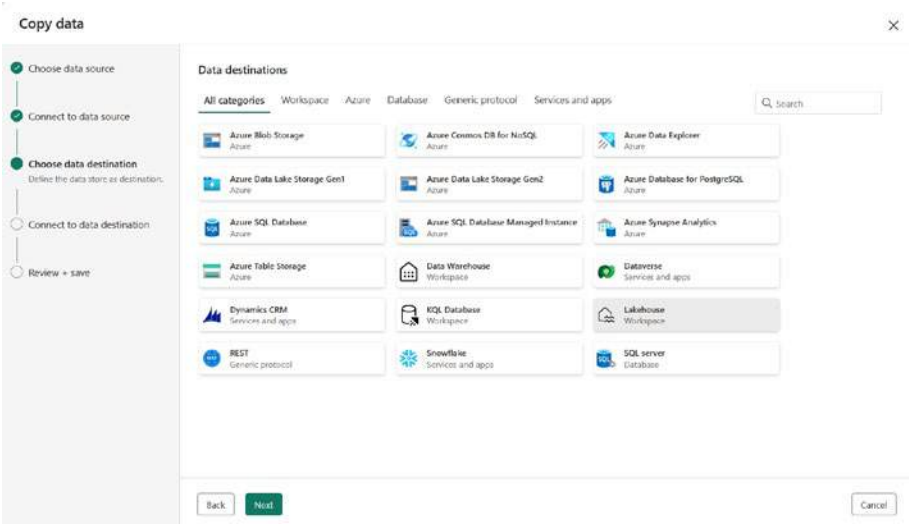


Figure 3-15. *"Choose data destination" canvas in the "Copy data" tool*

7. After you click Next, choose "Lakehouse details." For the existing lakehouse selection, you need to click the radio button Existing Lakehouse and then select the existing lakehouse from the drop-down list. Figure 3-16 shows the "Copy data" canvas and its user experience.

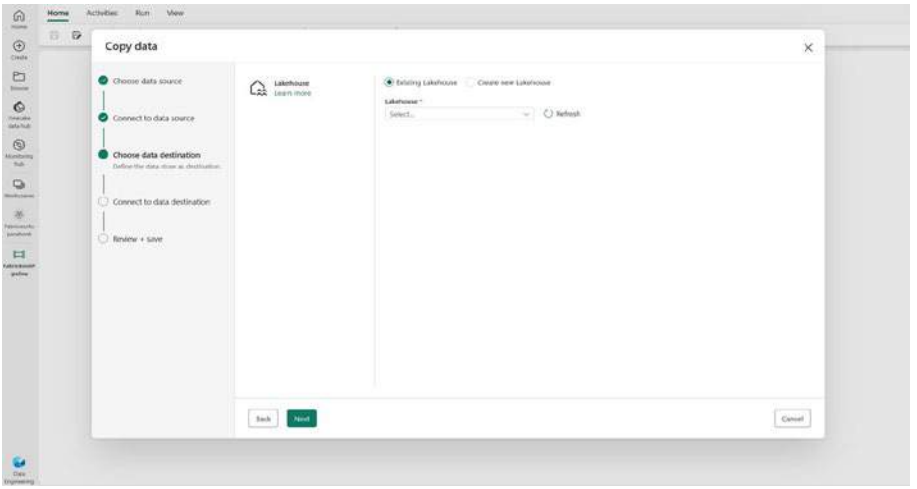


Figure 3-16. *"Copy data" canvas*

8. After you click Next, a new canvas appears to provide a table name (`bing_covid_19_data` in this case), as shown in Figure 3-17.

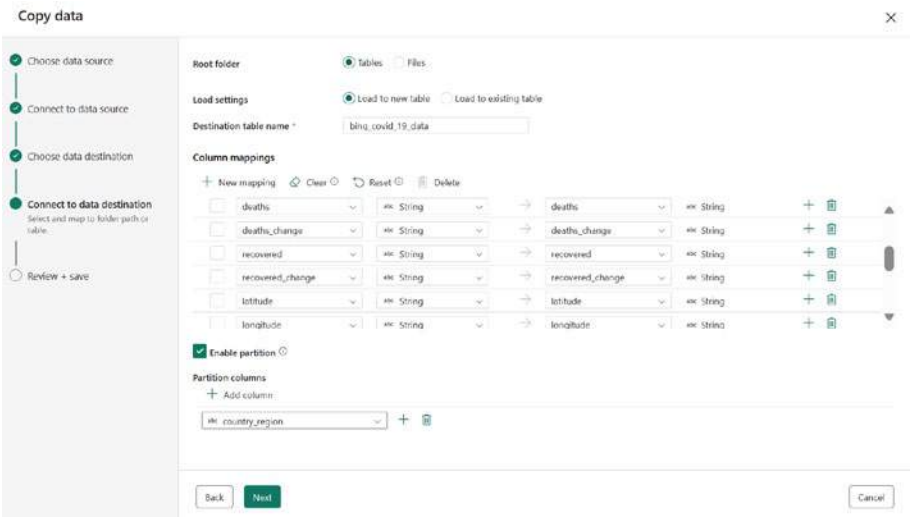


Figure 3-17. *Naming the table*

9. You can do column mapping, data type conversion, and partition enablement based on your technical needs. In Figure 3-17, we chose `country_region` as the partitioned column. After you click Next, the "Copy data" pipeline will start running. Figure 3-18 shows that FabricBookPipeline is successfully running.

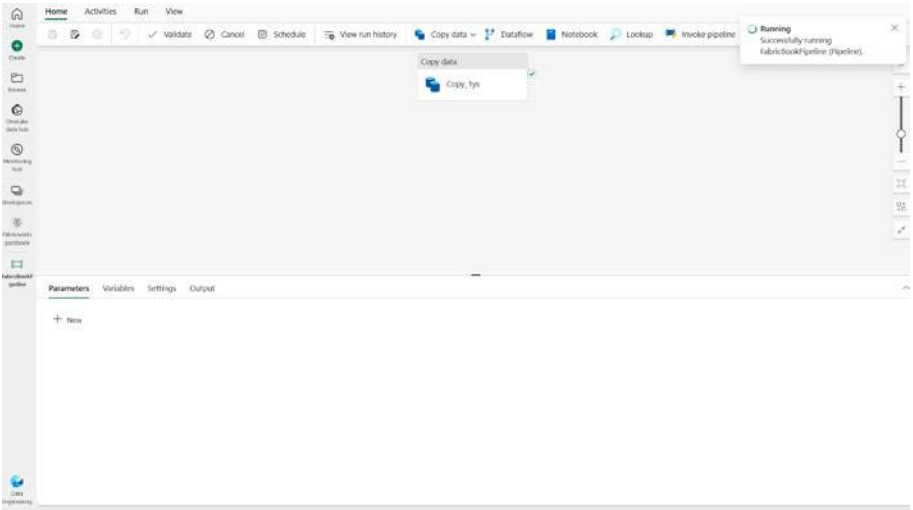


Figure 3-18. *FabricBookPipeline is successfully running.*

10. Now click the details icon beside the activity name (Copy_1ys) to see the copy data details. The "Copy data details" canvas shows the data read, data written size, rows written, throughput, and other details between the source and destination. Figure 3-19 shows the "Copy data details" canvas with activity details.

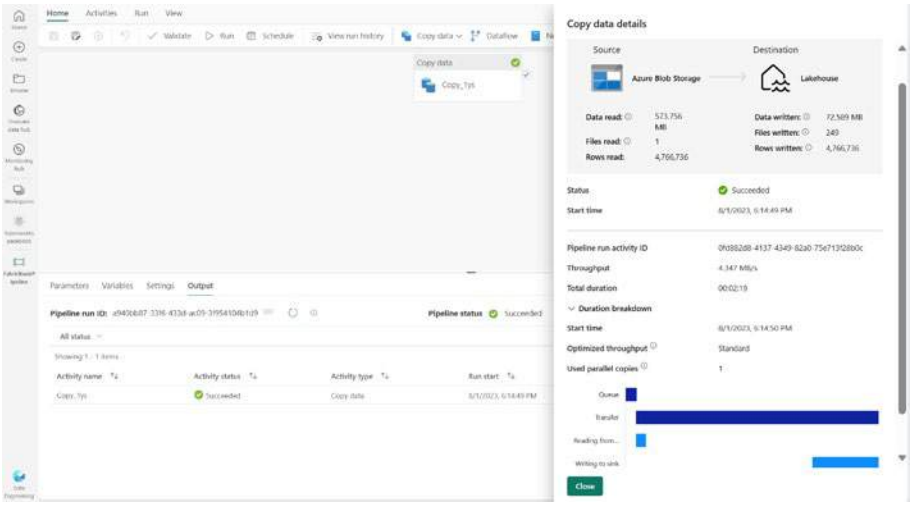


Figure 3-19. *"Copy data details" canvas with activity details*

To recap, in the previous few steps you copied a sample dataset from blob storage to lakehouse storage (OneLake) in Fabric. You executed steps that involved code-free tooling to achieve ingestion. Now let's move on to explore the dataset.

Lakehouse Explorer

The lakehouse explorer is the unified place where you can investigate the entire lakehouse. The lakehouse explorer has several sections such as Table, File, and Unidentified. When you ingest data into lakehouse, either automatically or manually, the metastore registration is done. Those automatically registered meta stores are shown in table format in the Table section. Unsupported files/folders inside the lakehouse managed area are shown in the Unidentified section inside the lakehouse explorer. For example, you can drop an audio file or image file inside a managed area, and the file will not be autodetected inside the managed area and will be

shown in the Unidentified section. The File section part of the lakehouse explorer is the landing zone for raw data. Figure 3-20 shows different sections within the lakehouse explorer.

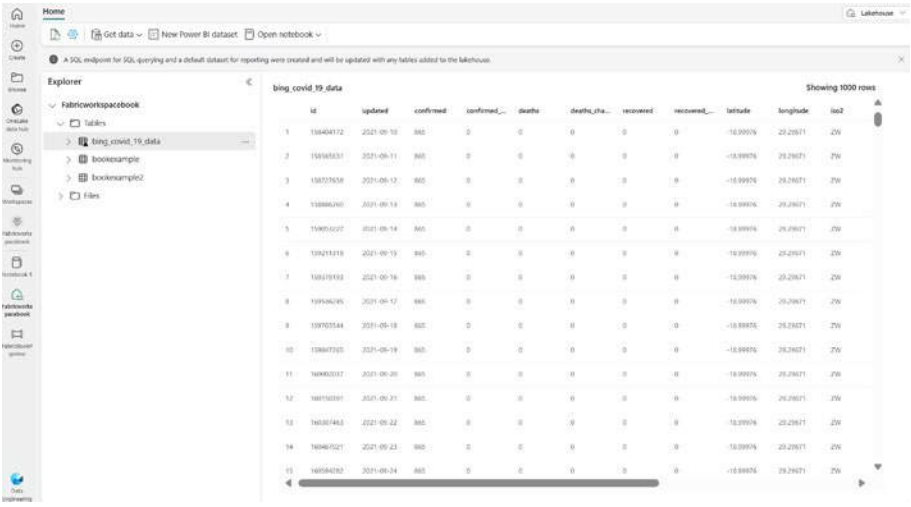


Figure 3-20. *Fabric lakehouse explorer*

To explore the dataset in Spark using a Fabric notebook, you need to click the three dots (...) and then select "Load data" and then Spark. Once you click Spark, the Fabric notebook populates with the following code to generate the data in the Spark data frame and display the dataset:

```
df = spark.sql("SELECT * FROM Fabricworkspacebook.bing_
covid_19_data")
display(df)
```

Figure 3-21 shows the outcome of executing the previous code snippet.

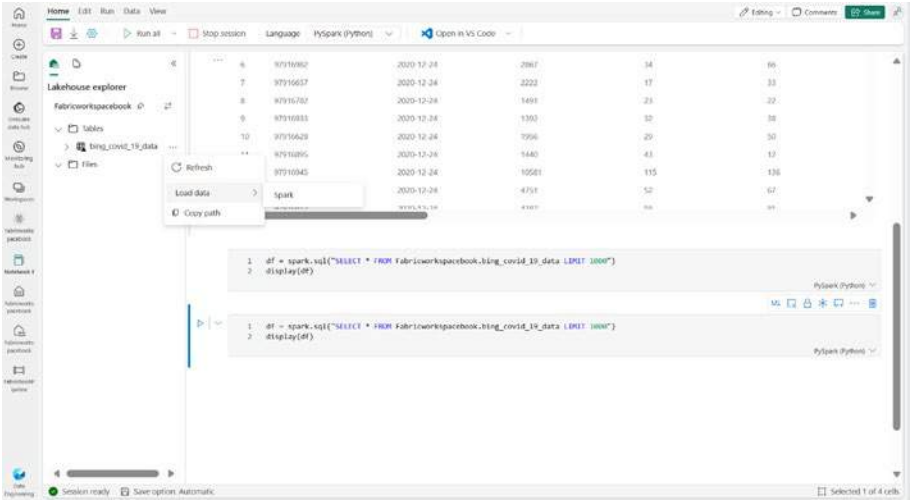


Figure 3-21. *explore the dataset in Spark using a Fabric notebook*

Now let's look at Fabric notebooks in more detail.

Data Preparation and Transformation Using a Notebook

You can access the Fabric notebook using other mechanisms. Once you go to the Fabric workspace (`https://msit.fabric.microsoft.com/home`) and click Synapse Data Engineering, you will see Notebook as well as "Import notebook" and "Use a sample," as shown in Figure 3-22. Click Notebook to start a new notebook.

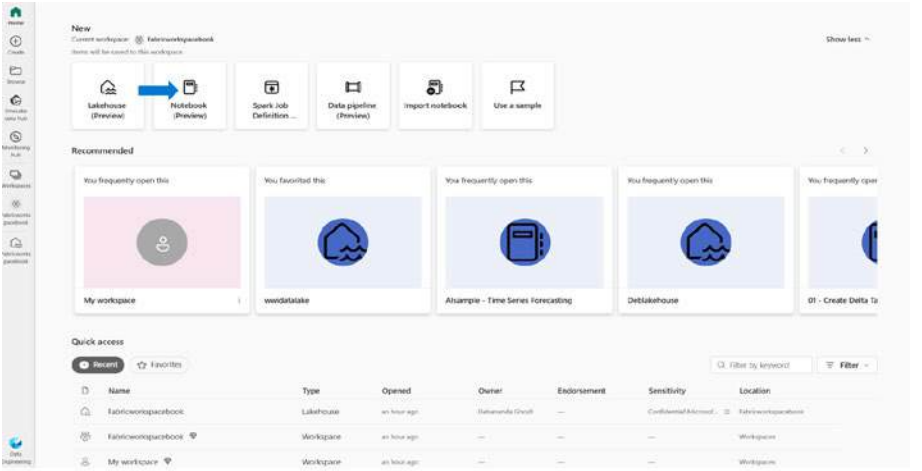


Figure 3-22. *Starting a new notebook*

After you click Notebook, you need to name the notebook. Subsequently, the workspace opens a new notebook in its lakehouse explorer. Execute the following steps to execute the Spark code:

1. Click Add to attach the lakehouse to a notebook.
2. Choose Existing Lakehouse and click Add.
3. Click the lakehouse that you want to attach (in this case Fabricworkspacebook).
4. Click Load Lakehouse.
5. Add the following code inside the Fabric notebook code cell:

#Group by covid data based on country

```
df = spark.sql("SELECT * FROM Fabricworkspacebook.bing_
covid_19_data")
display(df)
df2=df.groupby(['country_region']).count()
display(df2)
```

Once executed, the previous code produces the output shown in Figure 3-23.

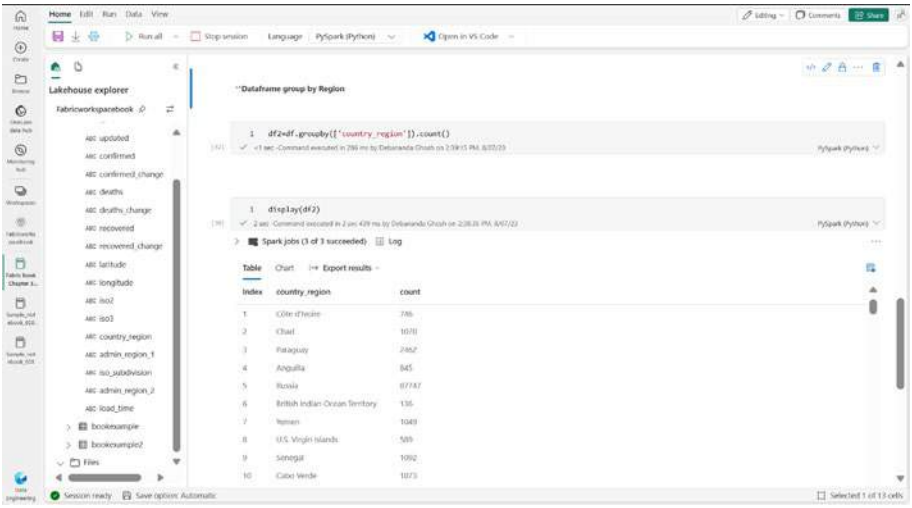


Figure 3-23. *Spark notebook-based data exploration activity*

Now you can use the PySpark language to do further development. Fabric provides the "Use a sample" option, as shown in Figure 3-24, that you can leverage for academic purposes. To do advance Python coding, go to the "Use a sample" canvas and the notebook named "A Data Engineering Starter Kit". Follow the self-explanatory steps for Python development using this starter kit.

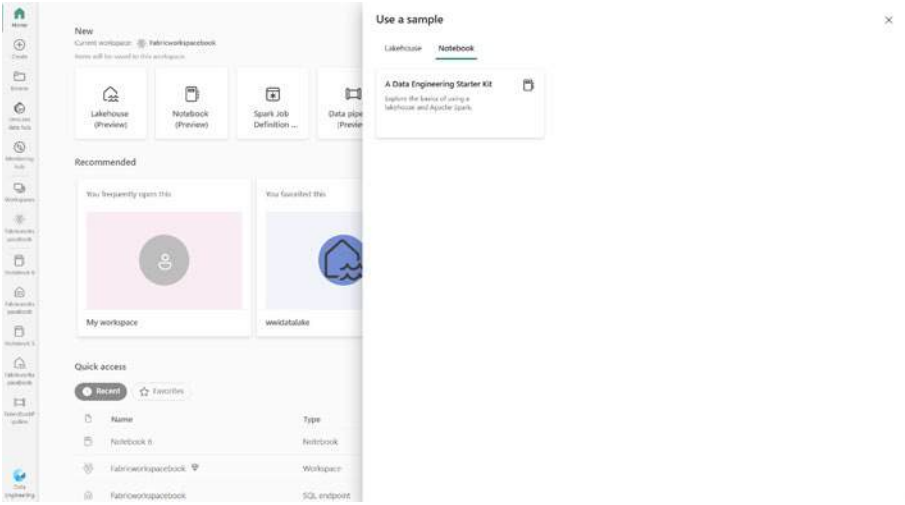


Figure 3-24. *Fabric sample notebook*

Let us focus on few development specific features. The Fabric notebook provides multiple-language support. You can choose a magic command like `%%spark`, or you can choose a language picker at the bottom of every cell to switch languages while doing development.

- PySpark (Python)
- Spark (Scala)
- Spark SQL
- SparkR

Another interesting feature of notebooks that you can leverage for quick development is drag-and-drop. You can click a table or file in the lakehouse explorer and drag it inside a notebook. In Figure 3-25, we have dragged the dataset `dimension_customer.csv` (another sample CSV file in the File section) and dropped it inside a notebook. The notebook populates the following code snippet automatically:

```
df = spark.read.format("csv").option("header","true").
load("Files/dimension_customer.csv")
# df now is a Spark DataFrame containing CSV data from "Files/
dimension_customer.csv".
display(df)
```

Figure 3-25 shows the outcome of executing the previous code.

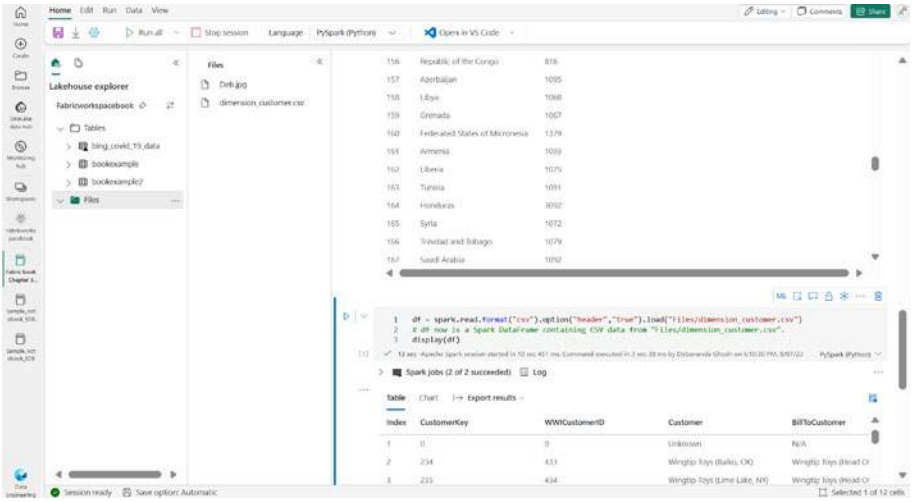


Figure 3-25. *Spark notebook drag-and-drop feature in Fabric*

Spark developers can continue to leverage the notebook capabilities, for example moving cells up and down, deleting cells, and so on. The Fabric notebook provides a Variables feature in the View section. You can see Variables inside the current Spark session in Figure 3-26.

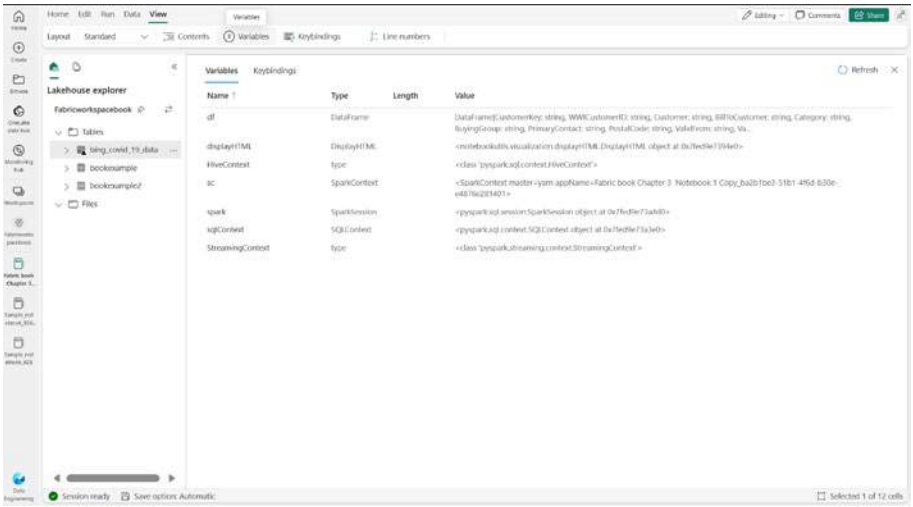


Figure 3-26. *Spark notebook Variables feature in Fabric*

You can also schedule notebook code for a production job by clicking the Schedule tab, as shown in Figure 3-27. Alternately, you can add it to a pipeline inside the Fabric workspace.

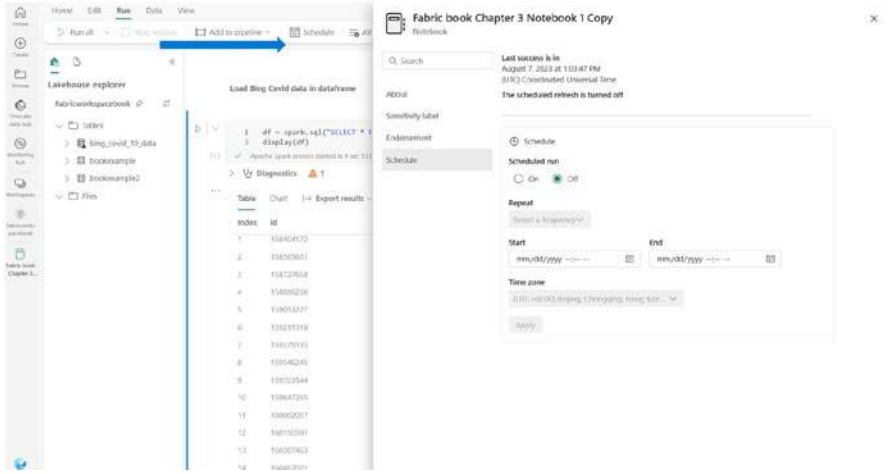


Figure 3-27. *Spark notebook scheduling in Fabric*

In this section, you learned about notebook-based features of Fabric and programming styles. In the next section, you will learn about the Spark batch job submission mechanism using compiled programming files.

Defining a Spark Job

In this section, we will focus on the Spark job definition features in Fabric. The Spark job definition capability allows users to upload a Python file, R file, or Scala/Java file, and create a Python/Scala/R Spark job definition per your technical needs. Using a Spark job definition, you can upload one main file and multiple reference files that are referenced by the main file. To define a job, you need to click Spark Job Definition, as shown in Figure 3-28, and then provide a Spark job name (Fabricbooksparkjob in this case).

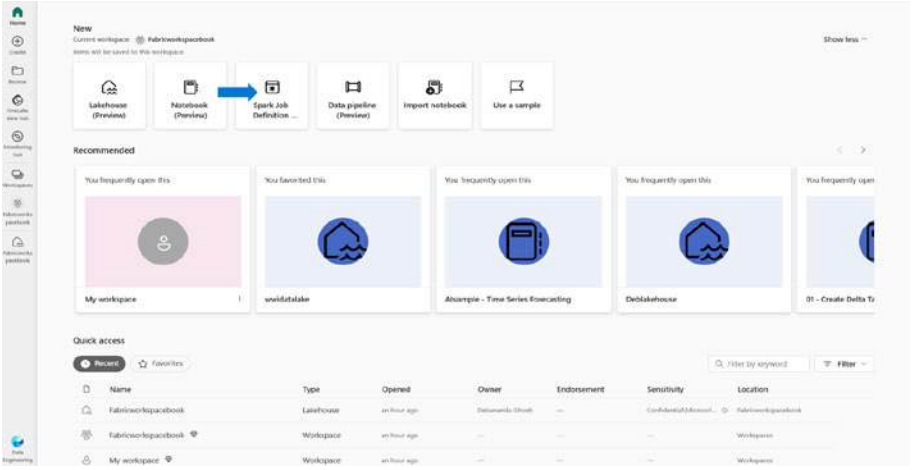


Figure 3-28. *Spark job definition in Fabric*

In Figure 3-29, we have uploaded two files. The main definition's file path consists of a word-counting program in the Python language. The reference file path is the input file path in ADLS Gen2 (Azure Data

Lake Gen2). Note that you also need to attach a lakehouse (in this case Fabricworkspacebook) for computation purposes, as shown in Figure 3-29. And make sure ADLS Gen2 has access to the user account that is used to submit the job.

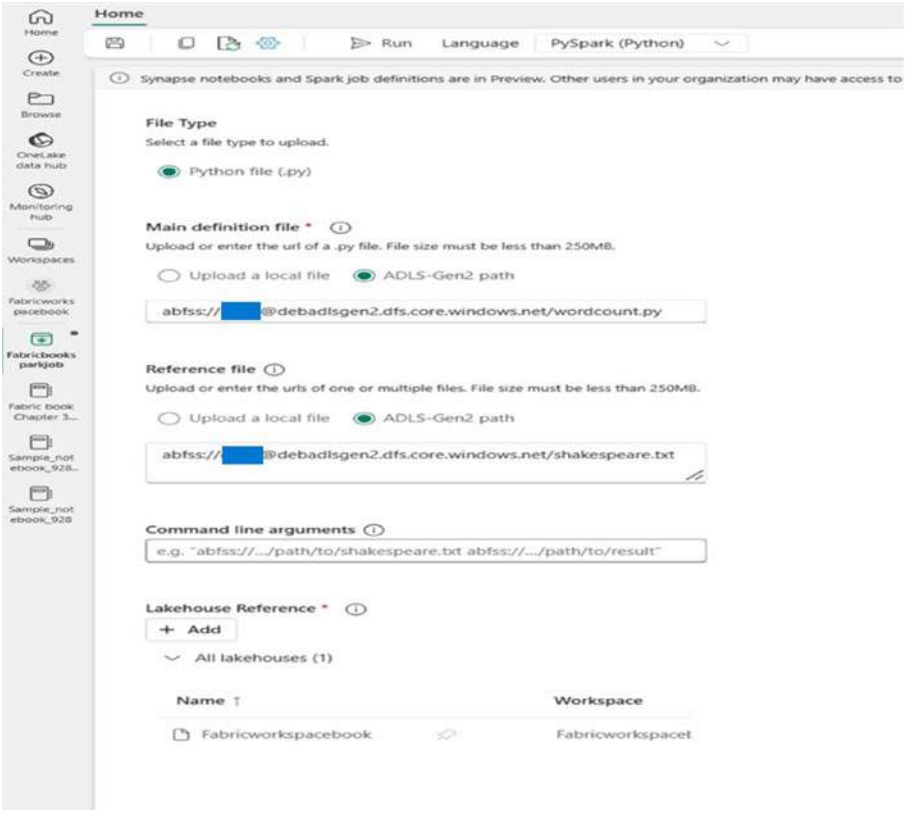


Figure 3-29. *Fabric Spark job definition*

Monitoring a Spark Job

Batch job monitoring is essential for any developer. Fabric provides diverse ways to do job monitoring.

- A notebook cell indicates the job progress when the jobs are running or have succeeded. In our previous example, when you were loading the `bin_covid_19_data` data frame, you could see the Spark job stages and tasks in the notebook cell indicator, as shown in Figure 3-30.

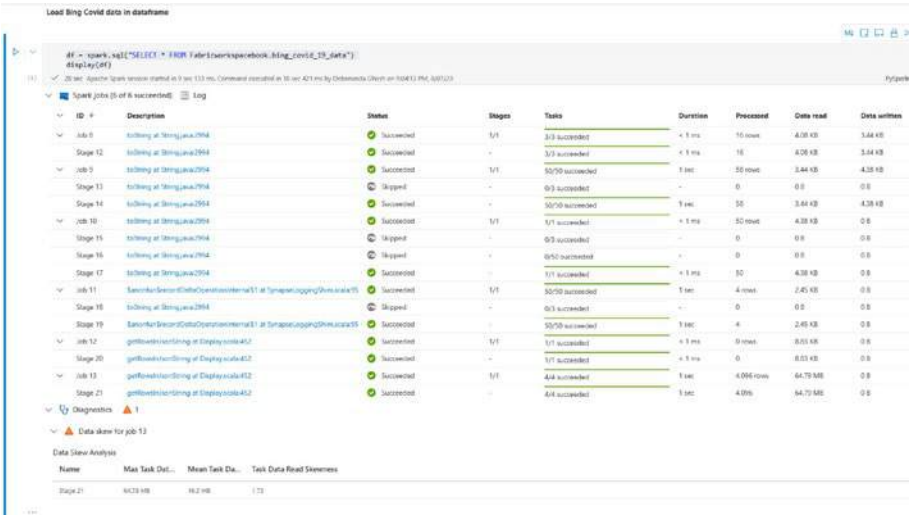


Figure 3-30. *Spark job status monitoring in Fabric*

Click "Job description," and it will take you to a Spark directed acyclic graph (DAG), as shown in Figure 3-31. This will be a familiar interface for Spark developers, and you can use it to monitor the Spark execution in detail.

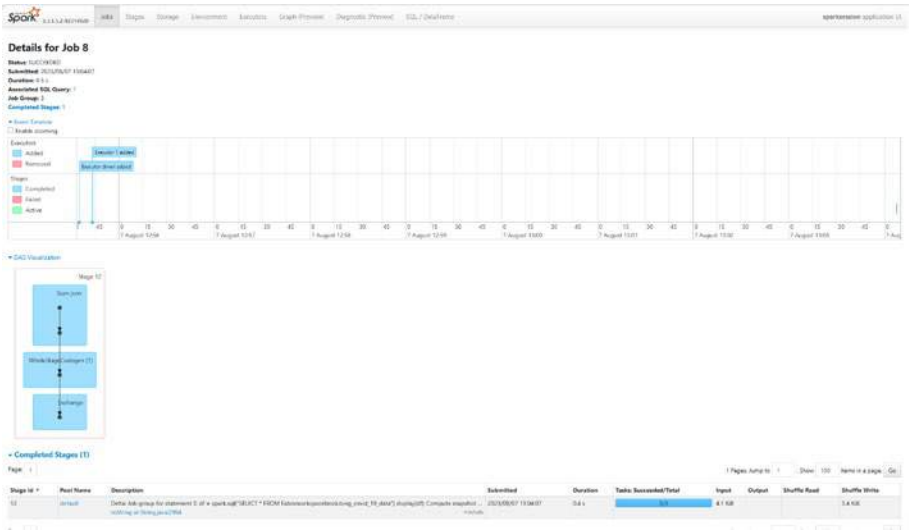


Figure 3-31. *Spark UI DAG*

- The Spark job can be monitored from the Fabric Monitoring hub, which is available on the left side, as shown in Figure 3-32. You can filter (right-side top) based on the Item type and other filter criteria to point to the right job and get the relevant details from the Monitoring hub.

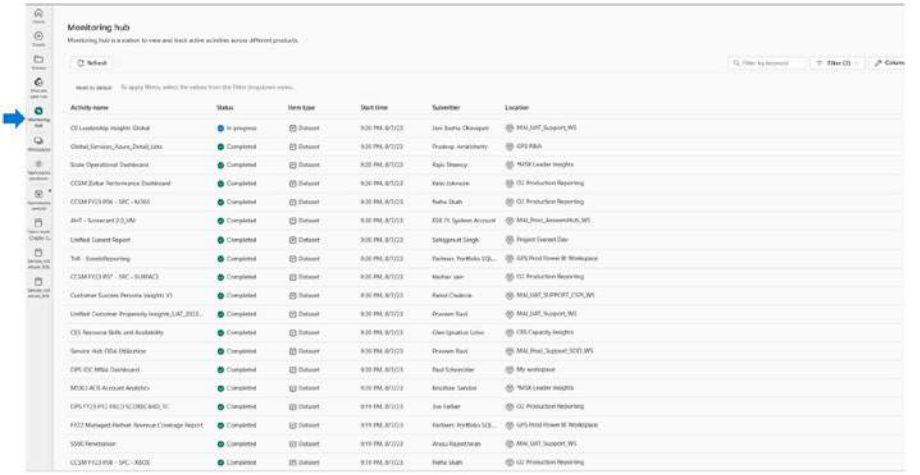


Figure 3-32. *Monitoring hub in Fabric*

The previous steps are a Spark-based experience for data engineers.

Another interesting point to note is that Fabric Spark provides a starter cluster. This starter pool helps to speed up Spark initialization time with no need for manual setup. In other words, it allows you to load the libraries quickly. Note that starter pools will continue to stay hydrated/ready for fast initialization always; however, you will be billed only when you use the actual Spark compute pool. Go to Workspace, click the three dots (…), click "Workspace settings," click Data Engineering/Science, and then click "Spark settings" to edit the default starter pool. Click the pencil icon to edit the Spark settings and environments, as you can see in Figure 3-33.

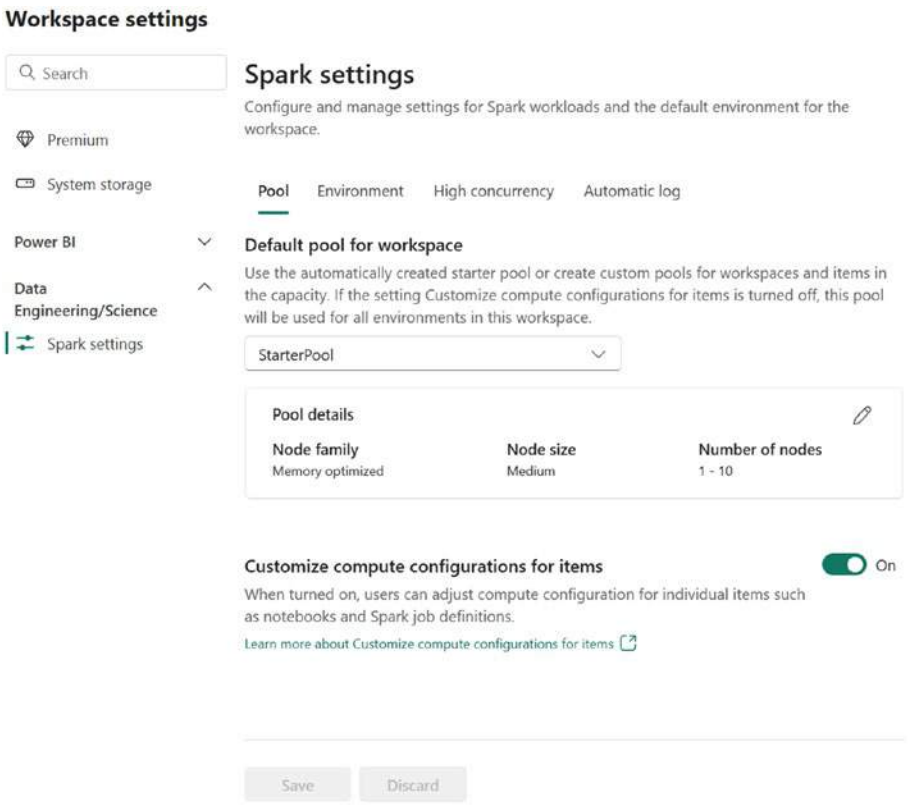


Figure 3-33. *Spark starter pool in Fabric*

Though Fabric follows a SaaS model, you have the flexibility to do certain configurations at the Spark computation and environment level. For example, you can create a customized Spark pool based on the Fabric capacity size. By creating a customized Spark pool, you enforce the workload with node requirements and scalability. Figure 3-34 shows the Spark pool customization features.

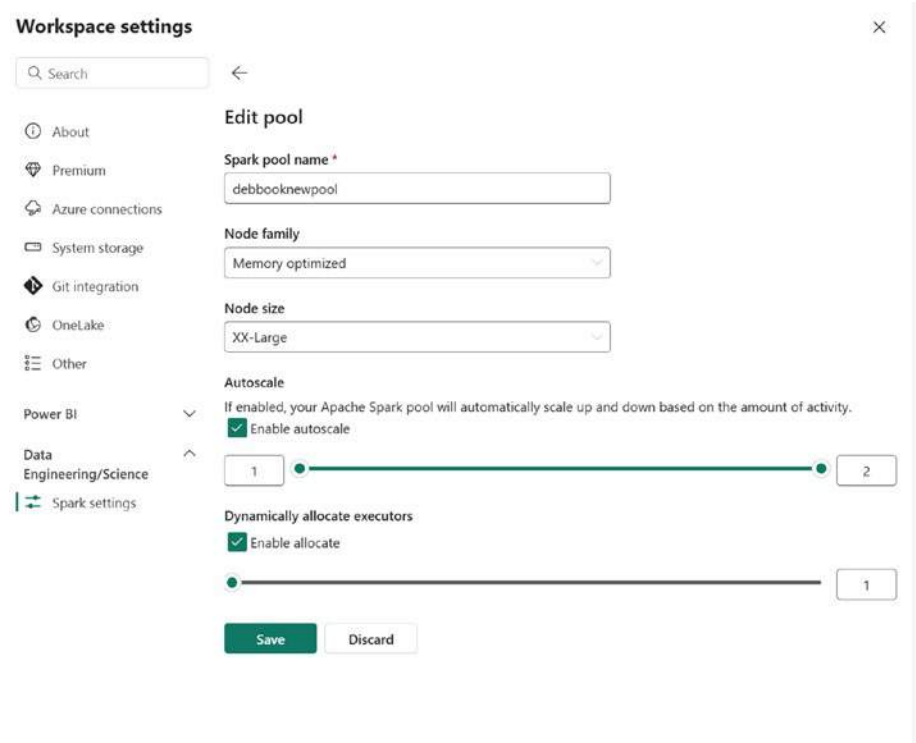


Figure 3-34. *Spark customized pool in Fabric*

In this case, we have changed the node size from Small to XX-Large. You may notice the maximum number of nodes is 2. Based on the Fabric capacity, you can allocate a certain number of maximum nodes. Refer to Table 3-2 to see an example of the maximum node mapping within the Fabric capacity based on node size.

Table 3-2. *Fabric Capacity, Node Size Mapping*

Fabric Capacity (F SKU)	Spark vCores	Node Size	Max Number of Nodes
F64	128	Small	32
F64	128	Medium	16
F64	128	Large	8
F64	128	X-Large	4
F64	128	XX-Large	2

To customize the Spark environment, follow these steps:

1. Click Environment in the Data Engineering interface and click New Environment, as you can see in Figure 3-35.

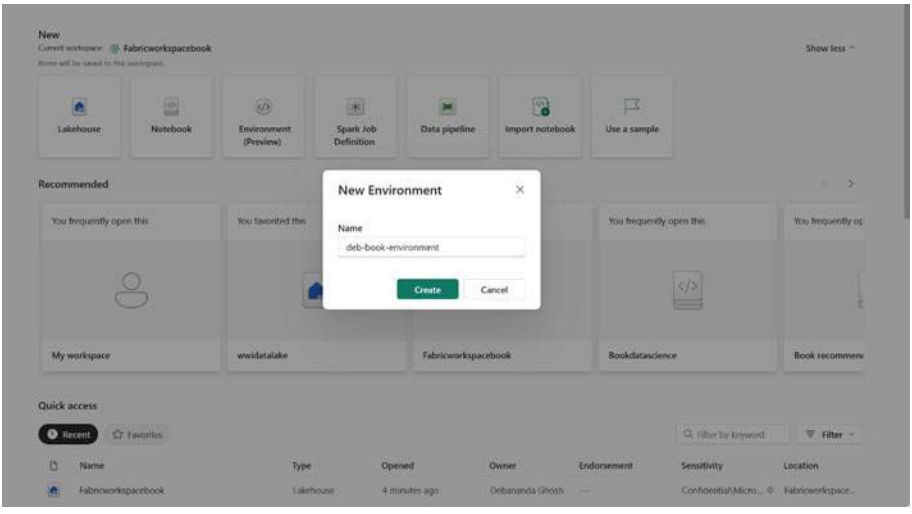


Figure 3-35. *Spark customized pool in Fabric*

2. Now you need to open the Environment canvas and can upload the Spark libraries, as shown in Figure 3-36.

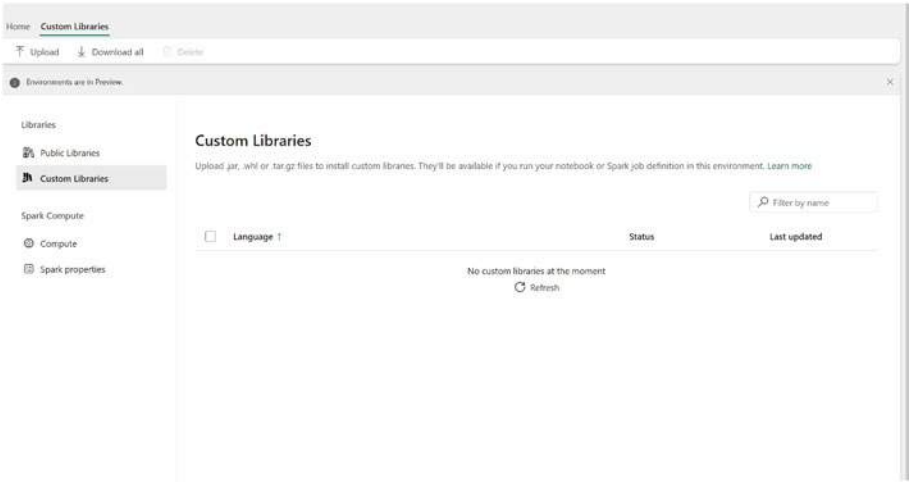


Figure 3-36. *Spark environment configuration in Fabric*

3. You need to attach the environment file that you created in the previous step (`deb-book-renvironment`) inside the Fabric notebook with the Fabric Workspace settings/customize spark pool item, as shown in Figure 3-37.

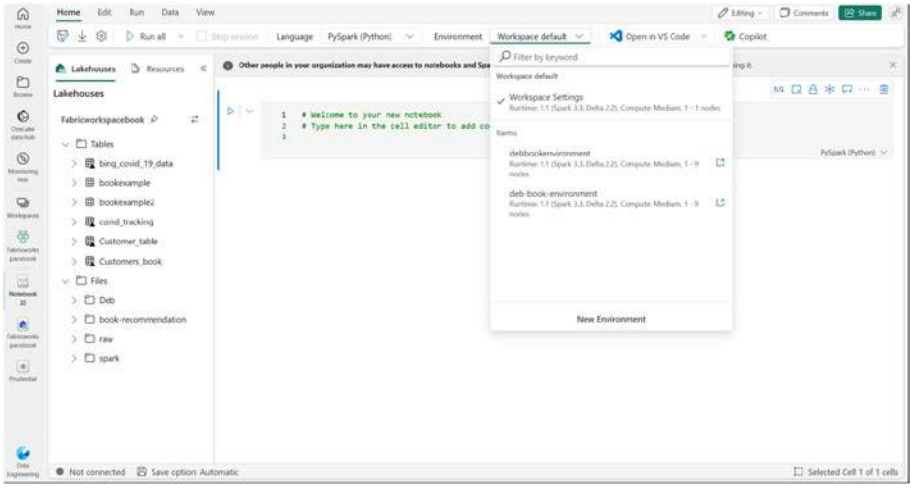


Figure 3-37. *Opening the file*

Now you have learnt Fabric spark configuration part and will now move on to SQL configurations. In the next section, you will see a SQL-based experience for the same lakehouse tables that we have preloaded using `bing_covid_19_data`.

SQL Endpoint of a Lakehouse

The Microsoft Fabric lakehouse explorer provides SQL-based experience for SQL developers. To leverage your SQL experience, you need to switch to the SQL endpoint from the lakehouse endpoint by clicking the lakehouse at the top-right corner, as shown in Figure 3-38. Note that SQL Endpoint of Lakehouse is read only mode for analyzing Delta lake data using T-SQL capability. For T-SQL developers who wants to do read-write operations using Delta lake file we will leverage Fabric Datawarehouse capability. We will discuss that in Chapter 5, 'Data warehousing in Microsoft Fabric.'

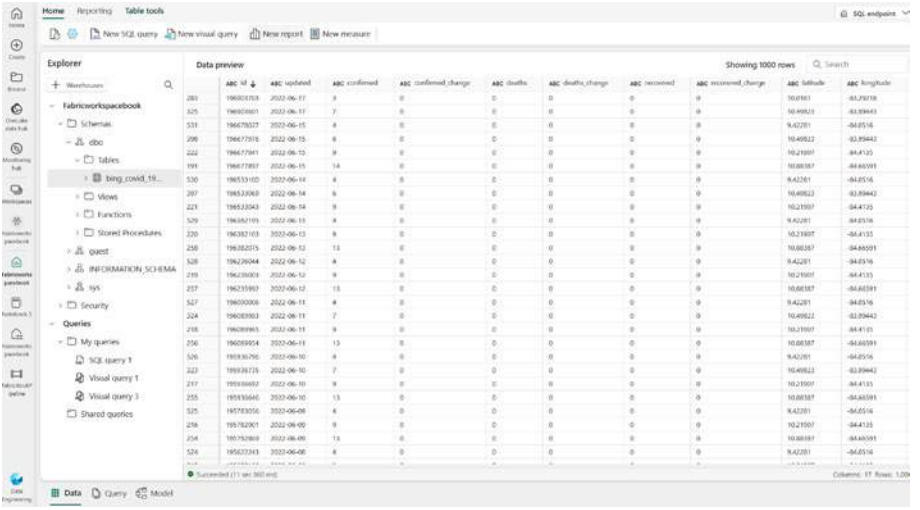


Figure 3-38. *SQL endpoint in Fabric*

In SQL endpoint mode, you can start exploring the same Fabric lakehouse managed table using a SQL endpoint. Right-click the table name under Workspace dboTables (in this case `bing_covid_19_data`), click New SQL query, and then Select Top 100 rows, as shown in Figure 3-39.

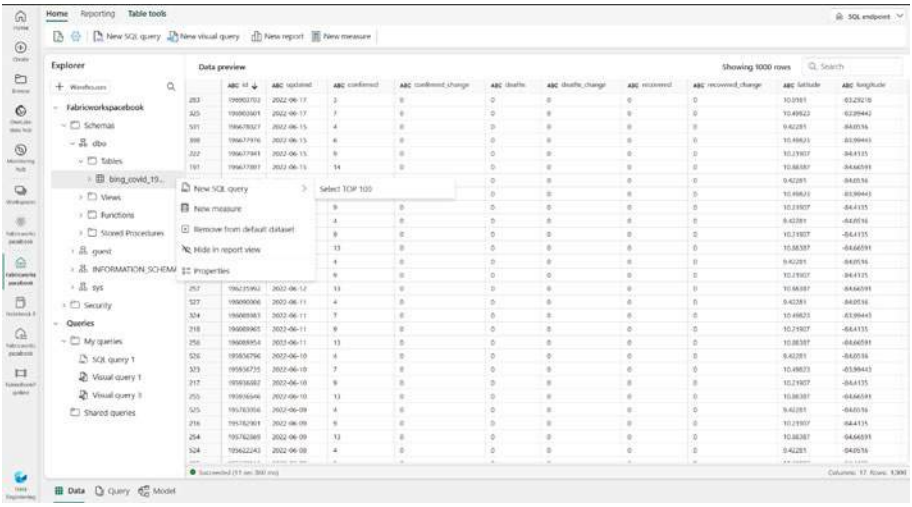


Figure 3-39. *Fabric lakehouse structure*

The following code is generated by the Fabric workspace SQL endpoint and fetches sample 100 rows of data from the table:

```
SELECT TOP (100) [admin_region_1]
              ,[admin_region_2]
              ,[confirmed]
              ,[confirmed_change]
              ,[country_region]
              ,[deaths]
              ,[deaths_change]
              ,[id]
              ,[iso2]
              ,[iso3]
              ,[iso_subdivision]
              ,[latitude]
              ,[load_time]
              ,[longitude]
              ,[recovered]
```

```
            ,[recovered_change]
            ,[updated]
FROM [Fabricworkspacebook].[dbo].[bing_covid_19_data]
```

Like wise we can create 'New measure' and create calculated columns using such one click features. In the next section, you will use previous query output and build a report with just a few clicks.

Data Visualization

To visualize the query output of the previous step, you need to click "Visualize results" (beside Results). Once you click, the "Visualize results" canvas will appear, as shown in Figure 3-40.

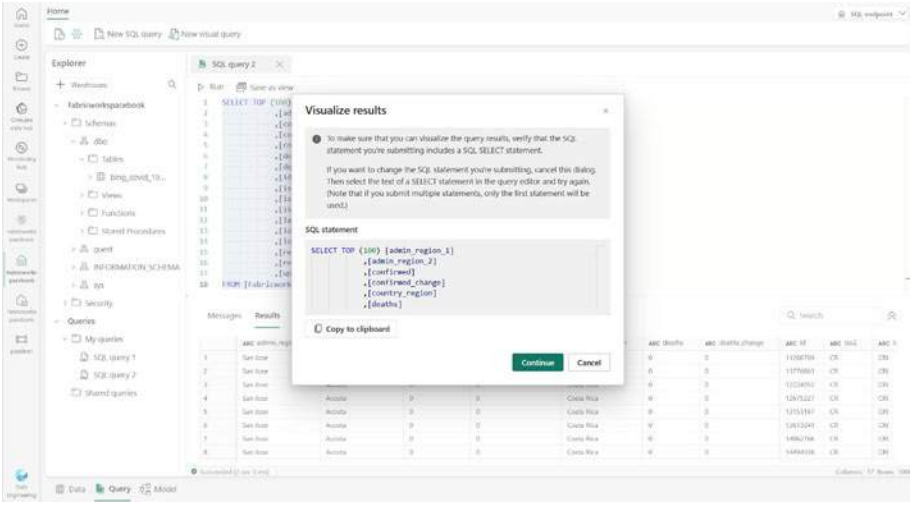


Figure 3-40. *"Visualize results" canvas*

After you click Continue, the Fabric workspace will provide a Power BI Visualization canvas with plugins on the right side, as shown in Figure 3-41. The Power BI Data section on the right will contain the dataset output of the previous SQL query.

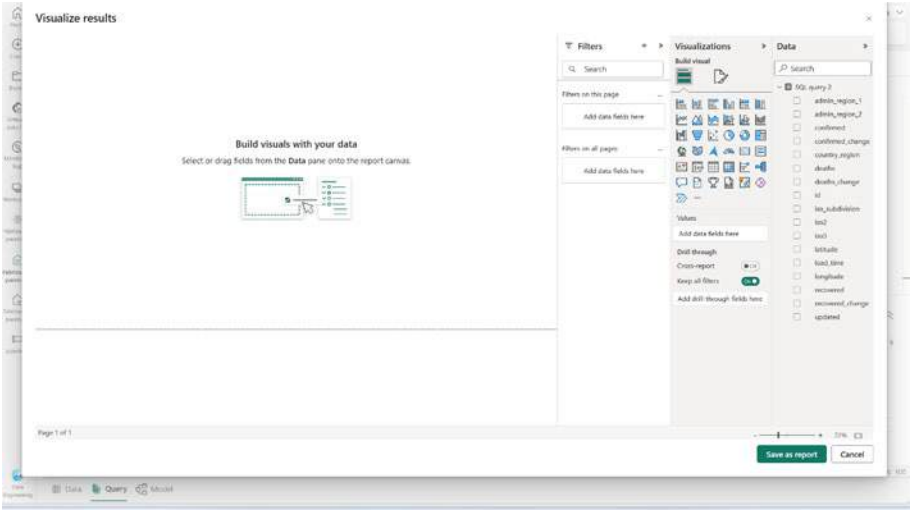


Figure 3-41. *Visualization of the Fabric dataset*

Now you are ready to build a report using the query results. When you use the geospatial visualization plugin, set Location to "country_region," and set "Bubble size" to "Count of deaths," the geospatial chart shown in Figure 3-42 will appear.

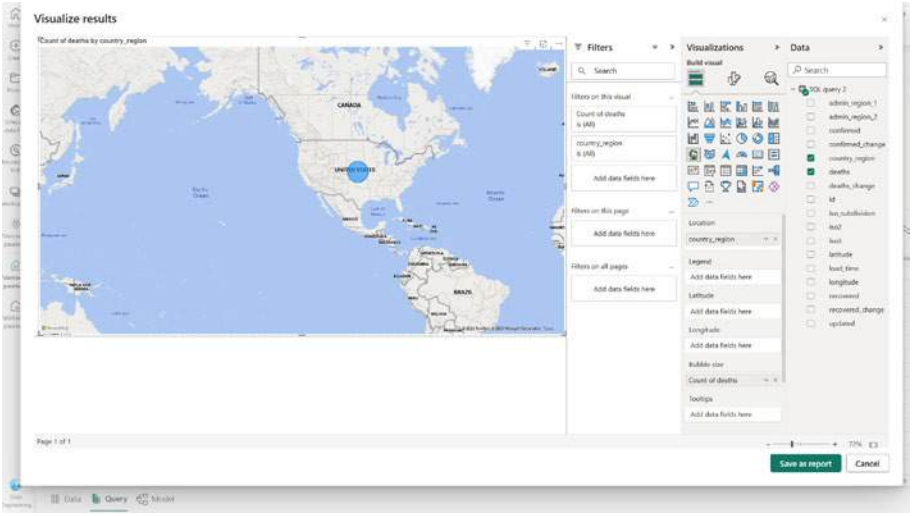


Figure 3-42. *Visualization of the Fabric dataset*

You can now click Save as Report. This report on the number of deaths across the country will be now saved in the Fabric workspace under the Report type for future reference.

Summary

In this chapter, we covered the Data Engineering capability of Microsoft Fabric. We started by introducing the delta lakehouse foundation and its benefits.

You then learned how to create and deploy a Microsoft Fabric lakehouse. You ingested sample data using lakehouse features. You then moved to a Spark-based transformation. You learned how to configure a Spark environment. We then walked you through a notebook-based visualization. Figure 3-43 shows the data architecture features we covered.

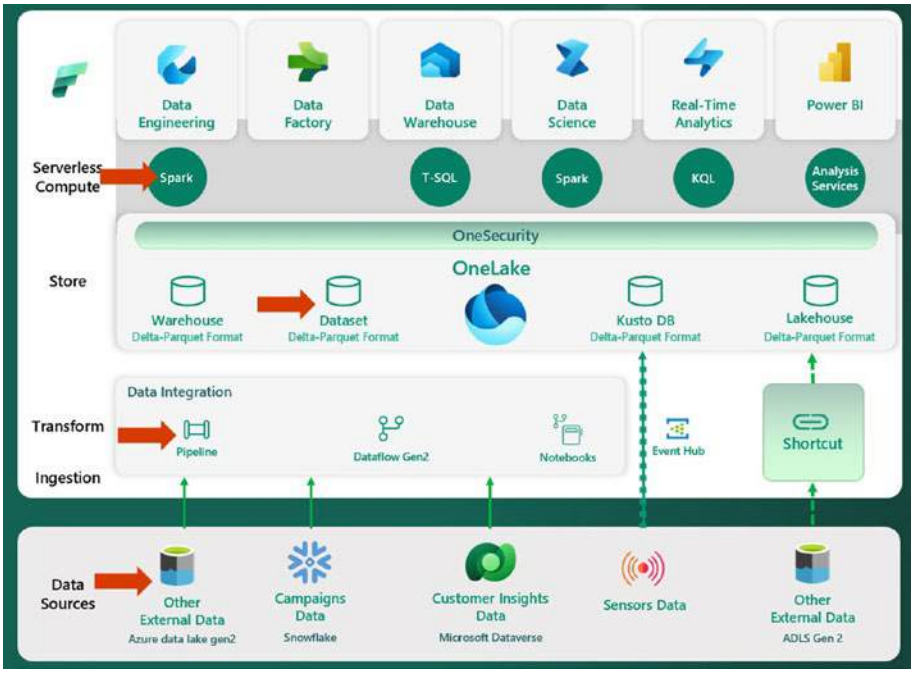


Figure 3-43. *Data engineering recap*

In the next chapter, you will focus on the data warehouse details of Fabric.

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric delta lake V-order:** https://learn.microsoft.com/en-us/fabric/data-engineering/delta-optimization-and-v-order?tabs=pyspark#what-is-v-order

- **Low-shuffle merge:** https://learn.microsoft.com/en-us/azure/synapse-analytics/spark/low-shuffle-merge-for-apache-spark
- **Fabric copy activity:** https://learn.microsoft.com/en-us/fabric/data-factory/copy-data-activity
- **Fabric lakehouse:** https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview

CHAPTER 4

Microsoft Fabric for Data Scientists

End-to-end data science development requires data scientists, domain scientists (industry experts), data analysts, data engineers, and database administrators. Also, the right tools for each data science activity is essential for AI experimentation. Microsoft Fabric provides role-specific user experiences for experimenting with machine learning and AI.

In this chapter, we will delve into the Fabric capabilities specific to data science. Data science uses statistical and mathematical algorithms to extrapolate knowledge from datasets. As per Microsoft,

> *Data science is a scientific study of data to gain knowledge. This field combines multiple disciplines to extract knowledge from massive datasets for the purpose of making informed decisions and predictions.*

A data science experiment consists of the following steps:

1. Think of a business use case.
2. Identify the relevant datasets.
3. Cleanse, wrangle, and preprocess the data.
4. Develop a model and experiment.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_4

5. Enrich the model.
6. Put the model to use.
7. Build insights.

Figure 4-1 shows the data science process steps.

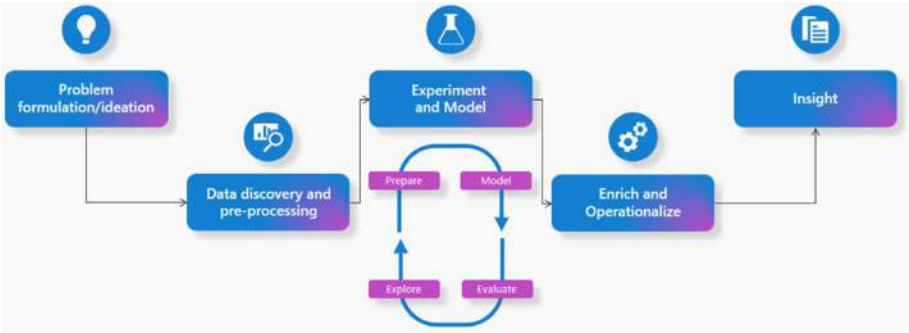


Figure 4-1. *Data science process overview from Microsoft*

Fabric offers end-to-end data science features for data scientists. Figure 4-2 depicts the Synapse Data Science canvas in the Fabric workspace.

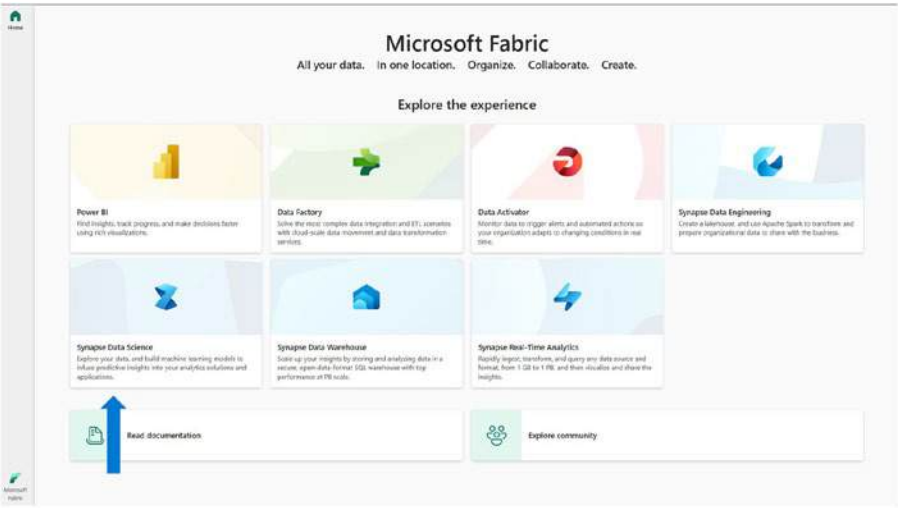


Figure 4-2. *Accessing the Synapse Data Science canvas in the Fabric workspace*

Note that data science is a big subject area, and we do not intend to walk through data science–related all topics. This chapter will focus only on the Fabric-specific capabilities that enhance the machine learning experience for data scientists. Specifically, you will learn about the following topics in this chapter:

- Fabric Data Science canvas overview
- Ingesting exploratory data analysis (EDA) and preparing data in the Fabric workspace
- Machine learning model development and experimentation tracking
- Large language modeling (LLM) with Microsoft Fabric

By the end of this chapter, you will have a good understanding of the Fabric-based data science features. We will begin with an overview of the Data Science canvas.

Fabric Data Science Overview

Microsoft Fabric offers both code and no-code capabilities. Fabric users can leverage the home page to explore data analysis, data wrangling, data preparation, and modeling, as well as experimentation tracking. The Synapse Data Science canvas in the Fabric workspace provides the following user interface (UI) features:

- **Model (Preview):** Model registration and reuse framework
- **Experiment (Preview):** Machine learning experimentation tracking framework
- **Notebook (Preview):** Code-free data wrangler capability.
- **Import notebook:** Notebooks (.ipynb files)
- **Use a sample:** Sample notebook for academic purposes

To access the Data Science canvas, go to the home page of Fabric (`https://msit.fabric.microsoft.com/home`) and then click Synapse Data Science, as shown in Figure 4-3.

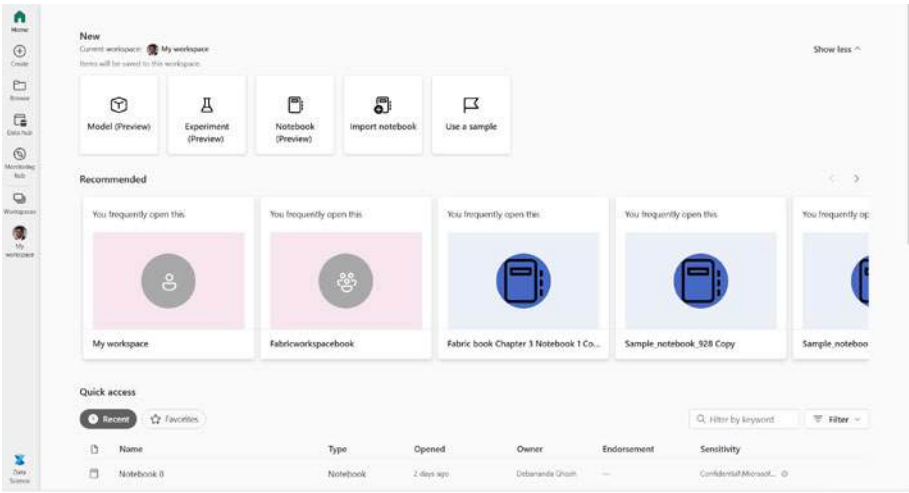


Figure 4-3. *Synapse Data Science canvas*

To learn more about the Synapse Data Science canvas, we will go through the following topics in the next section:

- Ingesting, exploring, and preparing data for data science experimentation using a Fabric notebook
- Using the native code-free Fabric data wrangling capabilities inside a notebook
- Doing data science coding using a notebook
- Using VS Code for data science development
- Doing data visualization using Fabric-native capabilities and open-source libraries
- Doing machine learning tracking using a Fabric-native user interface
- Registering and tracking models using Fabric capabilities

Exploring, Ingesting, and Preparing Data

Data ingestion in Microsoft Fabric can be done in numerous ways. One way is to use the copy tool discussed in Chapter 3. In this section, we will leverage a Fabric notebook to directly access the OneLake file path and external path to load the data in a OneLake/Fabric notebook session.

Ingesting Data

We recommend bringing any external file into OneLake prior to any data science experimentation. To download the file from the external ADLS Gen2 and upload it to OneLake, you need to use the following statement:

```
# (Replace adls path, lakehouse path)
import os, requests
# Welcome to your new notebook
# Type here in the cell editor to add code!
remote_url = "https://storageaccountname.blob.core.windows.net/
public/Credit_Card_Fraud_Detection"
fname = "creditcard.csv"
download_path = f"/lakehouse/default/Files/Deb"

if not os.path.exists("/lakehouse/default"):
        raise FileNotFoundError("Default lakehouse not found,
please add a lakehouse and restart the session.")
os.makedirs(download_path, exist_ok=True)
if not os.path.exists(f"{download_path}/{fname}"):
        r = requests.get(f"{remote_url}/{fname}", timeout=30)
        with open(f"{download_path}/{fname}", "wb") as f:
            f.write(r.content)
print("Downloaded demo data files into lakehouse.")
```

There might be scenarios where data already resides in a data lake. In such cases, you need to use this relative path or Azure blob file system driver (ABFS) path for the following scripts. To read the CSV file, you can write the following code to load the data into a Pandas dataframe:

```
#(Replace workspace name,path)

import pandas as pd
df = pandas.read_parquet("/LAKEHOUSE_PATH/Files/FILENAME.csv")
display(df)
```

For writing CSV file, we can leverage the following code:

```
#(Replace workspace name,path)
import pandas as pd
# Write a Pandas DataFrame into a CSV file in your Lakehouse
# Replace LAKEHOUSE_PATH and FILENAME with your own values
df.to_csv("/LAKEHOUSE_PATH/Files/FILENAME.csv")
```

To read from the Parquet file residing in the lakehouse path, you need to leverage the following code:

```
#(Replace workspace name,path)

import pandas as pd
df = pd.read_parquet("abfss://Fabricworkspacedeb@xxxx-onelake.
dfs.fabric.microsoft.com/Fabricworkspacedeb.Lakehouse/Tables/
bing_covid_19_data/part-00000-b65e8379-7231-4630-9526-
abe0c69c81dd-c000.snappy.parquet")
display(df)
```

Note that you must get the OneLake file `abfss` direct path or relative path for the previous steps. You can get this by clicking the three dots (...) beside the explorer table name or filename. Figure 4-4 shows how to get the ABFS path/relative path/file API path to the `book-recommendation` folder inside `Files` in the `Fabricworkspacebook` lakehouse.

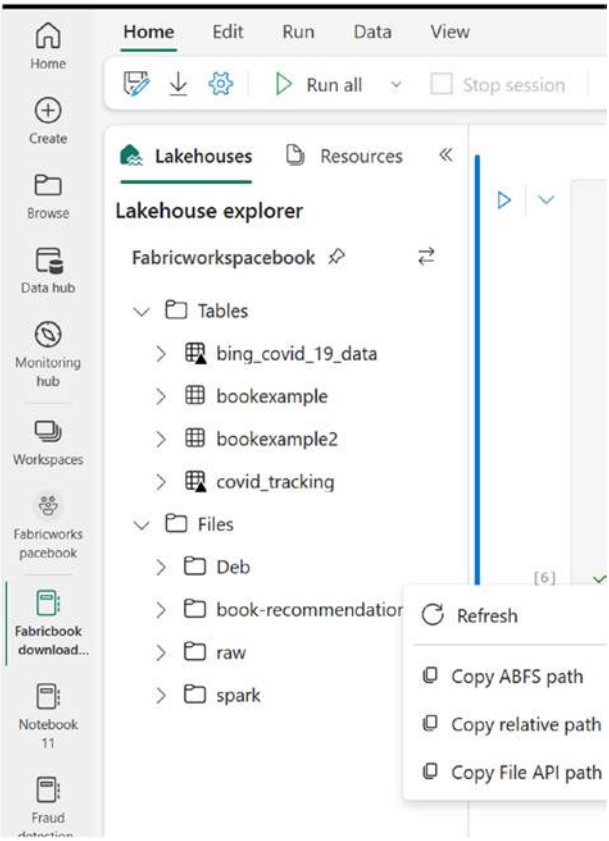


Figure 4-4. *book-recommendation folder*

Now the dataset is inside the OneLake storage account. There could be scenarios where you want to load the dataset into a dataframe directly from a remote location. You can leverage the following code snippet to load a CSV file from a remote location to a Pandas dataframe:

```
#(Replace path)
import pandas as pd

# Read a CSV into a Pandas DataFrame from e.g. a public blob store
df = pd.read_csv(" raw.githubusercontent.com/plotly/datasets/
master/polar_dataset.csv")
```

There are other Fabric tooling mechanisms you can use to ingest the data inside the OneLake storage account. We will cover them in Chapter 6. Since the data is ingested now, in the next section we will move on to exploring the data.

Exploring the Data

Microsoft Fabric allows you to explore any preloaded OneLake dataset easily. To browse a dataset, you need to go to the explorer view (under the lakehouse artifact) and click the three dots (…). Then click "Open in notebook" and "New notebook" to explore the dataset, as shown in Figure 4-5.

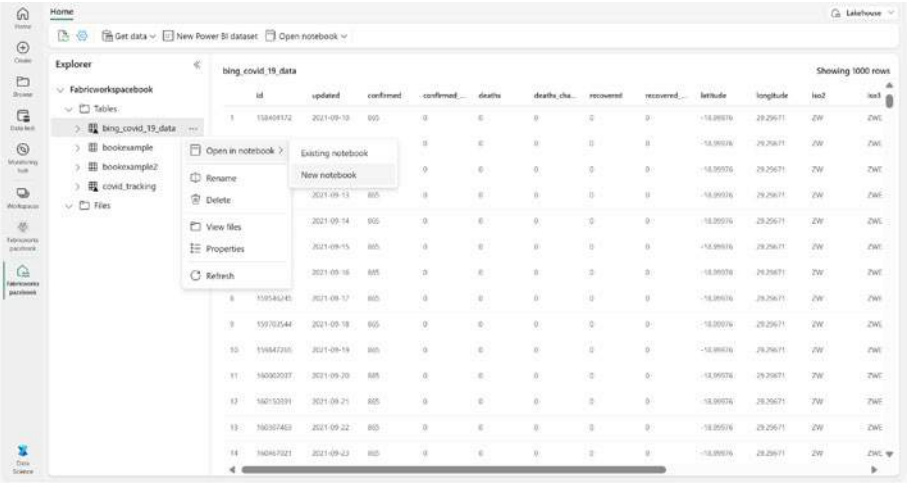


Figure 4-5. *Opening the notebook*

Once you click the dataset, it generates the Python code, as shown in Figure 4-6. This code fetches the sample data from the preloaded file or table (in this case the `bing_covid_19` table we loaded in Chapter 3). This code fetches 1,000 sample records and loads them into a Spark dataframe.

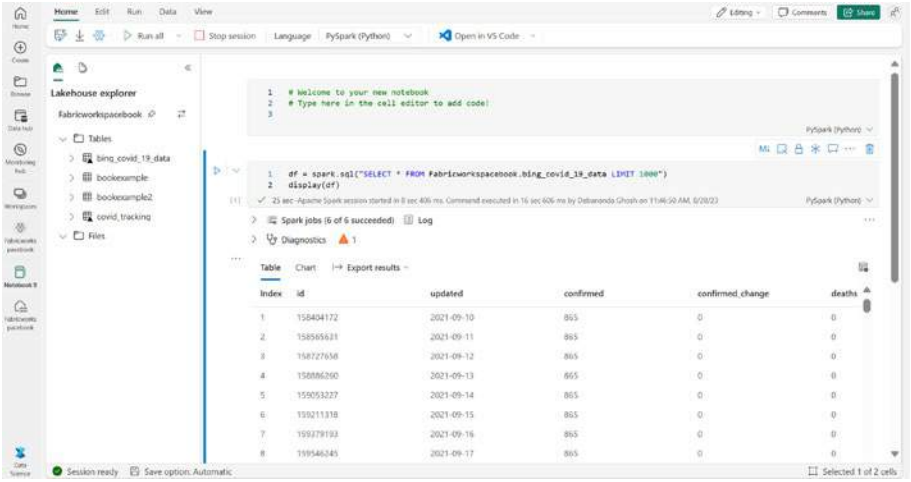


Figure 4-6. *Data exploration*

Once the dataset exploration is done using the previous techniques, the data scientist needs to move on to data preparation. In the following section, we will discuss how to leverage code-free tooling for certain data preparation tasks.

Preparing the Data Using a Data Wrangling Tool

Data Wrangler is a notebook-based experience that provides a user interface for accelerating data preparation. Data Wrangler has the following features:

- Summary of data statistics
- Searchable list for data wrangling operations
- Easy-to-use user interface

- Can generate corresponding code and append it inside the current notebook as a new cell

To leverage Data Wrangler, you need to go to the Data tab under the notebook ribbon. On the Data tab, you select Launch Data Wrangler in the drop-down. Once the cell execution is finished in previous step (Refer Figure 4-6, Data exploration), you will be able to see the dataframe (`df` in this case) is loaded in the drop-down, as shown in Figure 4-7. This dataframe is the outcome of the previous step.

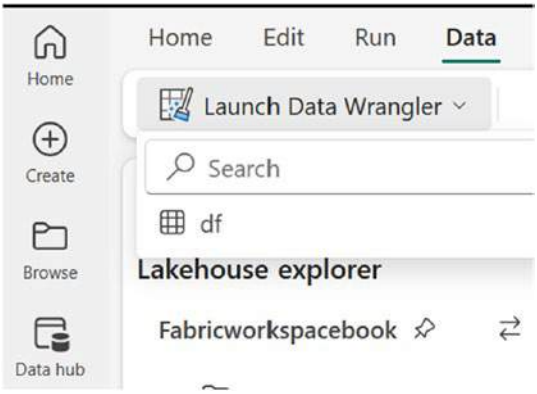


Figure 4-7. *Code-free data wrangling experience*

As you click `df`, the Data Wrangler canvas opens, as shown in Figure 4-8.

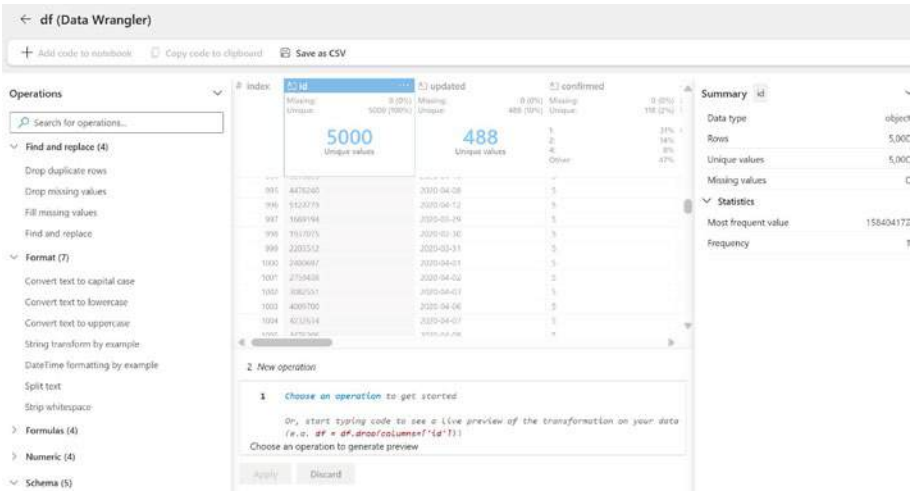


Figure 4-8. *Data Wrangler tool user interface*

The following operations are currently supported through the user interface of Data Wrangler under Operations on the left. You can do the following:

- Find and replace (drop duplicate rows, drop missing values, fill missing values, find and replace)
- Format (convert text to uppercase, middle case, lowercase, string transformation, split text, strip whitespace)
- Formula (one hot encoding, create column for formula)
- Numeric (round up, round down)
- Schema (change column type, drop columns)
- Sort and filter (group by, aggregate)

To drop a column, you need to search for *Drop column* in the Operations search bar on the left. Then select "Target columns," as shown in Figure 4-9.

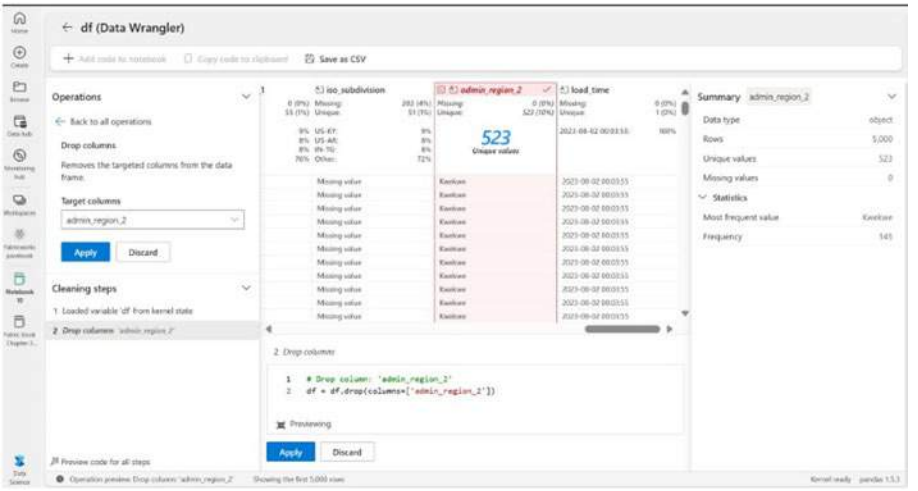


Figure 4-9. *Drop column operation using Data Wrangler*

The previous step generates the following code snippet. Once you apply this code, it is appended in the original notebook where you invoked the dataframe-reading code.

```
# Code generated by Data Wrangler for pandas DataFrame
def clean_data(df):
    # Drop column: 'admin_region_2'
    df = df.drop(columns=['admin_region_2'])
    return df

df_clean = clean_data(df.copy())
df_clean.head()
```

At the time of this writing, the Pandas dataframe is fully supported for data wrangling via the user interface. However, Spark dataframe support is in progress. To load a dataset inside the Pandas dataframe with just a few clicks, you can also explore the Fabric-provided user interface. Select Workspace, Files, Folder, and File; right-click the filename and then Copy

ABFS Path; and then select Pandas to load it into the Pandas dataframe, as shown in Figure 4-10.

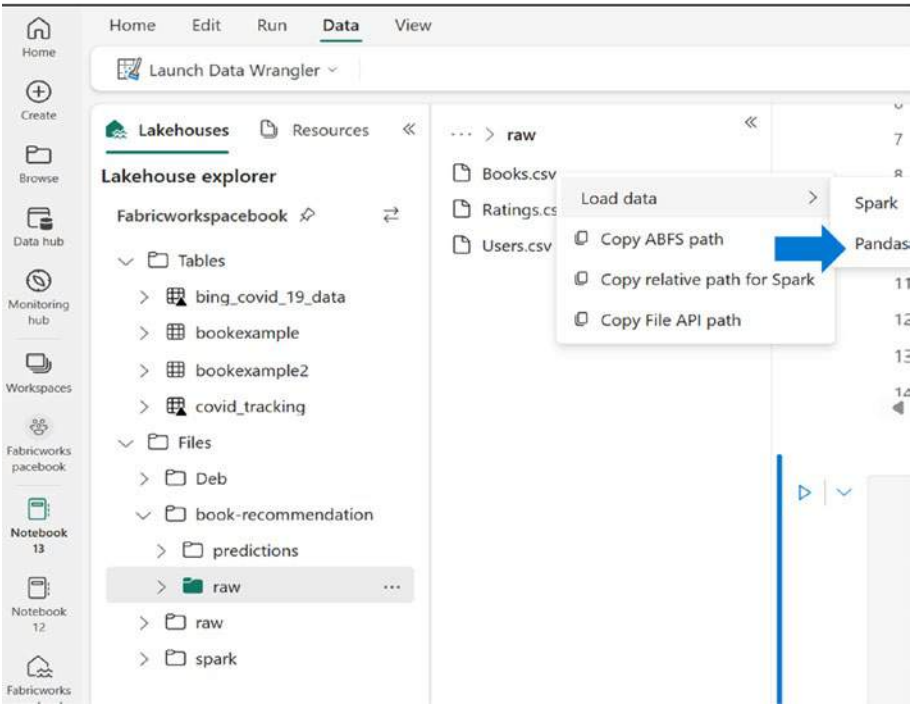


Figure 4-10. *Loading data into the Pandas dataframe with just a few clicks*

Exploratory Data Analysis (EDA) and Data Visualization

You can visualize data in Fabric with three different tools.

- The built-in Power BI–based visualization tool
- The built-in Fabric notebook visualization capabilities
- Custom-based data science visualization libraries and frameworks

You will explore the Power BI–based visualization in Chapter 9. Figure 4-11 shows the native visualization capability.

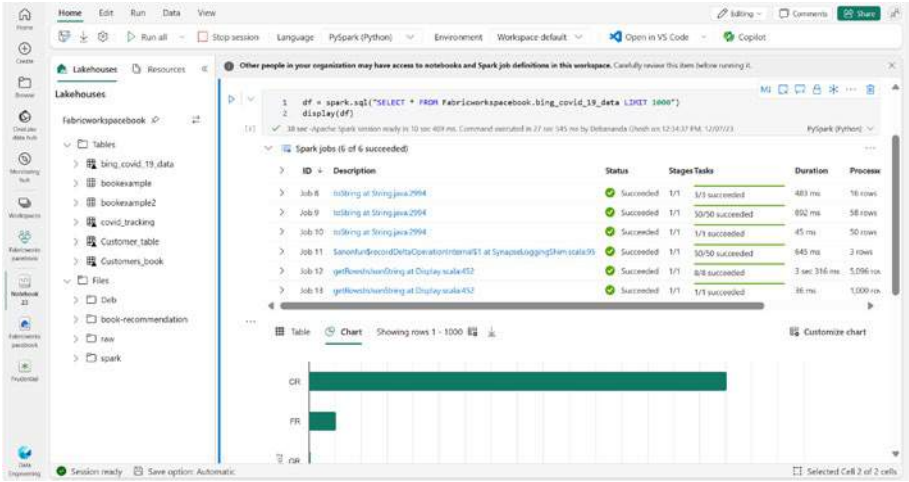


Figure 4-11. *Notebook built-in visualization*

For EDA-based visualizations, prepackaged libraries of open-source programming languages are popular. For example, in the Python runtime environment, you can use Matplotlib, Seaborn, Plotly, and many more. The following code snippet uses the Pandas library to create a simple bar chart using the dataframe prepopulated earlier:

```
df3.plot.bar(x='country_region',y='id_count',figsize=(40, 10,))
```

Figure 4-12 shows the outcome of the previous code snippet.

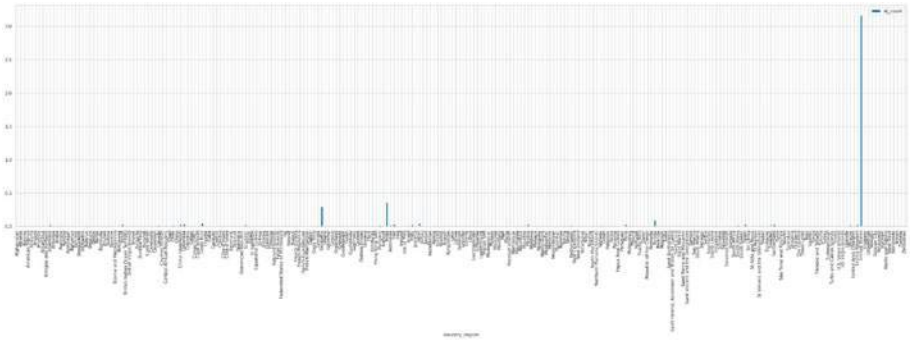


Figure 4-12. *Visualization of the dataset*

In the following section, you will focus on data preparation using code.

Preparing the Data with Notebook Code

You have explored the data wrangling tooling, which is a code-free experience. Fabric notebooks also support programming using the following Spark languages:

- PySpark (Python)
- Spark (Scala)
- Spark SQL
- SparkR

Developers have the flexibility to enjoy any of these programming languages to do further development. The following code is an example of Python-based sample data preparation:

```
df2 = df.groupby(['country_region'])['country_region'].count()
display(df2)
```

The output of the previous code is shown here:

```
country_region
Afghanistan       1099
Albania           1089
Algeria           1099
American Samoa     988
Andorra           1092
                  ...
West Bank         1025
Worldwide          975
Yemen             1049
Zambia            1078
Zimbabwe          1693
Name: country_region, Length: 246, dtype: int64
```

Here is the code to store the data in another dataframe:

```
def clean_data(df):
    # Performed 1 aggregation grouped on column: 'country_region'
    df = df.groupby(['country_region']).agg(id_count=('id',
    'count')).reset_index()
    return df

df3 = clean_data(df.copy())
df3.head()
```

Likewise, you can use PySpark, Spark SQL, R/SparkR, or Scala inside the same notebook to continue preparing the data.

Data Science with VS Code

Visual Studio Code (VS Code) is a lightweight source code editor that runs on Windows, macOS, and Linux. To run, author, and debug a Fabric notebook locally, you can use VS Code. To download VS Code, go to https://code.visualstudio.com/download. After you download VS Code, you need to have the Synapse VS Code extension installed locally. The prerequisites for the Synapse VS Code extension are as follows; you can find detailed instructions at the previous link:

- Java 1.8
- Conda
- Jupyter extension for VS Code

Please read https://learn.microsoft.com/en-us/fabric/data-engineering/setup-vs-code-extension for all prerequisite details and complete the VS Code extension installation instructions. Once you download the extensions, you are ready to do notebook CRUD operations (Create, Read, Update, Delete) using VS Code locally. To download a notebook locally, you need to click Open in VS Code, as shown in Figure 4-13.

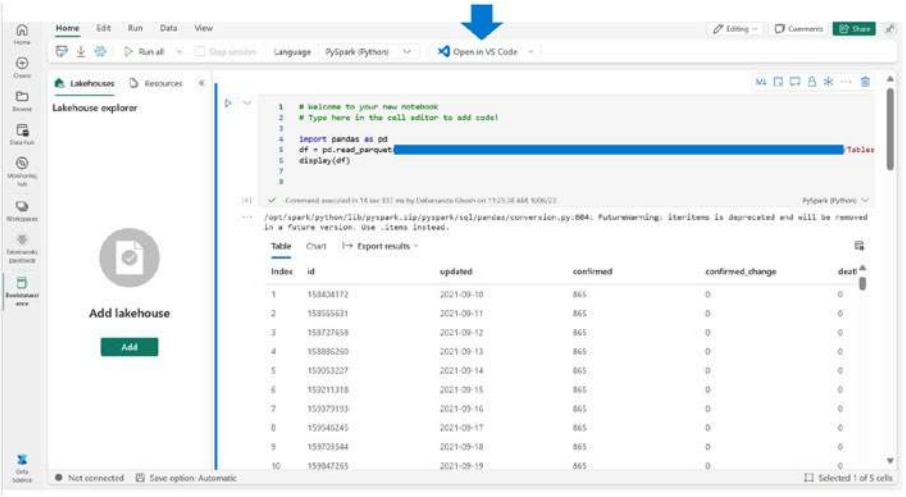


Figure 4-13. *Notebook and VS Code integration*

As you follow the previous steps, VS Code downloads the notebook and opens it in VS Code locally, as shown in Figure 4-14.

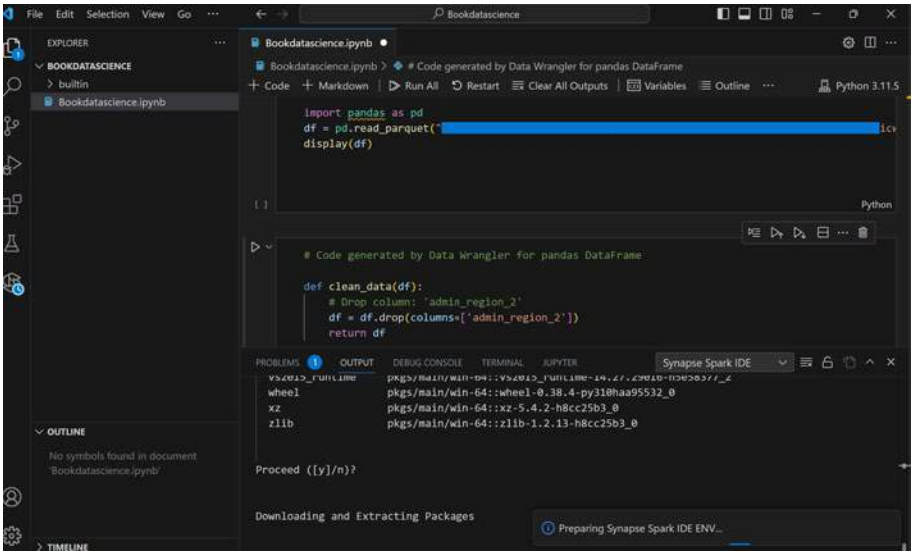


Figure 4-14. *VS Code setup for Synapse Spark*

Now you need to select the Synapse-Spark kernel, as shown in Figure 4-15, to execute the Spark notebook locally for debugging purposes.

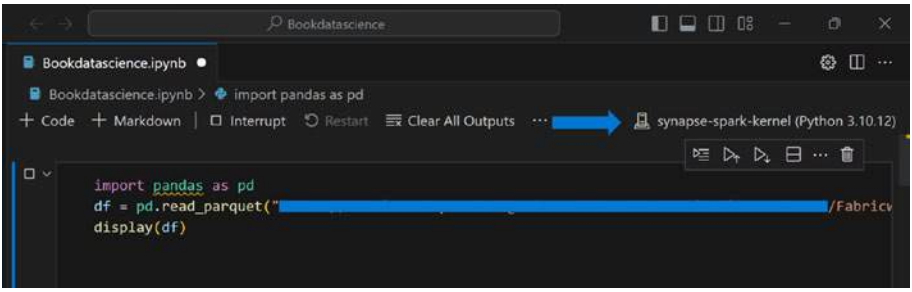


Figure 4-15. *Synapse Spark kernel selection for VS Code*

While doing this setup, there are some common errors you may face. To mitigate such issues, visit the following web pages:

- **Panda setup "Conda not recognized error":** https://stackoverflow.com/questions/44515769/conda-is-not-recognized-as-internal-or-external-command
- **Parquet read issues from VS Code:** https://stackoverflow.com/questions/50760351/how-to-identify-pandas-backend-for-parquet. Install the following packages:

  ```
  pip install pyarrow
  pip install fastparquet
  pip install adlfs
  ```

- **For permission errors:** https://github.com/conda/conda/issues/4644

To publish local changes to a remote workspace, click the icon shown in Figure 4-16.

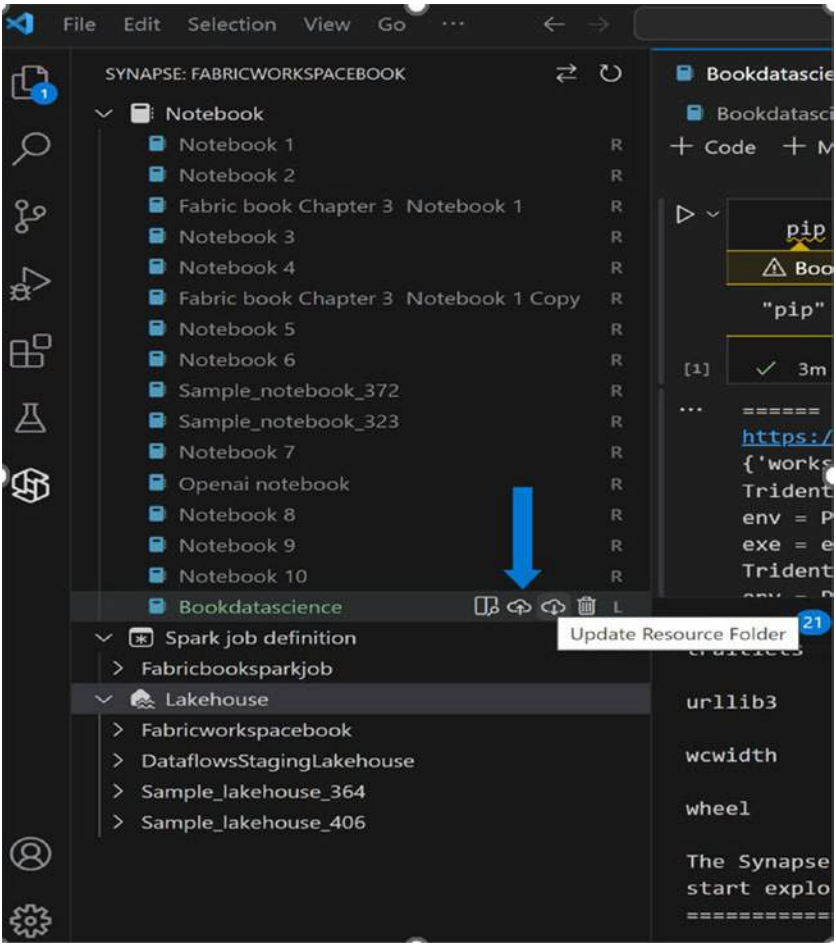


Figure 4-16. *The Fabric workspace integration*

Well, you are now familiar with both Fabric notebook-based development and VS Code development. In the next section, you will focus on model development related to the Fabric capabilities.

Developing the Model

In the following steps, you will focus on machine learning development related to Fabric. For academic purposes, we recommend reusing any of the sample dataset notebooks, as shown in Figure 4-17. In this section, we will refer to the "Book recommendation" sample notebook provided by Fabric to walk through the advanced features (the code sample is also available with our book).

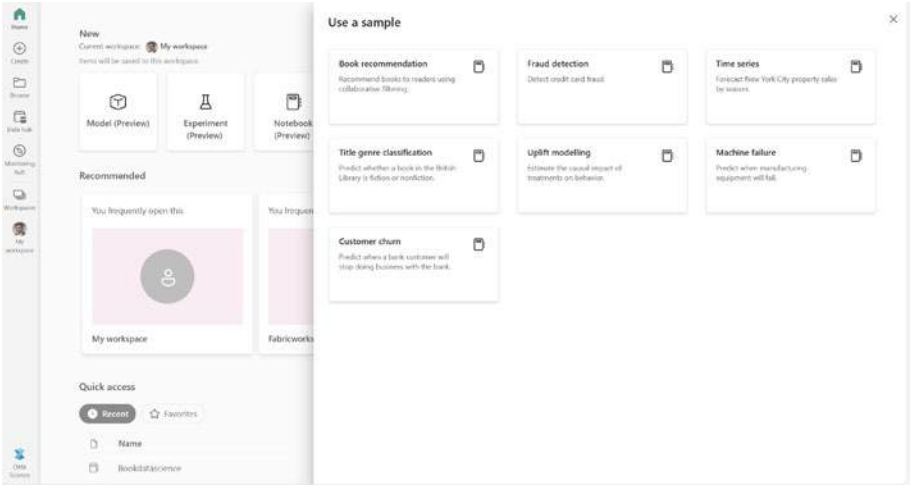


Figure 4-17. *Fabric-provided sample machine learning model*

By the way, what exactly is a recommendation system in machine learning? We can define it as follows:

> *A set of machine learning algorithm that deals with ranking or ratings of the products and eventually offers suggestions to users.*

You might have experienced personalized recommendation systems in your day-to-day life in different intelligent applications such as Netflix, Amazon, and Spotify. Building such a recommendation system needs the

right skill set, tooling, and domain knowledge. Figure 4-18 shows some machine learning modeling techniques.

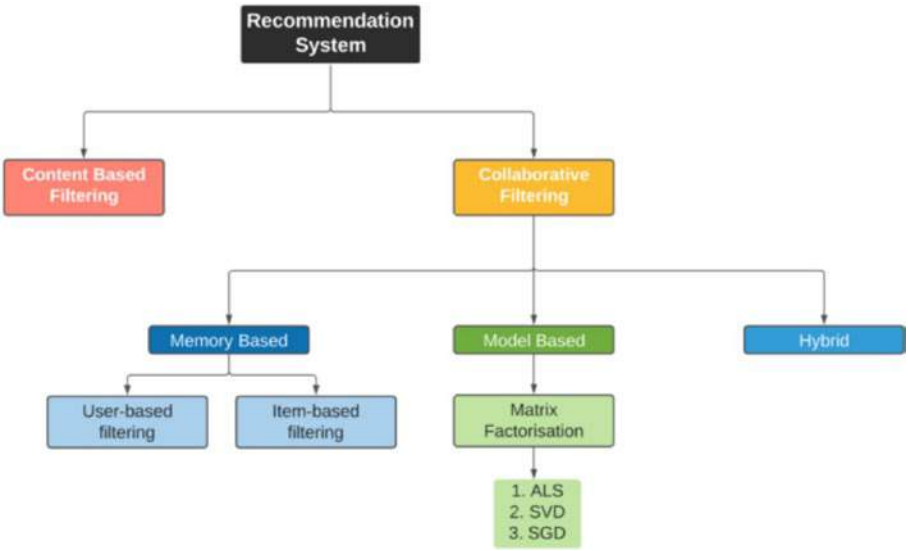


Figure 4-18. *Recommendation systems of notebook*

In this section, we are leveraging "Book recommendation," which recommends books to target users and leverages an ALS-based algorithm. The Fabric-provided "Book recommendation" notebook demonstrates how to create, evaluate, and deploy a recommendation system. This notebook executes the following steps:

1. Download the Users, Books and Book ratings dataset from the public storage account to the Fabric lakehouse default path.
2. **Set up MLflow-based experiment tracking.**
3. Read the dataset into a Spark dataframe.
4. Do exploratory data analysis (plotting top authors, visualizing most popular books).

5. Prepare the dataset with certain transformations.
6. Split the dataset for training and test purposes (80-20).
7. Import the machine learning libraries (regression evaluator/alternating least square).
8. Specify the training parameters of the model and build the recommendation system with ALS.
9. Tune the model with hyperparameters.
10. Evaluate the model.
11. **Experiment tracking with MLflow.**
12. **Register the model for tracking.**
13. Do the final predictions.

In the following section, we will focus on steps 2, 11, and 12. A data scientist gets extra miles by using these capabilities during operationalization of the machine learning models.

Setting Up Experiment Tracking and Registering the Model

The machine learning experimentation tracking capabilities help data scientists and machine learning engineers to maintain data science code versioning, parameter logging, metrics logging, and many more features. An experimentation in Fabric is a wrapper that controls and tracks all runs related to the machine learning experimentation. To set up experiment tracking, you need to use the following code snippet:

```
# Setup mlflow for experiment tracking
import mlflow
mlflow.set_experiment(EXPERIMENT_NAME)
```

In the "Book recommendation" notebook, you will tweak the previous code to use the new EXPERIMENT_NAME parameter value. This will create a new experiment in the Fabric workspace.

```
EXPERIMENT_NAME = "Fabric-book-aisample-recommendation" #
mlflow new experiment name appended with  Fabric-book here
# Setup mlflow for experiment tracking
import mlflow
mlflow.set_experiment(EXPERIMENT_NAME)
#mlflow.autolog(disable=True)  # disable mlflow autologging
```

As you run all the notebook cells, the entire experimentation will be logged under the new experiment name. In this case, the experimentation is logged inside Fabric-book-aisample-recommendation, as shown in Figure 4-19.

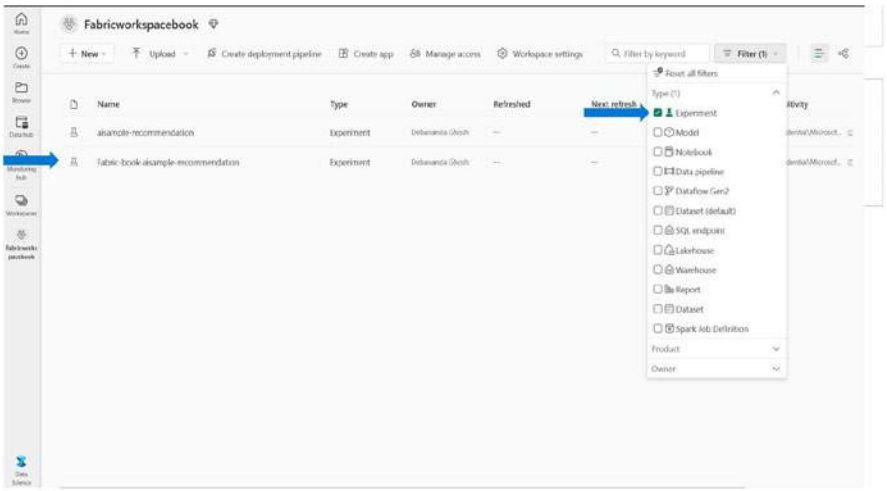


Figure 4-19. *Fabric item selection*

As you click the experiment name (here Fabric-book-aisample-recommendation), you get the experiment tracking canvas. This canvas logs the following metrics of the model experimentation:

- Run Name
- Start Date
- Duration
- Status
- Run ID
- Created by
- Source
- Experiment Name
- Model versions
- Run Metrics (R2, RMSE, MAE, Explained Variance)

Figure 4-20 shows the Fabric-book-aisample-recommendation experiment metrics canvas details.

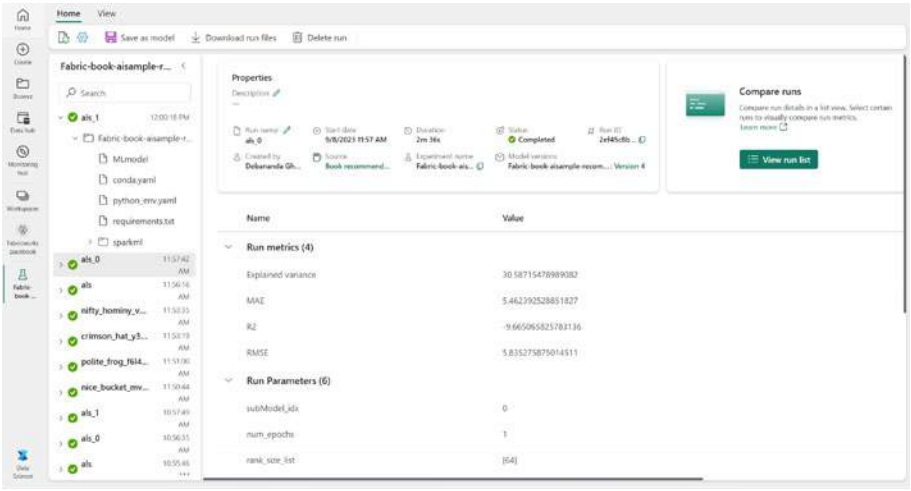


Figure 4-20. *Machine learning experiment comparison*

The Fabric workspace provides auto-logging capabilities to reduce a data scientist's development work and facilitate logging the parameters and metrics. The following code snippet is used to leverage `mlflow` auto-logging capabilities inside Fabric:

```
mlflow.autolog(
    log_input_examples=False,
    log_model_signatures=True,
    log_models=True,
    disable=False,
    exclusive=True,
    disable_for_unsupported_versions=True,
    silent=True)
```

When you run an experiment multiple times by changing the model parameters, it is essential that you study the model performance very carefully. The Fabric canvas provides a user interface to compare metrics.

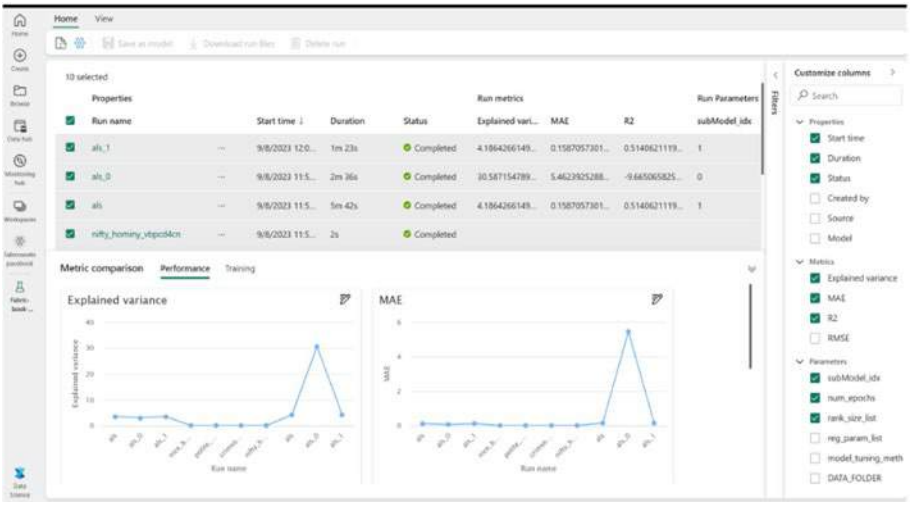


Figure 4-21. *Experiment comparison*

The following code snippet from the notebook logs/registers {EXPERIMENT_NAME}-alsmodel" as the model in the workspace.

```
# Log best model and related metrics and parameters to the
parent run
mlflow.spark.log_model(
    models.subModels[best_index],
    f"{EXPERIMENT_NAME}-alsmodel",
    signature=signature,
    registered_model_name=f"{EXPERIMENT_NAME}-alsmodel",
    dfs_tmpdir="Files/spark",
)
mlflow.log_metrics(best_metrics)
mlflow.log_params(
    {
        "subModel_idx": idx,
        "num_epochs": num_epochs,
        "rank_size_list": rank_size_list,
        "reg_param_list": reg_param_list,
        "model_tuning_method": model_tuning_method,
        "DATA_FOLDER": DATA_FOLDER,
    }
)
```

In our case, it's Fabric-book-aisample-recommendation-alsmodel, as shown in Figure 4-22.

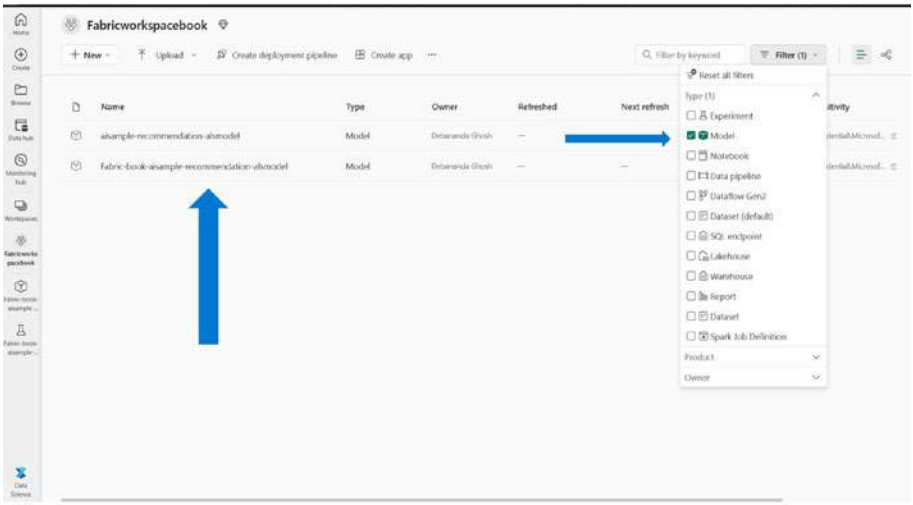


Figure 4-22. *Registered model inside workspace*

To compare the models, please refer to Figure 4-23. You need to select "Model list" and then select the Version check box. In Figure 4-23 we are comparing Version 5 and Version 6. On the right side you can select the necessary metrics and parameters that you want to compare.

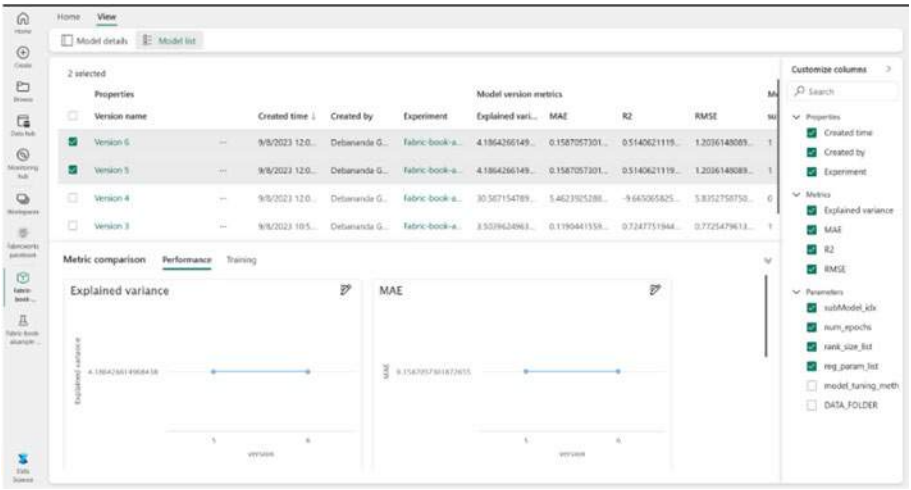


Figure 4-23. *Model performance comparison*

Now you have learned how to log the machine learning experimentation details and register the model. We recommend studying the provided notebook "Book recommendation" further to see the development patterns used inside the Fabric notebook.

Large Language Modeling and Azure OpenAI Integration

By now, every one of us has heard, explored, and delved into ChatGPT. With the rise of ChatGPT capabilities, large language modeling has also gained immense popularity. And what exactly is LLM? As per the Wiki definition,

> *A large language model (LLM) is a language model characterized by its large size. Their size is enabled by AI accelerators, which are able to process vast amounts of text data, mostly scraped from the Internet.*

Today's large language models help real-world problems by providing meaningful answers, extracting relevant information, and understanding context from vast information. Microsoft provides the Azure OpenAI service as an enterprise-grade OpenAI capability to leverage large language models. In this section, we will focus on the Microsoft Fabric capabilities integrated with the Azure OpenAI service.

Imagine you have a business use case where you want to build a knowledge portal. We want to feed the knowledge portal a huge number of PDF documents. As the outcome, you need to have a question-and-answer framework using the PDF knowledge.

Microsoft has provided a GitHub content address as a type of use-case solutioning, and you will study that for our understanding purpose of Fabric OpenAI integration piece, available here:

https://github.com/microsoft/SynapseML/blob/fa497f09b58a462a4ca47c14cbe4bfd12231f1b2/docs/Explore%20Algorithms/AI%20Services/Quickstart%20-%20Document%20Question%20and%20Answering%20with%20PDFs.ipynb

This notebook does the following. We need to focus on the bold highlighted code, which highlights invoking the OpenAI API:

1. Load the sample PDF documents into a Spark dataframe.
2. Read the documents using the Azure AI Document Intelligence in Azure AI Services.
3. Leverage Synapse ML to split the documents into chunks.
4. **Generate embeddings for the chunks using the SynapseML and Azure OpenAI services.**
5. Store the embeddings in a vector store using Azure Cognitive Search.

6. Search for the vector store to answer the user's question.
7. Retrieve the relevant document based on the user's question and provide the answer using Langchain.

This content leverages the Azure OpenAI API, the Synapse ML framework, and the Azure cognitive search capability. To integrate OpenAI, you need to execute the following code snippet. Note that you need to replace the endpoint and secret key to integrate with the Azure OpenAI service.

```
from pyspark.sql import SparkSession
from synapse.ml.core.platform import find_secret
# Fill in the following lines with your Azure service
information
aoai_service_name = "synapseml-openai"
aoai_endpoint = f"https://{aoai_service_name}.openai.azure.com/"
aoai_key = "openai-api-key"
aoai_deployment_name_embeddings = "text-embedding-ada-002"
aoai_deployment_name_query = "text-davinci-003"
aoai_model_name_query = "text-davinci-003"
```

Note that Azure OpenAI is a separate service (and subject area), beyond the scope of this book. At the time of writing, Aure OpenAI needs separate access through this form: https://azure.microsoft.com/en-us/products/ai-services/openai-service. To deploy the Azure OpenAI service, please follow this documentation. Also, read the charging and pricing part carefully while using the API:

https://learn.microsoft.com/en-us/azure/ai-services/openai/how-to/create-resource?pivots=web-portal

To get the Azure OpenAI endpoint and key, you need to go to the Azure OpenAI service and retrieve the key and endpoint, as shown in Figure 4-24.

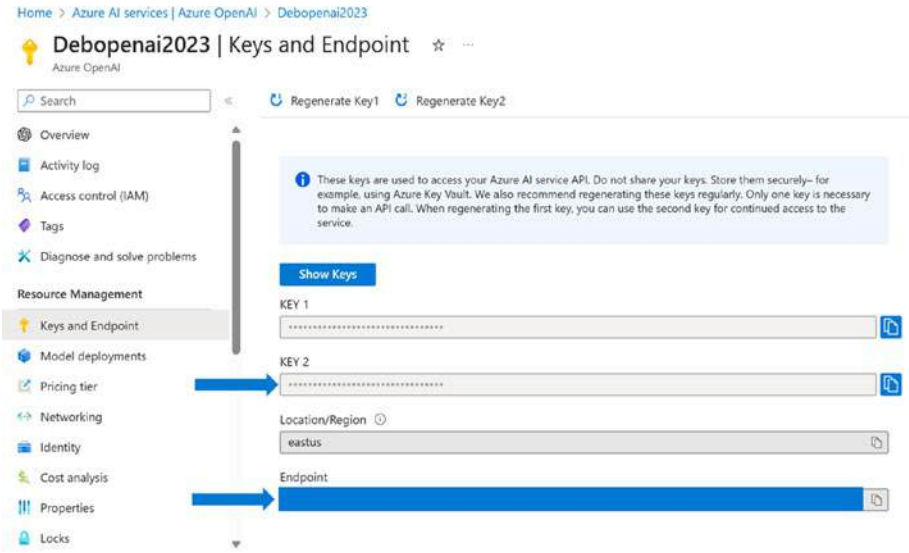


Figure 4-24. *Azure OpenAI endpoint and key retrieval*

Once you replace the API and endpoint, the Spark notebook in Fabric is now ready to invoke the OpenAI capability as an API. Likewise, you need to change the AI service and the cognitive search endpoint to leverage this notebook.

Summary

in this chapter, we introduced the data science workflow. You learned how to leverage the Microsoft Fabric data science persona and how to use the Fabric workspace for preparing and wrangling data. You also learned how to do machine learning operationalization using experimentation logging and model registration. Finally, you learned how to integrate with the Azure OpenAI service. In a nutshell, we covered the topics of the Fabric ecosystem that are shown in Figure 4-25. In the next chapter, you will move on to data warehouses.

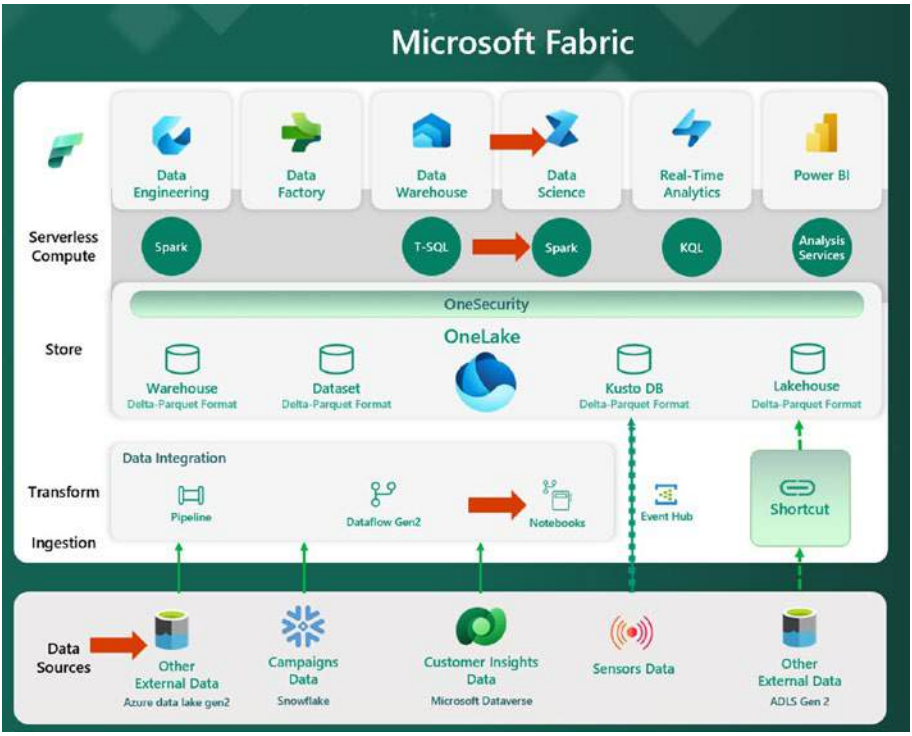


Figure 4-25. *Fabric data science recap*

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric Data science overview:** `https://learn.microsoft.com/en-us/fabric/data-science/data-science-overview`
- **VS Code extension:** `https://learn.microsoft.com/en-us/fabric/data-engineering/author-notebook-with-vs-code`

- **Sample data science notebooks:** https://github.com/microsoft/fabric-samples/tree/main/docs-samples/data-science/data-science-tutorial
- **Fabric and OpenAI integration:** https://blog.fabric.microsoft.com/en-us/blog/unleashing-the-power-of-synapseml-and-microsoft-fabric-a-guide-to-qa-on-pdf-documents-2?ft=All
- **Document summarization use case:** https://blog.fabric.microsoft.com/en-us/blog/harness-the-power-of-langchain-in-microsoft-fabric-for-advanced-document-summarization/

CHAPTER 5

Data Warehousing in Microsoft Fabric

We are living in an era of advanced lakehouse platforms. Lakehouses have distinctive capabilities when compared to data warehouses and data lakes. Previously, to build the right data architecture, data professionals had to decide whether they wanted to use a data warehouse or a data lake or a combination of both. For each workload, they would need to consider the data volume, transaction logs, and business intelligence needed before choosing a platform.

In Chapter 1, we discussed that lakehouses come with advantages over both data lakes and data warehouses. Hence, most developers want to modernize their analytics projects by using a lakehouse. Microsoft Fabric embraces lakehouses as well.

So, why are we discussing data warehouses (DWs) again in this chapter?

Because of several factors, such as developers' skill sets, ease of migration from legacy data warehouses, and more, developers want a DW-like development experience, and Microsoft Fabric provides end-to-end data warehousing capabilities. Thus, data professionals can leverage both lakehouse development and data warehouse development in the same Fabric workspace. This simplifies the architecture and is cost effective. In this chapter, we will cover the Synapse data warehouse capability of Microsoft Fabric, as shown in Figure 5-1.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_5

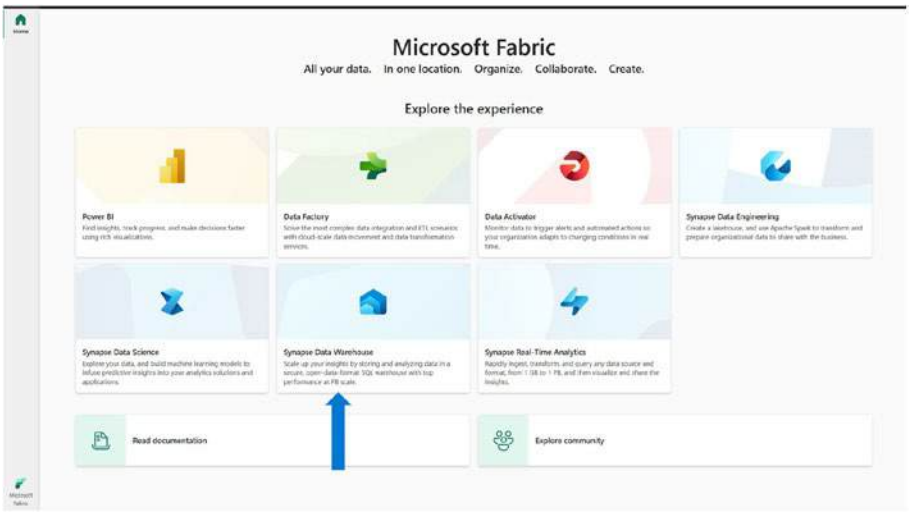


Figure 5-1. *Synapse Data Warehouse canvas in the Fabric workspace*

Specifically, you will learn about the following topics in this chapter:

- Fabric data warehouse concept
- How to provision a Fabric data warehouse
- Data warehouse development tasks
- Cross-database and virtual data warehouse queries
- SQL queries and SQL Server Management Studio connectivity
- Advanced DW capabilities such as workload management and performance tuning
- Integration with Power BI and Microsoft Office tools
- How to monitor a Fabric DW pipeline

Fabric Data Warehouse Introduction

The Fabric data warehouse simplifies enterprise-grade data warehousing workloads for every user by providing a Microsoft Office 365–like user interface. This SaaS experience makes adoption easier for any skill level, from beginning developer to professional developer (such as a data engineer). Table 5-1 shows the different features of data warehouses versus lakehouses in the Fabric workspace.

Table 5-1. *Data Warehouse vs. Lakehouse*

Criteria	Fabric Data Warehouse	Fabric Lakehouse/SQL Endpoint
Developer skill set	SQL	PySpark, SparkR, Scala, Spark SQL
Development interface	SQL editor/scripts	Spark notebook, job definition
Data format	Structured	Structured, semi-structured, and unstructured
Data organized	Schema, table based	Folders, file based
Multitable transaction	Appropriate for multitable transactions	SQL endpoint supports read-only multitable transaction

The following data warehouse capabilities of Fabric make it competitive as an end-to-end cloud data warehouse:

- Simple and intuitive user experience for users and professional developers
- Fully separate computation and storage layers
- Stores data in OneLake in an open delta format and thus provides openness

- Ease of data management using a SaaS experience
- Fully integrated with all Fabric and other Azure analytics workloads out of the box, with an integrated semantic layer
- Loads and transforms data with high-performance scalability
- Full multitable transactional guarantees provided by the SQL engine
- Visual query development experience; choice of no-code, low-code, or T-SQL for transformations
- Virtual warehouses with cross-database querying
- Enterprise-ready platform with end-to-end performance and usage visibility, with built-in governance and security

To learn about such capabilities in more detail, you will load a dataset in the Fabric workspace in this chapter. As a prerequisite of loading, you need to create a Fabric data warehouse serverless computation. We will focus on that in the next section.

Provisioning a Data Warehouse

In this section, you will learn how to provision a new warehouse in the Fabric workspace. Because of the SaaSification nature of Fabric provisioning components is quite simple. We will use the portal to provision a Fabric DW computation.

First select "Data warehouse" on the bottom left, then click "New warehouse," and finally click "Create" to create Fabric Serverless DW computation. This will create a new warehouse (Fabricbookdw in this scenario), as shown in Figure 5-2.

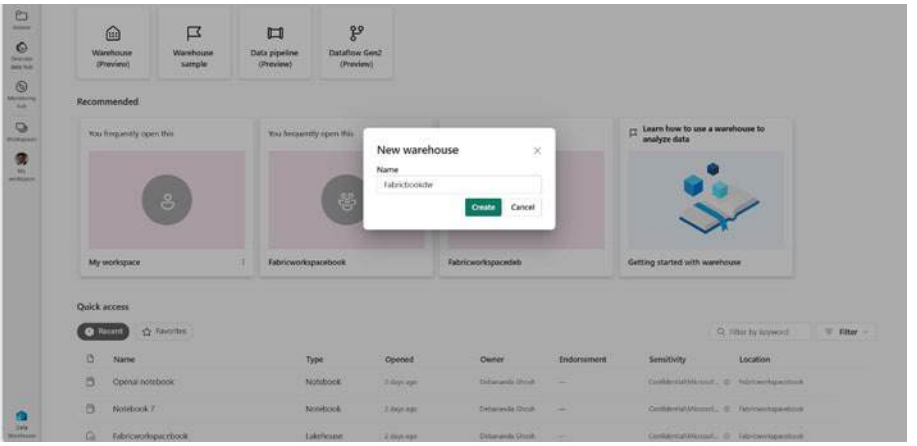


Figure 5-2. *Creating a data warehouse*

Figure 5-3 shows the data warehouse "explorer" user interface in the Fabric workspace. Fabric data warehouse supports full T-SQL (transactional SQL) like any data warehouse appliance. Unlike SQL Analytics endpoint you have full control of the data warehouse which includes create, load, transform operations. In the data warehouse explorer, you will be able to see the ingested data in database, schema, and table formats. Also, you can run SQL and build visual queries in this canvas.

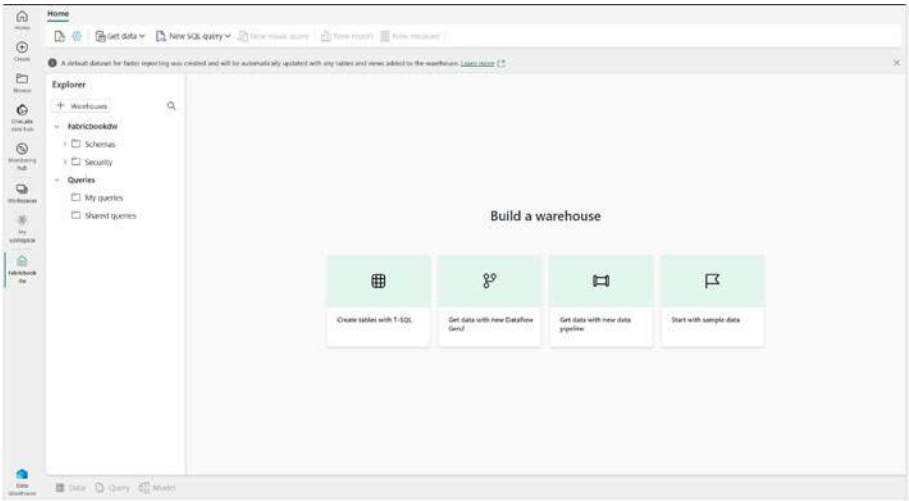


Figure 5-3. *Data warehouse explorer*

To load the data in the data warehouse, you can leverage T-SQL statements, Data Flow Gen2, or data pipelines. In the following section, you will use the "Get data with a new data pipeline" capability to ingest some sample data into the Fabric DW.

Ingesting Data into a Data Warehouse

In this section, you will ingest a sample dataset known as "Retail data mode from Wide World Importers" like you did in Chapter 3. Note that this "Get data with new data pipeline" feature leverages Azure Data Factory features from the pipeline canvas.

Let's perform the following steps to ingest the sample data in the data warehouse:

1. Click "Data pipeline" and give it a name in the "New pipeline" box. In this case, we are using fabricdwpipeline, as shown in Figure 5-4.

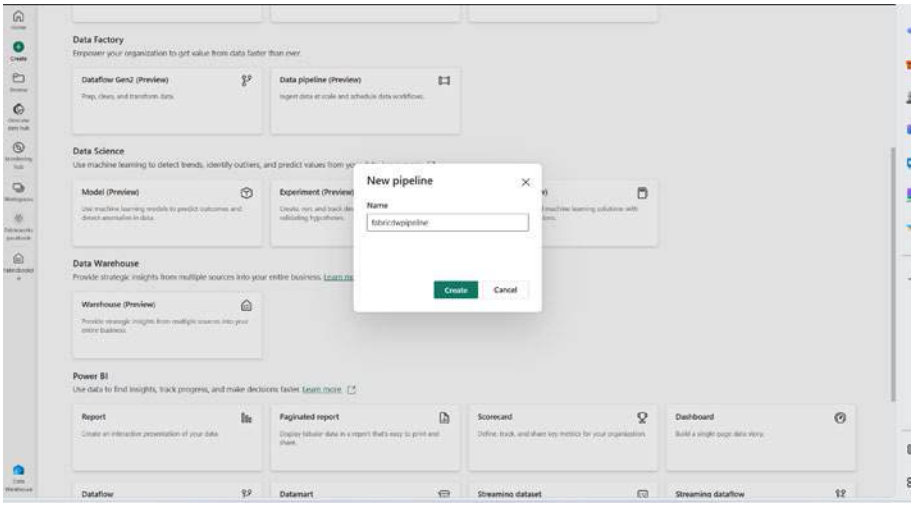


Figure 5-4. *Creating fabricdwpipeline*

2. Select "Copy data" and then "Sample data," as shown in Figure 5-5.

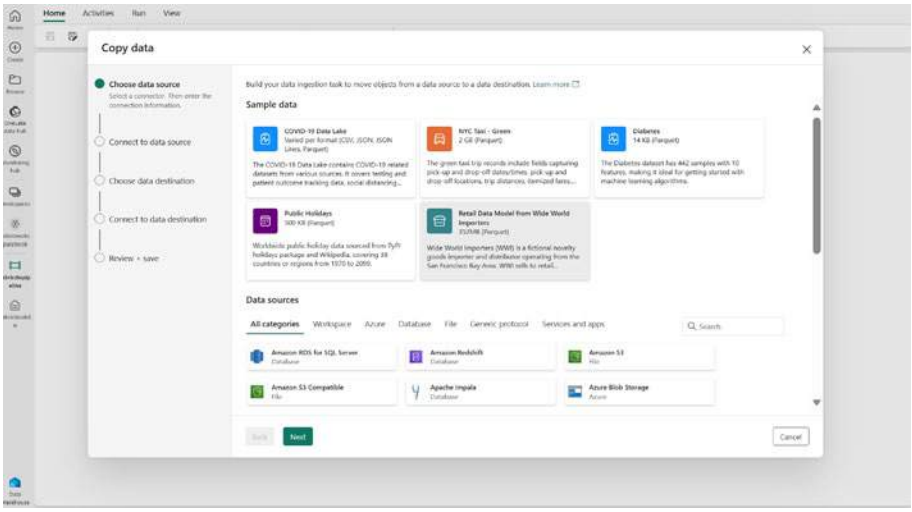


Figure 5-5. *Copying data*

3. Continue to the "Choose data destinations" tab and then click Data Warehouse, as shown in Figure 5-6.

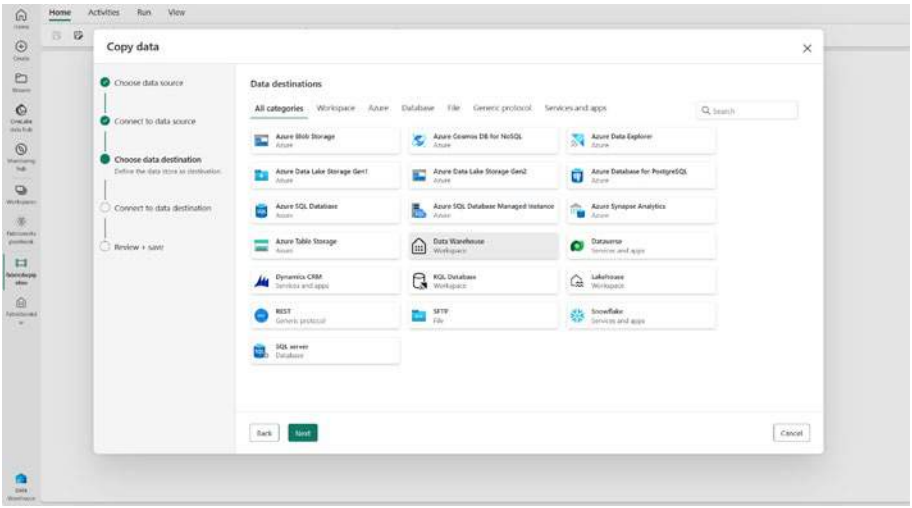


Figure 5-6. *Selecting the Data Warehouse tab*

4. Continue to click Next and follow the steps like you did in Chapter 3 to complete the ingestion. Note that you can ingest multiple tables in the same step, as shown in Figure 5-7. To initiate bulk ingestion, select the database name (Retail Data Model from Wide World Importers in this case), and all the tables will be selected as part of the bulk copy activity.

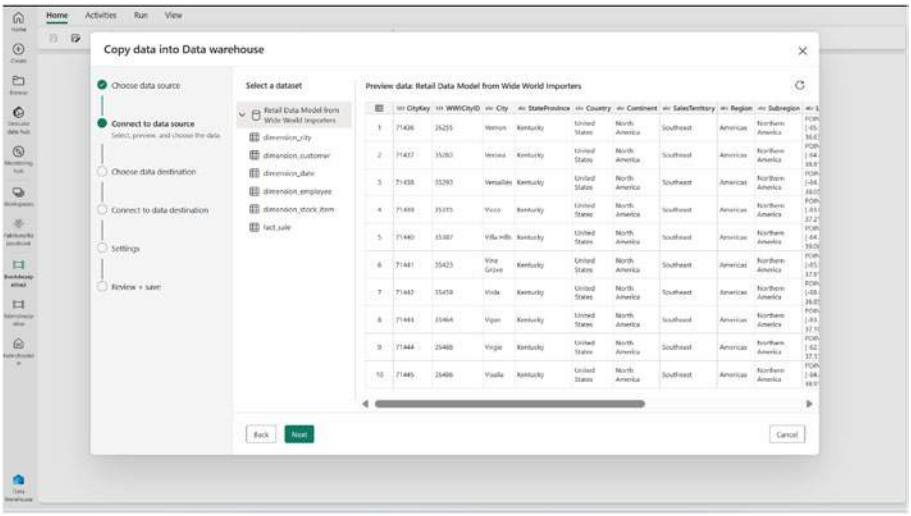


Figure 5-7. *Ingesting multiple tables*

Figure 5-8 shows that the parallel copy task is triggered after completing the previous steps. That step involved copying data into multiple data warehouse tables. Note that the activity status will change as the job status progresses.

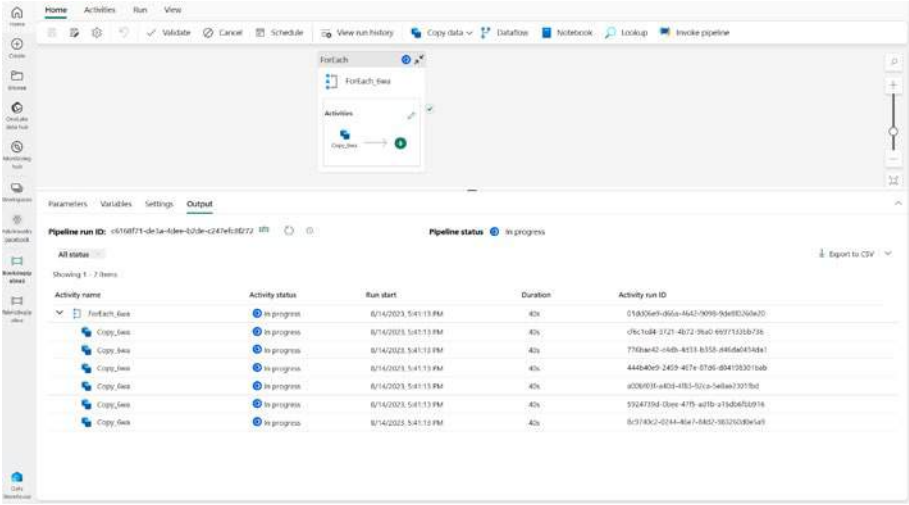


Figure 5-8. *Fabric DW pipeline status*

Once these files are ingested, they are visible as data warehouse tables in the explorer view. Figure 5-9 shows the tables under Warehouses/Schemas/dbo/Tables in the OneLake explorer.

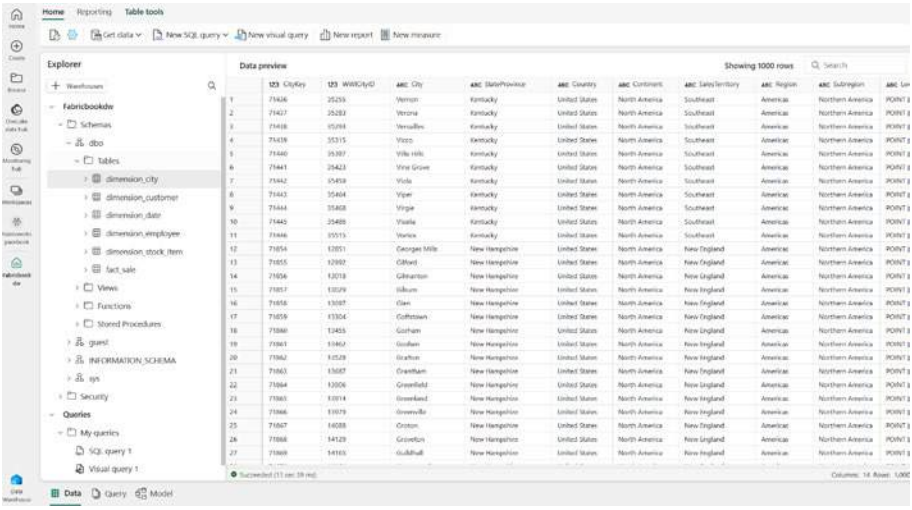


Figure 5-9. *Viewing the tables*

The OneLake client provides a similar experience to Microsoft Office 365 OneDrive. Note that you can find the underlying raw Parquet files of the table data under the OneLake name. In this case, it's `C:\Users\xxxxxx\OneLake - Microsoft\Fabricworkspacedeb\Debfabricwarehouse.Datawarehouse\Files`, as shown in Figure 5-10.

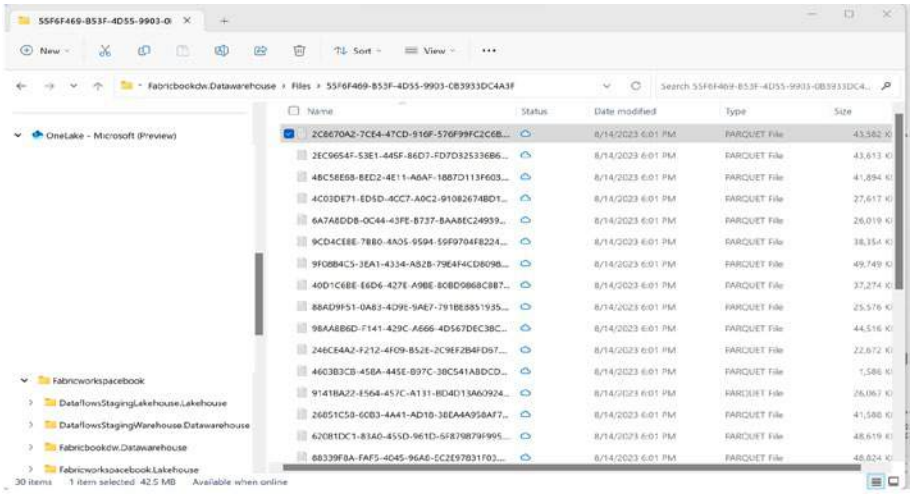


Figure 5-10. *Viewing the data warehouse file using the OneLake explorer*

Now since some data is ingested inside the data warehouse, let's explore the developer tooling and features available.

Data Warehouse Development

Because of the nature of SaaS, the Fabric DW provides both code-free and with-code tooling. Using the visual query graphic user interface (GUI) tools, you can drag and drop tables and build queries in the canvas without writing any code. In addition, for professional developers, there are SQL query interfaces and SQL client connectivity mechanisms. You can also leverage Transact-SQL (T-SQL) using the Fabric SQL query editor or SQL clients like SQL Server Management Studio (SSMS). In the following sections, we will go through both experiences.

Visual Interface Transformation

Fabric provides a visual query editor for those without coding experience. To leverage this code-free query development mechanism, click New Visual Query in the Home ribbon within the Explorer canvas, as shown in Figure 5-11. You can drag and drop a table from the left side inside the right pane. In this case, we have dragged and dropped the prepopulated table `dimension_customer`.

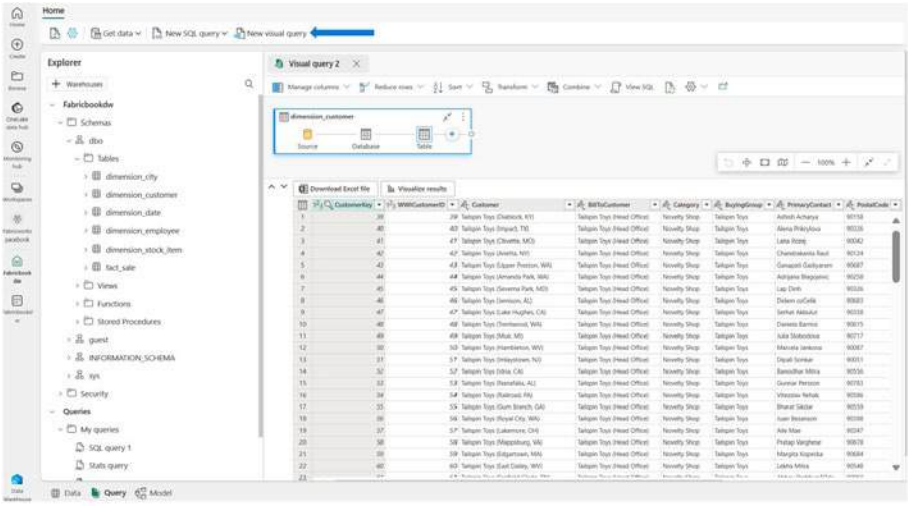


Figure 5-11. *Visual query transformation*

Now you can use the visual query editor for development using the prepopulated table. The visual query editor's top middle ribbon provides the following GUI capabilities:

- **Manage columns:** Keep and remove columns of a prepopulated table.
- **Reduce rows:** Remove certain rows such as top rows and bottom rows, and use advanced filter criterion without any coding.

- **Sort:** Sort the dataset in ascending or descending order.
- **Transform:** Use a visual interface to perform "group by" SQL statements.
- **Combine:** Use a "join" SQL statement.

It is recommended for users to play around with the options on this user-friendly ribbon. You can do simple column profiling using the Data view and then click "Enable column profile, show column quality details and other details" to check the column-related data demographics. Figure 5-12 illustrates these features in the visual query editor.

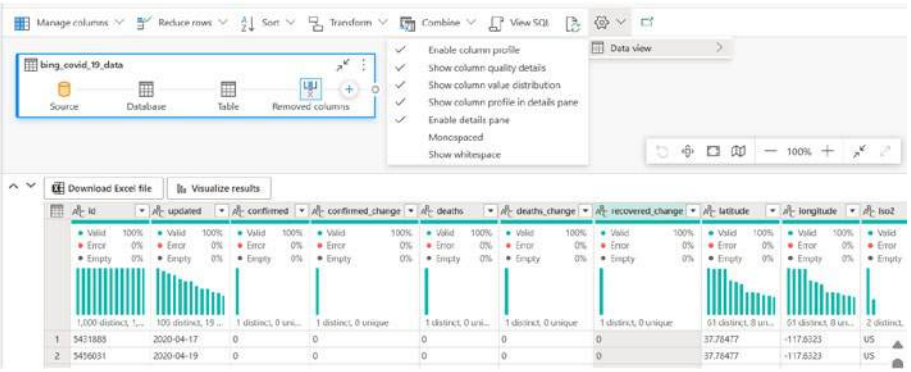


Figure 5-12. *Column profiling features*

To leverage the advanced visual editor, you need to click the right-side top of the ribbon. This will open the Power Query editor, as shown in Figure 5-13. The Power Query editor is used in Power BI Desktop as well, and you can learn more about Power Query editor at `https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-query-overview#the-query-ribbon`.

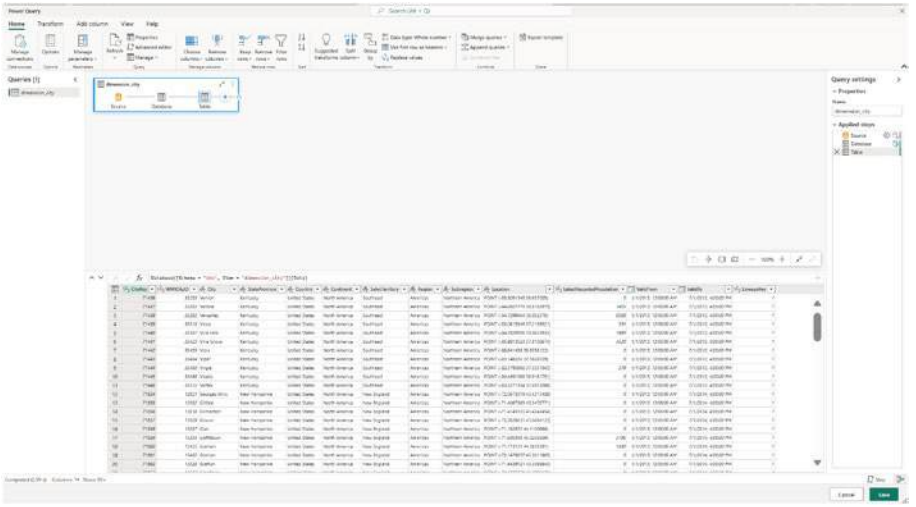


Figure 5-13. *Power Query editor*

Let's close the Power Query editor and continue working in the original workspace canvas. When you click the three dots on the right, the canvas expands with diverse options, as shown in Figure 5-14.

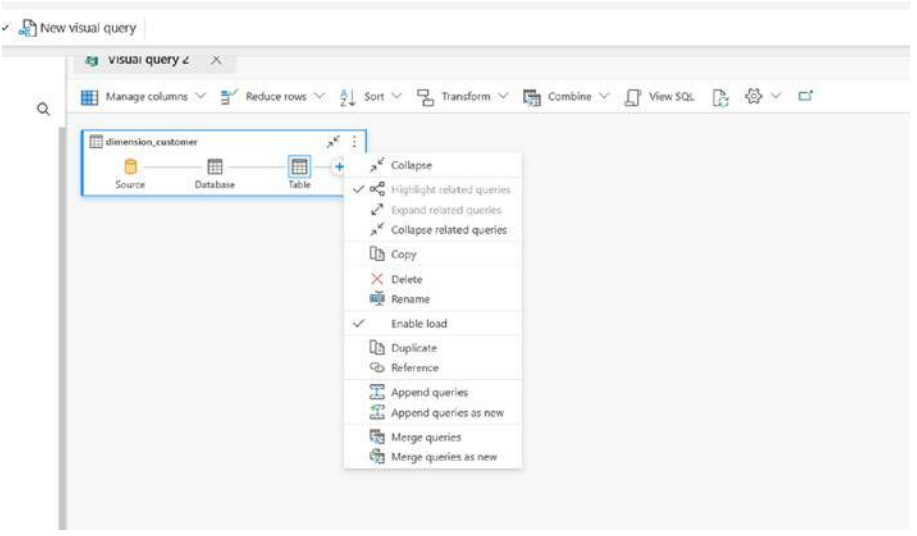


Figure 5-14. *Visual editor canvas*

Let's now attempt to join two tables using a visual query. You must drag over another table (here fact_sale) to do this. Then you can click Merge to perform a join step. As the Merge canvas appears, you select the tables that you need to merge/join. Select the join kind (left outer join, right outer join, inner join) and merge the key names, as shown in Figure 5-15.

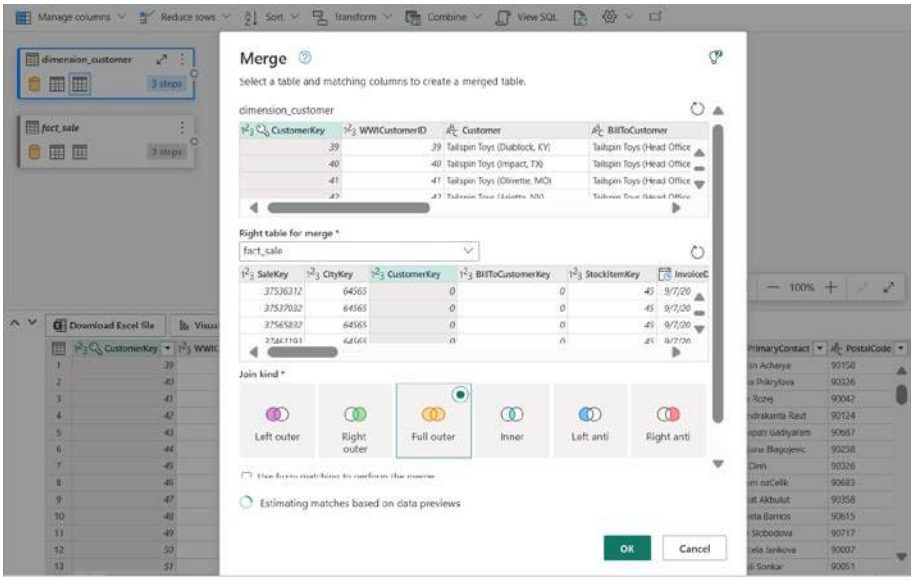


Figure 5-15. *Joining two tables*

As you click OK in the canvas, the visual query will produce the relevant output. You can see (and edit) the underlying SQL as well as use the View SQL feature. Note that this step joins two tables in the same database. In the next step, you will learn how to join two tables in different databases (and outside the current Fabric data warehouse).

Cross-Database and Virtual Datawarehouse Queries

The Fabric DW provides table joining features across different databases. This database can be inside a data warehouse or inside a lakehouse within the Fabric workspace. To create a cross-database query, you need to add a warehouse or lakehouse SQL endpoint by clicking the +Warehouse button, as shown in Figure 5-16.

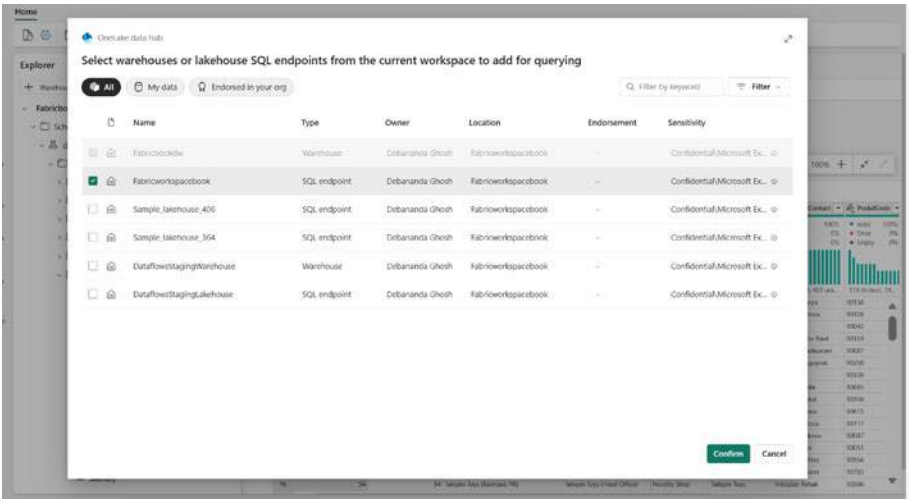


Figure 5-16. *Adding a new warehouse inside the datawarehouse explorer*

After you select the database, you will be able to see all the tables inside that database within the Explorer canvas. After you select the lakehouse SQL endpoint (Fabricworkspacebook), you can see the database inside the explorer, as shown in Figure 5-17. You can now execute the visual query in same way as explained in the previous section to join tables across this database. Thus cross database querying is now available as one of the key offerings of the Fabric SaaS foundation.

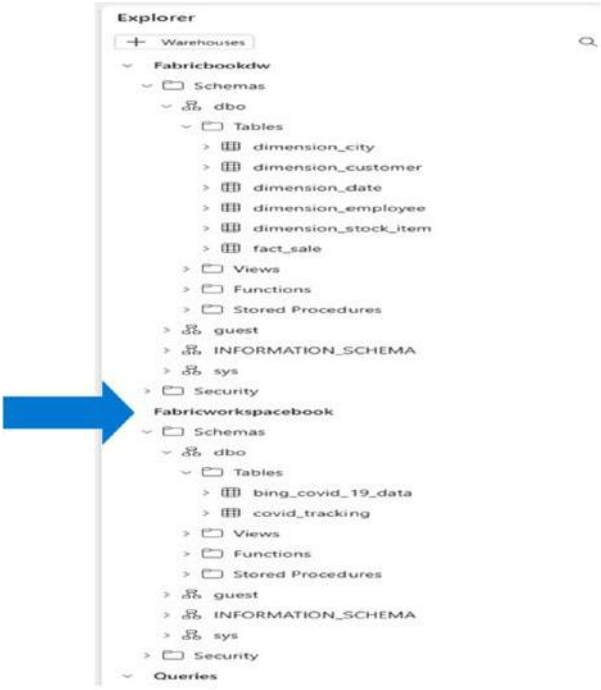


Figure 5-17. *Virtual data warehouse*

We have discussed GUI-based data transformation. In the next section, we will move on to the SQL development experience and tooling.

SQL Query

For advanced T-SQL developers and professional data engineers, you need to use a SQL editor. To open the SQL query editor in the Fabric workspace, you need to click New SQL Query beside New Visual Query inside the ribbon, as shown in Figure 5-18. Subsequently, the SQL query editor appears for our T-SQL development purposes. We can right-click and use features such as auto-populate SQL scripts (for example, select New SQL Query ➤ Select Top100). Alternately, you can start writing T-SQL scripts from scratch in this editor.

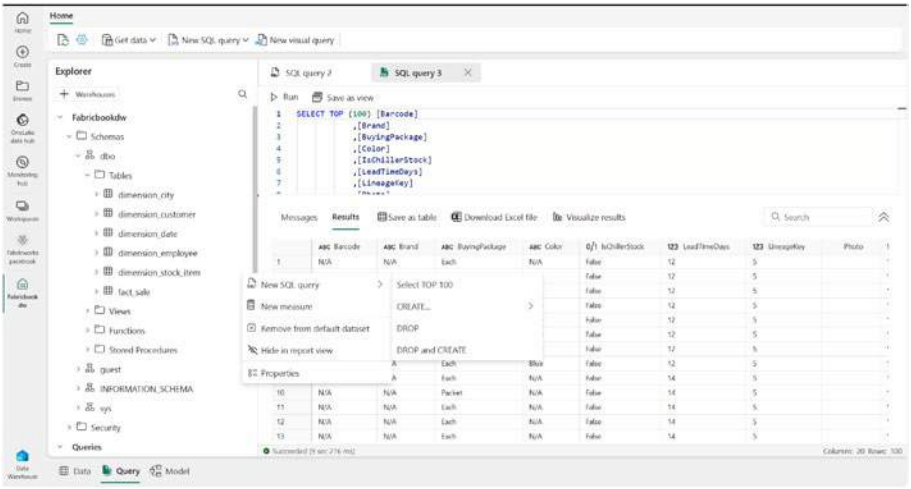


Figure 5-18. *SQL editor*

In the next section, we will focus on how SQL experts can use SQL desktop clients using Microsoft Fabric DW endpoint.

SQL Server Management Studio Connectivity

In this section, you will learn how to connect SQL Server Management Studio with the Fabric DW and subsequently execute a SQL statement. To download SQL Server Management Studio, go to `https://learn.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms?view=sql-server-ver16`. As a prerequisite of SSMS connectivity, you need to get the SQL connection string from the Fabric workspace.

To get the SQL connectivity string, you need to perform the following steps:

1. Click the Settings icon (below the Home tab).
2. Once you click the Settings icon, the data warehouse information appears, as shown in Figure 5-19.
3. Copy the SQL connection string for connectivity from SSMS.

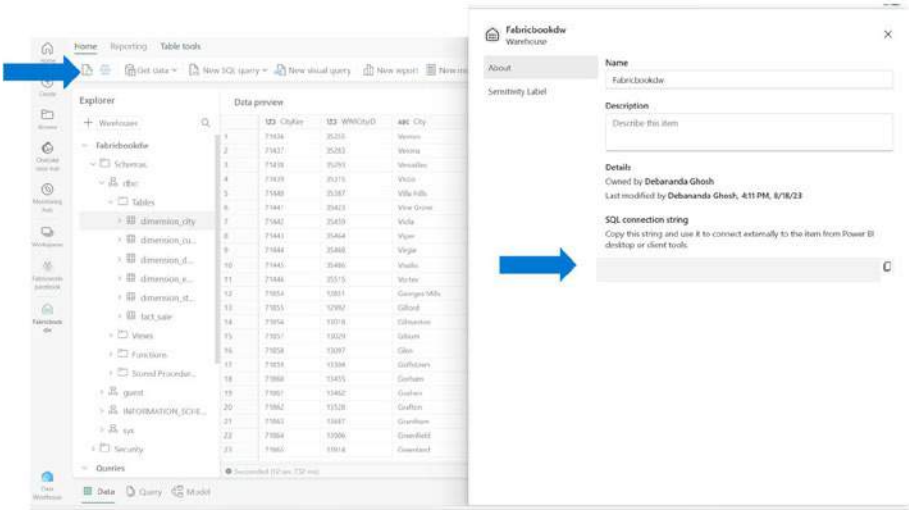


Figure 5-19. *SQL connection string connectivity*

Perform the following steps to connect with SQL Server Management Studio (SSMS):

1. Open the SSMS tool and click the File tab in the top-left corner.
2. Click Connect Object Explorer.

3. Provide the connection string in the server's name. We copied it in the previous steps; it should be in the format *xxxyyyyyyyyyyyyxz.msit-datawarehouse.pbidedicated.windows.net*.

4. Change the username to the organization user ID (the same domain used for the Fabric tenant and user ID who has access to the data warehouse), as shown in Figure 5-20.

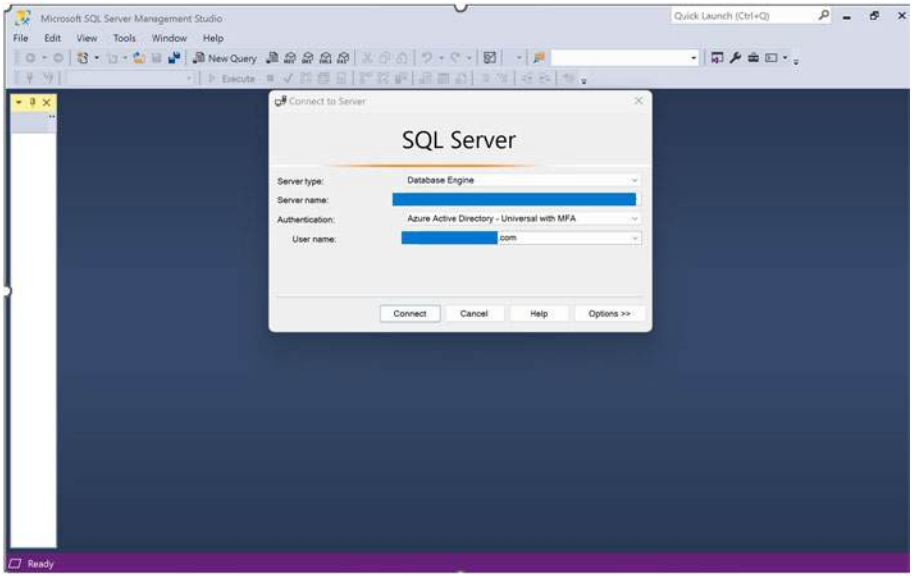


Figure 5-20. *SQL Server Management Studio connectivity*

Once you perform the preceding steps, the workspace databases appear inside Object Explorer. You can now run the following scripts to explore the database and assess the SSMS connectivity with Fabric:

```
SELECT TOP (100) [Barcode]
            ,[Brand]
            ,[BuyingPackage]
            ,[Color]
            ,[IsChillerStock]
            ,[LeadTimeDays]
            ,[LineageKey]
            ,[Photo]
            ,[QuantityPerOuter]
            ,[RecommendedRetailPrice]
            ,[SellingPackage]
            ,[Size]
            ,[StockItem]
            ,[StockItemKey]
            ,[TaxRate]
            ,[TypicalWeightPerUnit]
            ,[UnitPrice]
            ,[ValidFrom]
            ,[ValidTo]
            ,[WWIStockItemID]
FROM [Fabricbookdw].[dbo].[dimension_stock_item]
```

Figure 5-21 shows the outcome of executing the previous code snippet inside SSMS Studio.

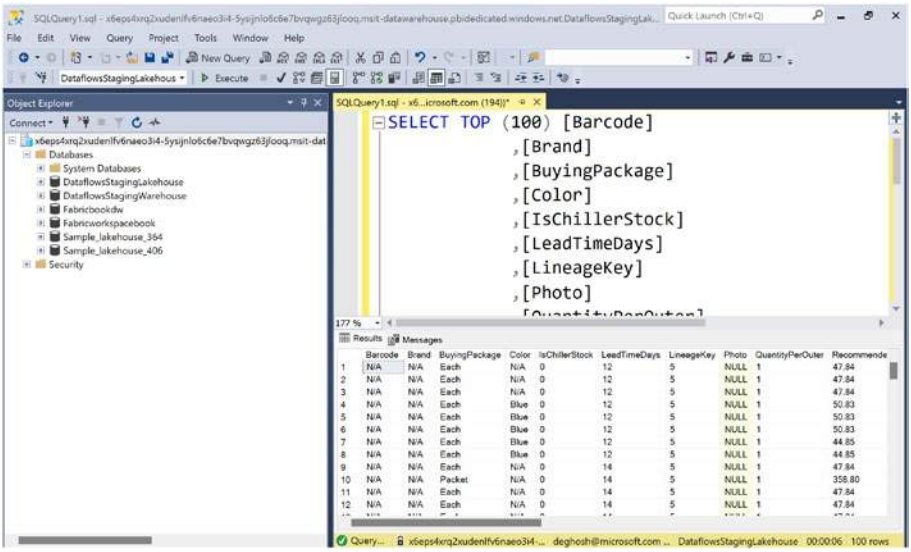


Figure 5-21. *T-SQL development inside SSMS*

Now T-SQL developers can leverage the SSMS tool features for advanced DW SQL development. In the next section, we will discuss a few advanced topics related to data warehouses.

Advanced DW Capabilities

In this section, we will focus on how the underlying technology of Fabric serverless DW computation works. Also, we will focus on other topics such as workload management and performance optimization. Note that the lakehouse SQL endpoint (covered in Chapter 3) and the SQL DW endpoint share the same architecture. Figure 5-22 explains the SQL endpoint front and SQL DW back-end architecture. When consumers (the T-SQL query) leverage the data warehouse or SQL endpoint, the SQL front-end computation engine works on query optimization. The query plan of the SQL statement is generated by the Distributed Query Processing (DQP) component of Fabric DW. The query plan executes multiple smaller

queries (known as *tasks*) and distributes them in the back-end compute pool. Based on the nature of the queries, tasks will read from OneLake and join the data with other tasks and then return the results to the SQL front end or write them back to OneLake.

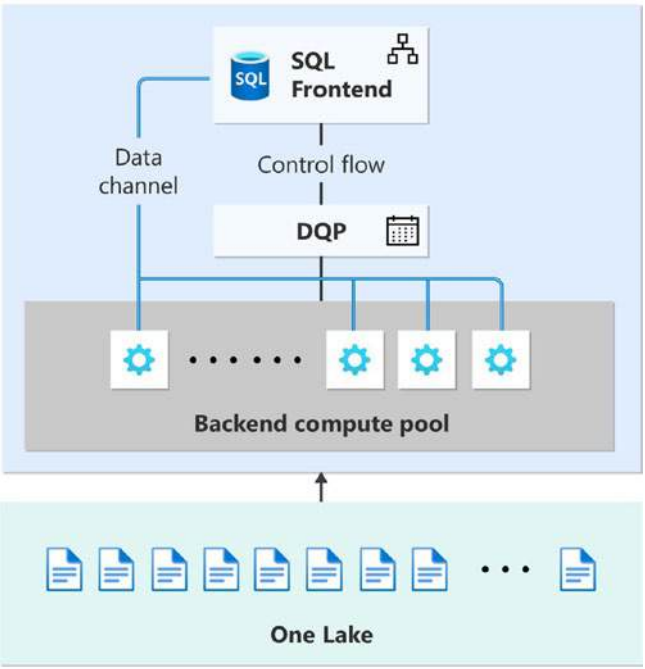


Figure 5-22. *SQL compute pool architecture*

To summarize, Fabric SQL queries execute the following tasks:

- Fetch metadata
- Simplify queries
- Execute automated stats
- Estimate costs and optimize queries based on the query cost
- Plan and search space size

Workload Management

The Fabric DW offers autonomous workload management; hence, this is no longer a developer's task. The Fabric DW engine automatically detects the SQL statements and isolates the capacity based on the SQL type. The Fabric DW can isolate the ingestion workload and queries; thus, isolated computing resources provide a predictable performance. Also, the SQL back end compute pool can scale up or scale down autonomously based on the resources needed by the query. Because of this fast-provisioning capability of back-end pools, the Fabric DW serverless computation has an online scaling capability. Figure 5-23 depicts the query and ingestion isolation.



Figure 5-23. *Fabric computation isolation*

Automated Multitiered Coaching and Query Optimization

The Fabric DW provides a few built-in query optimization techniques. There are a few caching mechanisms available by default in multiple layers of the compute pool.

- Result set caching in the SQL front-end layer
- In-memory catching
- SSD caching in the OneLake layer

Data warehouse computation leverages the power of the SQL Server Query Optimizer (QO), Power BI Veripaq analyzer, and the Polaris capability to optimize T-SQL queries.

- The SQL server QO optimizer provides a query optimization capability.
- Vertipaq provides fast columnar query processing, which is inherited from Power BI and the Analysis server.
- The Polaris engine provides a petabyte-scale distributed processing capability (see https://www.microsoft.com/en-us/research/publication/polaris-the-distributed-sql-engine-in-azure-synapse/).

For any SQL statement query plan, each step produces efficient resource consumption techniques and optimized node usage. Subsequently, automatic statistics are generated at query time. Note that users can manually create statistics. The following are sample code snippets for creating and checking table statistics:

```
--To create statistics
CREATE STATISTICS  DimCustomer_Customer_stats_book
ON  [Fabricbookdw].[dbo].[dimension_customer](CustomerKey) WITH
FULLSCAN;

-To check statistics created
DBCC SHOW_STATISTICS ("dimension_customer", "DimCustomer_
Customer_stats_book");

-To check statistics created (Automatically or manual)

select
    object_name(s.object_id) AS [object_name],
```

```
    c.name AS [column_name],
    s.name AS [stats_name],
    s.stats_id,
    STATS_DATE(s.object_id, s.stats_id) AS [stats_update_date],
    s.auto_created,
    s.user_created,
    s.stats_generation_method_desc
FROM sys.stats AS s
INNER JOIN sys.objects AS o
ON o.object_id = s.object_id
INNER JOIN sys.stats_columns AS sc
ON s.object_id = sc.object_id
AND s.stats_id = sc.stats_id
INNER JOIN sys.columns AS c
ON sc.object_id = c.object_id
AND c.column_id = sc.column_id
WHERE o.type = “U” -- Only check for stats on user-tables
    AND s.auto_created = 1
    AND o.name = “<YOUR_TABLE_NAME>'
ORDER BY object_name, column_name;
```

You can also leverage `sys.stats` to understand multiple types of stats available in Fabric DW tables.

Fabric DW Transaction Support

The Microsoft Fabric DW supports transaction capabilities via snapshot isolation. For example, you can commit inserts and changes to multiples tables. If you’re changing details in sales-related information that affects five tables, you can group those changes into a single transaction. This is

a common best practice, and it means that when those tables are queried, either they all have the changes or none of them do. Table 5-2 shows a mapping between statement types and locks.

Table 5-2. *Fabric SQL Statement and Native Lock Mapping*

Fabric SQL Statement	Locks
SELECT	Schema-Stability (Sch-S)
INSERT	Intent Exclusive (IX)
DELETE	Intent Exclusive (IX)
UPDATE	Intent Exclusive (IX)
COPY INTO	Intent Exclusive (IX)
DDL	Schema-Modification (Sch-M)

Data Models with Fabric DWs

Building a good data model and relationship definition is essential for a successful DW project. Logical data modeling (LDM) is fundamental before you start any analytics use case. Fabric helps you do data modeling inside the workspace. The Model view helps to define the logical and physical relationships of entities. To view or create referential integrity, you need to go to the Model view section, as shown in Figure 5-24.

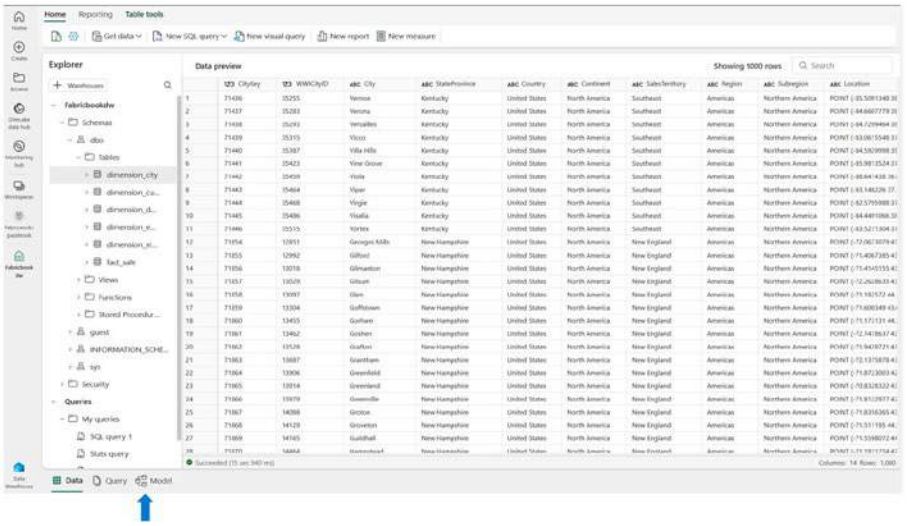


Figure 5-24. *Data model in Fabric DW*

To create a relationship, drag the Customer Key column from `fact_Sale` to the customerKey of `dimension_customer`, as shown in Figure 5-25.

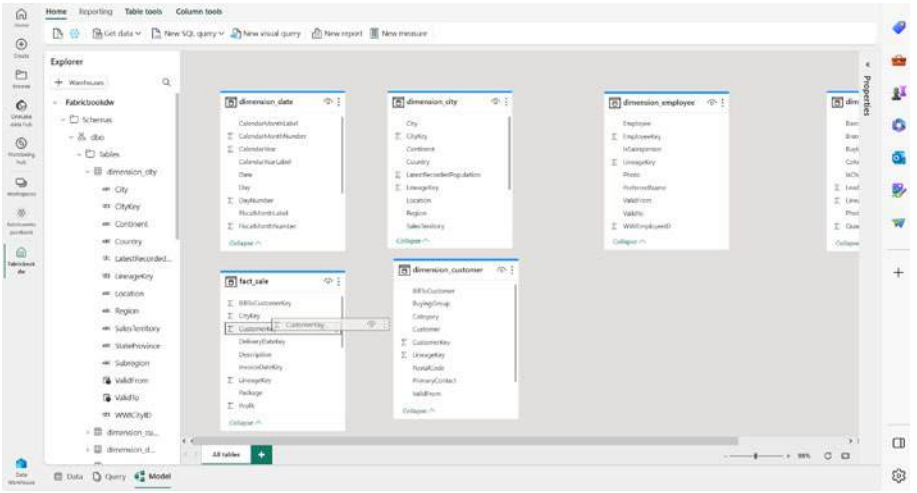


Figure 5-25. *Creating a relationship*

Now you can create relationship by selecting Cardinality, as shown in Figure 5-26.

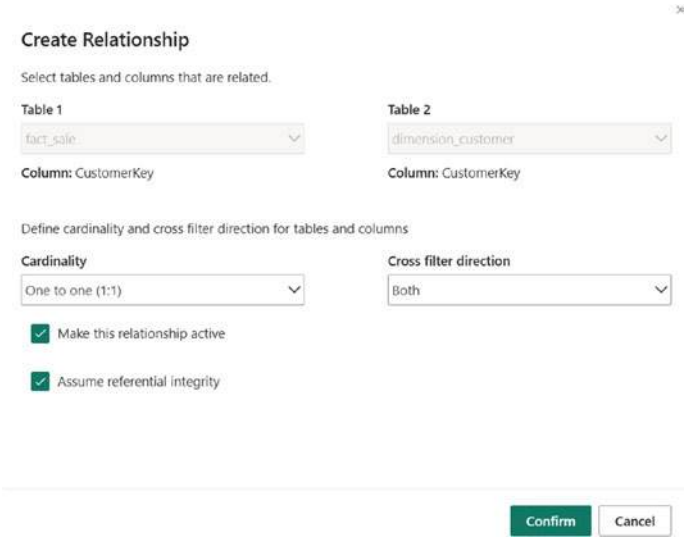


Figure 5-26. *Selecting Cardinality*

To recap, you have now done some data modeling in the Fabric workspace. In the next section, you will learn how to do reporting using the native Fabric DW functionalities.

Integration with Power BI and Microsoft Office Tools

The Fabric DW provides built-in integration with Microsoft Excel, Power BI, and SQL Server Reporting Service (SSRS). Once you select the Power BI Dataset or Dataset (Default) item type, you can easily analyze the dataset in Excel or create a visualization or paginated report. However, let's take a step back and understand the Power BI Dataset or Dataset (Default) item.

Whenever you create a lakehouse or warehouse, a default dataset is created automatically. It represents a semantic layer and inherits the business transformation logic. It allows you to analyze, slice, and dice the dataset. It synchronizes the data in the back end with the lakehouse or warehouse tables. This dataset is provided for SQL queries using SQL endpoints and DW endpoints. Also, this dataset is used for Power BI analysis. The default Power BI dataset also inherits the data modeling and entity relationship.

You need to click the three dots (…) beside the default dataset to leverage the powerful reporting features provided by the dataset (default), as shown in Figure 5-27. "Create paginated report" provides a report like the SQL server reporting service would or a paginated report like you can create in Power BI. "Analyze in Excel" takes the dataset and opens it in a Microsoft Excel pivot page. "Create report" provides a Power BI visualization capability using this dataset.

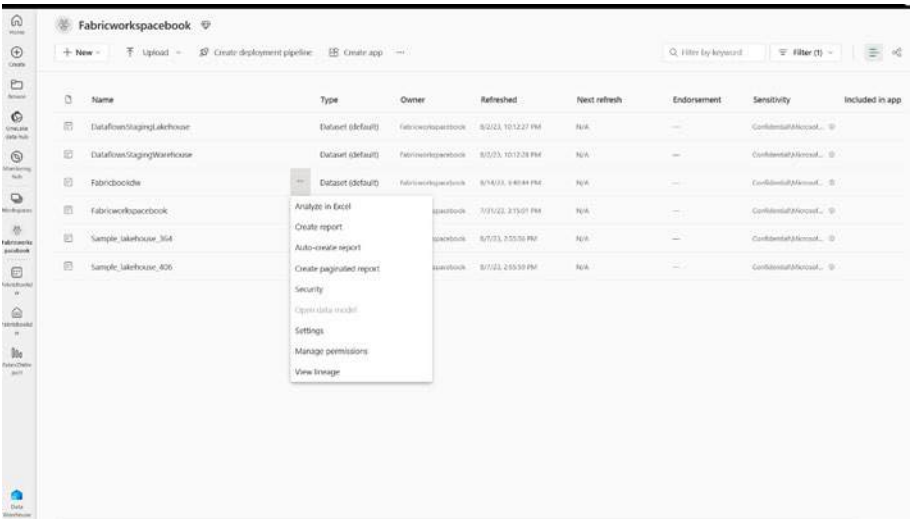


Figure 5-27. *Analyzing a DW dataset using Excel*

Once you click the dataset, the canvas provides a "+Create a report" button, as shown in Figure 5-28. Using this feature, you can create a visualization using a Fabric dataset.

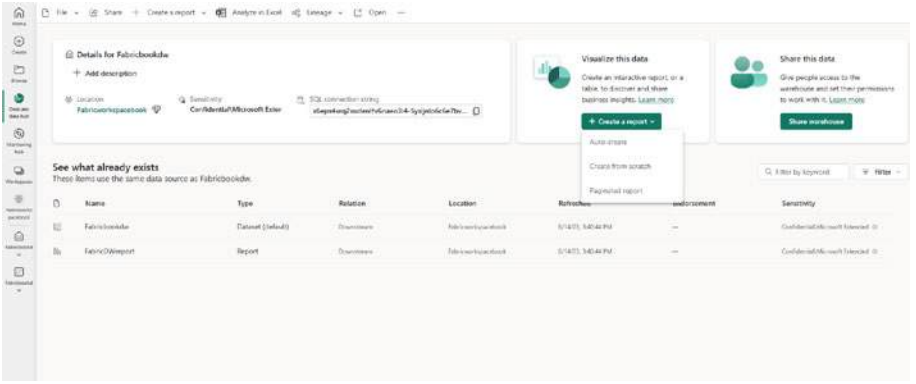


Figure 5-28. *Creating a report with just a few clicks*

When you click "Auto-create" from the list provided in the "+Create a report" feature, the Fabric data warehouse generates a visualization, as shown in Figure 5-29.

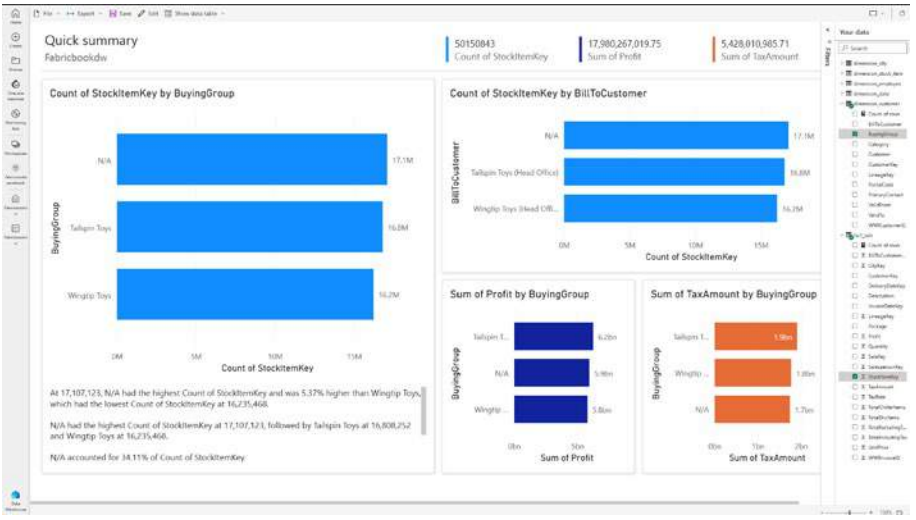


Figure 5-29. *Auto-creating a report*

We will discuss more details of Power BI related to Fabric in Chapter 9. In the next section, we will discuss the pipeline monitoring capabilities in Fabric.

Monitoring a Fabric DW Pipeline

The Fabric Monitor hub is a visual interface to monitor all activities in the Fabric workspace including pipelines, Spark jobs, notebooks, and more. You need to use the Filter condition in the top-right corner and select Submitter as the developer ID. Figure 5-30 shows all the tasks submitted by our ID in the Fabric workspace. To search for specific item types, you need to use that filter condition. For example, for data pipeline monitoring, use a filter condition as a data pipeline.

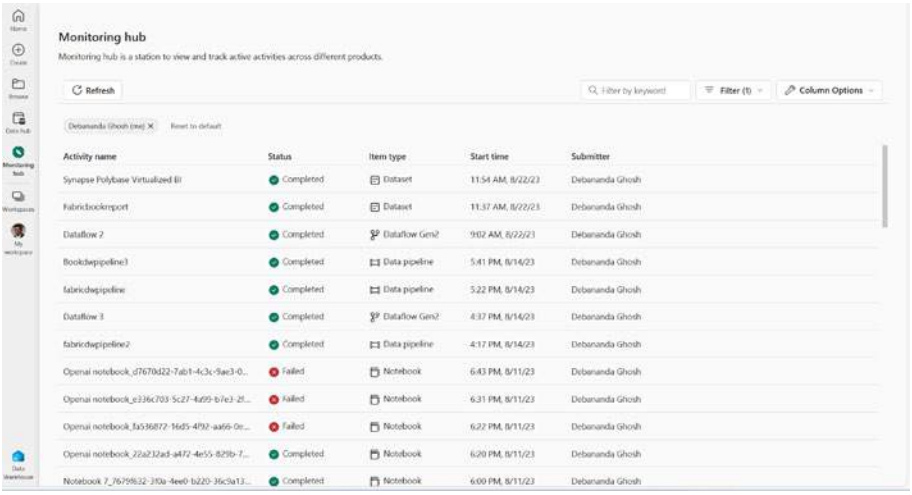


Figure 5-30. *Fabric Monitor hub*

Based on the status, you need to find your own job and then start looking into the details of each activity using "Pipeline run details for further debugging," as shown in Figure 5-31.

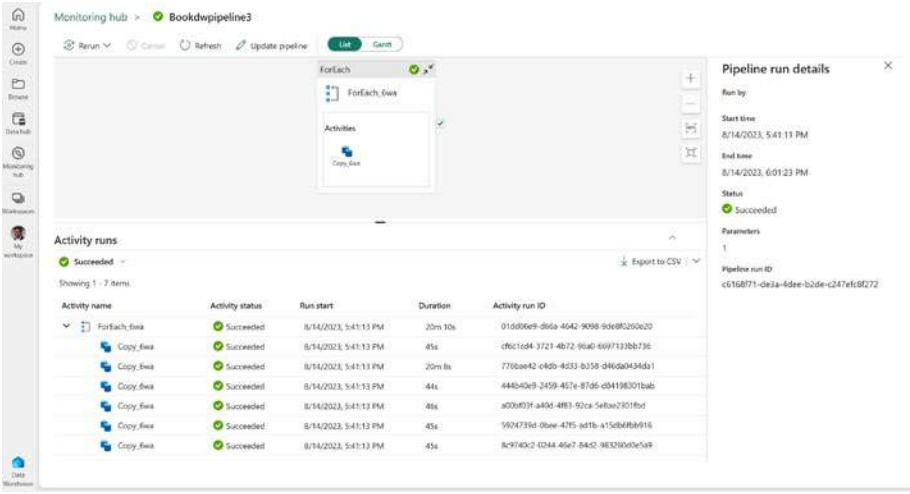


Figure 5-31. *Pipeline monitoring inside the Monitor hub*

The previous approach is the monitoring provided in the interface for data warehouse pipelines. You can also monitor Fabric DW SQL using SQL queries. Just execute the following example scripts in the SQL editor of the Fabric workspace:

```
SELECT r.request_id, r.session_id, r.start_time, r.total_
elapsed_time, s.login_name
FROM sys.dm_exec_requests AS r
JOIN sys.dm_exec_sessions AS s
ON s.session_id = r.session_id
WHERE r.status = “running'
ORDER BY r.total_elapsed_time DESC
```

Figure 5-32 shows the outcome of the previous query execution. This query provides the login name, details, session ID, and elapsed time details of a session.

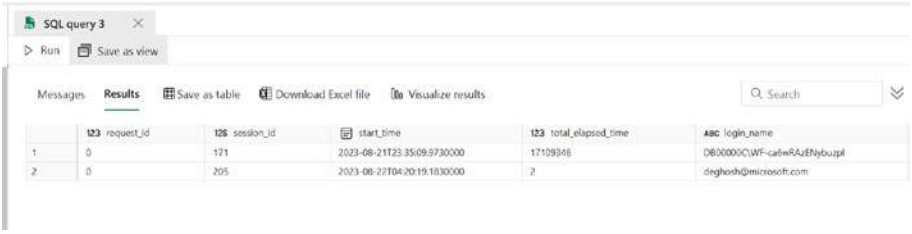


Figure 5-32. *Monitoring the Fabric DW SQL using SQL queries*

Summary

The main objective of this chapter was to introduce the Fabric data warehouse service and its features. You saw how to create a data warehouse instance, configure it, load data into it, and query it using SQL. You also explored some of the advanced options such as partitioning, compression, encryption, and caching. You looked at how to use the Monitor hub to track and analyze the performance and health of your data warehouse pipelines.

Additionally, we discussed some of the best practices and tips for optimizing our data warehouse design and usage. To conclude, you learned about the Fabric data warehouse concepts, provisioning, development, performance tuning capabilities, monitoring, and other relevant details. Refer to Figure 5-33 for a recap.

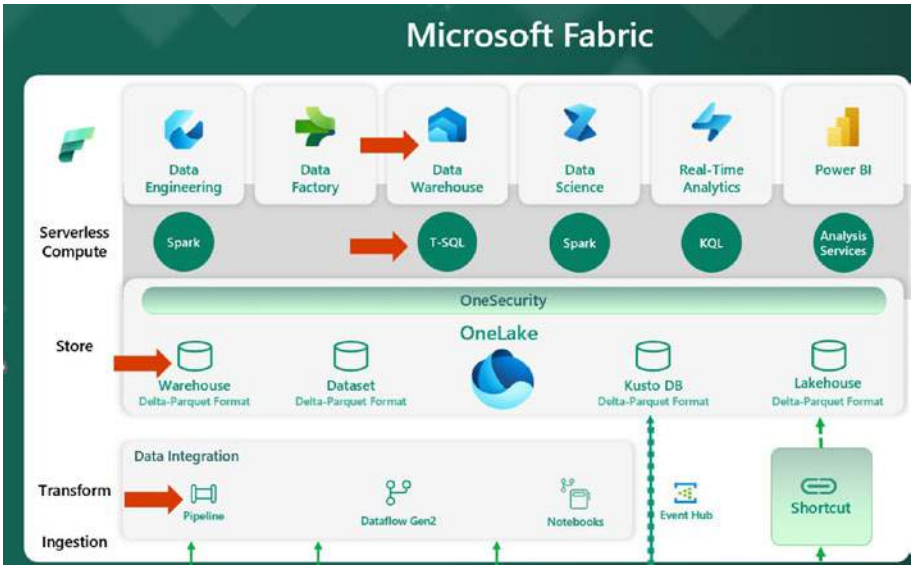


Figure 5-33. *Recap of data architecture*

In the next chapter, we will focus on more details related to data integrations.

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric DW SQL architecture:** https://learn.microsoft.com/en-us/fabric/data-warehouse/workload-management
- **Fabric DW SQL stats:** https://learn.microsoft.com/en-us/fabric/data-warehouse/statistics
- **T-SQL:** https://learn.microsoft.com/en-us/sql/t-sql/language-reference?view=sql-server-ver16

CHAPTER 6

Data Integration for Office Users

Our day-to-day activities generate a ton of data from many types of data sources. For example, when consumers go to an e-commerce platform and buy products, their website browsing activity generates some data. Likewise, when we go to brick-and-mortar shops and use our smartwatch to pay the bills, that activity generates data.

All these platforms, such as point-of-sale (POS) terminals and finance gateways, smart watches, and e-commerce websites generate data related to sales transactions. This sales data resides in the underlying transactional database of the e-commerce platform. When businesses want to build a sales analytics platform for business decision purposes, this data needs to be ingested into the e-commerce analytics platform.

To achieve this, the technology platform needs data integration capabilities to transform and import the data from various sources. Data integration features help to integrate data from various sources and consolidate it into a single platform. In this chapter, you will learn about the modern data integration capabilities of Microsoft Fabric.

Microsoft Fabric offers modern data integration capabilities that empower users to perform complex data integration tasks using easy-to-use code-free or code-based methods. Users can choose from a rich set of built-in Fabric connectors to access data from various sources, such as relational databases, cloud storage, web services, file systems, and more.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_6

Users can also use the graphical interface to design data flows, orchestrate pipelines, monitor activities, and debug errors. Alternatively, users can use code-based methods to write custom scripts, functions, or expressions to perform complex data transformations, validations, and calculations.

Figure 6-1 shows the modern integration experience in Fabric known as Data Factory. Note that the Data Factory capability is based on a software-as-a-service (SaaS) foundation and brings in the best features from Azure Data Factory (another Azure service) and Power BI.

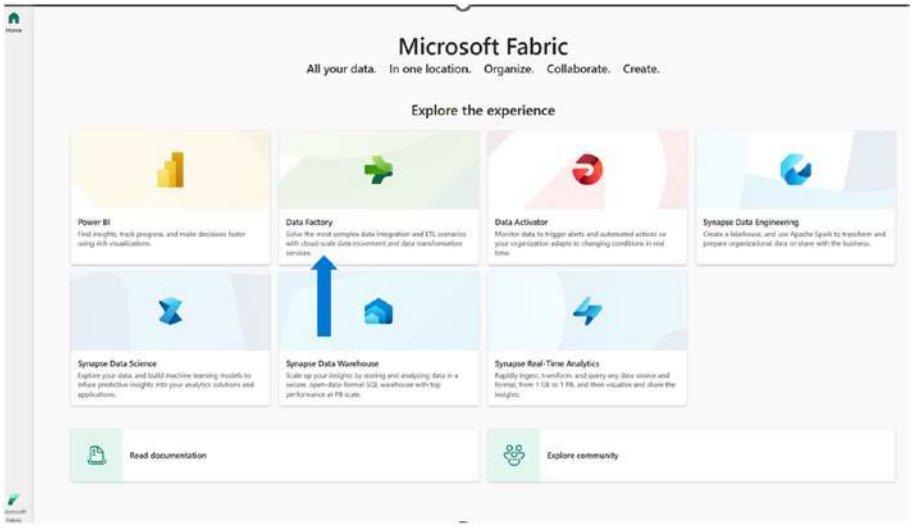


Figure 6-1. *Data Factory canvas*

Let's take a step back and understand how modern SaaS data integration helps developers today. Previously data integration tooling required significant programming skills. Additionally, it required server provisioning, budgeting, and other maintenance-related overhead. Microsoft Fabric Data Factory reduces the organization overhead of the managing extract/transform/load (ETL) infrastructure. By using the modern data integration capabilities of Microsoft Fabric, you get an intuitive user experience and a serverless and scalable data integration computation.

Specifically, Microsoft Fabric provides the following business benefits:

- Code-free ELT/ETL
- Both pay-as-you-go and provisioned licensing models
- 100+ native connectors
- Orchestration at scale
- Intuitive monitoring

To understand the anatomy of the Fabric Data Factory capability, let's do a deep dive into the UI canvas. The Data Factory capability comes with these three main features, as you can see in Figure 6-2:

- **Data Flow Gen2:** This leverages Power-based next-gen capabilities including a data flow designer and AI-based transformation tools mostly for business users.
- **Data pipeline:** This is the next generation of Azure data factory features and comes with rich orchestration tooling and workflows suitable for enterprise data orchestration needs.
- **Data Factory Mount:** This is primarily for migrating Azure data factory workloads created earlier in the organization.

You will learn about each of these features in detail in the following sections.

Data Flow Gen2

Data Flow is a visual querying tool for Power users and comes from the Microsoft Power platform and Dynamics 365 insights application. Power users can leverage the Data Flow features to build end-to-end data transformations and write the data in a lakehouse. Figure 6-2 shows the Data Flow Gen2 capability in the Microsoft Fabric workspace.

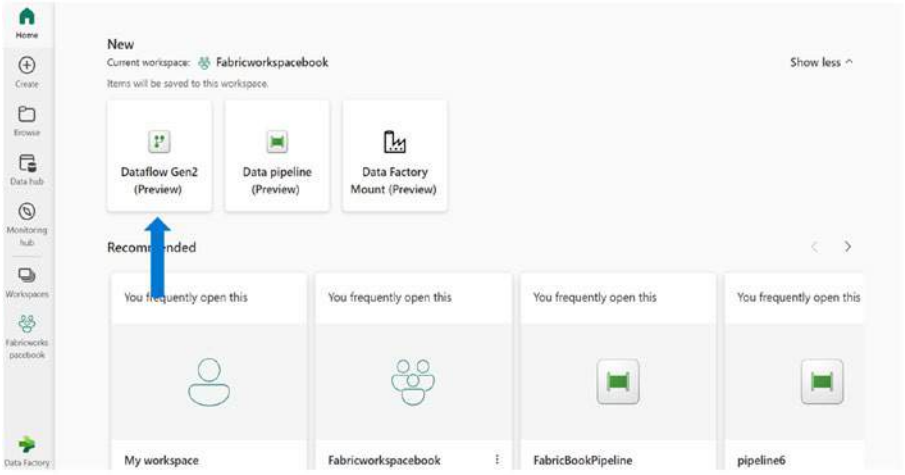


Figure 6-2. *Data Flow Gen2*

Data Flow Gen2 in Microsoft Fabric comes with the following features:

- Intuitive user experience
- Choice of Azure data destinations
- Monitoring capability
- Scalable computation

There are many more similar intuitive features that you can leverage to build an end-to-end data flow. Figure 6-3 shows the Power Query–based Data Flow Gen2 designer canvas. To explore more Data Flow Gen2 key features, follow these steps:

1. Create a data pipeline using Data Flow Gen2 artifacts.
2. Load a sample CSV file.
3. Transform the CSV file.
4. Write the CSV file in a Fabric lakehouse.
5. Deploy the newly created pipeline in Fabric.

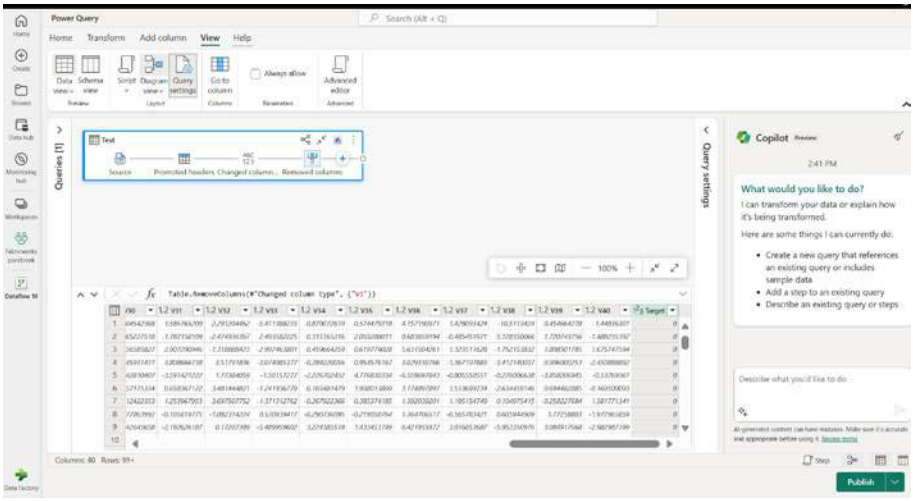


Figure 6-3. *Dataflow Gen2 designer canvas*

We started by loading a CSV file using Data Flow Gen2. After you click the Data Flow Gen2 feature, you will see the following tabs, as shown in Figure 6-4:

- Import from Excel
- Import from SQL Server
- Import from a Text/CSV file
- Import from dataflows

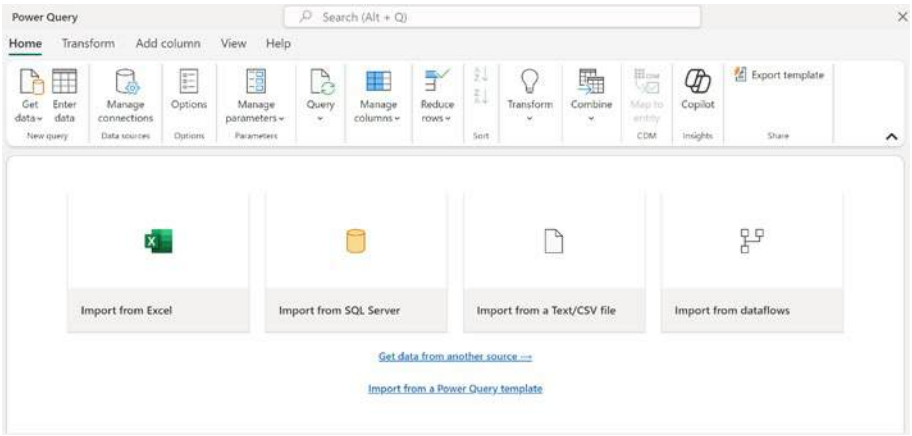


Figure 6-4. *Dataflow Gen2 Import mechanisms*

Note that the data flow connectors are not just restricted to CSV or Excel and can connect to hundreds of connectors. Click "Get data" (at the top left) to use these connectors. To see an updated connectors list, please refer to `https://learn.microsoft.com/en-us/fabric/data-factory/dataflow-support`. Figure 6-5 shows the sample Data Flow Gen2 connectors.

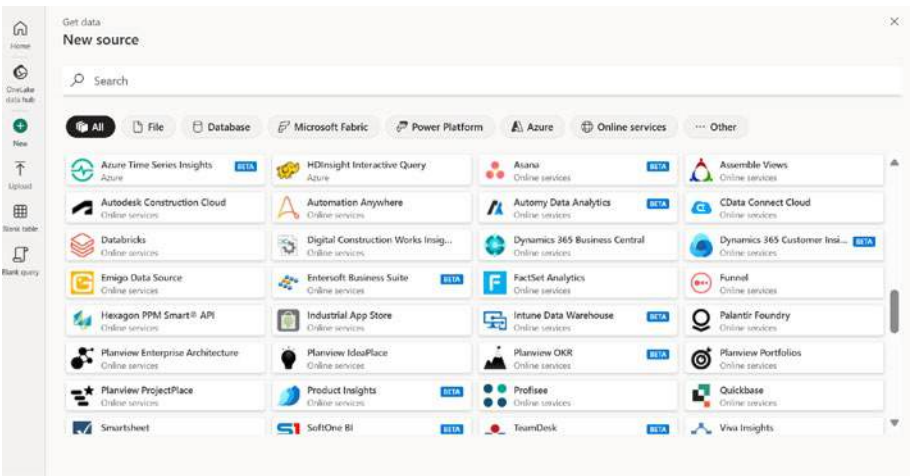


Figure 6-5. *Sample Data Flow Gen2 connectors*

Let's now continue to load the CSV data using the Dataflow Gen2 mechanism. Click "Import from a Text/CSV file" to start. The "Connect to data source" canvas opens, as shown in Figure 6-6.

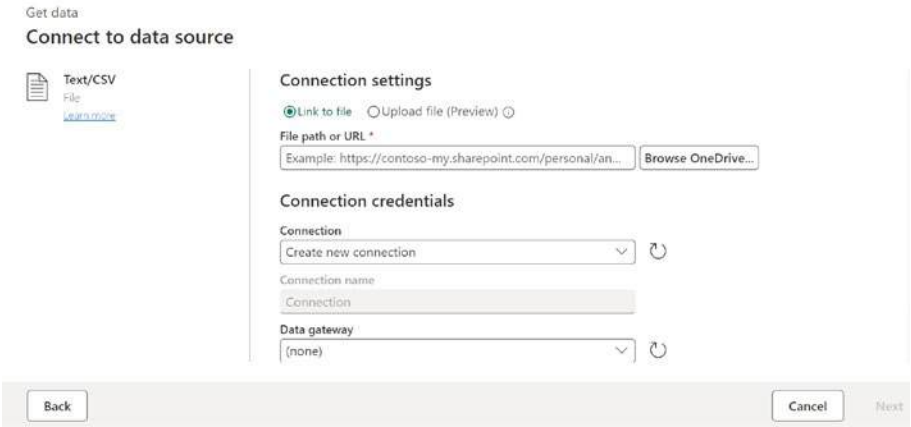


Figure 6-6. *"Connect to data source" canvas*

This is a user-friendly canvas, and you can leverage "Link to a File" or use the "Upload file" radio button to load the CSV file. In Figure 6-7, we are uploading the file `Customers1.csv` using the "Upload file" feature.

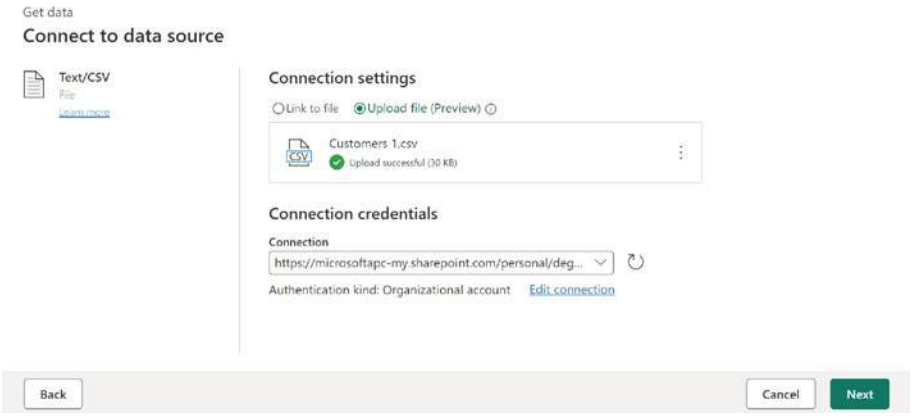


Figure 6-7. *Connecting to a data source*

As the next step, you need to provide the connection credentials. Note that there will be a few details like "Data gateway" and "Authentication kind" that you need to select. Let's understand these two features.

- **Data gateway:** For on-premises integration, Microsoft provides an on-premise data gateway for the secure data transfer of data between on-premises machines and the cloud. An on-premises data gateway (OPDG) is installed in an on-premises network machine. Go to https://learn.microsoft.com/en-us/data-integration/gateway/service-gateway-install for installation instructions. To avoid a single point of failure, you can install a second gateway on another computer within the same network. Leverage the on-premise gateway app for gateway management in the on-premise network computer; see https://learn.microsoft.com/en-us/data-integration/gateway/service-gateway-app. Once you configure OPDG, it will appear in the "Data gateway" drop-down in the Fabric connection credentials. This feature is recommended for hybrid data integration in an enterprise scenario.
- **Authentication kind:** The Fabric Data Factory feature provides diverse types of authentications. When you use "Link to file" for cloud-to-cloud data transfers, you get the following options:
  - **Basic:** Based on user ID and password
  - **Anonymous:** No authentication
  - **Organization (domain ID):** Advanced authentication of user ID and password within domain (recommended for interactive cases)

- **Service principle-based authentication:** For automated tools and applications (recommended for non-interactive cases)

For on-premises, Fabric provides a Windows authentication mechanism as well. You need to choose the right data gateway strategy and authentication method based on your data integration needs and move to the next step. See Figure 6-8.

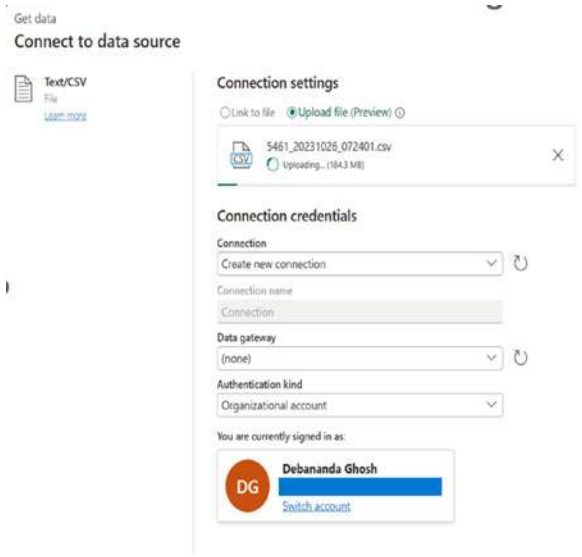


Figure 6-8. *Data Flow Gen2 authentication and data gateway*

When you go to the next canvas, you will be able to preview the sample data that you are loading, as shown in Figure 6-9. Click Create to create a Data Flow Gen2 data pipeline in the Fabric workspace.

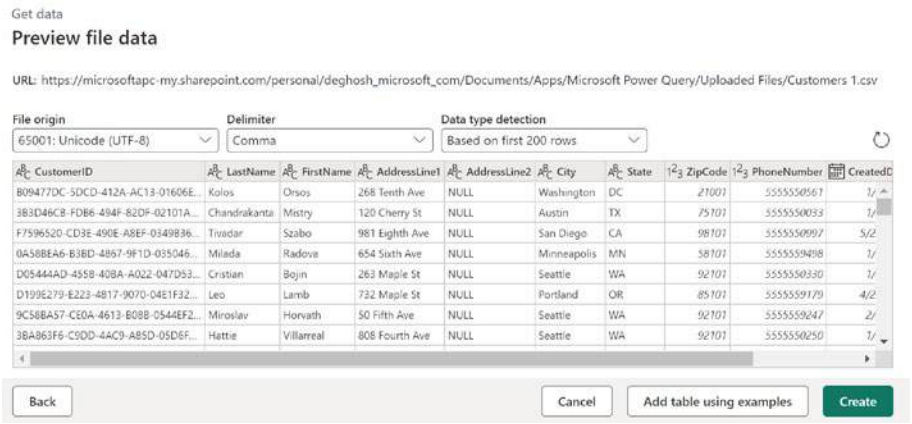


Figure 6-9. *Data Flow Gen2 data preview*

Note that until now we have connected our Fabric Data Flow Gen2 with a data source (in this case, using the local path of the CSV file). As the next step, you need to transform this data and load it into the destination (the Fabric lakehouse in this case). Power Query provides numerous transformation tools to do this. The following are some examples:

- **Group by:** Group rows of tables based on the selected column.
- **Use first row as headers:** Promote the first row as headers.
- **Transpose:** Transpose the table.
- **Reverse rows:** Invert the table rows.
- **Count rows:** Return the number of rows of a table.
- **Replace Values:** Replace existing values with new values.
- **Data type:** Convert the data type.
- **Mark as a key:** Mark the identified column as the key.

- **Rename:** Change the name of the selected column.
- **Pivot columns:** Use the currently selected column names to create a new column.
- **Fill up and Down:** Fill up the values with the neighboring empty cells.
- **Split column:** Split columns using a delimiter.
- **Format:** Format in uppercase, lowercase, and other case formatting.
- **Extract:** Extract text between delimiters.
- **Statistics:** Get the statistics of the data.
- **Index column:** Create a new column with an index value.
- **Rank column:** Use for the ranking column.

Figure 6-10 shows Power Query's Transform ribbon and its features.



Figure 6-10. *Data Flow Gen2 Power Query ribbon*

To start transforming the uploaded/ingested dataset, you need to click the Power Query ribbon. For example, if in the ingested dataset you need to replace an integer values with another integer value, you need to click "Replace values," as shown in Figure 6-11.

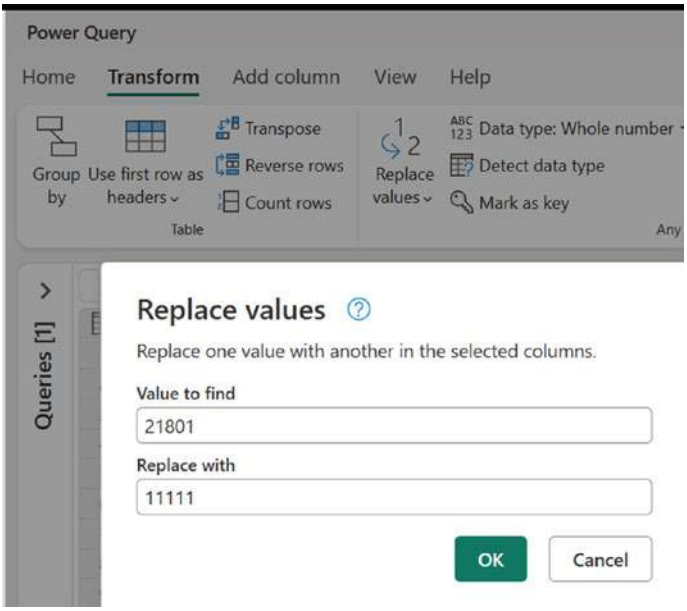


Figure 6-11. *Replacing values*

Likewise, we are using the "Remove column" and "Split column" transformation steps, as shown in Figure 6-12. You should play around with the Transform ribbon and explore the code-free transformation features.

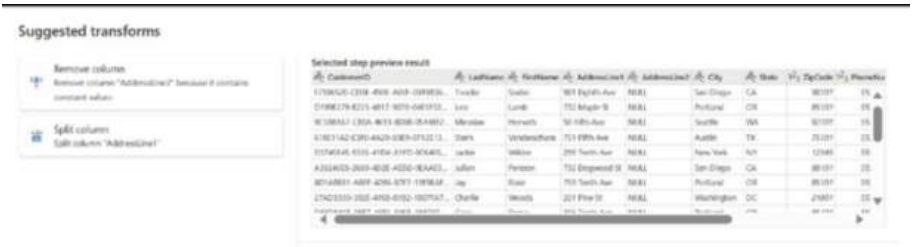


Figure 6-12. *Data transformation example*

In Data Flow Gen2 you can leverage the graphical user interface (GUI) as well. To get to the GUI-based designer, you need to go to View and click Diagram View. Once you are in the dataflow visual designer pane, you can leverage the user interface and start building the transformation steps with just a few clicks.

This is another way to develop a data integration module using Data Flow Gen2 without using its ribbon features. In Figure 6-13, we clicked the + sign, which opens the code-free transformation features with a search pane. Note that every query step that you develop in Data Flow Gen2 is also logged under "Applied steps."

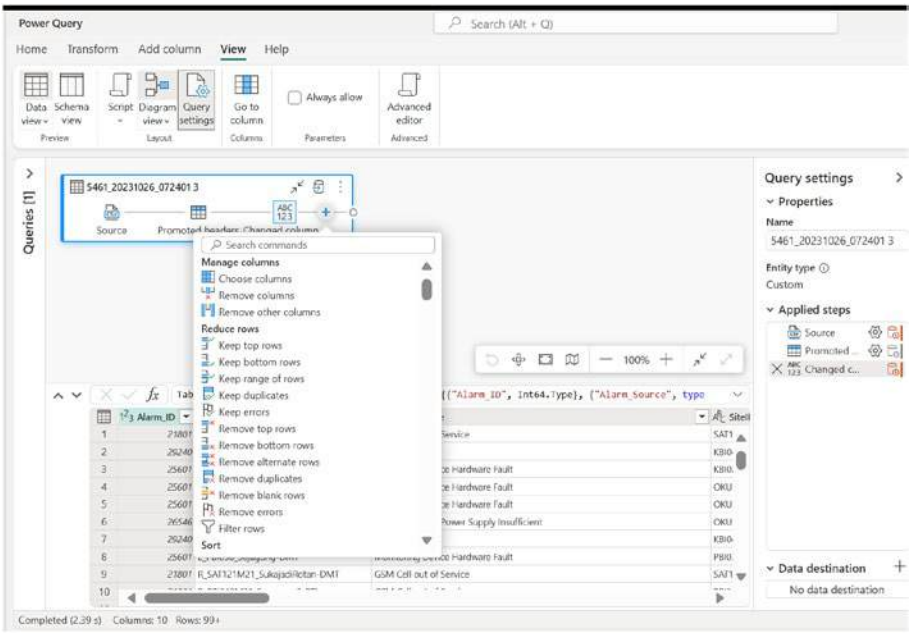


Figure 6-13. *Code-free transformation features and a search pane*

Once you complete the transformation activity, you need to write the data in the desired destination. To add a data destination, you will click Home and then the "Add data destination" tab in the middle of the

ribbon, as you can see in Figure 6-14. At the time of this writing, "Add data destination" supports the following destinations including the Fabric lakehouse:

- Azure SQL Database
- Lakehouse (Fabric)
- Azure data explorer
- Azure synapse analytics
- Warehouse (Fabric)



Figure 6-14. *Data Flow Gen2's "Add data destination" tab*

In this scenario, we want to write the ingested data into a Fabric lakehouse. Hence, you need to select Lakehouse on the "Add data destination" tab (Figure 6-14 middle) and then follow the steps to connect with the lakehouse. Create a new connection unless there is already a precreated connection. See Figure 6-15.

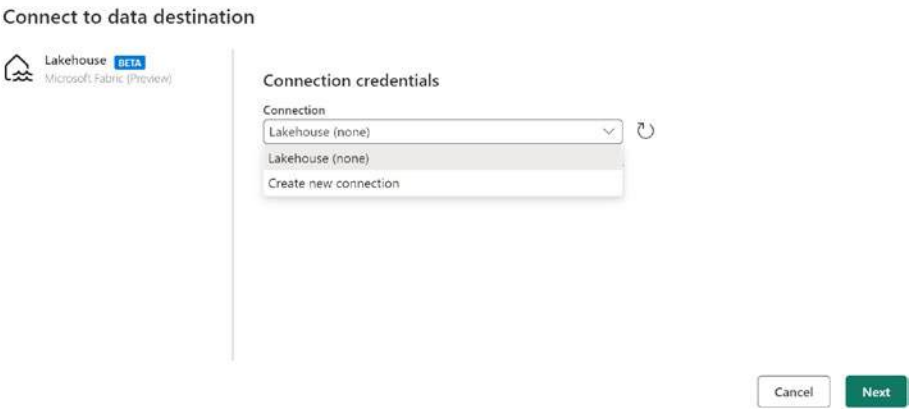


Figure 6-15. *Connecting to a data destination*

Select "Create new connection" and provide the "Data gateway" and "Authentication kind" details, as shown in Figure 6-16. This experience is similar to the steps in the previous section.

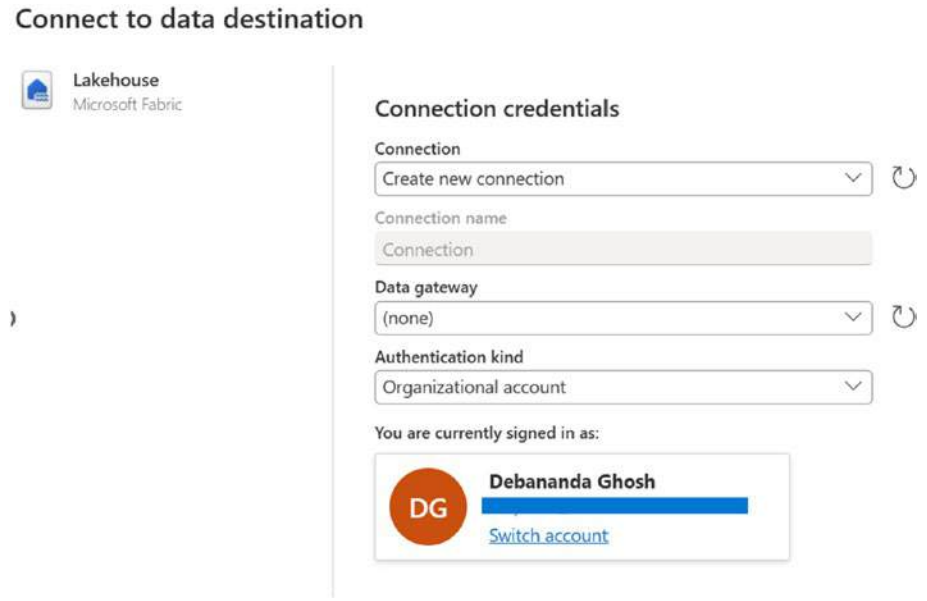


Figure 6-16. *Providing the credentials*

Once you sign up using an organization account, you will be able to see workspaces across the tenant, as shown in Figure 6-17. You need to click the Lakehouse folder where you want to move the data. You need to provide a table name for a new table. Alternately, for a precreated table, you need to select "Existing table.'

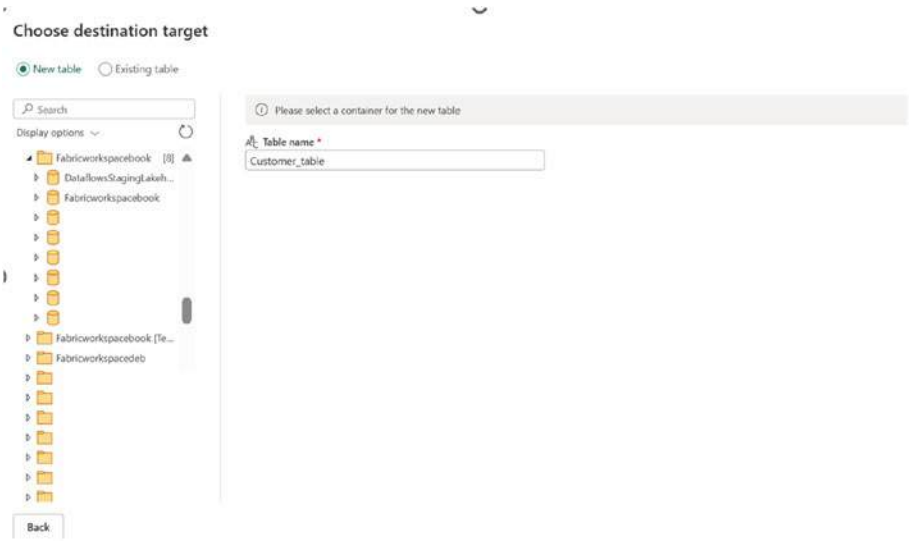


Figure 6-17. *Naming the table*

As you move to the next step, the canvas provides data destination settings to append and replace the ingested data in target destination, as shown in Figure 6-18. Note that the Append option is not Update Insert/Upsert in the existing table but Insert Only. Also, you can map the source with the source type and then click "Save settings."

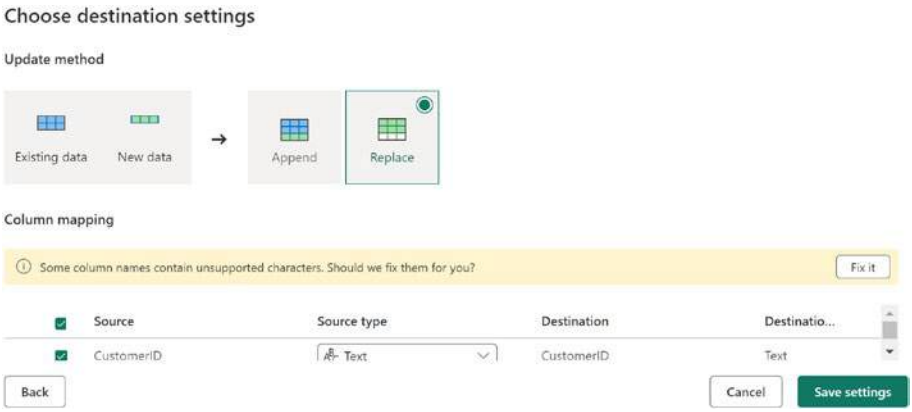


Figure 6-18. *Append and Replace capability*

After you click "Save settings," you can load the full CSV file in a newly created table. However, for enterprise scenarios, you also need to incrementally refresh at the destination table. Currently, Data Flow Gen2 supports incremental changes (different than incremental refreshes) of the data. In incremental changes of data, you apply the "Filter rows" (under Home) on the data destination side. Note that besides using "Filter rows," you can leverage any other transformation query that does row subset selection, as well as attach notebook-based code snippets as part of incremental amassing. See Figure 6-19.

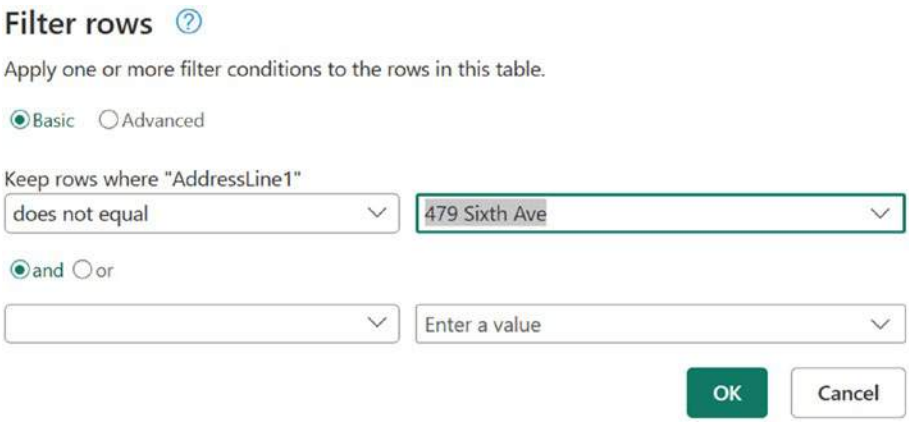


Figure 6-19. *Filtering rows*

You have integrated the data flow with the source and destination, so you will now learn how to apply the transformation step in between the source and destination targets. Click "Publish" and then "Publish now," as shown in Figure 6-20. A Fabric item with the type "dataflow" will be created and published in Fabric.

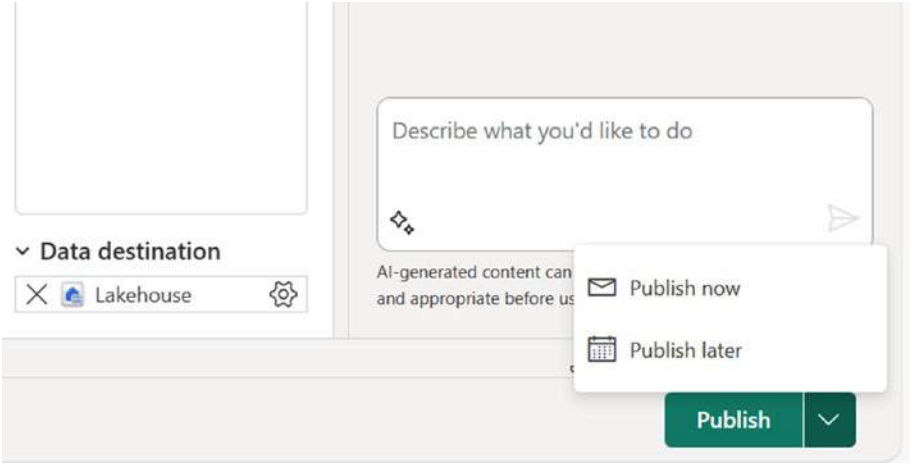


Figure 6-20. *Publishing the item*

To identify your recently published data flow, go back to the workspace's main page and then search using Filter set to Data Flow Gen2, as shown in Figure 6-21.

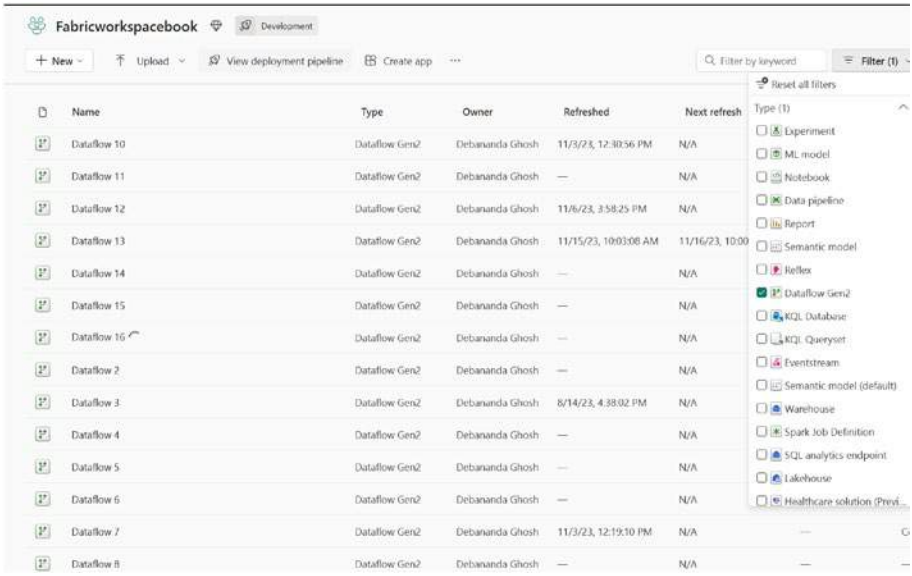


Figure 6-21. *Searching for our data flow*

Once the data integration pipeline is built, the next step of development is to set up the orchestration/scheduling window of the data integration batch job. In this step, you will set up the scheduling of this Dataflow Gen2 item. Click the three dots (…) beside Dataflow Gen2 (Data flow 10 in this case in Figure 6-22), which you recently published and then follow these steps:

1. Select Settings from the ribbon to set up the orchestration, as you can see under the three dots (…) in Figure 6-22.

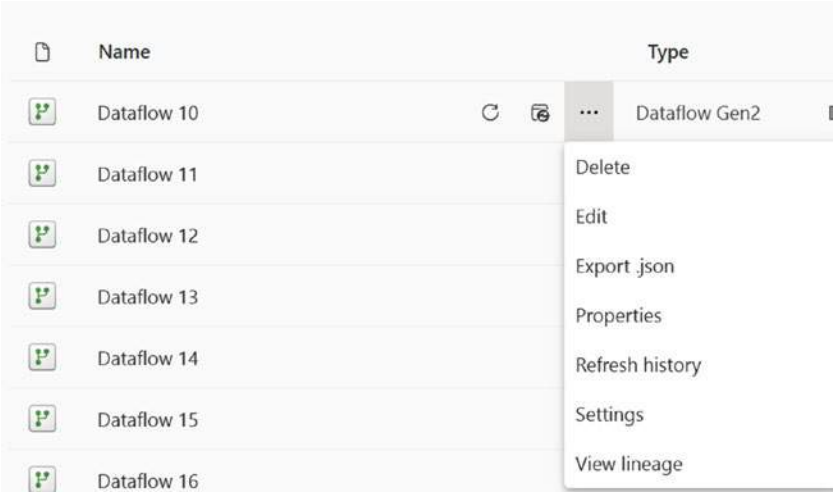


Figure 6-22. *Setting up the orchestration*

2. Select Refresh under Gateway Connection on the Dataflows tab to set up the scheduling window, as shown in Figure 6-23.

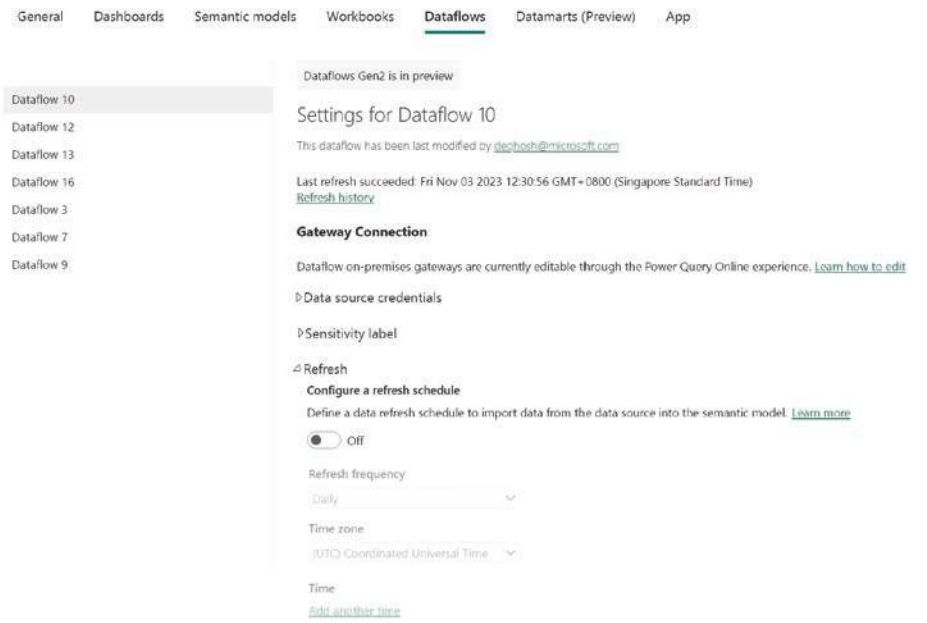


Figure 6-23. *Setting up the scheduling window*

3. Now in this canvas you can set up a refresh schedule and notification failures, as shown in Figure 6-24.

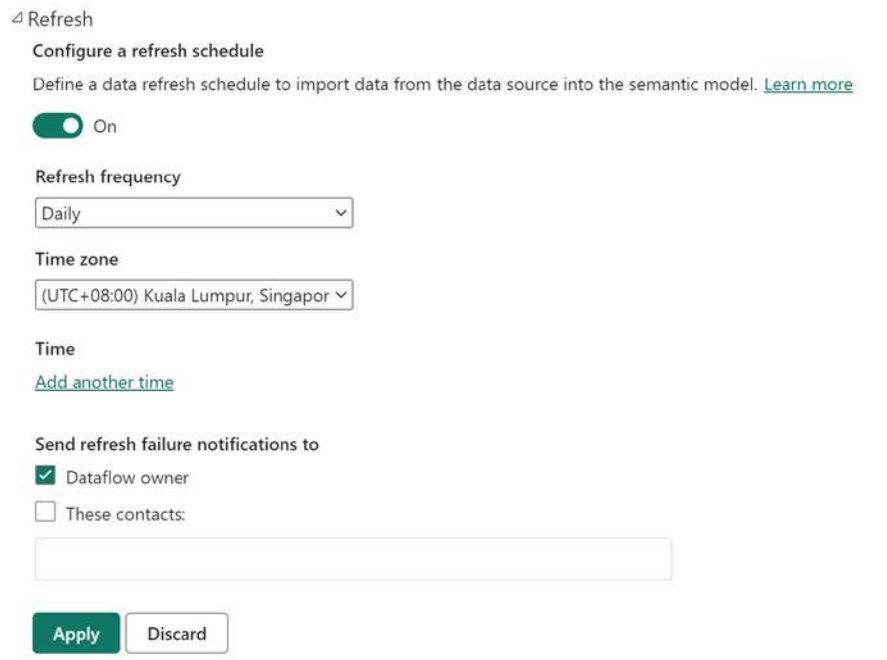


Figure 6-24. *Setting up a schedule*

You learned about Data Flow Gen2 in this section. Note that Power BI previously had a Data Flow Gen1 capability. To migrate Data Flow Gen1 to Data Flow Gen2, you need to leverage an export template.

Let's take a quick look at how exporting templates works for Power BI Data Flow Gen1. Click Edit and then click "Export template," as shown in Figure 6-25.

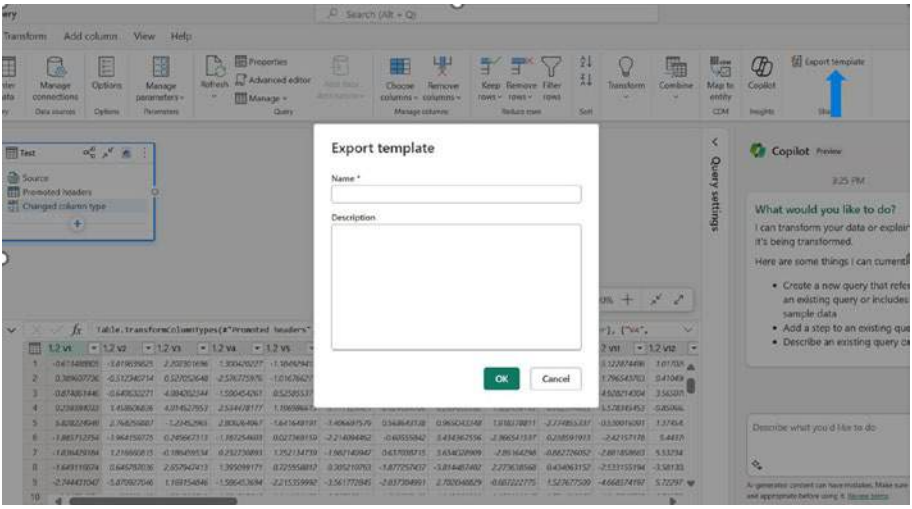


Figure 6-25. *Exporting a template*

After you fill in the "Name" and "Description" fields and click OK, the workspace will save this template in PQT format. Figure 6-26 shows the "Import from Power Query" template. Once you click and point to the predownloaded PQT file, it will load in Data Flow Gen2. You need to leverage this mechanism for a Gen1 to Gen2 migration.



Figure 6-26. *Importing from a Power Query template reference*

You have now completed the Data Flow Gen2 development techniques. Next you will learn about Data Flow Gen2 development using the Copilot capability.

Fabric Copilot Experience

Microsoft Copilot is a smart assistant that helps you with various tasks. The Copilot capability in the Microsoft stack helps you in the following ways:

- Setting up your project and tech environment
- Managing your users and permissions
- Monitoring your data pipelines and data quality
- Troubleshooting issues and errors
- Exploring and visualizing your data
- Discovering and sharing insights and best practices

In the Fabric Data Factory context, Copilot can help you create and manage data sources, data destinations, data pipelines, data transformations, data quality rules, and more. It can also give you insights and recommendations based on your data and your goals. Copilot is powered by artificial intelligence and natural language processing. It can understand your queries and commands and generate code and scripts for you. It can also learn from your feedback and preferences and improve its performance over time.

You can access Copilot from the Fabric UI or from the command line. You can use voice or text to interact with it. You can also ask it questions or give it instructions in plain English or in SQL. For example, you can ask Copilot to do the following:

- "Show me the data sources available in my project."
- "Create a data destination for Amazon S3 with these credentials."
- "Build a data pipeline that extracts data from Oracle, transforms it using Spark, and loads it to Snowflake."

- "Generate a data quality rule that checks for null values in this column."
- "Explain what this data transformation does."
- "Optimize this data pipeline for performance and cost."
- "Run this data pipeline and send me a notification when it is done."

Copilot is designed to make your data work easier and faster. It can help you with simple and complex tasks and guide you through the Fabric features and functions. It can also collaborate with other Copilot users and leverage the collective knowledge and experience of the Fabric community. In Figure 6-27, we have asked Copilot to "generate a data quality rule that checks for null values in this column." We have selected "Zip code" to apply the same data quality rule as shown.

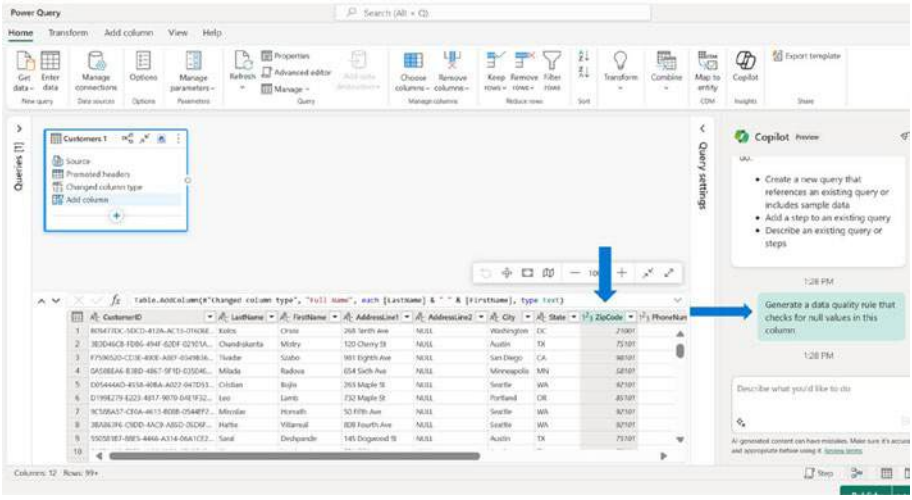


Figure 6-27. *Fabric Data Factory Copilot*

Copilot creates a new data quality rule-based customer function using its artificial intelligence capability and applies that in "Zip code" column. Copilot automatically creates another column with the outcome FALSE or TRUE, as shown in Figure 6-28.

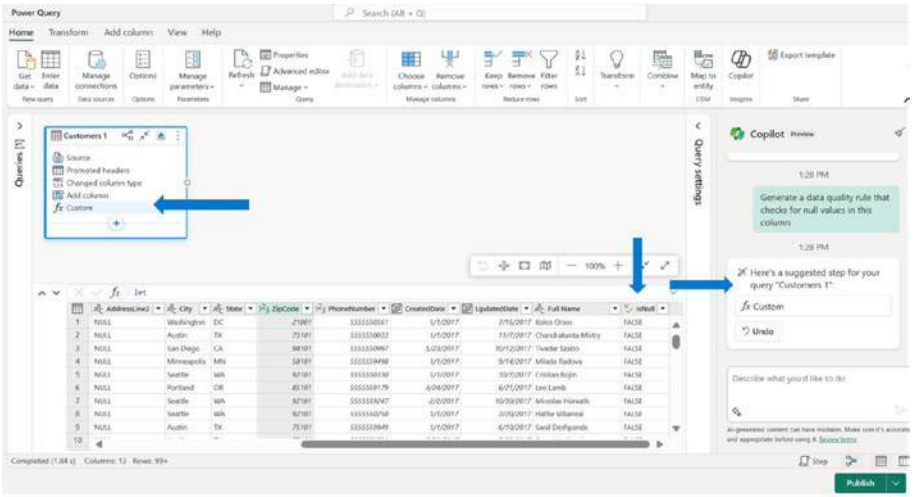


Figure 6-28. *Creating another column*

In Figure 6-29 we are using Copilot to create a new column that merges the first name and last name. The outcome is a new attribute called Full Name.

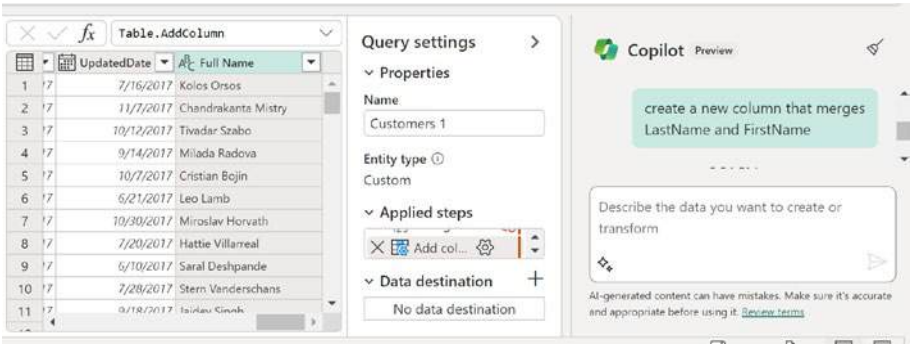


Figure 6-29. *Creating a new attribute*

As a quick recap, you have learned how to use Data Flow Gen2 for data integration purposes. You have also learned how to get Copilot assistance to do this. Now you will learn another way of doing data integration with the Fabric based on Azure Data Factory (separate azure service) capability.

Data Pipeline

In this section, you will learn about the Microsoft Fabric "Data pipeline (Preview)" capability. This is another data integration capability in the Microsoft Fabric Data Factory capability for end-to-end data ingestion, transformation, and loading. The "Data pipeline (Preview)" feature experience in Microsoft Fabric is similar to Azure Data Factory (ADF, https://azure.microsoft.com/en-us/products/data-factory) as it is the next generation of ADF based on a SaaS foundation. Figure 6-30 shows the "Data pipeline (Preview)" feature in Fabric.

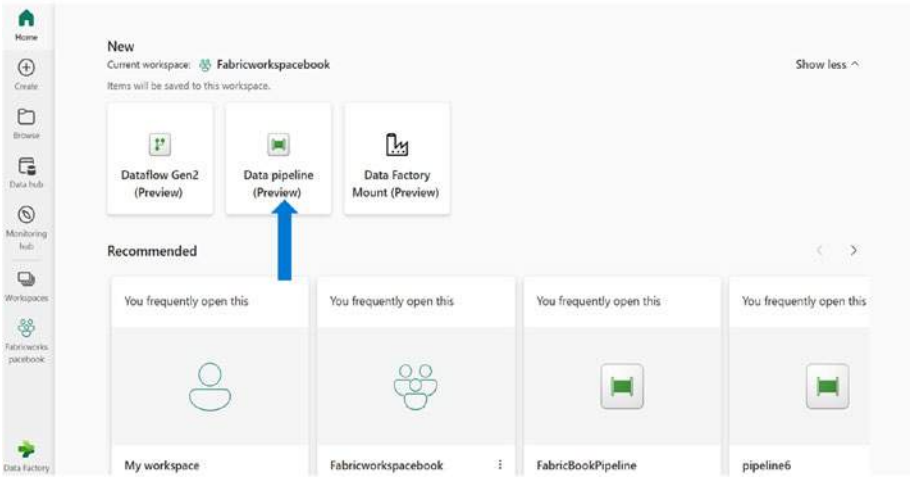


Figure 6-30. *"Data pipeline (Preview)" feature*

We will go through the steps to create a data pipeline and explain the configuration options and properties of each component. Let's start creating a data pipeline by first clicking the + New drop-down and then selecting "Data pipeline," as shown in Figure 6-31.

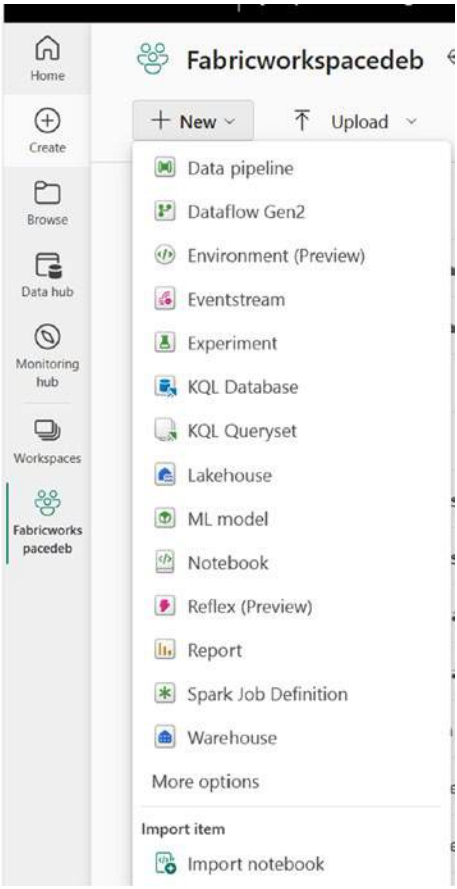


Figure 6-31. *Creating a data pipeline*

Once you provide a suitable name (in this case "Deb book pipeline"), you need to click Create, as shown in Figure 6-32.

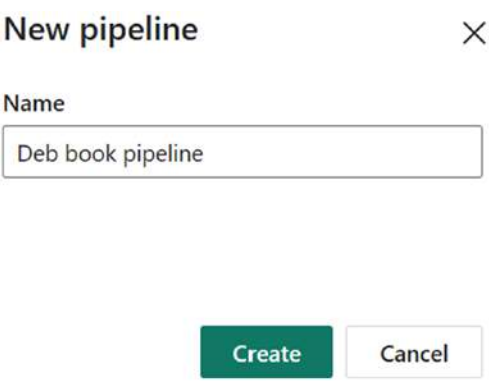


Figure 6-32. *Naming the pipeline*

Once the data pipeline is created, you will see these three tabs, as shown in Figure 6-33:

- **Add pipeline activity:** Code-free tooling for extensive data orchestration, validation, control flow
- **Copy data:** Code-free tooling for primarily moving the data from one place to another place
- **Choose a task to start:** Contains reusable template to increase developer productivity

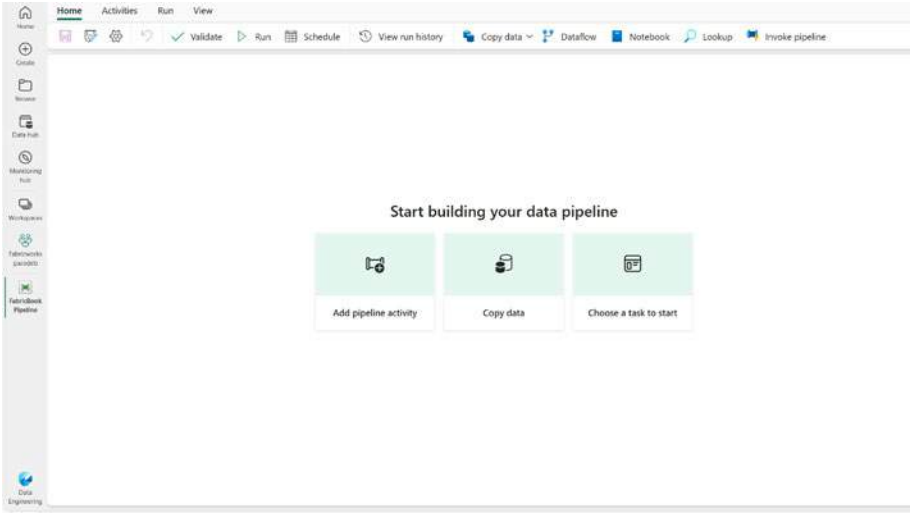


Figure 6-33. *Pipeline tabs*

Let's first see what is in the reusable gallery. Click "Choose a task to start," as shown in Figure 6-34. You can use a reusable template and start developing your data integration module if the template is aligned with your enterprise requirements. These templates are self-explanatory and help to increase developer productivity by providing reusable and relevant code artifacts.

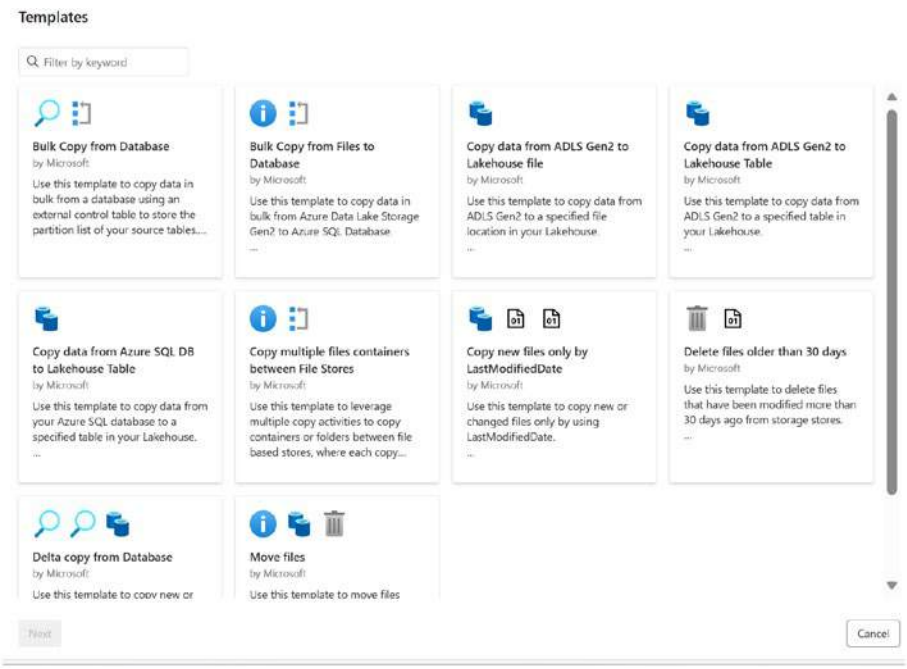


Figure 6-34. *Template gallery*

To explore this template gallery, click "Copy new files only by LastModifiedDate." Once you click that template, it opens the canvas shown in Figure 6-35. You need to create the source and destination connection. Subsequently this template generates a pipeline artifact that you can use for further development.

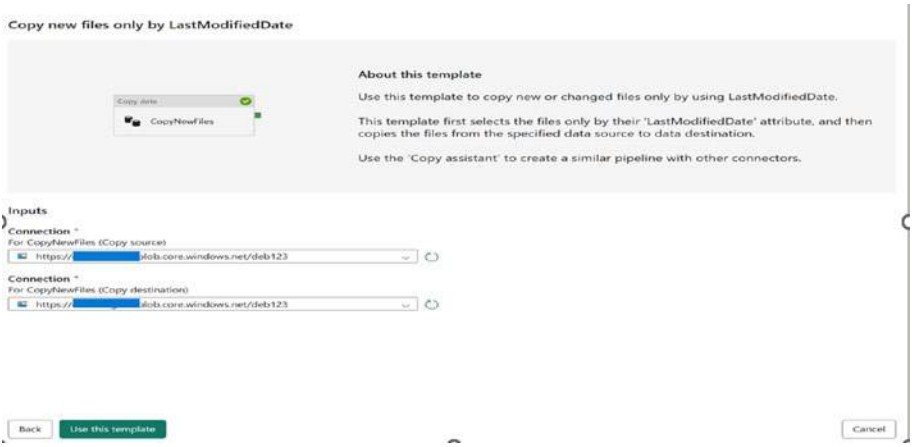


Figure 6-35. *Fabric data pipeline template experience*

Figure 6-36 shows the new files that are created in the data pipeline.

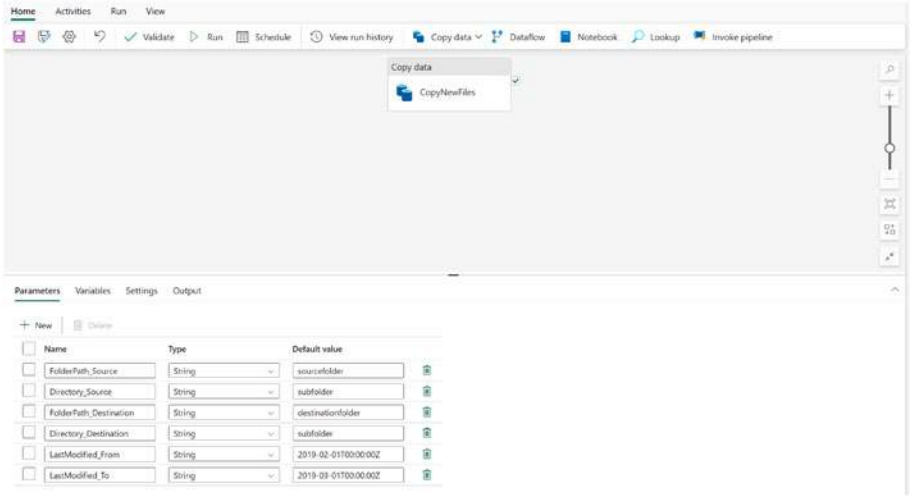


Figure 6-36. *Creating new files via a template*

You will now start developing the data pipeline. Click "Add pipeline activity." You will then see the various data integration code-free activities, as shown in Figure 6-37, with a search pane to choose the appropriate tooling.

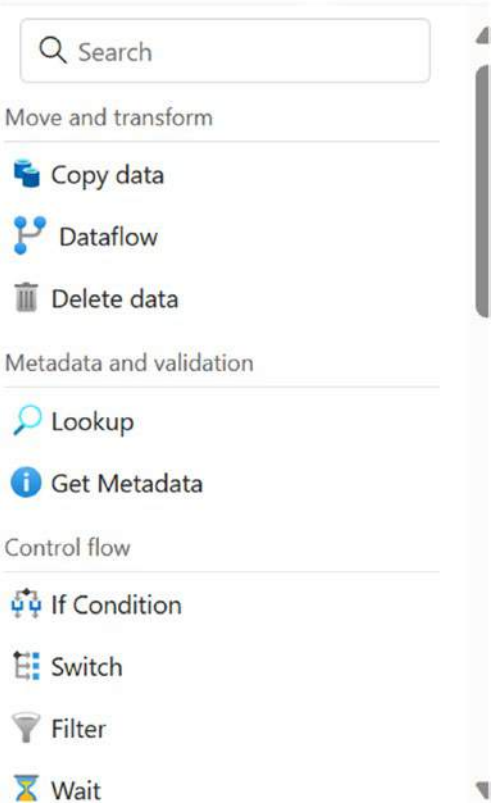


Figure 6-37. *Searching for a tool*

You will build a small utility here using the Fabric Data Factory "Data pipeline" capability by performing the following steps:

1. Execute a Spark notebook.
2. Upon Spark notebook execution, the pipeline sends an alert to Microsoft Teams (a Microsoft 365 business communication platform).

3. From the search pane, select the Notebook activity and then select a precreated notebook. You can leverage the one created in Chapter 3, as shown in Figure 6-38.

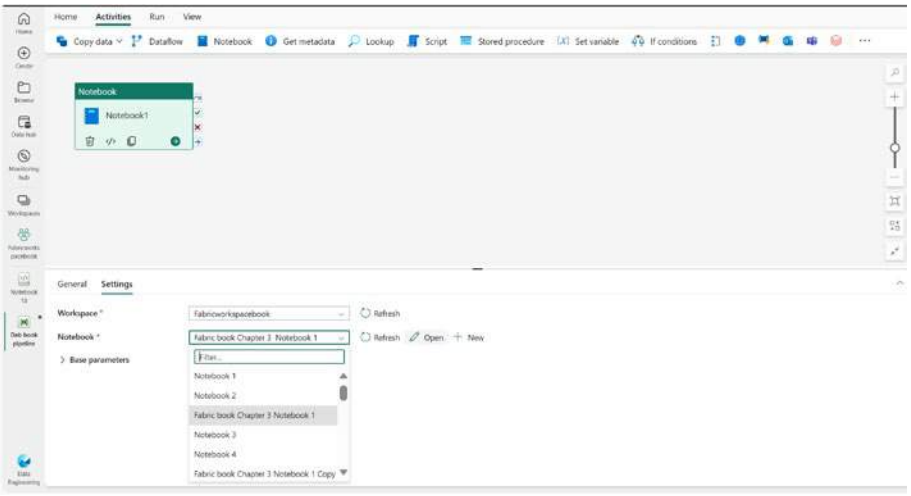


Figure 6-38. *Selecting a precreated notebook*

4. Go to the Settings tab and select Open to select the precreated notebook. Alternately, you can click + New and create a new notebook and start doing notebook development there.

As the notebook is invoked, you need to establish a downstream activity for notebook execution failure, as shown in Figure 6-39.

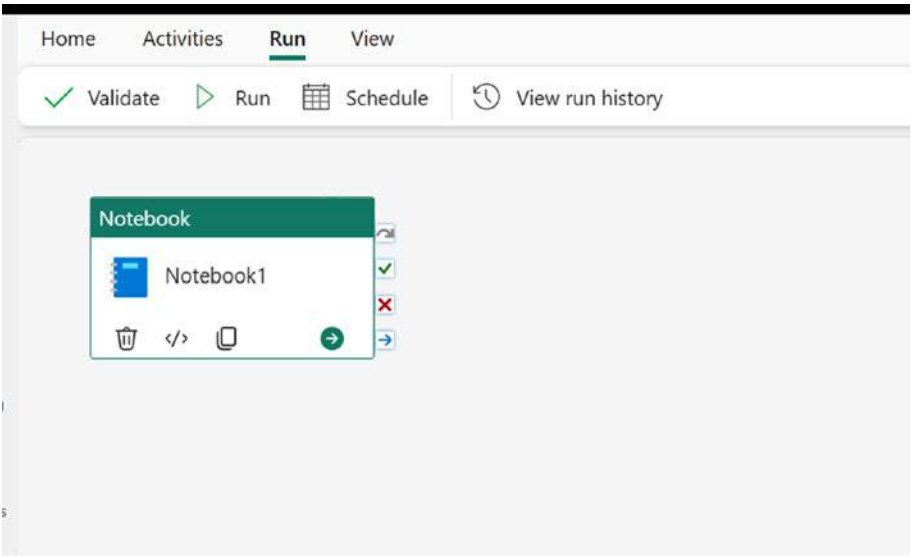


Figure 6-39. *In case of notebook execution failure*

To achieve an "on success" (successful execution of the job step) notification, you need to click the check mark beside the Notebook activity, as shown in Figure 6-40.



Figure 6-40. *Fabric data factory condition execution: On skip, On success, On failure, On completion*

Now select Teams from Notifications, as shown in Figure 6-41. We are selecting the channel here where the notification of job completion will be published. Note that similarly you can select Office 365 Outlook to send an email notification.

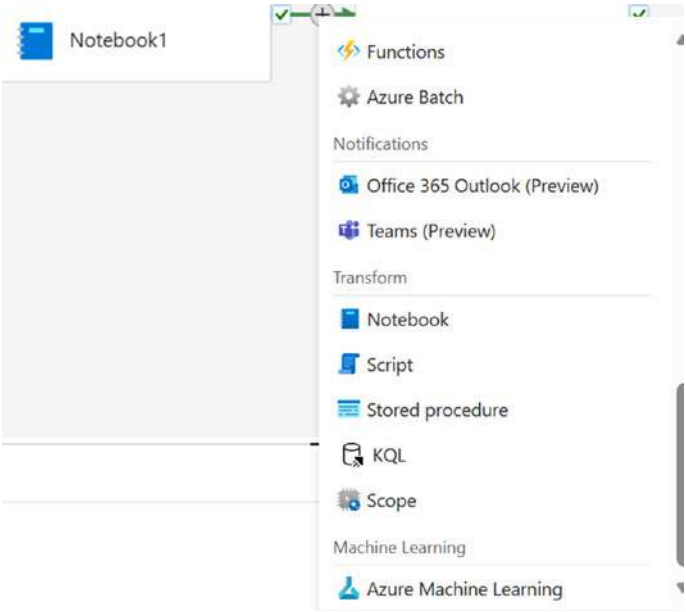


Figure 6-41. *Selecting a channel*

We have now attached Teams as the downstream step. Go to Settings to sign into the Teams account, as shown in Figure 6-42.

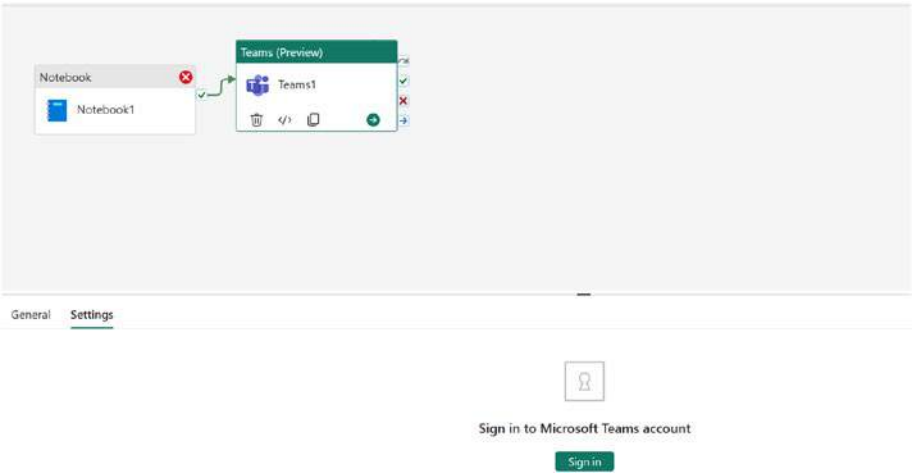


Figure 6-42. *Integrating Teams with the notebook*

Once you provide the authentication information and sign in, you need to select "Group chat" in the "Post in" tab, as shown in Figure 6-43. Subsequently select the Teams group you need to post to. By doing that, you select the Teams group chat where the job success notification needs to be posted. Notification-related messaging details need to be added in the Message section.

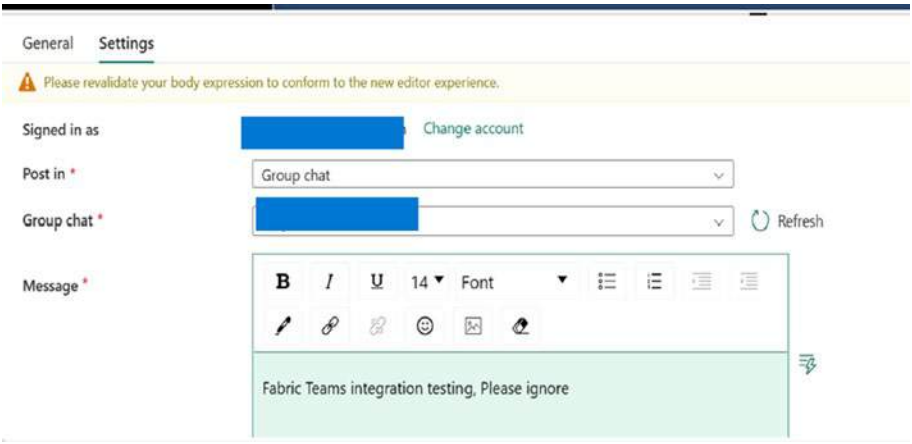


Figure 6-43. *Selecting the group chat*

Now that our pipeline is, you can click Run, as shown in Figure 6-44, to evaluate the notification message.

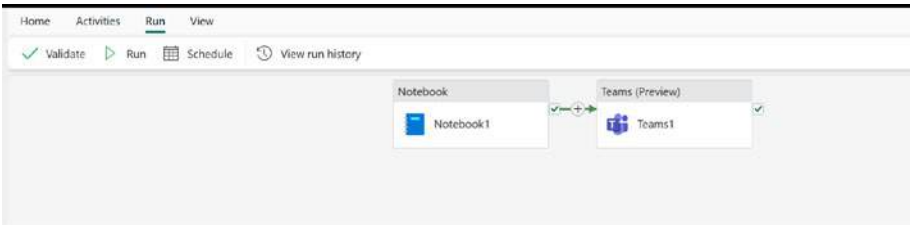


Figure 6-44. *Executing the data pipeline*

Subsequently, the pipeline run starts, as shown in Figure 6-45.

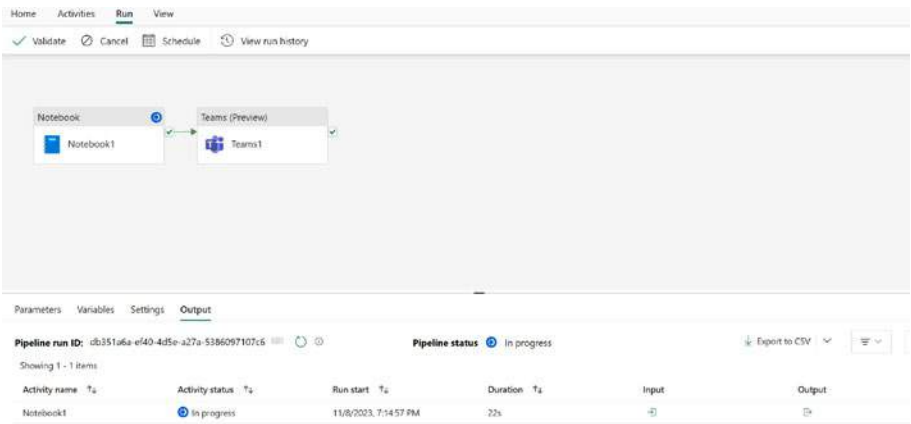


Figure 6-45. *The pipeline is running*

Once the job is completed, you will be able to see desired message is posted in the Teams chat group. See Figure 6-46.

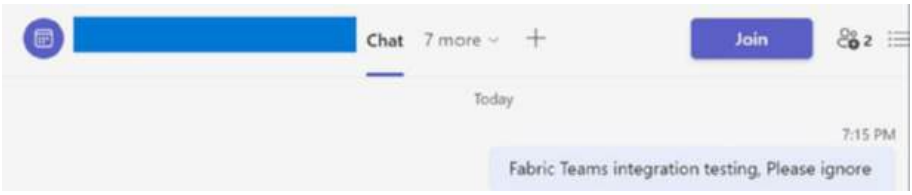


Figure 6-46. *Team chat*

Data Factory Mount

In this section, you will go through Data Factory Mount capability. Before we dive into the details of Data Factory Mount, let's first understand what it is and why it is useful. Data Factory Mount is a feature of Fabric that allows you to run a precreated Azure data factory workload in Fabric. Using Data Factory Mount, you can avoid any immediate upgrades of the Azure Data Factory to Fabric. Figure 6-47 shows the Data Factory Mount capability in Fabric.

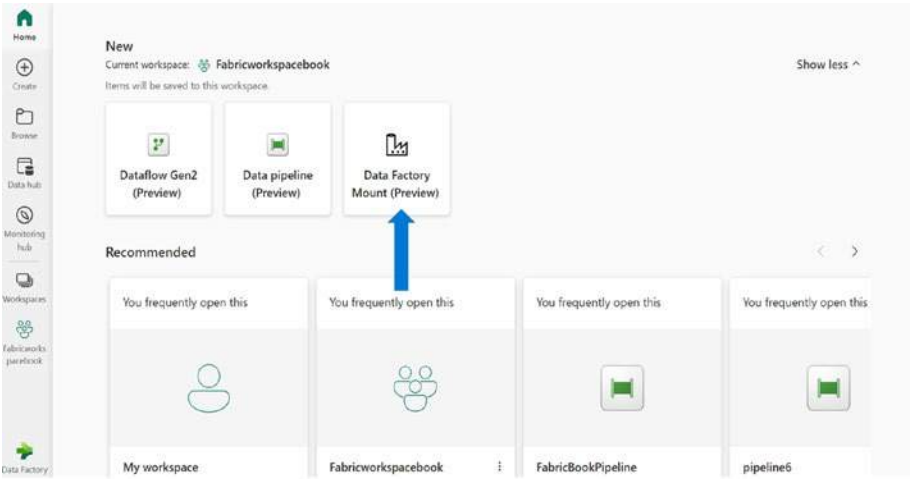


Figure 6-47. *Microsoft Fabric Data factory mounting*

To explore this feature, you need to click the Data Factory Mount (Preview) tab. The tab will list the subscriptions that you have access to and the corresponding data factory (ADF) services, as shown in Figure 6-48. You need to select the ADF service to mount to the Fabric workspace.

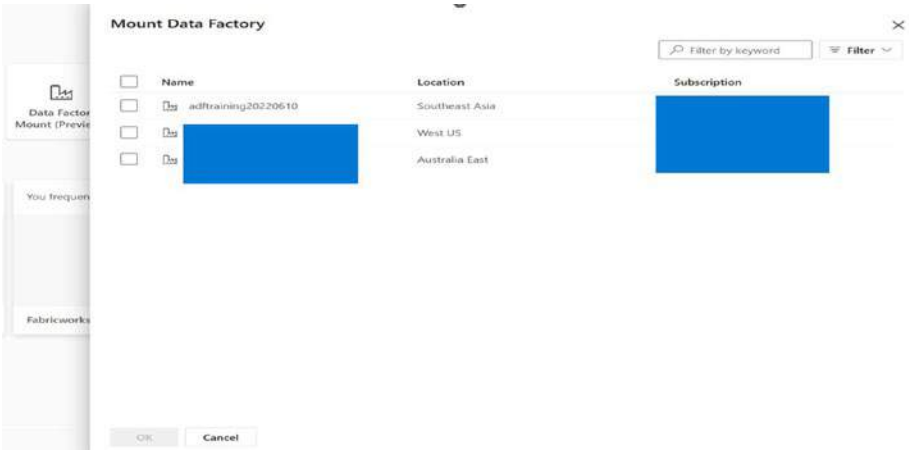


Figure 6-48. *Subscriptions available*

Now you can see the ADF is mounted in Fabric, as shown in Figure 6-49.

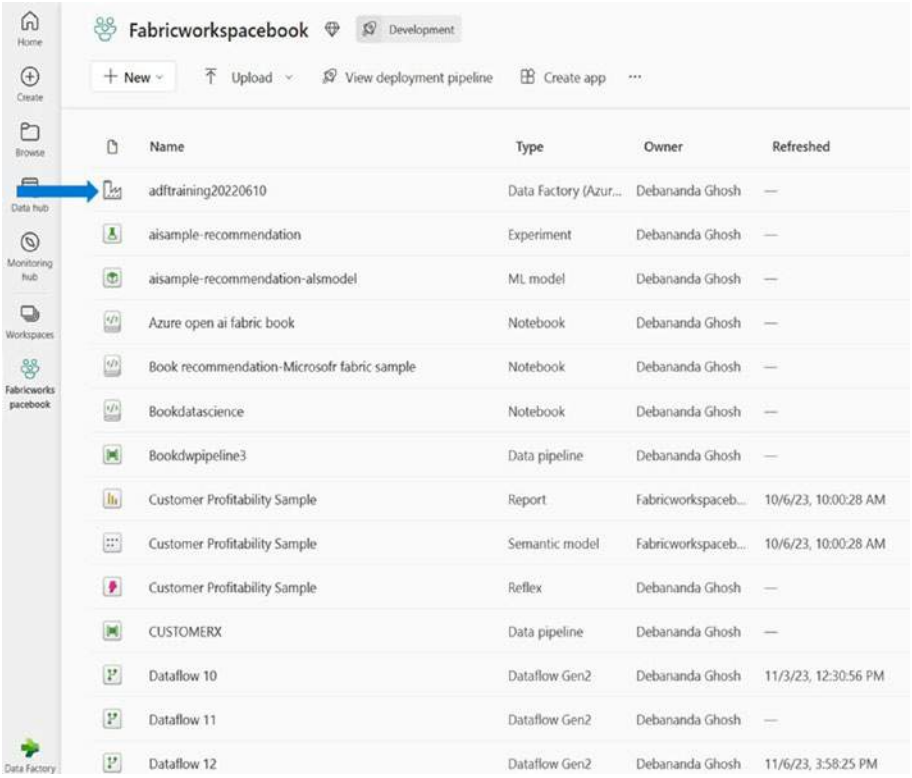


Figure 6-49. *Mounting ADF*

Note that Data Factory Mount primarily is designed for the following three purposes:

- Smoothly transitioning from Data Factory to the Fabric workspace
- Leveraging the modernized user interface for a preexisting data factory workload
- Leveraging the new capabilities of Fabric with a preexisting data factory

Summary

In this chapter, you learned that Microsoft Fabric is an end-to-end data platform architecture. We covered the Fabric Data Factory feature and data integration techniques, including the pipeline and Dataflow Gen2 features, as highlighted in Figure 6-50. You also learned about the Copilot capability as related to the Microsoft Fabric data factory features. Finally, we briefly covered how to migrate Power BI Data Flow Gen1 and the Azure Data Factory workload.

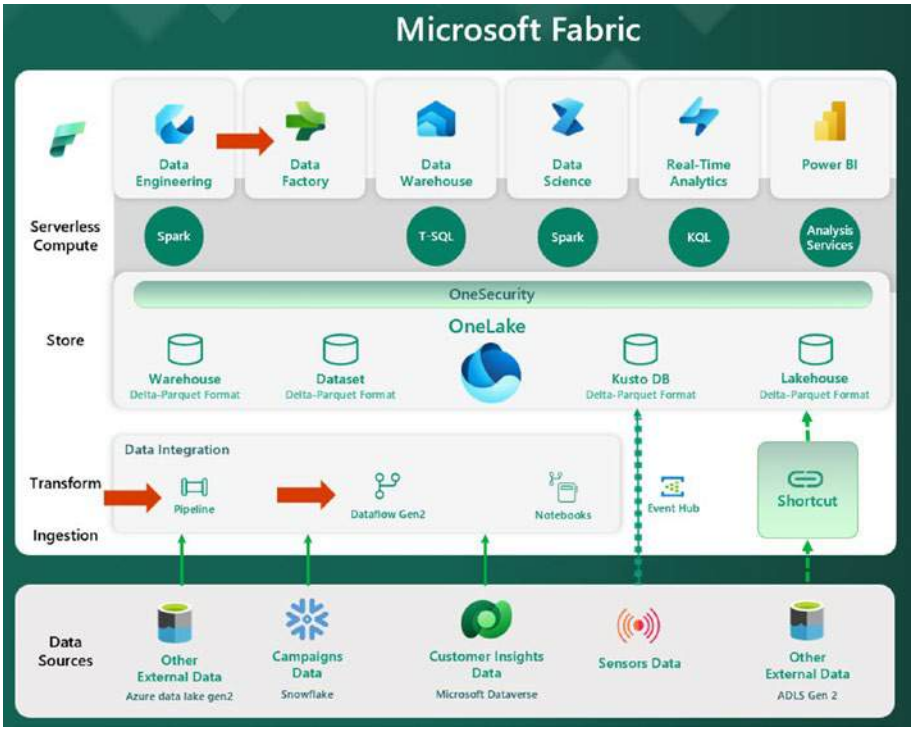


Figure 6-50. *Chapter recap*

In the next chapter, we will move on to real-time streaming analytics.

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric Data pipeline:** https://learn.microsoft.com/en-us/fabric/data-factory/
- **Fabric data factory tutorial:** https://learn.microsoft.com/en-us/fabric/data-factory/tutorial-end-to-end-introduction
- **Migrating your workload from Azure Data Factory:** https://learn.microsoft.com/en-us/fabric/data-factory/upgrade-paths
- **Data transformation with data profile:** https://learn.microsoft.com/en-us/fabric/data-factory/tutorial-end-to-end-dataflow
- **Incremental amassing of data:** https://blog.fabric.microsoft.com/en-us/blog/wondering-how-to-incrementally-amass-data-in-your-data-destination-this-is-how/

CHAPTER 7

Real-Time Analytics with Microsoft Fabric

In today's world we are surrounded by intelligent applications and real-time data management. When you wear a smartwatch, you measure health-related insights like physical workouts and calorie details. When you drive a smart car, you can see your fuel usage analysis during the trip. These are outcomes of real-time analytics. Smart applications and devices are continuously generating tons of data that needs to be analyzed in near real-time for better decisions. Real-time analytics is the ability to provide insights as quickly as possible once data is generated. As per Gartner,

Real-time analytics is the discipline that applies logic and mathematics to data to provide insights for making better decisions quickly. For some use cases, real time simply means the analytics is completed within a few seconds or minutes after the arrival of new data.

Figure 7-1 shows a few near real-time smart applications that you probably use every day.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_7

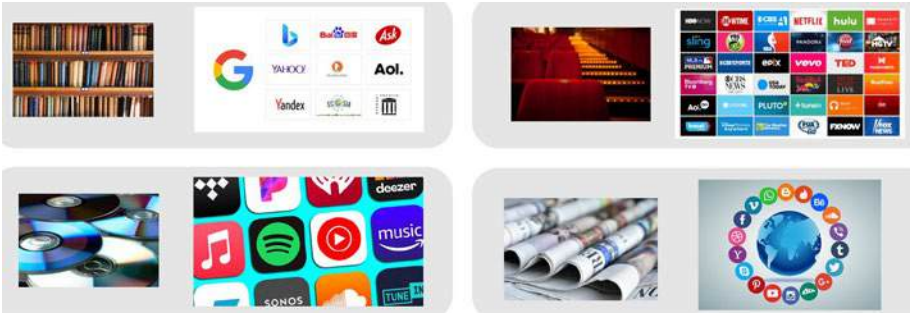


Figure 7-1. *Real-time applications*

Most of the time, organizations want to solve the following challenges:

- Get immediate insights once data is generated
- Build real-time analytics without investing a lot into skillsets and tooling
- Build a real-time analytics ecosystem

Building a real-time data pipeline usually comprises the following capabilities:

- Ingesting, transforming, and writing/serving to the consumer application
- Running analytical queries in near real time
- Scaling the underlying infrastructure in real time
- Doing high-performance, low-latency data management

In this chapter, you will learn how to build a real-time streaming data pipeline and build real-time analytics features seamlessly using Microsoft Fabric. Figure 7-2 shows the Synapse Real-Time Analytics canvas in the Fabric workspace.

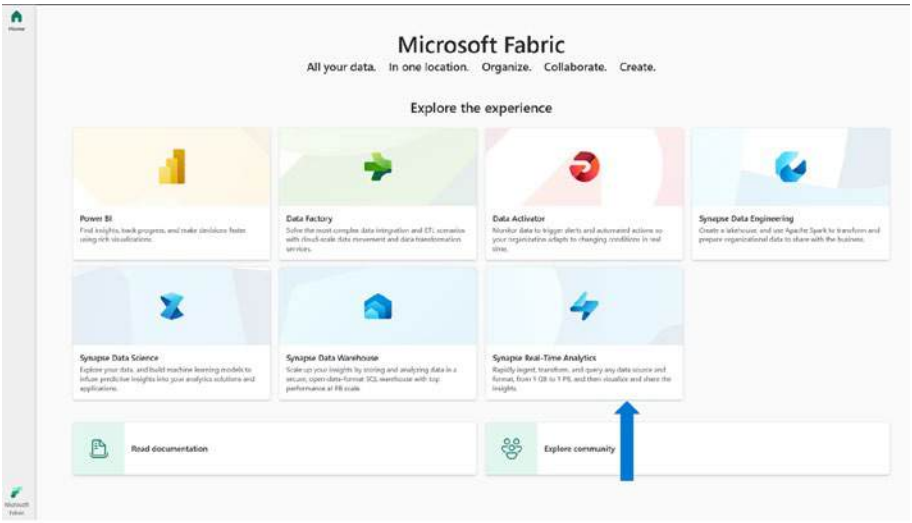


Figure 7-2. *Synapse Real-Time Analytics canvas*

Fabric provides real-time analytics (RTA) persona experiences in the Synapse Real-Time Analytics canvas. The Fabric RTA capability evolved from the Azure Data Explorer capability and embraces software-as-a-service (SaaS) foundation.

Specifically, in this chapter, you will learn about the following topics:

- Real-time analytics anatomy
- KQL database queries
- OneLake integration with Fabric RTA
- Event streams

Fabric Real-Time Analytics

Let's start with the key pillars for the Fabric real-time analytics capability. Fabric RTA comprises these three main components:

- **Kusto query database:** High-performance database for low-latency streaming dataset
- **KQL query component:** Fabric item for developers to query the KQL database
- **Event stream:** No-code tooling to build streaming pipeline, ingest, query, and serve the data

Once you click Synapse Real-Time Analytics, the RTA canvas appears. You will be able to see the following tabs in the RTA canvas, as shown in Figure 7-3:

- KQL Database
- KQL Queryset
- Eventstream
- Use a sample

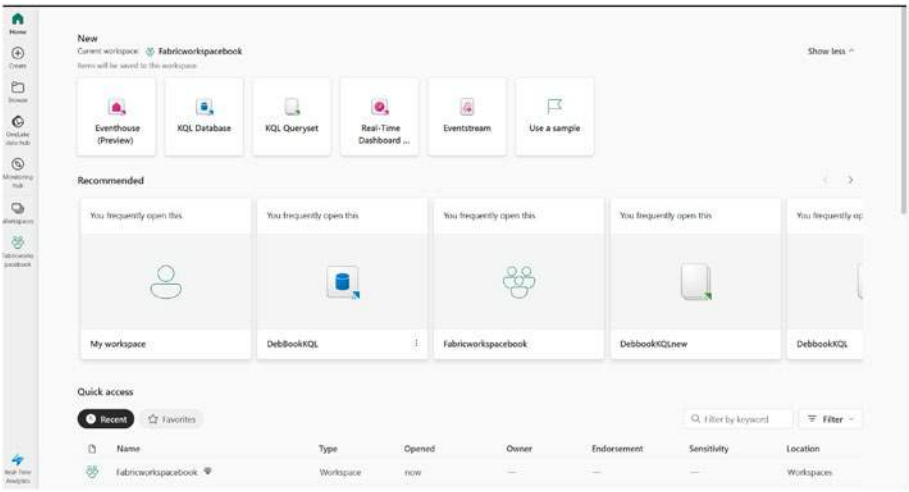


Figure 7-3. *RTA canvas*

Let's now start using these intuitive tab to create real-time analytics. Perform the following simple steps:

1. Create a Kusto (KQL) database to store a real-time streaming dataset as a prerequisite to using the UI.
2. Learn how to ingest a sample file/data into the database.
3. Run a sample query in the database.
4. Create a streaming data pipeline using the intuitive UI via an event stream.
5. Write a streaming data pipeline in the Kusto database.

Create a Kusto Database

To store real-time analytics, you will first create a Kusto database. KQL database are fast, fully managed databases that come integrated within the Fabric workspace. They are used for interactive queries and analysis on high-volume and high-velocity scenarios. Streaming data from web applications, mobile devices, websites, and IoT devices can be analyzed to understand the data trends and eventually build a cost-optimized, high-performing streaming analytics platform.

To provision a new KQL database, you need to click KQL Database. The canvas shown in Figure 7-4 will open; provide a name for the KQL database for provisioning purposes and then click the Create button.

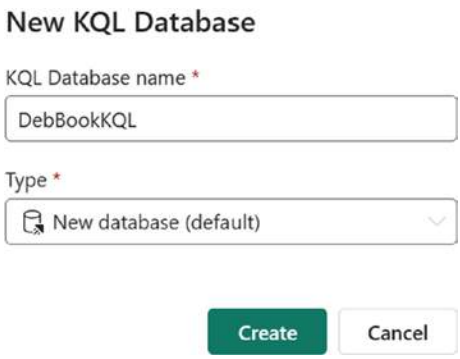


Figure 7-4. *Creating a Fabric KQL database*

You have now created an empty database. Once it's created, you will be able to see the following details of the database, as shown in Figure 7-5:

- **Created by:** This is the username of who created the database.
- **Region:** This is the service's hosted region.
- **Created on:** This is the date the database was created.
- **Last Ingestion:** This is the last timestamp for the data ingestion.
- **Query URI:** This is the URI to run management commands; it's in this format: `https://abc-xyzrabcpqrtyuzio.z6.kusto.fabric.microsoft.com`.
- **Ingestion URI:** This is the URI that can be used in the application to ingest the data; see `https://ingest-abc-xyzrabcpqrtyuzio.kusto.z6.kusto.fabric.microsoft.com`.

- **OneLake URI:** This is the path for the OneLake folder. Here is the sample format of the URI: `https://onelakemsit.pbidedicated.windows.net/b58wftewwee-f06e-4fbc-86fegsdgrwtrweterty/fdsgsdgsdgssdgsddg-4491-aa7b-d7bd8f5634fd`.
- **Compressed:** This is the total volume of compressed data.
- **Original Size:** This is the total size of the data uncompressed.
- **Compression ratio:** This is the compression ratio of the compressed data to the total size.
- **Most active users:** This is the username of the most active users in the database.
- **Queries ran last month:** This is the number of queries run per user in the last month.
- **Recently updated function:** This is the last function name that was updated.
- **Recently used query sets:** This is the recently used KQL query set.
- **Recently created data connections:** This is the data connection name and the time it was created.

Figure 7-5 shows the Fabric KQL database details and data tree containing the database and table hierarchy.

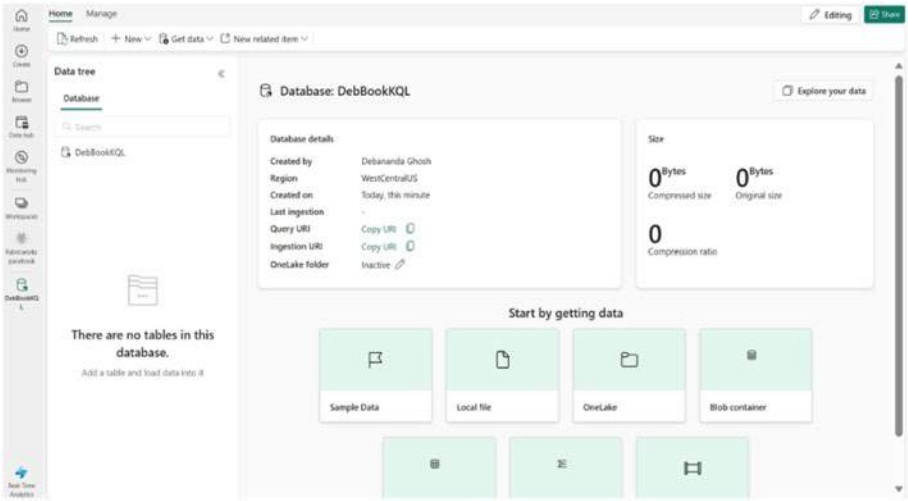


Figure 7-5. *Fabric KQL Database Home page*

Ingesting Data into the Kusto Database

Once the database is created, you need to load a sample dataset into the database. To ingest data with just a few clicks, you can leverage the "Get data" drop-down, as shown in Figure 7-6. Note that there are other ways to leverage this feature. For example, you can click the three dots beside the database name.

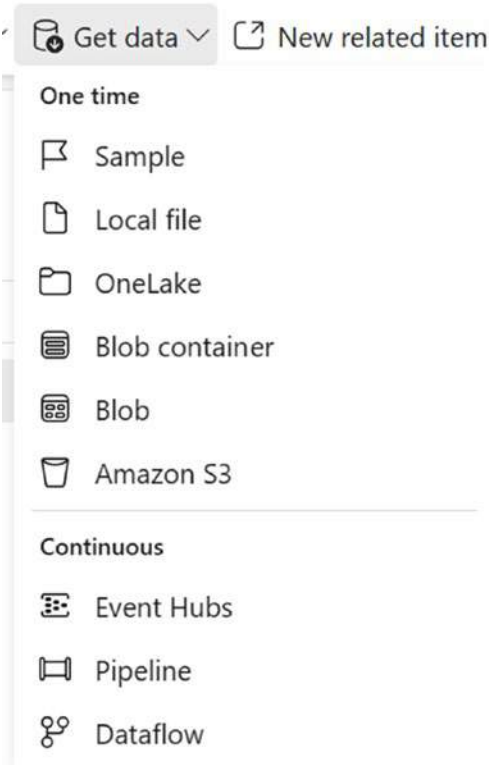


Figure 7-6. *Fabric KQL database ingestion options*

The Fabric RTA "Get data" tab also provides a few sample datasets for academic and product test-driving purposes. For our purposes, click Sample under "One-time ingestion" on the "Get data" tab to get some sample data, as shown in Figure 7-7. Select a sample dataset from the gallery and continue with the self-explanatory next steps to ingest the sample data.

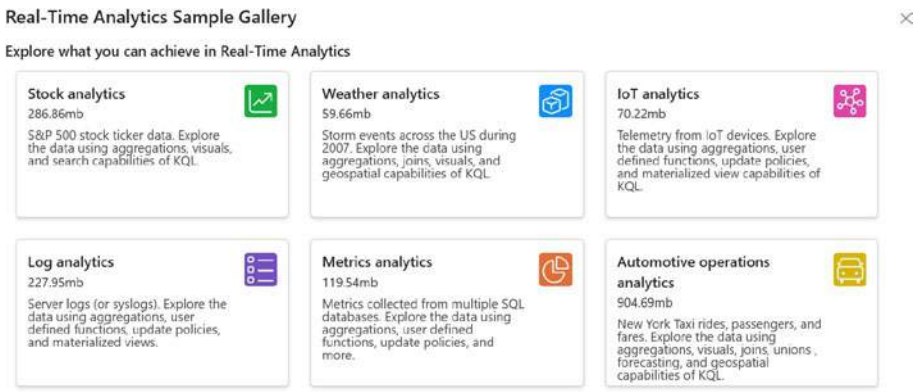


Figure 7-7. *Fabric KQL gallery*

The "Get data" feature is a no-code mechanism for loading datasets inside the KQL database. Next click Next Source and then provide a new table name or select "Existing table," as shown in Figure 7-8.

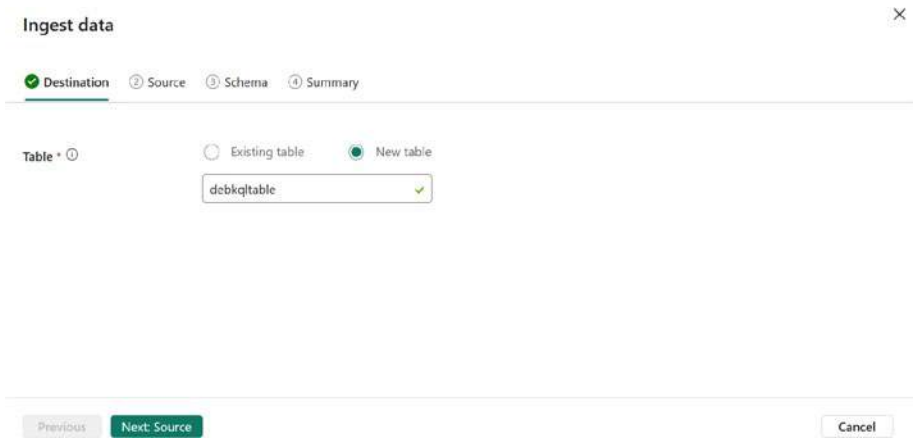


Figure 7-8. *Naming the table*

You have now loaded a sample dataset inside the KQL database. Continue exploring the other native integrations such as a local file, OneLake, Blob container, AmazonS3, event hub, pipeline, or data flow,

which all provide an easy interface to load data into the KQL database. As of now, you have completed the first and second steps. You will now move on to the third step and learn how to use the Kusto database.

1. Create a Kusto (KQL) database to store real-time streaming datasets as a prerequisite for using the UI.
2. Learn how to ingest a sample file/ data into the database.
3. Run a sample query in the database.

KQL Database Query

You can use the KQL database code-free tooling, such as the KQL query set and SQL and Python extensions to run Kusto, SQL, and Python code in the database. Programmers who want to interact with database programmatically can leverage Python, Node.js, Go, .NET, the Java SDK, and APIs.

To query the loaded table in a code-free manner, you need to click the three dots (…) beside the table name and click "Show any 100 records." Likewise, you can use similar code-free tooling by clicking the Query Table tab on the top left and then clicking "Query table," as shown in Figure 7-9.

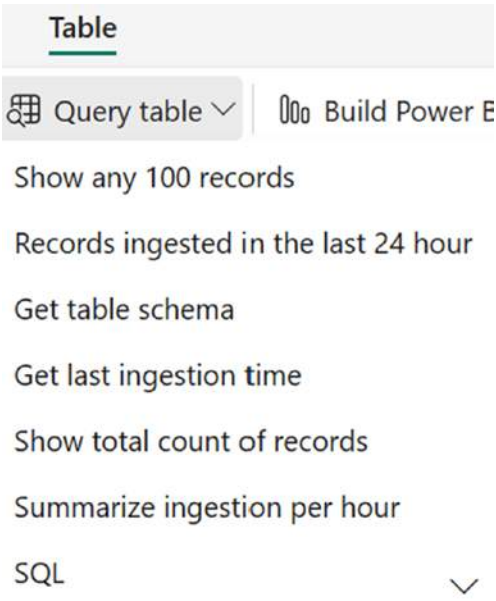


Figure 7-9. *Creating a Fabric KQL database code-free query*

The following code snippet is generated when you click the "Show any 100 records":

```
//**************************************************************
*******************************************
// Here are two articles to help you get started with KQL:
// KQL reference guide - https://aka.ms/KQLguide
// SQL - KQL conversions - https://aka.ms/sqlcheatsheet
//**************************************************************
*******************************************

// Use "take" to view a sample number of records in the table
and check the data.
YOUR_TABLE_HERE
| take 100
```

```
// See how many records are in the table.
YOUR_TABLE_HERE
| count

// This query returns the number of ingestions per hour in the
given table.
YOUR_TABLE_HERE
| summarize IngestionCount = count() by bin(ingestion_
time(), 1h)

// Use take to view a sample number of records in the table and
check the data.
RawSysLogs
| take 100
```

Figure 7-10 shows the outcome of the previous code snippet.

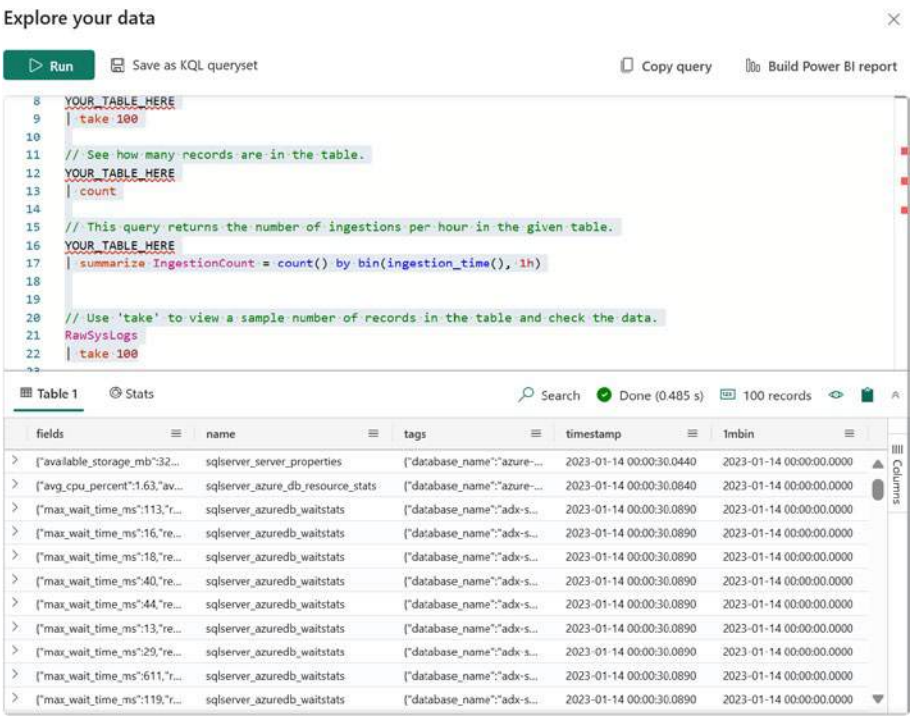


Figure 7-10. *Fabric KQL query execution and sample outcome*

For business intelligence developers, a visualization development option is provided. Click "Build Power BI report" to visualize the dataset, as shown in Figure 7-11.

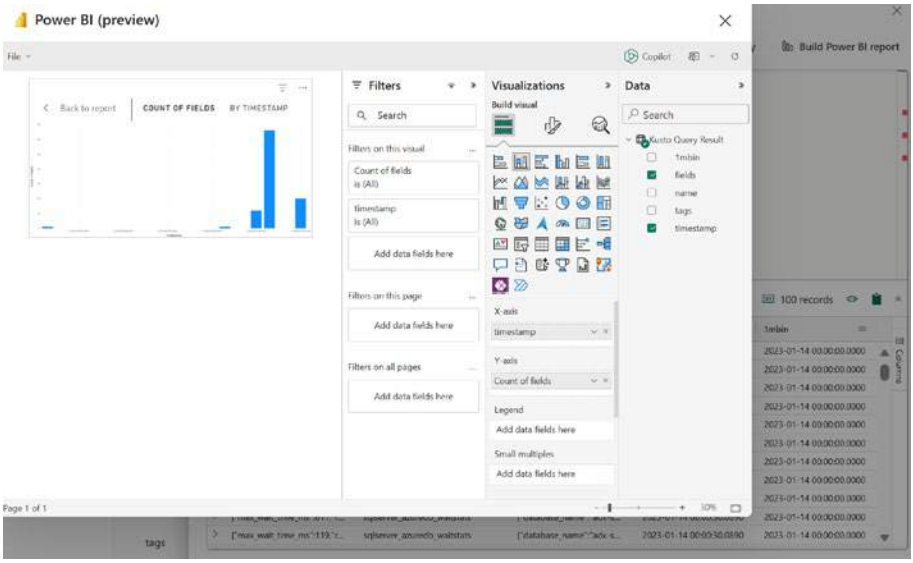


Figure 7-11. *Fabric BI visualization using a KQL dataset*

Kusto Query Development

The Kusto Query Language is an effective query language for exploring and discovering data. It is used to identify data anomalies and outliers, create statistical modeling, and do much more. A Kusto query follows a similar pattern as a SQL query.

Overall, Kusto is simple and powerful and provides the following benefits:

- Rich query language (filter, aggregate, join, calculated columns, and more)
- Query language–based syntax
- Hierarchical schema
- Built-in visualization

- Built-in full-text search, time series, user analytics, geospatial, and machine learning operators
- Extensible
- In-line Python

To write a KQL query, you need to click "New related item," as shown in Figure 7-12. Subsequently provide the new KQL query set name (in this case DebbookKQLnew) and then click the Create button.

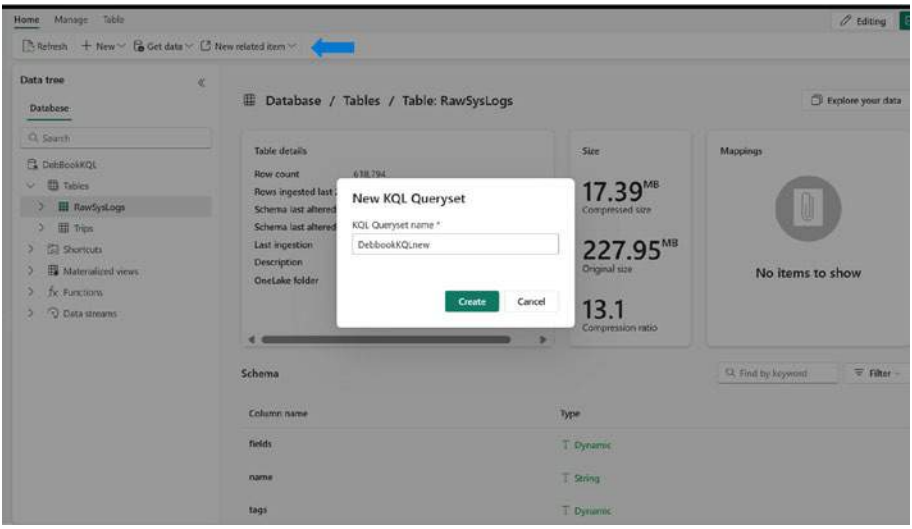


Figure 7-12. *Creating a KQL query set*

You can also start doing KQL script development using other UI features. For example, you can click +New at the top left below the workspace name and then select KQL Queryset, as shown in Figure 7-13.

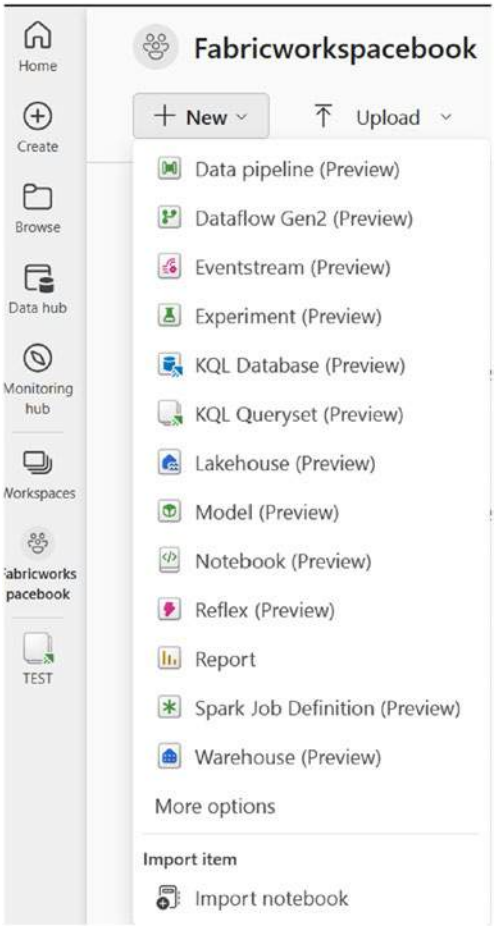


Figure 7-13. *Using the menu to create a query set*

The following query set is generated inside the KQL query set artifact:

```
// See how many records are in the table.
YOUR_TABLE_HERE
| count

// This query returns the number of ingestions per hour in the
given table.
```

```
YOUR_TABLE_HERE
| summarize IngestionCount = count() by bin(ingestion_
time(), 1h)
```

You need to tweak the table name to get the `count()` of the table. Note that KQL query operators are used in queries and indicated by the pipe (|) character.

```
 RawSysLogs
 | summarize IngestionCount = count() by bin(ingestion_
time(), 1h)
```

Now let's look at a few more examples of KQL queries. Here are sample queries for analysis purposes:

```
// Find count of number of rows in a table-
RawSysLogs | count

// Retrieve selective columns from the table

RawSysLogs
| take 500
|  project fields,name

// Retrieve selective columns from the table and sort


RawSysLogs
|  sort by fields
| take 500
|  project  fields,name

// Retrieve selective column and aggregate
RawSysLogs
|   summarize tags = count() by timestamp
```

Note that there are different types of KQL statements. These statements are used by both user queries and application queries and are delimited by semicolons (;).

- **Alias statement:** This sets the alias of a database.
- **Pattern statement:** This helps to define the mapping between tuples and tabular data.
- **Query parameter statement:** This sets the name or value pairs, for example, to pass user ID passwords.
- **Restrict statement:** This sets a restriction to table view and hence can be used for entity security purposes.
- **Let statement:** This is used primarily to set a variable name.
- **Set statement:** This is for query execution and query return control.
- **Tabular expression statement:** This is the most common part of query for both input and output for a tabular dataset.

Here are some examples of KQL statements:

```
// Pattern statement
declare pattern book = (name:string)[state:string]
{
  ("Fabric").["spark"] = { print Capital = "lakehouse" };
  ("Fabric").["dataexplorer"] = { print Capital = "Real time
analytics" };
  ("Fabric").["datawarehouse"] = { print Capital =
"datawarehouse" };
};
book("Fabric").dataexplorer
```

```
// Set statement
set querytrace;
RawSysLogs | take 100

//Tabular statement
RawSysLogs
| where name =='sqlserver_server_properties'
```

In the next section, you will learn how you can leverage the Python extension to explore the Synapse RTA database.

Python Plugin

Note that the KQL query set is not limited to Kusto programming development. You can leverage Python as well. The KQL database plugin provides a Python sandbox and helps Kusto scripts to invoke Python user-defined functions (UDFs). To turn on the Python extension, click Manage in the database tab and then click the Plugin feature. The canvas shown in Figure 7-14 will open to enable the Python plugin feature.

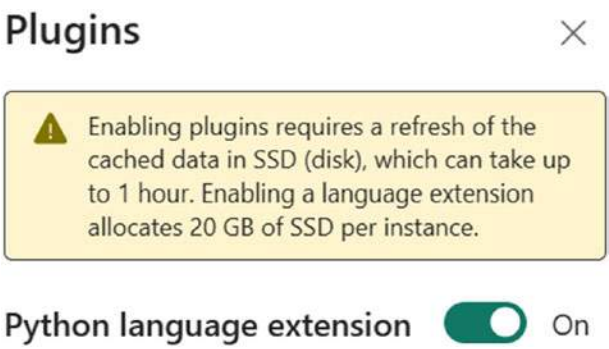


Figure 7-14. *Fabric Python language extension*

Now let's run some extensible Python code using a KQL query. In the following code, you are executing Python UDF code from a KQL script:

```
range y from 1 to 720 step 1
| evaluate python(
//
typeof(*, fy:double),
'''
result = df
n = df.shape[0]
g = kargs["gain"]
f = kargs["cycles"]
result["fy"] = g * np.cos(df["y"]/n*2*np.pi*f)
'''
, pack('gain', 100, cycles', 4)
)
| render areachart
```

The outcome of the previous code snippet renders the area chart, as shown in Figure 7-15.

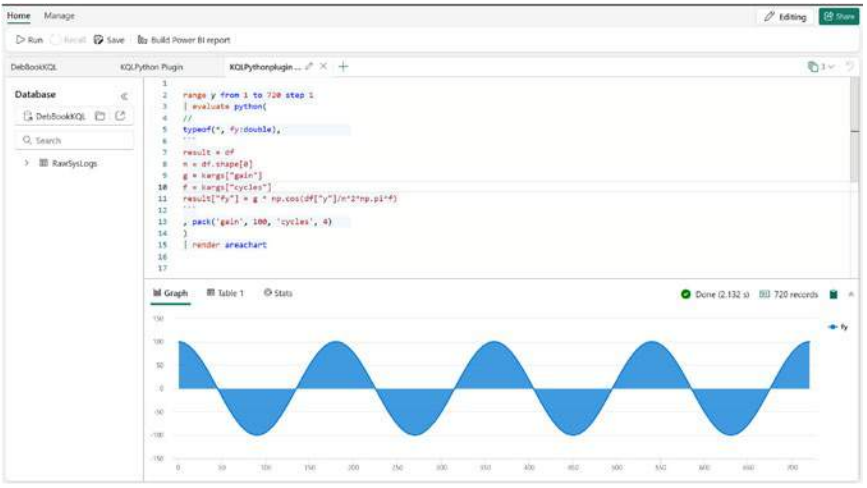


Figure 7-15. *Rending using the Python language extension*

Likewise, you can leverage a Python extension for similar analysis in the KQL query set.

Kusto API and SDK

Kusto offers an API, Python SDK, .NET SDK, Node SDK, Go SDK, and Java SDK for programmers to connect with a Kusto cluster. For example, the Python SDK mechanism allows you to leverage the Python client library and interact with the Fabric RTA through such libraries. The following are some useful packages for Python developers:

- **azure-kusto-data:** This library provides query capabilities in the Kusto cluster.
- **azure-kusto-ingest:** This library helps to ingest data inside a Kusto cluster.
- **azure-mgmt-kusto:** This library is used for Kusto management purposes.

The following code snippet shows how to install Python packages from the Python integrated development environment (IDE):

```
%pip install azure-kusto-data
from azure.kusto.data import KustoClient,
KustoConnectionStringBuilder
from azure.kusto.data.exceptions import KustoServiceError
from azure.kusto.data.helpers import dataframe_from_
result_table
import pandas as pd
```

Subsequently you need to leverage the Kusto Ingestion URI in this format: `https://ingest-abc-xyzrabcpqrtyuzio.kusto.z6.kusto.fabric.microsoft.com`. You can leverage sample Microsoft-provided GitHub projects for further development purposes; see `https://github.com/Azure/azure-kusto-python/tree/master/quick_start`.

Data Retention, Caching, and OneLake Integration

Once the data is ingested and stored in database, you need to build a data retention policy and catching policy for optimizing costs and achieving high performance. This is done using a simple interface inside the KQL database canvas. Under Manage Table, you have data policies and plugins, as shown in Figure 7-16. The data retention policy is when data will be removed from the table and materialized view. Likewise, the caching policy governs how long the data will stay in the processing nodes.

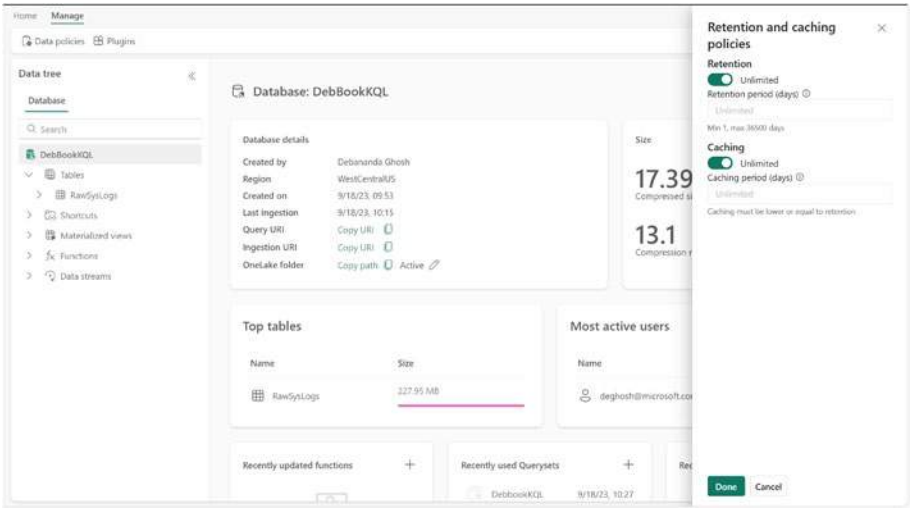


Figure 7-16. *Setting up a retention and caching policy*

To activate the KQL database with the OneLake filesystem, you need to turn on the Active toggle, as shown in Figure 7-17. This feature helps to synchronize the KQL database and OneLake filesystem.

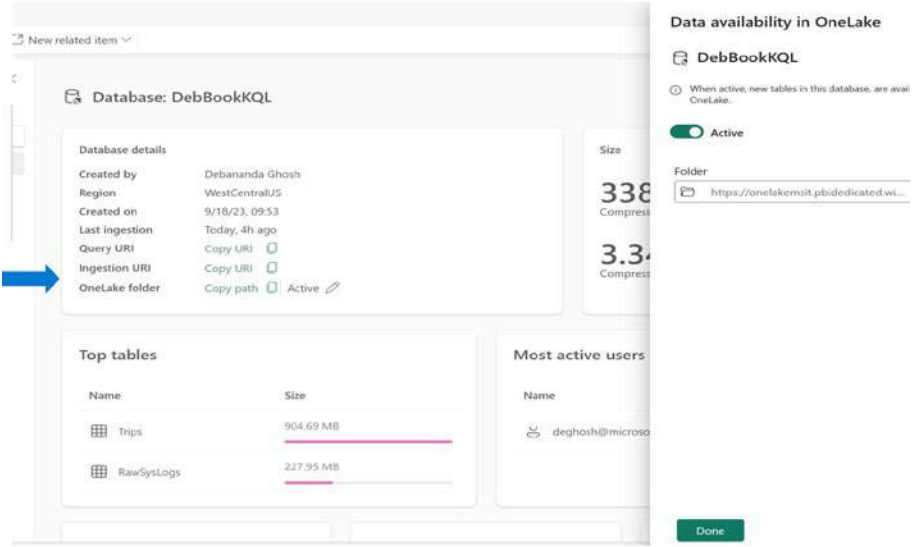


Figure 7-17. *OneLake integration*

In the next section, you will learn how to build a streaming pipeline. Specifically, you'll follow these steps:

1. Create a streaming data pipeline using the intuitive UI via an event stream.
2. Write a streaming data pipeline in the Kusto database.

Using an Event Stream

Fabric's event stream capability is a centralized place to build a streaming pipeline using the workspace UI and to manage the streaming ingestion. The Fabric event stream capability helps to capture a transform and write the real-time transaction data using a code-free tooling mechanism. This experience provides the following capabilities:

- Using a native streaming connector via a no-code experience
- Integrating with the Azure event hub
- Building a custom application and integrating with the Fabric event stream
- Connecting with multiple destination sources using a no-code experience

Figure 7-18 depicts the event stream feature in the Fabric RTA canvas.

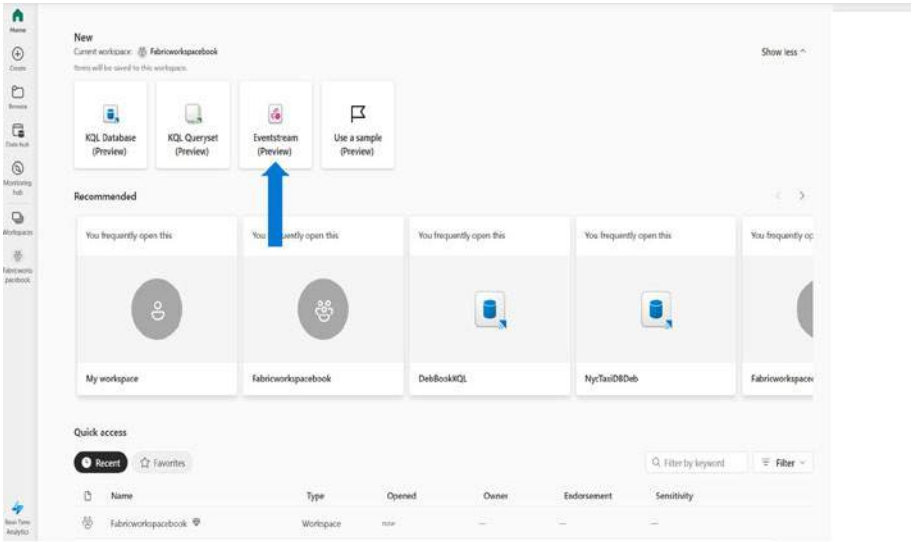


Figure 7-18. *Fabric Eventstream (Preview) feature*

You will execute the following steps to explore the event stream feature further:

1. Create a streaming pipeline using the UI.
2. Read the Fabric-provided sample streaming data using that pipeline.
3. Write it in the previously created KQL database.

To create a streaming pipeline, you need to click "Event stream." Subsequently, provide the new event stream's name, as shown in Figure 7-19.

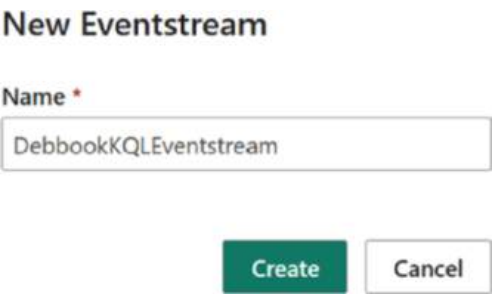


Figure 7-19. *Creating an event stream*

After you click the Create button, the empty event stream artifact will be created, as shown in Figure 7-20.

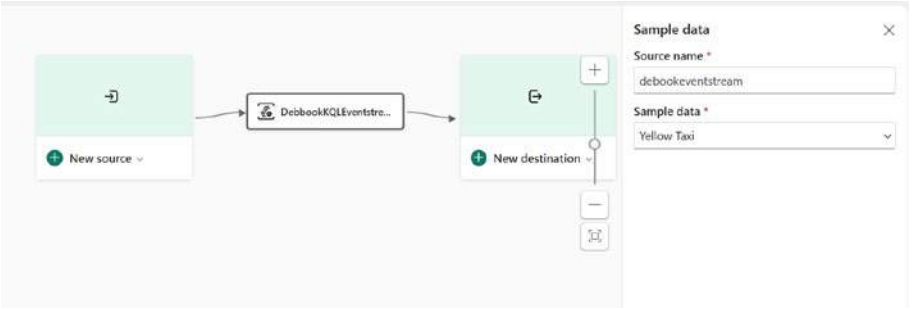


Figure 7-20. *Event stream empty canvas*

As the next step, you must configure the origin and destination of the event stream. Once you have configured the event stream destination, you can start sending data to the event stream. Note that you can natively connect with Event Hubs, IOT Hub, and custom apps via the UI. Using a custom app, you can use any of the supported sources to send data in

JSON format. For example, you can use a custom app to send simulated data using the REST API endpoint provided by the event stream. The sources shown in Figure 7-21 are natively supported via the canvas.

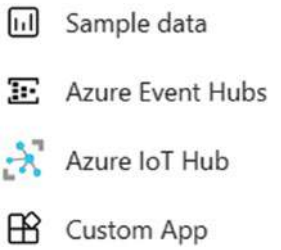


Figure 7-21. *Event stream sources*

Likewise, you need to click "New destination" and select the destination. The destinations shown in Figure 7-22 are supported via the UI canvas.

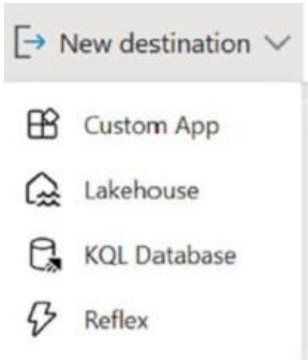


Figure 7-22. *Fabric Event destination sources*

For academic purposes, we have selected the sample data provided by the Fabric workspace. We also selected a new table inside the KQL database to load the sample dataset. Figure 7-23 depicts the fully developed event stream pipeline.

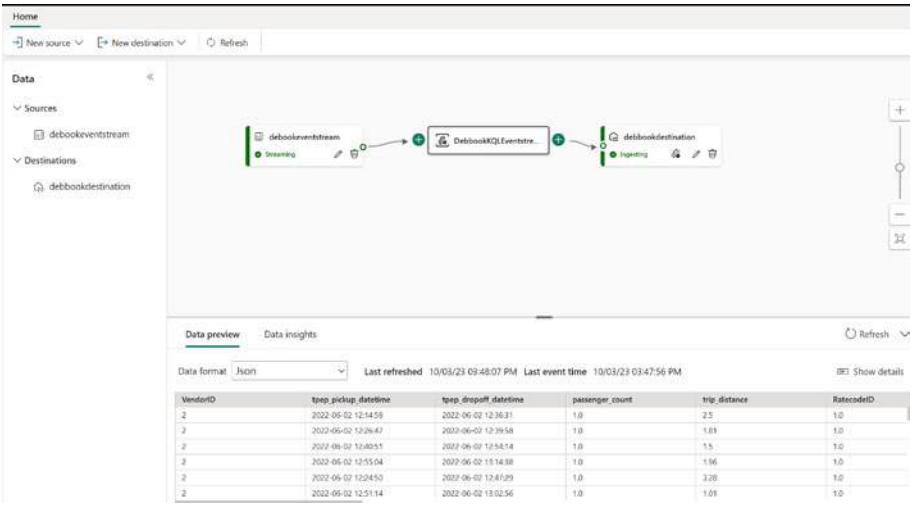


Figure 7-23. *Full event stream pipeline*

For applications where you need to leverage the Kafka application, you need to first fetch the connection key string from the Details section, as shown in Figure 7-24. Then look at the sample code to see how to use the key in a Kafka application.



Figure 7-24. *Custom apps development*

In our scenario connection string, the sample format looks like this: `Endpoint=sb://ppppppp.servicebus.windows.net/;SharedAccessKeyName=key_aaaaaaaaaaaaa;SharedAccessKey=bbbbbbb=;EntityPath=es_ccccccc`.

The following Java code snippet for the Kafka application is generated by the "Sample code" section:

```
// Copyright (c) Microsoft Corporation. All rights reserved.
// Licensed under the MIT License.

    package EventStream.Sample;

    import org.apache.kafka.clients.producer.KafkaProducer;
    import org.apache.kafka.clients.producer.Producer;
    import org.apache.kafka.clients.producer.ProducerConfig;
    import org.apache.kafka.clients.producer.ProducerRecord;
    import org.apache.kafka.common.serialization.
    LongSerializer;
    import org.apache.kafka.common.serialization.
    StringSerializer;

    import java.util.Properties;

    public class KafkaSender {
        private static final int NUM_MESSAGES = 100;
        private static final String connectionString =
        "Endpoint=sb://ppppppp.servicebus.windows.
        net/;SharedAccessKeyName=key_aaaaaaaaaaaaa;SharedAccess
        Key=bbbbbbb=;EntityPath=es_ccccccc
";
        private static final String eventHubName = "es_xxxxx";

        public static void main(String[] args) {
            // Initialize the Properties
```

```
        Properties props = getProperties();

        // Create the Producer and send the message
        try (Producer<Long, String> producer = new
        KafkaProducer<Long, String>(props)) {
            publishEvents(producer);
        }
    }

    private static Properties getProperties() {
        Properties props = new Properties();
        String namespace = connectionString.
        substring(connectionString.indexOf("/") + 2,
        connectionString.indexOf("."));
        props.put("bootstrap.servers", String.format("%s.
        servicebus.windows.net:9093", namespace));
        props.put("security.protocol", "SASL_SSL");
        props.put("sasl.mechanism", "PLAIN");
        props.put("sasl.jaas.config", String.format("org.
        apache.kafka.common.security.plain.PlainLoginModule
        required username="$ConnectionString"
        password="%s";", connectionString));
        props.put(ProducerConfig.CLIENT_ID_CONFIG,
        "KafkaExampleProducer");
        props.put(ProducerConfig.KEY_SERIALIZER_CLASS_
        CONFIG, LongSerializer.class.getName());
        props.put(ProducerConfig.VALUE_SERIALIZER_CLASS_
        CONFIG, StringSerializer.class.getName());
        return props;
    }
```

```
    private static void publishEvents (Producer<Long,
    String> producer) {
        for(int i = 0; i < NUM_MESSAGES; i++) {
            long time = System.currentTimeMillis();
            System.out.println("Test Data #" + i + " from
            thread #" + Thread.currentThread().getId());

            final ProducerRecord<Long, String> record =
            new ProducerRecord<Long, String>(eventHubName,
            time, "Test Data #" + i);
            producer.send(record, (metadata,
            exception) -> {
                if (exception != null) {
                    System.out.println(exception);
                    System.exit(1);
                }
            });
        }
        System.out.println("Sent " + NUM_MESSAGES + "
        messages");
    }
}
```

Likewise, you can leverage AMQP, the Azure Event hub-based integration using the custom application feature.

Summary

In this chapter, you learned about the various KQL development and integration features. Refer to Figure 7-25 for a recap.

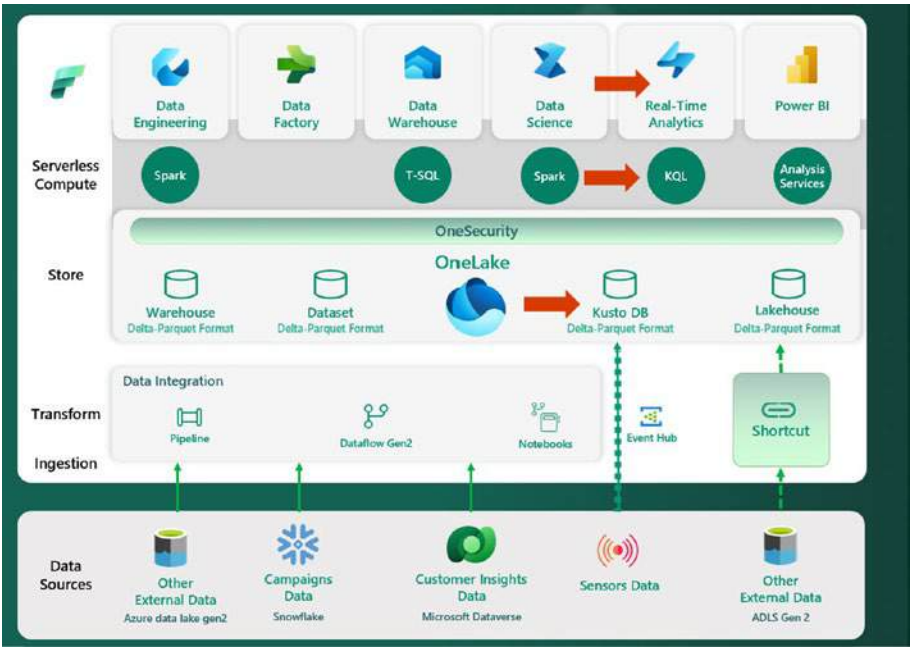


Figure 7-25. *KQL development and integration features*

In the next chapter, we will discuss in detail how to generate alerts from real-time events via the Microsoft Fabric activator capability.

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric real-time analytics:** `https://learn.microsoft.com/en-us/fabric/real-time-analytics/realtime-analytics-compare`
- **Fabric event streams:** `https://learn.microsoft.com/en-us/fabric/real-time-analytics/event-streams/overview`

- **Fabric real-time custom apps integration:** https://learn.microsoft.com/en-us/fabric/real-time-analytics/event-streams/stream-real-time-events-from-custom-app-to-kusto
- **Kusto SDK:** https://learn.microsoft.com/en-us/azure/data-explorer/kusto/api/python/kusto-python-client-library
- **Geospatial analytics using Kusto:** https://techcommunity.microsoft.com/t5/azure-data-explorer-blog/geospatial-analytics-for-connected-vehicle-with-synapse-data/ba-p/3566185
- **Aviation analytics using Kusto:** https://techcommunity.microsoft.com/t5/azure-data-explorer-blog/aviation-flight-data-analytics-with-azure-synapse-analytics/ba-p/3566273

CHAPTER 8

Microsoft Fabric Activator Real-Time Monitoring and Alerts

In the previous chapter, you learned how to implement real-time analytics (RTA) for organizations using Microsoft Fabric. In this chapter, you will focus on the next step in real-time analytics once the RTA foundation is built.

Note that businesses want real-time alerts to prevent any incidents. For example, a banking institution needs to monitor credit card transactions in near real-time to find suspicious transactions and subsequently act on them. Once a bank detects such activity, it needs to be alerted instantaneously so it can mitigate the downstream business impacts.

For such use cases, an enterprise needs to take the following steps at a high level:

1. Monitor and detect real-time analytics using RTA (for example, monitor and detect suspicious transactions).
2. After incident detection, generate alerts in real time (for example, send SMS messages to credit card owners and enterprise application stakeholders).

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_8

3. Trigger an automated workflow process (for example, an application trigger to block a credit card).

In this chapter, you will focus on all these subject areas. By the end of this chapter, you will be able to create a real-time alert and triggering mechanism for enterprise deployment purposes. Figure 8-1 shows the real-time monitoring and alert/trigger generation capability in RTA.

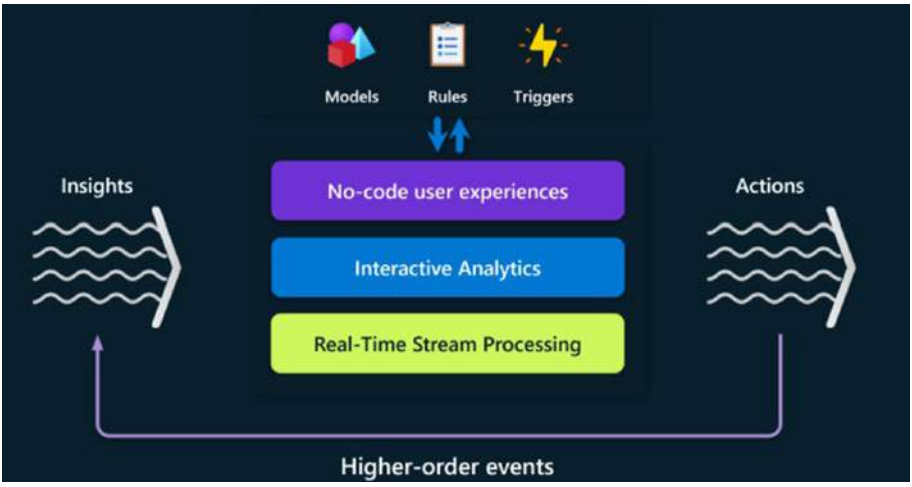


Figure 8-1. *Real-time trigger concepts*

Note that to implement this real-time monitoring, alerting, and trigger mechanism, an organization faces the following challenges:

- **Multiple monitoring tools:** The enterprise needs to engage with multiple open-source software tools/vendors for the real-time monitoring activity.
- **Development effort:** The end-to-end real-time alert and trigger generation requires development effort and a certain skillset.

- **Generating the alert in near real time:** Organizations need to generate alerts on data in transit with an accepted latency (usually less than 10 seconds).
- **Stakeholder management:** Organizations need to send alerts to the right channels and stakeholders.
- **Downstream integration process:** Integration effort is required to trigger downstream process.

All such challenges lead to increased IT costs and less business agility. Microsoft Fabric eliminates such challenges and overhead drastically with the Data Activator capability. The Data Activator capability provides a no-code user experience to trigger alerts and trigger downstream processes with just a few clicks. Figure 8-2 shows the Data Activator feature on the home page of Fabric.

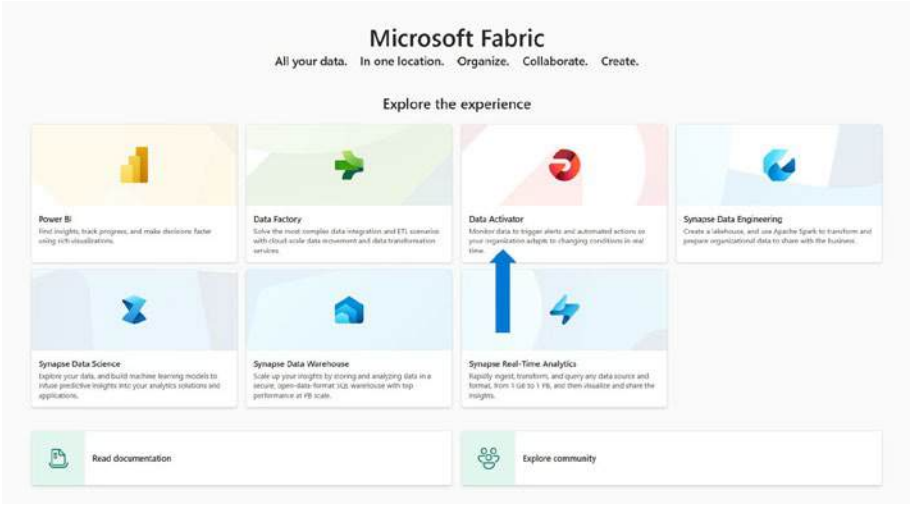


Figure 8-2. *Data Activator offering in Fabric*

Data Activator Anatomy

To learn more about Data Activator, let's click the Data Activator tab. We will use the Data Activator UI to create a simple real-time monitoring and alert generation feature with just a few click. We will first walk you through a Fabric-provided sample for learning purposes.

In this scenario, click "Reflex sample" in the workspace where you need to start development. This step will ingest the simulated event data and then deploy an end-to-end Reflex object (a Data Activator–based Fabric item) for academic purposes. Figure 8-3 shows the landing page of Data Activator (also known as Reflex).

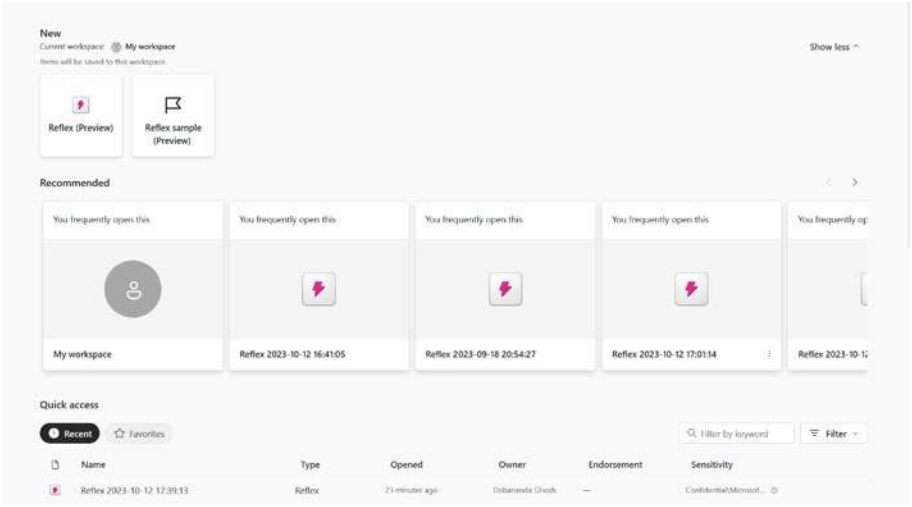


Figure 8-3. *Fabric Data Activator canvas*

You can start playing with this tool by clicking "Reflex sample." Fabric will create some Reflex objects for learning purposes. Data Activator consists of a few core parts, as explained in the following points:

- **Objects:** An object is the business object/physical object that you need to monitor and generate a trigger on if required.
- **Events:** An event in Data Activator is the observation of the field value that you need to monitor.
- **Triggers:** This allows you to define a pattern and act based on the pattern detection.
- **Properties:** Properties are provided for logic reusability purposes.

In Figure 8-4 you can see a business object named Package (on the left side below Objects). This object consists of triggers, properties, and events, which are core pillars of the Reflex object. This Reflex object deploys sample event data related to the subject area Package.

In this scenario, you have the Package Delivery Attempt, Package In Transit, and Package Shipped events prepopulated as part of the built-in sample, as shown in Figure 8-4.

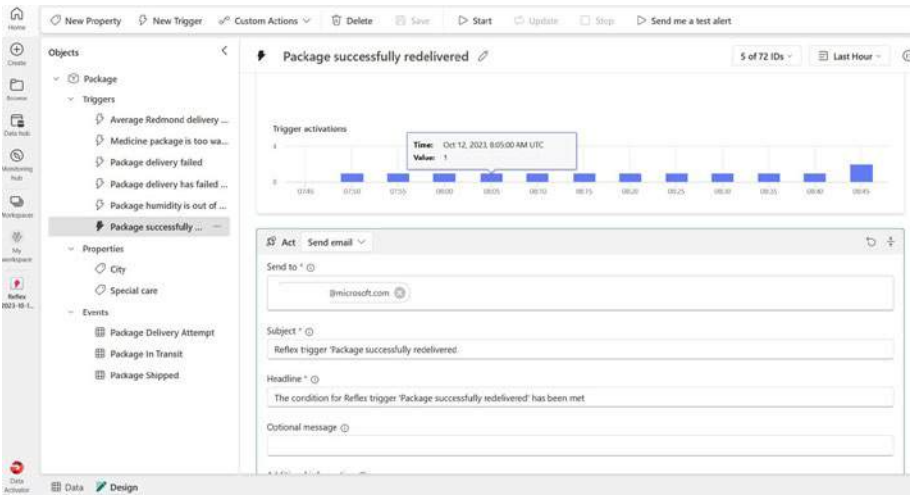


Figure 8-4. *Fabric Data Activator sample Reflex objects*

To understand the underlying data events, you need to click Data in the bottom-left corner. Figure 8-5 shows the Package Delivery Attempt, Package In Transit, and Package Shipped dataset records and related visualizations.

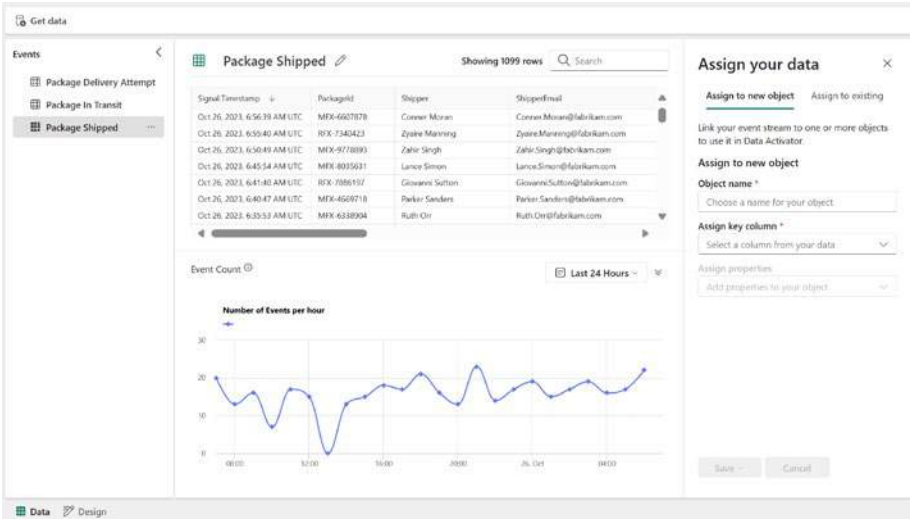


Figure 8-5. *Reflex event data and visualizations*

We will now switch back from the Data tab to the Design tab again (bottom-left corner) and focus on the Triggers field, as shown in Figure 8-6. For example, "Average Redmond delivery time above target" is an event-based trigger that leverages the built-in event Package Delivery Attempt. This event-based trigger further detects anomalies by using a filter condition in the Detect field. Subsequently, this event-based trigger leverages the Act field and sends an email or team message to the right stakeholder. To initiate a trigger, you need to click Start, as shown in Figure 8-6.

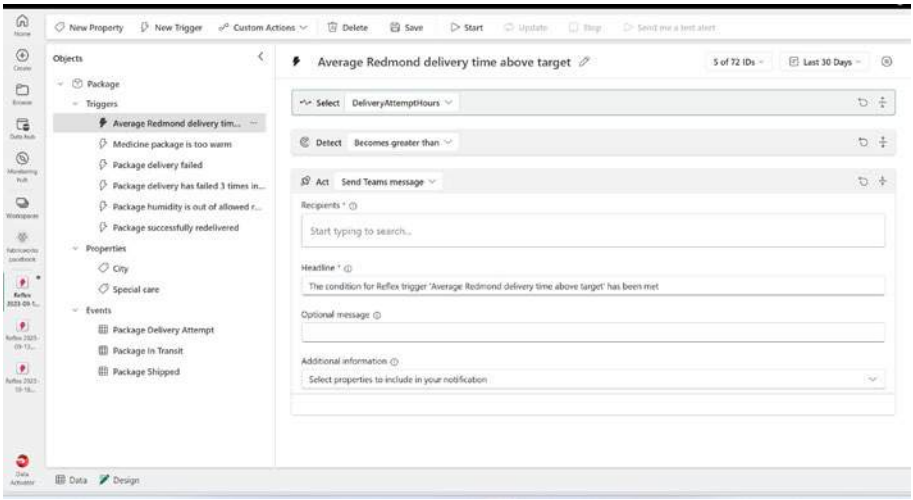


Figure 8-6. *Configuring a Reflex trigger*

Figure 8-7 represents the trigger-based visualization inside the canvas. The visualization depicts how many times "Average Redmond delivery time above target" was activated during a time window.

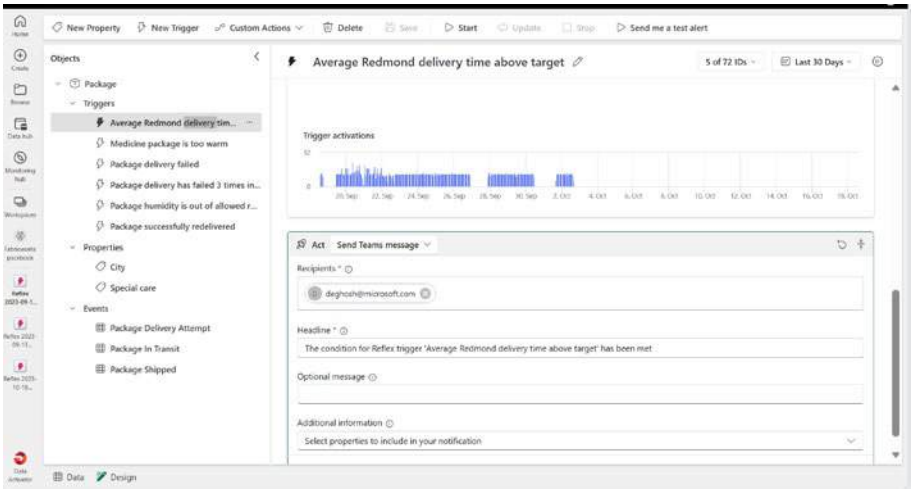


Figure 8-7. *Reflex object trigger visualization*

The previous example was all about using the prebuilt sample. However, for project development purposes, you need to learn more about how you can work on such configuration to build a similar Reflex object. To start designing a reflex workflow for events, you need to first connect with some data sources/real-time KPIs. Designing a Data Activator primary workflow consists of these three steps:

1. Connect with data sources like a Power BI dataset or event stream.
2. Define and detect actionable patterns.
3. Trigger actions based on the detected pattern.

Let's now double-click each of these three parts separately in the following sections.

Connect with Data Sources

To build an event alert, you need to first integrate Fabric with the event. Before looking at this, you will first start creating a new Reflex object. Click Reflex, as you can see in Figure 8-8.

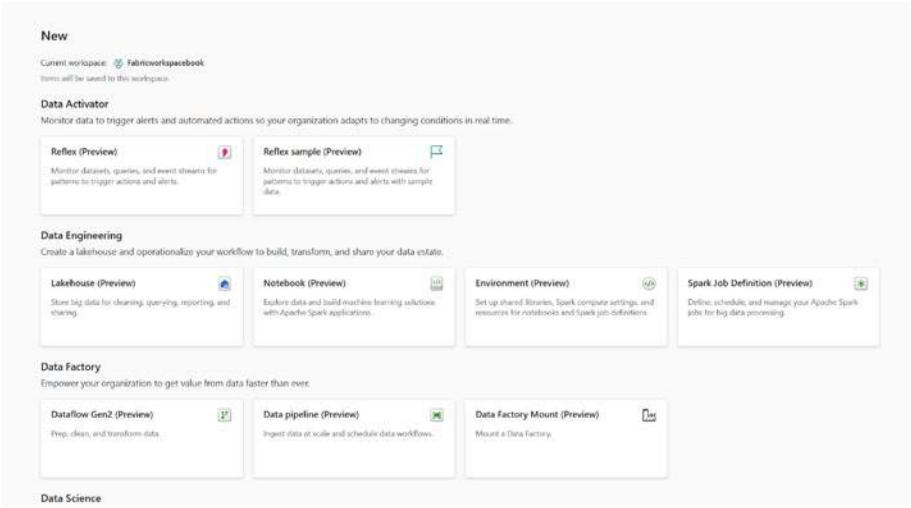


Figure 8-8. *Creating a Reflex object*

Once the object is created, as the next step you need to connect the Reflex object with various data sources. Here you will click Get Data for the Reflex object, as shown in Figure 8-9. After seeing the "Reflex get data instruction for Power BI " message, you need to open the Power BI report visual and then integrate the visual with Reflex.

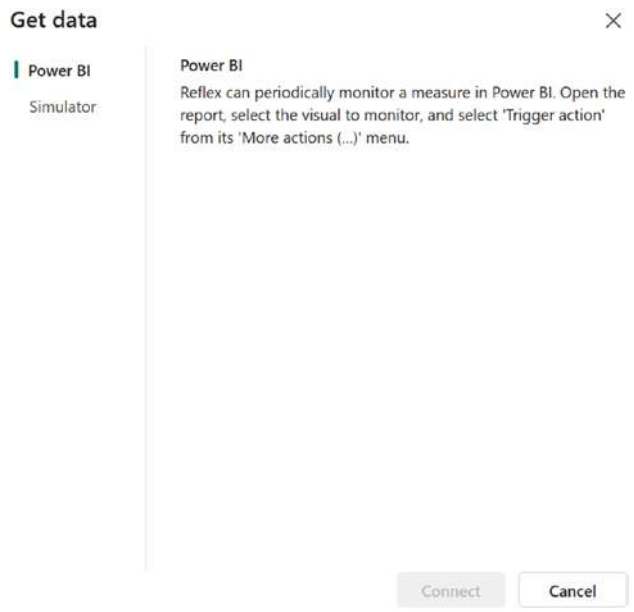


Figure 8-9. *Getting data instructions for Power BI*

At the time of this writing, Power BI and Event Hub support Reflex object connectivity through Power BI visuals and the Eventstream canvas, respectively.

Power BI as the Data Source

In this section, you will learn how to monitor a Power BI visual and then generate an alert when the visual crosses some threshold. Let's learn how to do that with just a few clicks in the Microsoft Fabric activator. Go back to Power BI (the Fabric item) and start with a sample report that already exists in the workspace. Figure 8-10 shows that we are leveraging the Customer Profitability Sample report. You can get the corresponding `.pbix` file at `https://learn.microsoft.com/en-us/power-bi/create-reports/sample-customer-profitability`.

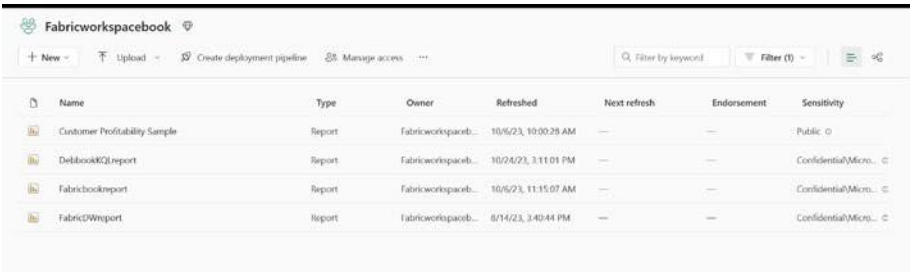


Figure 8-10. *Power BI sample report*

Note that in Chapter 9 we will discuss Power BI in more detail. To go to a Power BI report, go to the Filter section and select Report.

The Customer Profitability Sample report has an Execution Scorecard page, as shown in Figure 8-11. Now you want to monitor the Power BI measure called Revenue TY and generate an alert when that value exceeds a certain value. To generate an alert, you will leverage Fabric reflex item with the following steps:

1. Click Alerts + Power Automate, as shown in Figure 8-11.

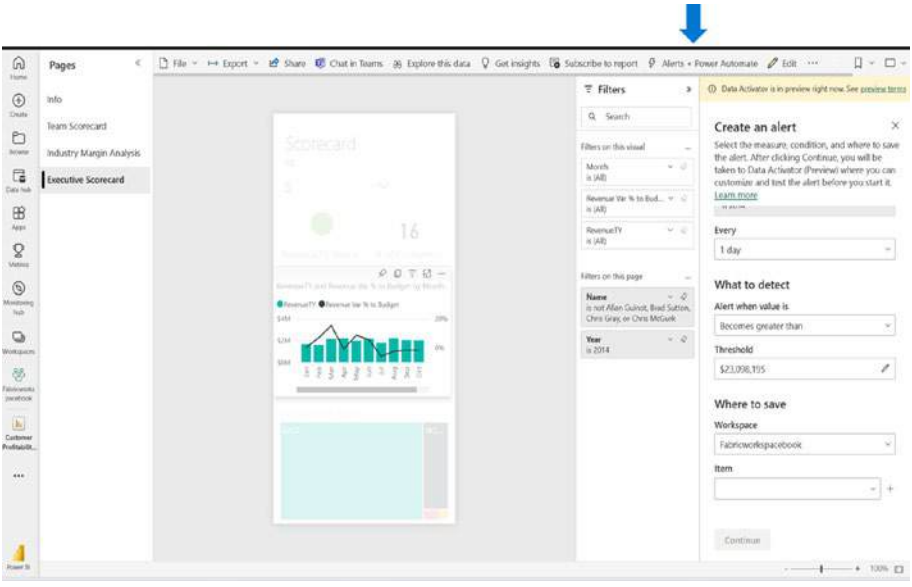


Figure 8-11. *Alert and Power Automate*

2. Set the detect conditions using Threshold for "Alert when value is."

3. Tag the 'Item' with an existing or new Fabric reflex item (in this case we have created a new Reflex item named "Customer profitability sample"), as shown in Figure 8-12.

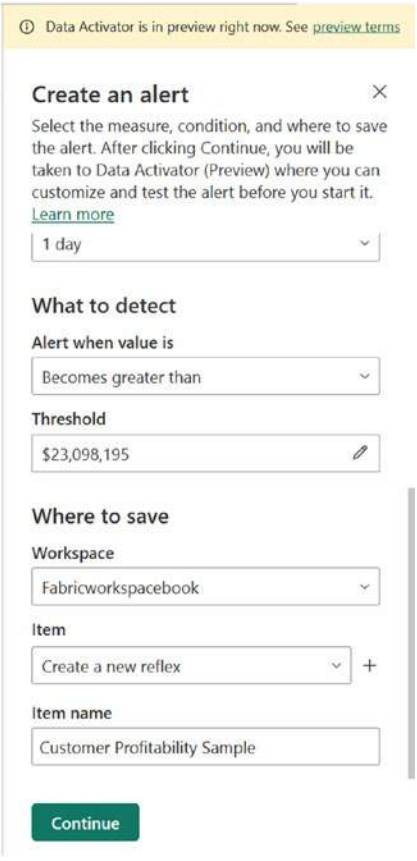


Figure 8-12. *Detecting and generating an alert*

4. Once a trigger is created, click Start Trigger.

This step will create a Reflex object, as shown in Figure 8-13. This object includes a trigger named "Revenue TY becomes greater than 23098195.236." Within this trigger, you can see that the Select, Detect, and Act tabs are prepopulated by the canvas itself, and thus you do not need any further manual configuration. While the previously mentioned steps were about generating triggers via Power VI visuals, you will now move on to streaming sources.

Event Stream as a Source

In this section, you will learn how to generate alerts and trigger events via a streaming pipeline. To achieve this, you will use a Fabric streaming pipeline (known as an *event stream*) as the source for the Reflex object. You will integrate the Reflex object as the destination of the event stream event. Subsequently, you will be able to monitor and detect an actionable pattern and trigger an alert/custom action. To explore the Activator UI experience, you need to go back to an event stream (created in Chapter 7). To go back to the previously created event stream, you need to click Filter in the Fabric workspace's home page and search for *event stream*, as shown in Figure 8-13.

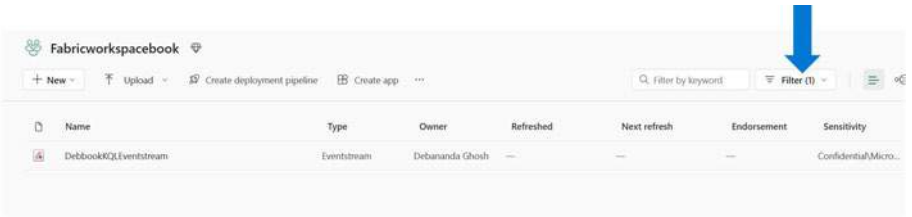


Figure 8-13. *Fabric workspace filter*

Once you click the pre-created event stream, the corresponding canvas appears, as shown in Figure 8-14.

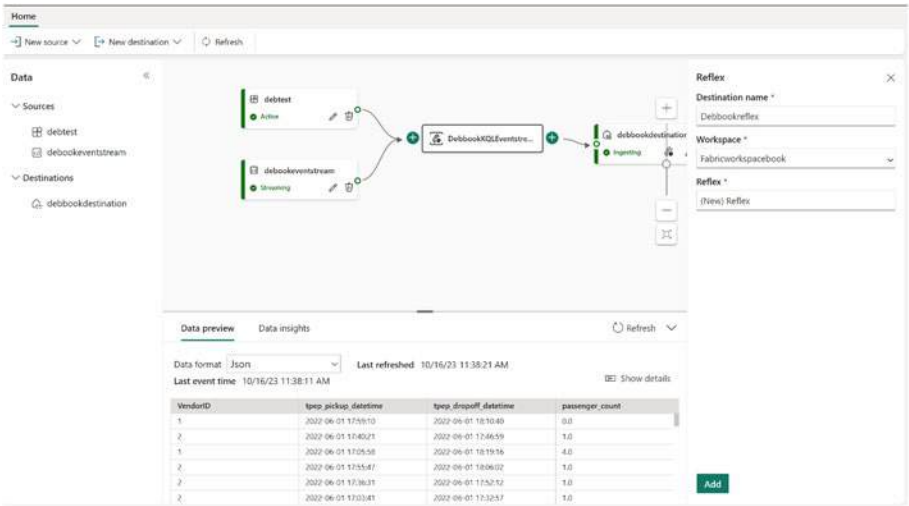


Figure 8-14. *Synapse Eventstream canvas*

Recall that in Chapter 7 we discussed that new sources of event streams can be EventHub streams or custom apps. Additionally, in the "New destination" field of the event stream, you will see Reflex as one of the available features, as shown in Figure 8-15.

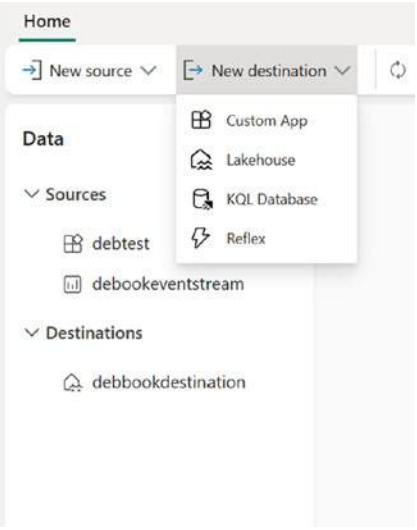


Figure 8-15. *Reflex destination*

Now you want to integrate an event stream with a Reflex action. To do this, click Reflex as the new destination. Once you click, the screen shown in Figure 8-16 appears.

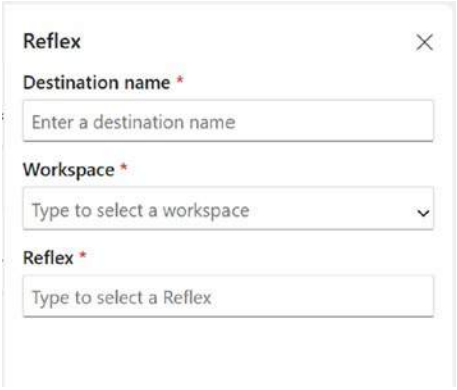


Figure 8-16. *Reflex integration*

We have now created a new Reflex destination linked with our existing streaming pipeline named "Debbookreflex, Related item name Reflex_2023-10-16T03:40:03," as shown in Figure 8-17.

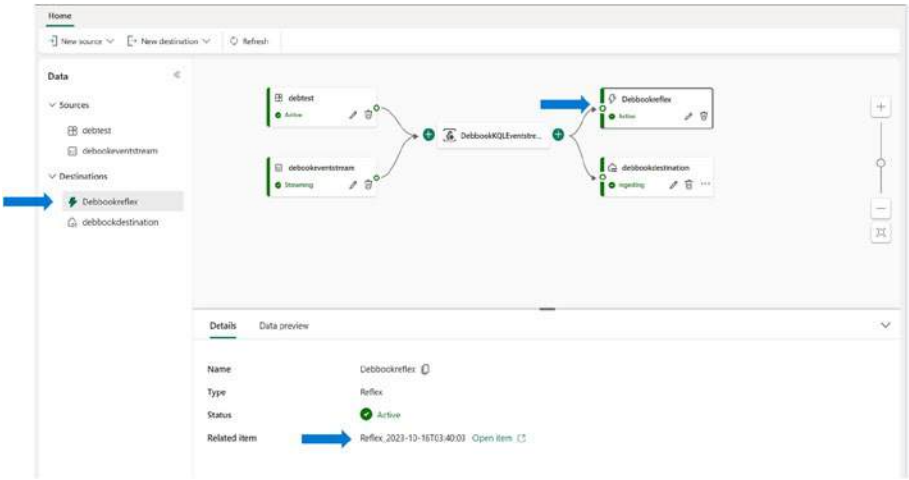


Figure 8-17. *Linking the Reflex object*

Now let's go to the Fabric workspace filter and search for Reflex items. Spot the newly created Reflex item, as shown in Figure 8-18, and then open it by clicking beside it.

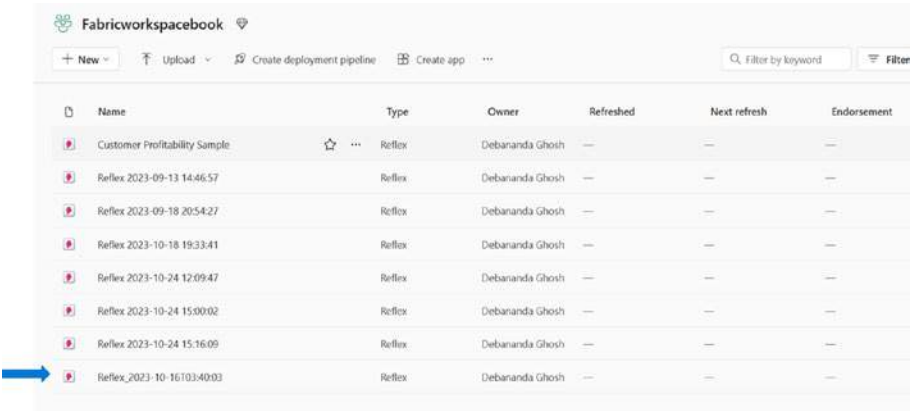


Figure 8-18. *Opening the new Reflux item*

You can now start observing the data by clicking Data on the bottom-left side. Notice that you have only events visible in the Reflex object in Figure 8-19.

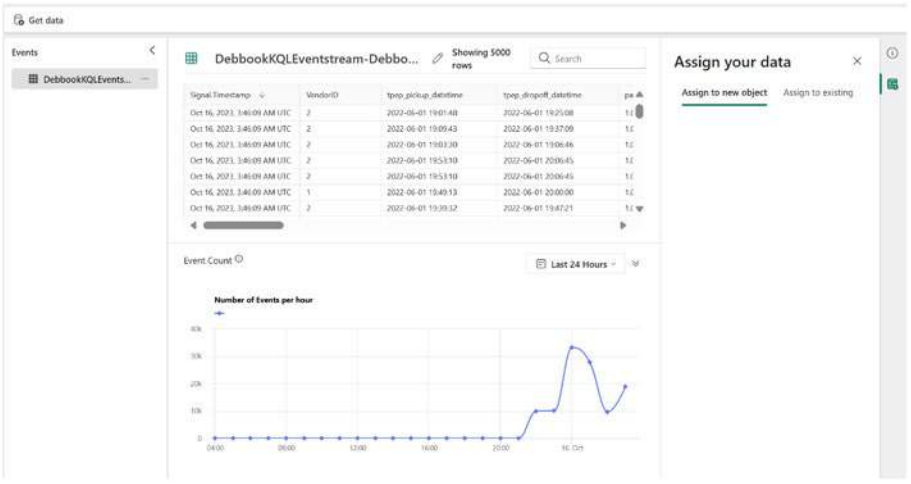


Figure 8-19. *Reflex object assignment with events*

Now you want to associate the streaming data with a new object by clicking "Assign to new object." To attach to an existing object, you can click "Assign to existing object," as shown in Figure 8-20.

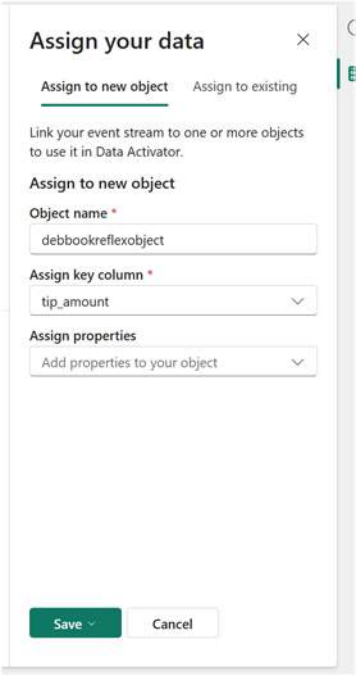


Figure 8-20. *Reflex object, key column, and properties*

Once the event is associated with a Reflex object, you can start building a code-free trigger using the Reflex design canvas. You will learn about that in the next section.

Define and Detect Actionable Patterns

In this section, you will focus on how to define and detect an actionable pattern through the Fabric Activator UI. You will click the Design canvas of the previously created Reflex object, as shown in Figure 8-21. Once you are in the Design canvas, you will click New Triggers to select and detect actions and trigger alerts.

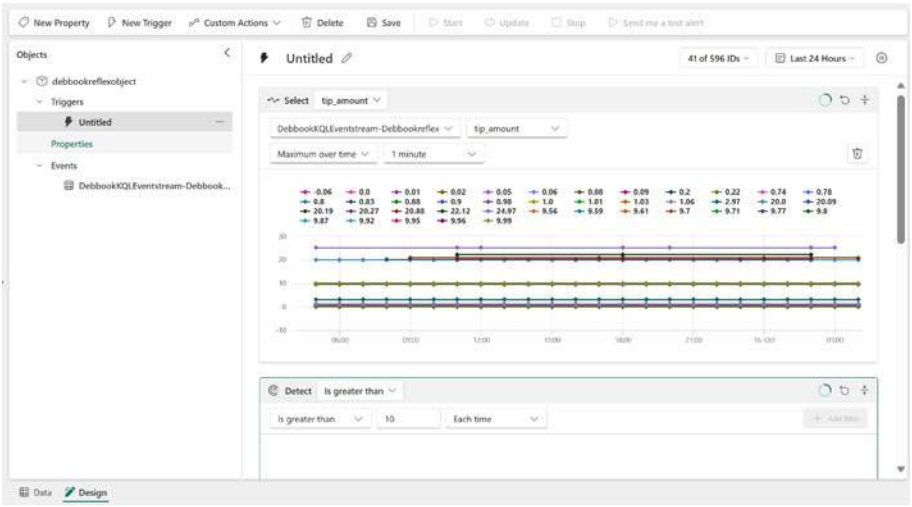


Figure 8-21. *Creating a new trigger*

After you click New Triggers, the canvas shown in Figure 8-22 opens. In this scenario, you are building a simple alert mechanism that will be triggered whenever a high tip amount is detected. As the next step, you need to select a tip amount as the Select attribute. These fields are self-explanatory. You need to then provide the conditions through the UI, as shown in Figure 8-22.

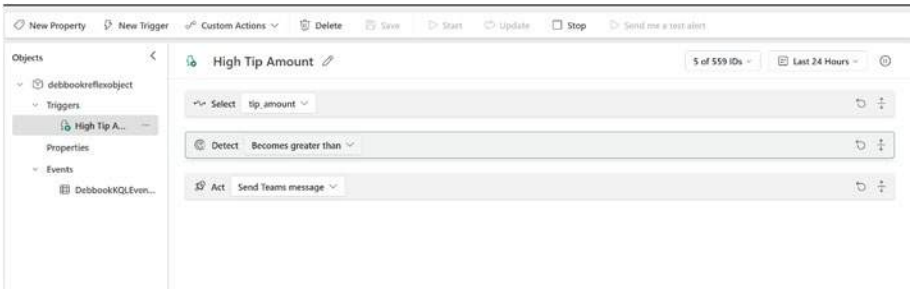


Figure 8-22. *Selecting, detecting, and acting*

This canvas focuses on the Act feature, which can trigger the following actions, as shown in Figure 8-23:

- Send a Teams message to an email
- Send an email
- Trigger a custom action



Figure 8-23. *Reflex act*

Once you provide the Detect and Act field details, click "Start trigger" to start this trigger for High Tip Amount.

Trigger an Automated Workflow: Custom Action

The previous example is about alert generation and sending a message to some recipients via Teams or email. Note you can leverage the Act capability by further triggering a custom application action based on the Microsoft Power Automate activity. To trigger an automated workflow, you need to choose Custom Action. This helps to invoke a Power Automate job. This was previously known as Microsoft Flow and helps to automate the workflow using an intuitive UI, as shown in Figure 8-24.

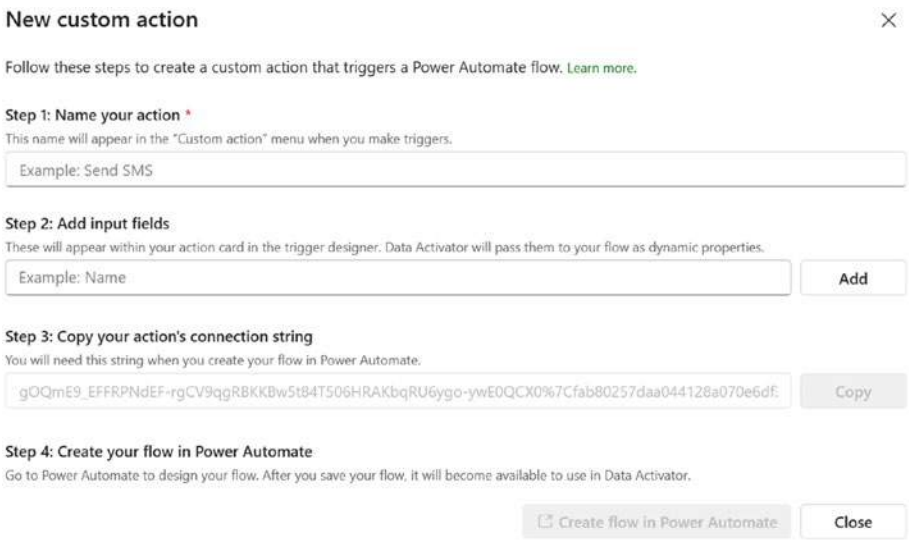


Figure 8-24. *Triggering a Power Automate job*

To learn more about Power Automate jobs, refer to `https://powerautomate.microsoft.com/en-sg/`. It is part of the Power platform family and not within the scope of this book.

Summary

In this chapter, you learned how to use Reflex triggers to create rules that monitor the data and events in your organization. We explored the different types of triggers, such as threshold, anomaly, goal, and custom triggers, and showed how to set up their conditions and parameters. You saw how Reflex helps you to identify and respond to important situations and trends in your data.

You learned how to configure the action settings and select the appropriate recipients or channels for notifications. We also discussed how to use Power Automate to invoke custom workflows that can enhance the business value of Reflex. By using Reflex and Power Automate together, you can create powerful and automated solutions that respond to real-time events and insights. Figure 8-25 recaps what you learned in this chapter.

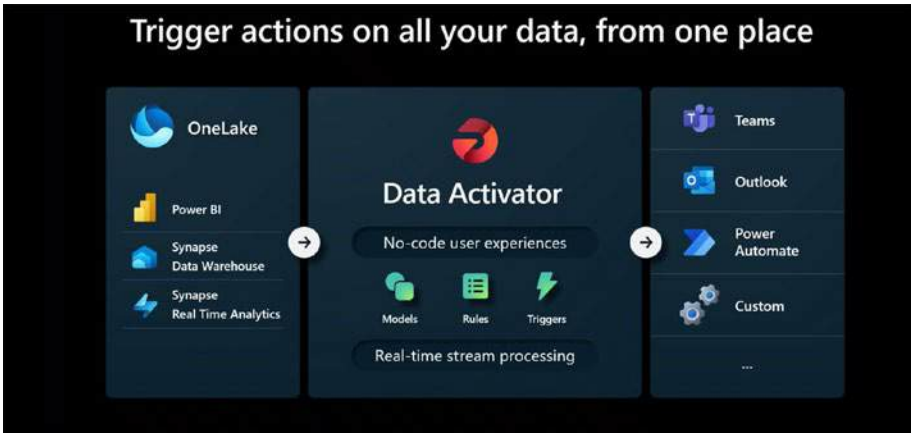


Figure 8-25. *Data Activator summary*

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric Data activator:** https://learn.microsoft.com/en-us/fabric/data-activator/data-activator-introduction
- **Activator blog:** https://blog.fabric.microsoft.com/en-US/blog/driving-actions-from-your-data-with-data-activator

CHAPTER 9

Power BI in the Microsoft Fabric Workspace

In this chapter, we will go through the Power BI capabilities of Fabric. Microsoft Power BI is an advanced data visualization tool that is part of the Microsoft Power platform; its primary focus is serving business intelligence users. Power BI was introduced in 2011 as business intelligence and interactive data visualization tool. Business intelligence tools help to analyze data and transform it into an actionable format.

Visualization is an integral part of any business intelligence tool that is used for enterprise reporting purposes. Data visualization is all about telling a story with your business data using visual representations such as charts and graphics. It is an effortless way to communicate information from a huge volume of data.

Power BI is a cloud-based business intelligence and analytics platform that enables users to create interactive visualizations, dashboards, and reports from various data sources. Power BI can connect to database Azure services as well as on-premises and third-party data sources. The tool provides advanced features such as natural language processing, artificial intelligence, and data modeling, along with business intelligence. Figure 9-1 illustrates the Power BI service landscape.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_9

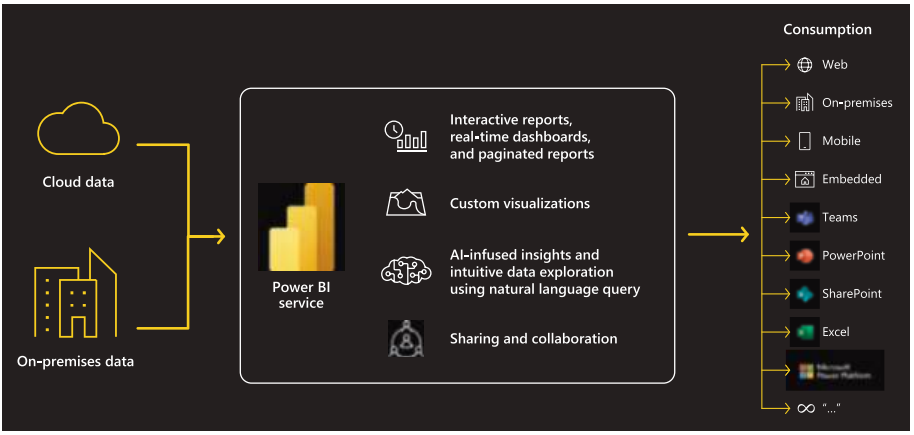


Figure 9-1. *Power BI concept*

Microsoft Power BI provides the following advanced capabilities:

- Getting started with visualizations with just a few clicks
- Asking questions about your data
- Automatically discovering insights from your enterprise data
- Governance, security, and compliance
- Real-time dashboards
- Visualizing insights in the context of your business

In the previous chapters, we discussed Microsoft Fabric's new features, which received Generally Available (GA) status in November 2023. In this chapter, we will discuss Power BI in a Fabric context. We will discuss the Power BI core concepts/fundamentals to start with and then move on to the new capabilities of Power BI with Fabric context. Figure 9-2 highlights the Power BI capability in Fabric.

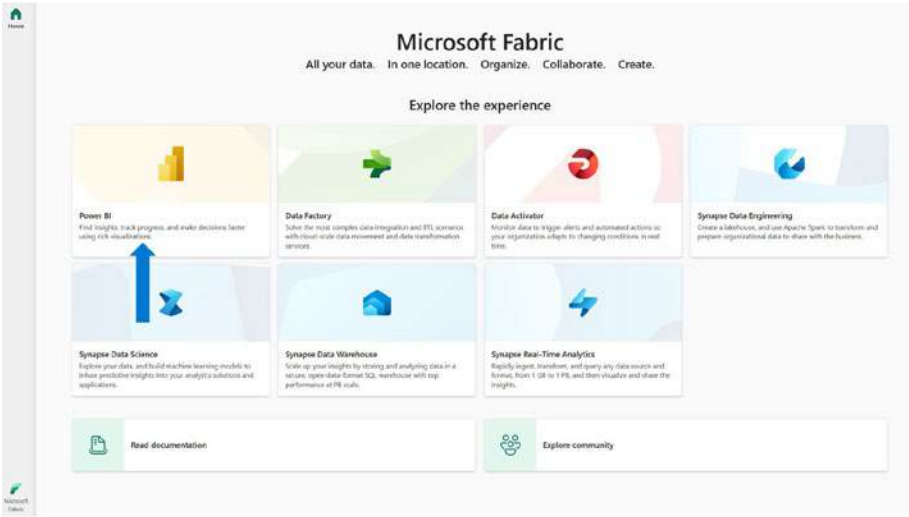


Figure 9-2. *Power BI capability in Fabric*

We will begin by covering the Power BI fundamental concepts; specifically, you will learn about the development tooling and collaboration techniques of Power BI.

Power BI Fundamentals

Power BI is a business analytics solution that lets you visualize your data and share insights across your organization or embed them into your app or website. Power BI can connect to hundreds of data sources and bring your data to life with live dashboards and reports. Specifically, with Power BI, you can do the following:

- Create stunning interactive reports
- Explore your data with intuitive visualizations
- Connect data from various sources, both on-premises and in the cloud

- Apply advanced analytics and AI to uncover insights
- Collaborate and share your findings with others
- Embed your reports and dashboards in other applications

Power BI consists of three primary parts.

- **Power BI Desktop and Power BI Report Builder:** These two tools are desktop-based thick clients for authoring Power BI visualization-based reports and paginated reports, respectively.
- **Power BI service:** This is an online software-as-a-service platform for sharing Power BI reports and for collaboration.
- **Power BI mobile apps:** This is for customizing app-based reports and sharing them with mobile devices.

Let's start developing and publishing a new report to understand more about these three components.

Power BI Desktop and Power BI Report Builder

Power BI Desktop is a free application that allows you to connect to, transform, model, and visualize data from various sources. You can use the Power BI Desktop to create interactive reports and dashboards that can be shared with others on the Power BI service or embedded in other applications. The Power BI Desktop tool consists of three primary areas: the ribbon, the canvas, and the development pane (Filter, Visualization and Fields tooling). You need to download Power BI Desktop to start Power BI development; see https://powerbi.microsoft.com/en-sg/downloads/.

Power BI Report Builder is another free application that allows you to create paginated reports that can be published to the Power BI service or printed. Paginated reports are formatted reports that have a fixed layout and can span multiple pages. You can use the Power BI Report Builder to create reports with tabular, matrix, chart, map, and other types of data regions, as well as parameters, expressions, and custom code. Power BI Report Builder uses the Report Definition Language (RDL), which is an XML-based standard for defining reports. You can download Power BI Report Builder from https://www.microsoft.com/en-us/download/details.aspx?id=58158.

Let's start by exploring the Power BI Desktop capability. We'll follow these steps:

1. Load the sample file (a financial sample) in the free Power BI Desktop on a PC.
2. Do the data transformation in Power BI Desktop locally.
3. Build a geospatial visualization in Power BI Desktop locally.
4. Save the report as a local file in `.pbix` format.
5. Publish the local file in the Power BI online service (license needed).
6. Walk through the Power BI mobile apps using the same report.

Once you download Power BI Desktop and open the tool locally, the screen shown in Figure 9-3 appears.

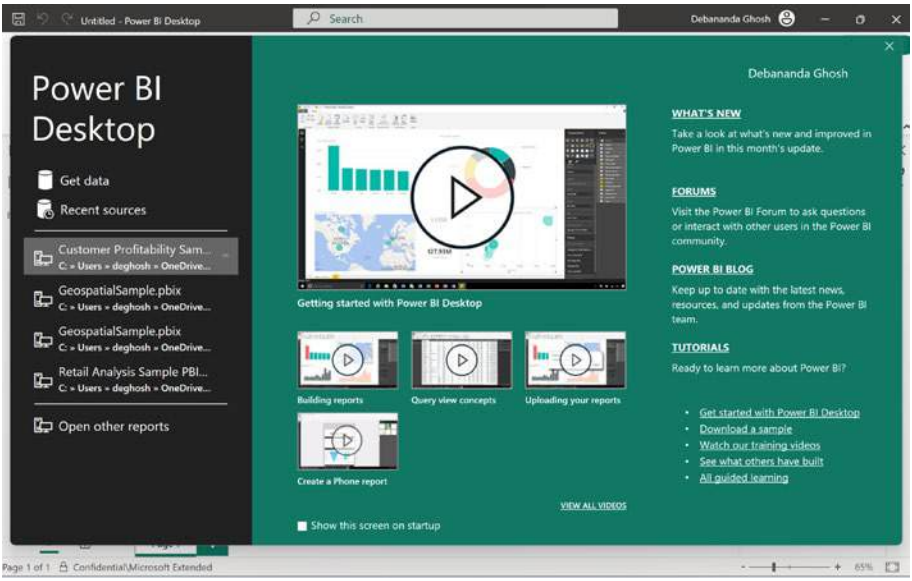


Figure 9-3. *Power BI Desktop*

We will now discuss the previously mentioned steps in more detail to learn how to use Power BI Desktop.

1. **Loading:** To load the data, you need to leverage the "Get data" tool, which provides numerous connectors. In this step, you are loading a sample file locally to create reports using that dataset. To make it simple, you will load a sample by clicking the "Try a sample dataset" feature. This will load one sample `.xlsx` file (`Financial sample.xlsx`) in Power BI Desktop. Eventually we want to build a geospatial visualization using this dataset. Figure 9-4 shows the Power BI Desktop canvas, which is similar to the Power Query–based Data Flow Gen2 that you used in Chapter 6.

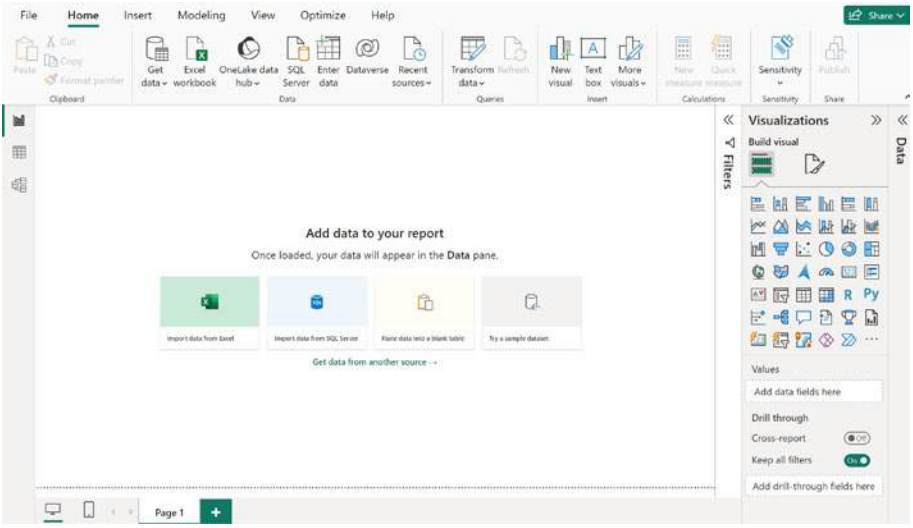


Figure 9-4. *Power BI Desktop canvas*

This is the canvas where you design your report by adding visualizations, filters, and slicers. You can manage your data sources, queries, relationships, and fields. The ribbon contains commands for working in the following views:

1. **Report view:** This is for creating reports in Power BI Desktop.
2. **Table view:** This helps you to understand, explore, interact with the data loaded in Power BI Desktop.
3. **Model view:** This is for creating semantic models and model relationships in Power BI Desktop.
4. **Dax Query view:** This is for Data Analysis Expressions (DAX) query-based development (in preview at this time).

The icons shown in Figure 9-5 open the Report, Table, and Model views, respectively, from top to bottom.



Figure 9-5. *Power BI Desktop ribbon icons to open different views*

Let's now focus on our loading process using the "Try a sample dataset" feature. Figure 9-6 shows a preview of the data before loading it in Power BI Desktop. You need to click Load to load it in the Power BI Desktop tool to complete the loading process.

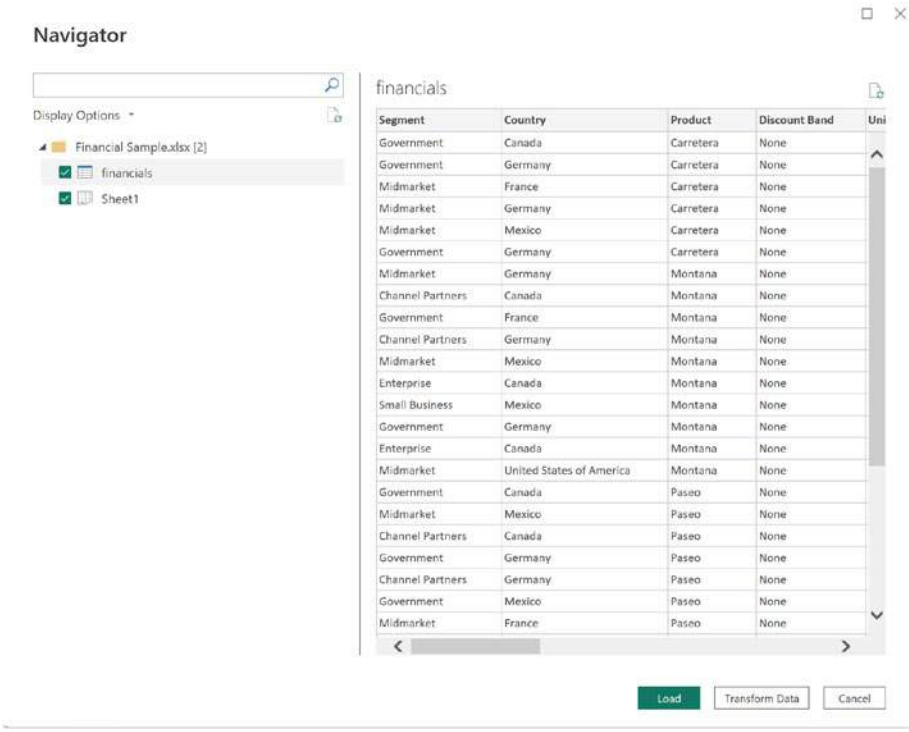


Figure 9-6. *Financial report Excel data preview*

2. **Transformation:** In this step, you will proceed with the UI-based transformation logic. Once you click Load, the Power Query editor opens, as you can see in Figure 9-7. This UI-based transformation experience is similar to Data Flow Gen2 described in Chapter 6.

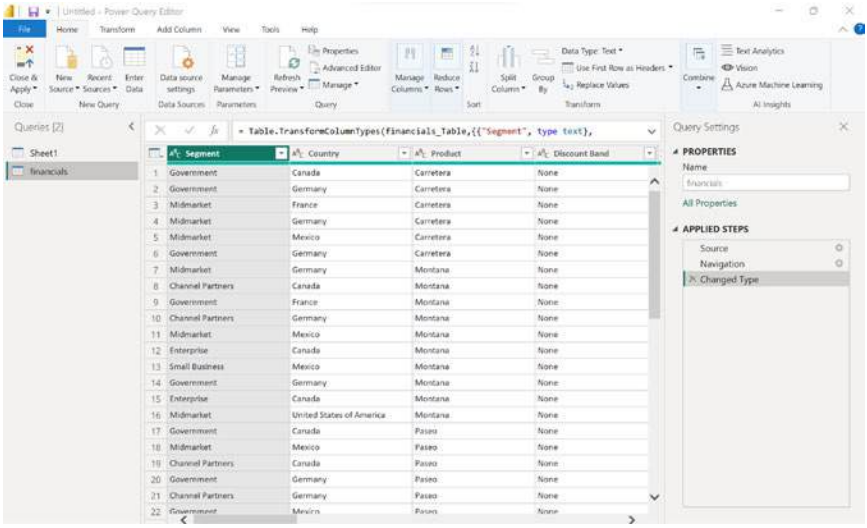


Figure 9-7. *Data transformation using Power Query*

In the next example, we have used the Replace Values UI feature (Beside Group By) to replace the None phrase with the Zero Discount phrase. Click OK to continue, as shown in Figure 9-8.

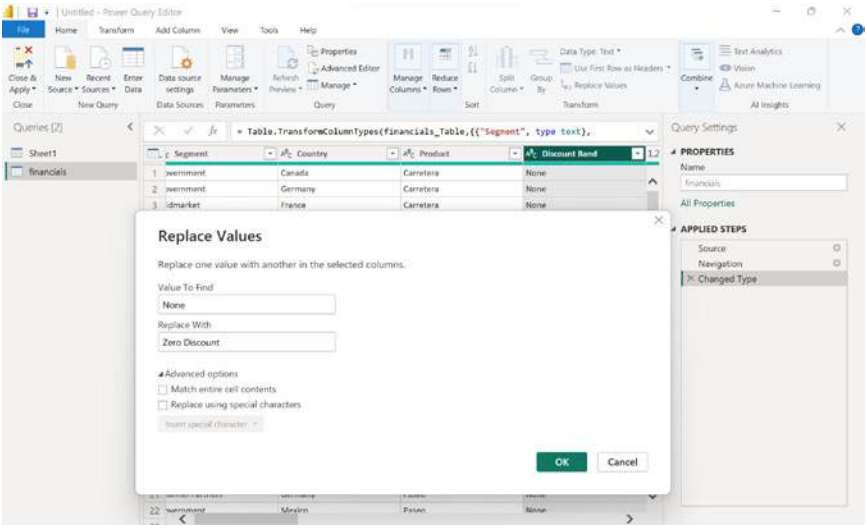


Figure 9-8. *Replacing the None phrase*

As the next step, click Close and Apply in the top-left toolbar, as shown in Figure 9-9. This loads and transforms the dataset. We will focus on developing visualization charts in the next step.

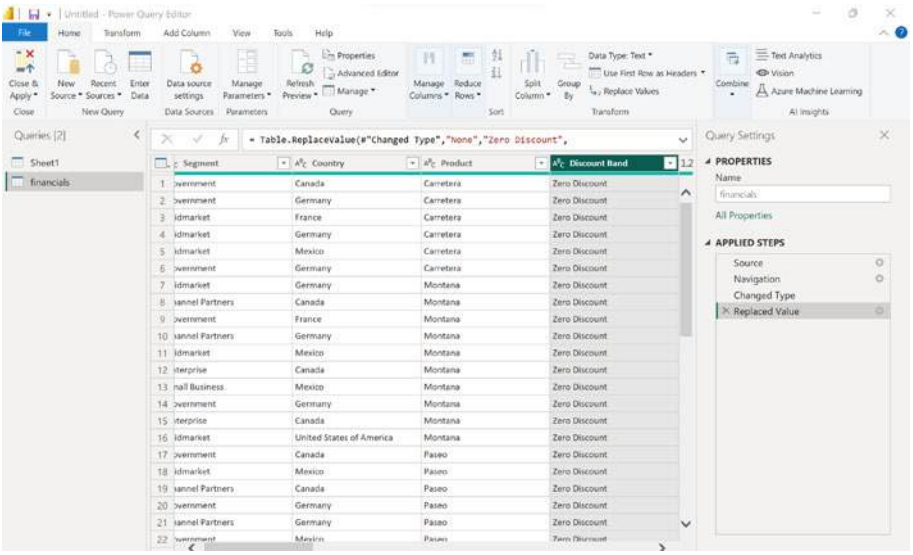


Figure 9-9. *Loading and transforming the dataset*

3. ***Visualization:*** Let's now focus on the visualization part. To do this, you need to move to Report view. You can see this view under Build Visuals in Figure 9-10. These are visualization plugins that you can drag and drop on the work area; they have some filter configurations for visualization purposes.

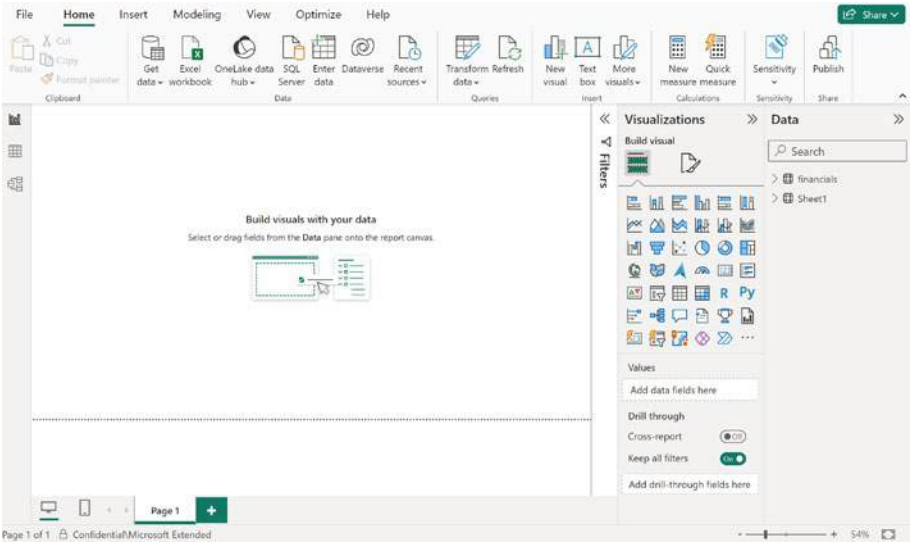


Figure 9-10. *Visualization canvas*

While Power BI Desktop provides native plugins, it is not restricted to only those visualization plugins. There are out-of-the-box visualizations that you can download from `https://appsource.microsoft.com/en-US/marketplace/apps?product=power-bi-visuals`. When you download a visual, it will be downloaded in a `*. pbiviz` file. You need to click the three dots (…) and then select "Import a visual from a file" to import the visualization plugin, as shown in Figure 9-11.

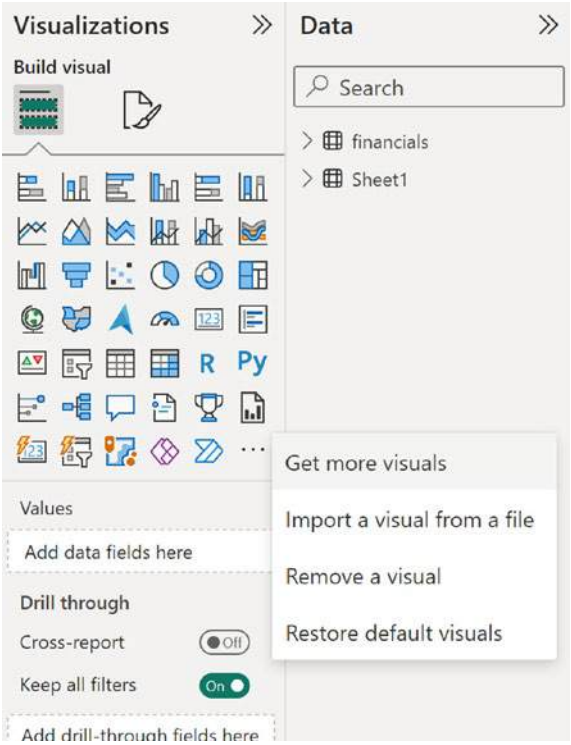


Figure 9-11. *Importing the data visualization plugin*

Prior to visualization, it's essential to create a relationship between the imported datasets. The sample file that you uploaded contains two worksheets (Financial and Country data), which are uploaded as two different datasets in the Power BI Data view. To create a relationship between the financial data and the country master data, you need to click "Manage relationships," as shown in Figure 9-12.

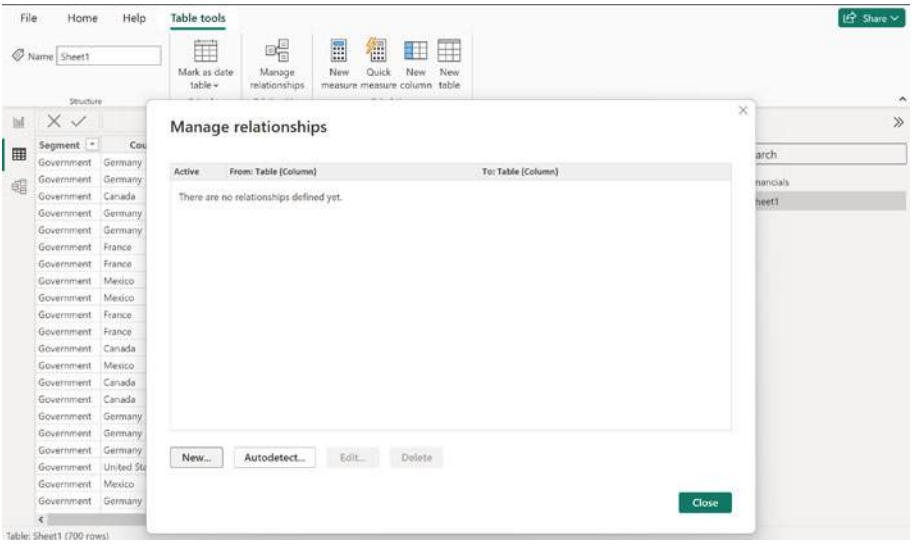


Figure 9-12. *Managing relationships in Power BI Desktop*

Click New. Then join the Segment column from both the worksheets (financials and Sheet1), as shown in Figure 9-13. Next, click Close to continue with our Power BI Desktop–based report development.

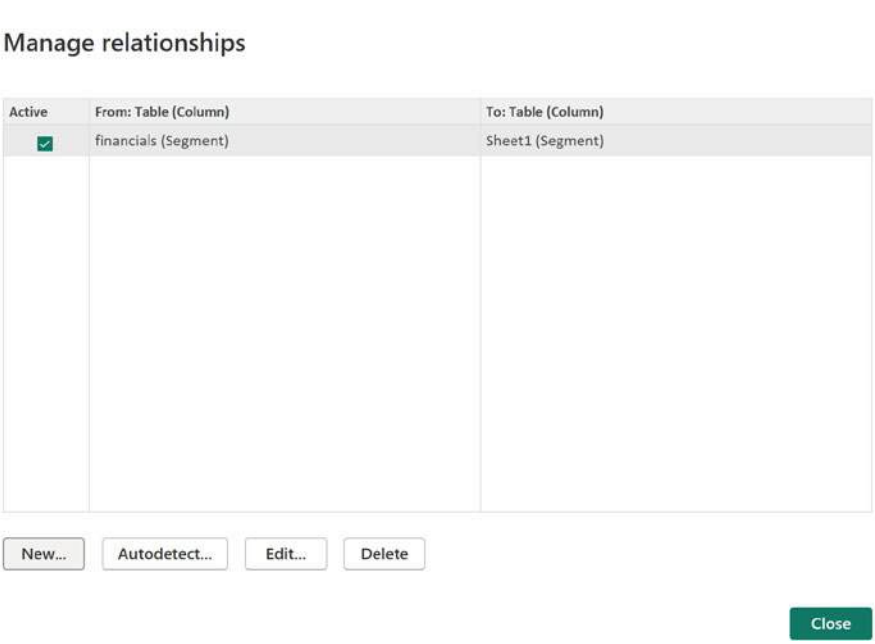


Figure 9-13. *Joining the worksheets*

Let's now move on to Report view for our visualization-related development purposes. Click Map and then select the Location field as the Country attribute, as shown in Figure 9-14, to configure a geospatial visualization.

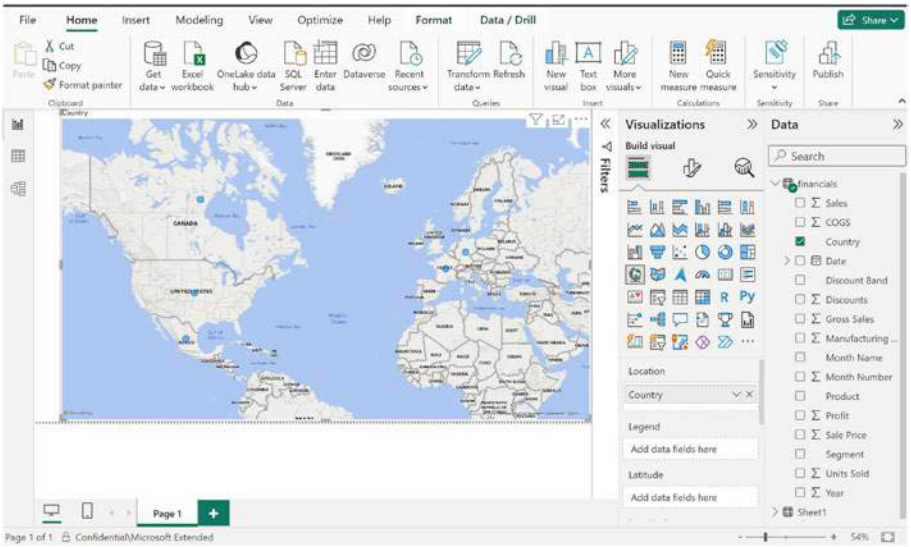


Figure 9-14. *Geospatial plugin of Power BI Desktop*

As the next step, select "Bubble size" for Sum of Sales to configure the map. You will now be able to reflect the sales data in a geospatial way, as shown in Figure 9-15.

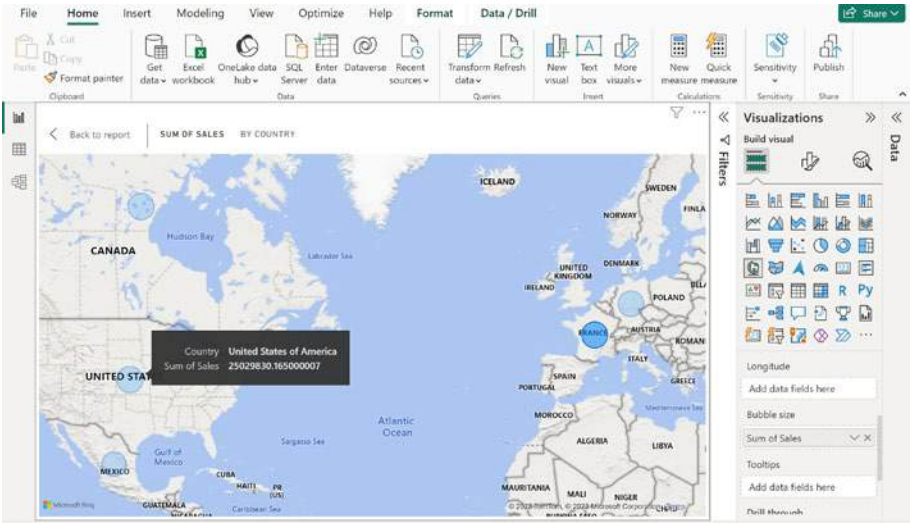


Figure 9-15. *Configuring the geospatial visualization*

Continue to explore the Filters and Format visual panes. These are available on the right side of the Visualization canvas. You can plug in specific visual formatting using the Format Visual toolbar. Figure 9-16 shows the features provided on the General tab in the "Format visual" capability. You are now done authoring this report.

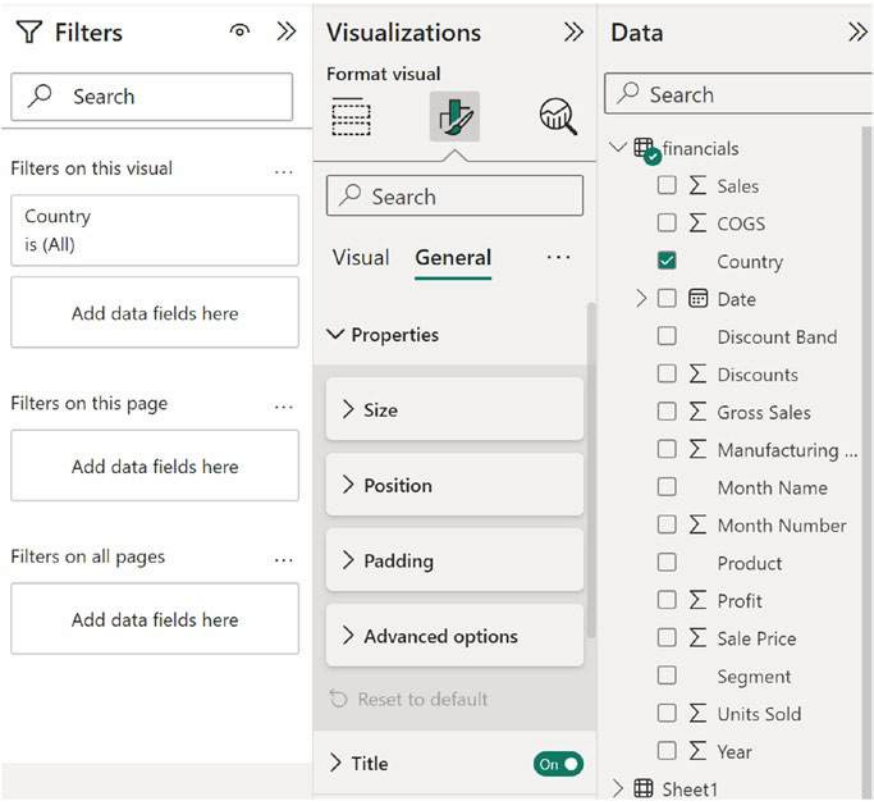


Figure 9-16. *Exploring the filter and formatting options*

4. ***Store the local .pbix file:*** Once your report is authored locally, you can save it as a `.pbix` file in your local desktop, as shown in Figure 9-17. You can share it with other users for co-development purposes.

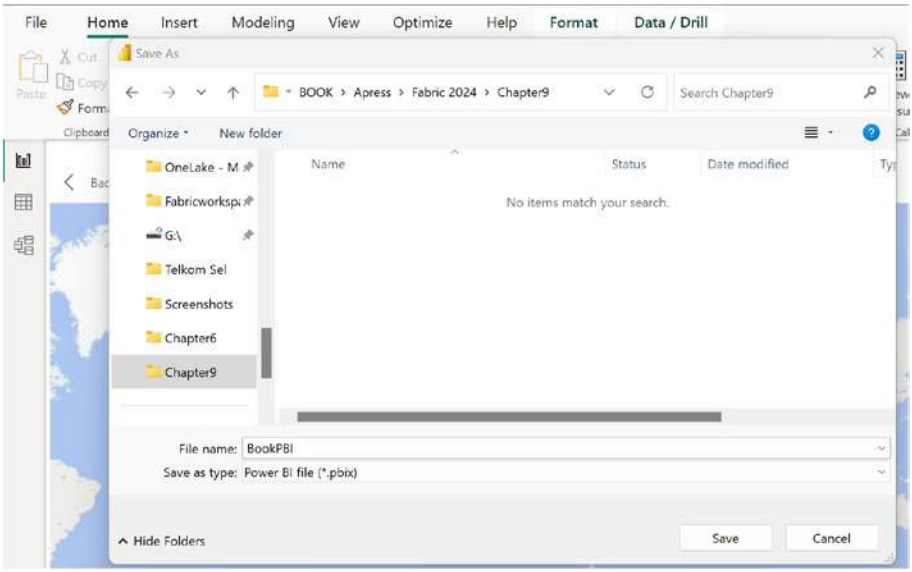


Figure 9-17. *Saving your file*

We will now move on from Power BI Desktop to the online tool. Then we will cover Power BI report builders later in this chapter.

Power BI Service

The Power BI service is the cloud service counterpart of the Power BI local tooling that allows users to create, view, share, and manage dashboards and reports based on their data sources. The Power BI service enables users to access their data and insights from anywhere, on any device, and with real-time updates.

The Power BI service also offers the following various features and capabilities:

- **Datasets:** Power BI online datasets are collections of data that can be used to create reports and dashboards in the Power BI service. Datasets can be created from data flows, imported from files or databases, or pushed from external applications or services. Datasets can also be shared with other users or groups and can be updated automatically or manually.
- **Reports:** Reports are interactive visualizations that display data and insights from datasets. Reports can be created using the Power BI Desktop application or the Power BI service web interface. Reports can include several types of visuals, such as charts, tables, maps, gauges, slicers, filters, and more. Reports can also be customized with themes, layouts, formatting, and interactions.
- **Dashboards:** Dashboards are single-page summaries that display key information and metrics from one or more reports. Dashboards can be created by pinning visuals from reports or by adding tiles from other sources, such as web content, images, videos, or text boxes. Dashboards can also be personalized with annotations, alerts, comments, and bookmarks.
- **Data flows (Gen1):** Data flows (Gen1) are a way of creating and managing data entities that can be reused across different reports and dashboards. Data flows allow users to define data transformations and

calculations using a graphical interface or the Power Query language and then to store the data in the Power BI service. Data flows can connect to various data sources, such as files, databases, web services, or other data flows, and can be refreshed on a schedule or on demand. We will not focus on this topic in this book since in Fabric we introduced Data Flow Gen2 in Chapter 6.

To use the Power BI service, users need to have a Power BI account and sign in with their Microsoft credentials. Note that to share and collaborate within/across the Power BI workspace securely, you need a Power BI/Fabric license. We will briefly discuss Fabric licensing in Chapter 10.

Now let's start pushing the local `.pbix` file created in the previous section online.

5. **Publishing to Power BI online:** Click Publish in Power BI Desktop to publish the developed report in web. Now Power BI Desktop will start communicating with the Power BI online service. Once you log in using your Entra ID/enterprise domain ID, you need to choose the right destination/workspace to host the report. In this scenario, the Fabricworkspacebook workspace becomes the host of the report, as shown in Figure 9-18.

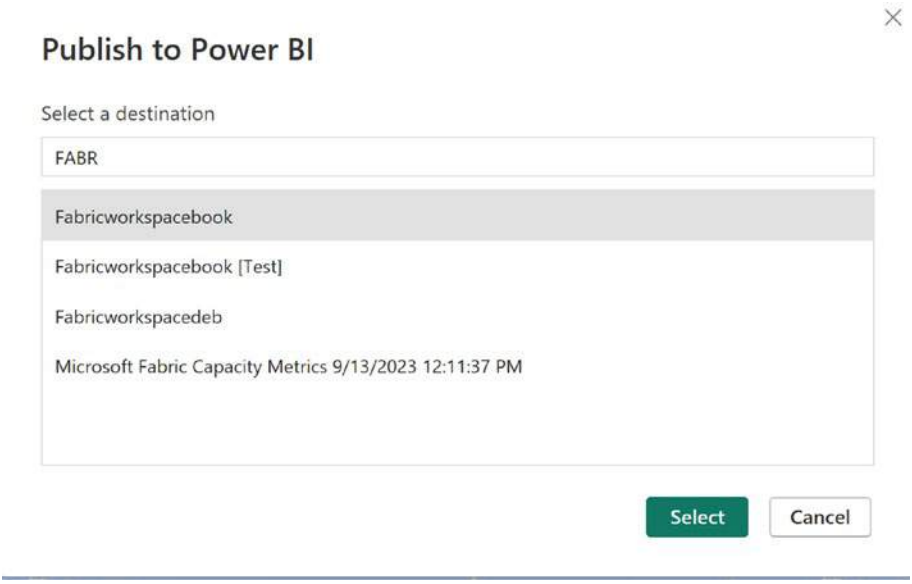


Figure 9-18. *Publishing Power BI reports to the online cloud workspace*

Once you click Select, this report is published in the Power BI online workspace. The notification shown in Figure 9-19 will appear while publishing the report.

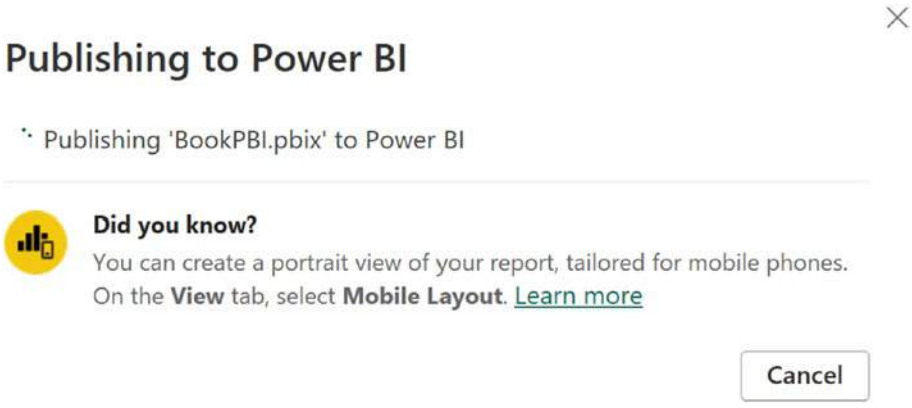


Figure 9-19. *The message shown while publishing*

Let's now go to the workspace and set Filter to Report. You will be able to see the recently published report called BookPBI in the workspace, as shown in Figure 9-20.



Figure 9-20. *Published report*

Now you have created and published a Fabric report in your workspace. You will now learn how to create a Fabric dashboard from a Fabric report. A Power BI dashboard is a single-page, often interactive, graphical display of key metrics and trends derived from one or more reports. Dashboards are designed to provide quick insights and enable users to drill down into the underlying data or reports for more details. To create a dashboard from the report you published, you need to follow these steps:

1. Open the report in Power BI service and pin the visuals that you want to include in the dashboard by clicking the pin icon at the top-right corner of each visual. You can choose to pin a visual to an existing dashboard or a new dashboard.
2. If you choose to create a new dashboard, you need to give it a name and a description. You can also specify the tile size and the theme for the dashboard.
3. Once you have pinned all the visuals you want, you can go to the dashboard by clicking the dashboard name at the top-left corner of the report page.

4. We can rearrange, resize, edit, or delete the tiles on the dashboard by using the options available on each tile. You can also add text boxes, images, videos, web content, or streaming data tiles to the dashboard by using the "Add tile" button at the top-right corner of the dashboard page.

5. You can also customize the dashboard settings by clicking the gear icon at the top-right corner of the dashboard page. You can change the name, description, theme, tile flow, featured status, or access permissions of the dashboard from the Settings menu.

Figure 9-21 shows the "Pin to dashboard" feature for creating a new dashboard or adding the report into an existing dashboard using an existing report.

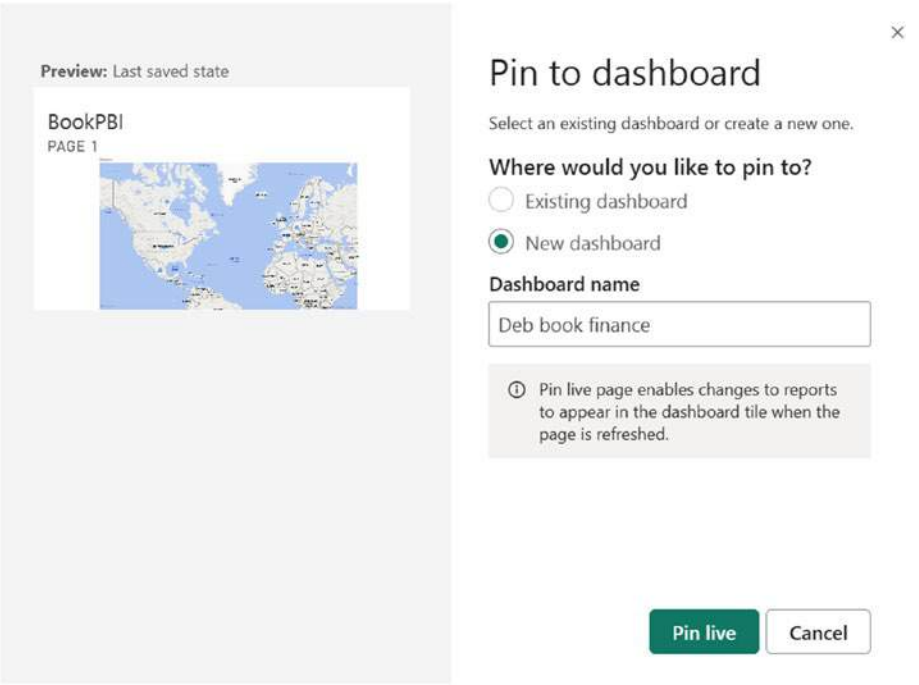


Figure 9-21. *Fabric dashboard creation*

Once you create a new dashboard, it will appear in the workspace under Fabric Item Type as Dashboard, as shown in Figure 9-22.



Figure 9-22. *Fabric dashboard Item*

Power BI Mobile Apps

In this section, we will discuss Power BI mobile apps and the related configuration steps. Power BI mobile apps enable users to access and interact with their dashboards and reports on their smartphones or tablets. Users can view, filter, sort, highlight, and drill down into the data, as well as share insights with others. Power BI mobile apps support both cloud and on-premises data sources and can work offline or online.

- They provide a consistent and familiar experience across different devices and platforms, such as iOS, Android, and Windows.
- They allow users to stay connected and informed with real-time data updates and notifications.
- They enable users to collaborate and communicate with their colleagues and stakeholders through annotations, comments, and sharing features.
- They offer security and governance features, such as biometric authentication, device management, and data encryption.

You now continue to leverage the previously created geospatial visualizations in mobile layout for learning purposes.

1. To start configuring the mobile layout, click the View tab and then select Mobile Layout. In the Power BI Desktop, open the report that you want to optimize for mobile devices.
2. Click the View tab and then select Mobile Layout. This will open a new canvas where you can drag and drop the visuals from your report, as shown in Figure 9-23.

3. To add a visual to the mobile layout, select it from the Visualizations pane and drag it to the canvas. You can resize and reposition the visual as needed. You can also use the Format pane to change the appearance and settings of the visual.
4. To remove a visual from the mobile layout, select it and press the Delete key or click the X icon in the top-right corner of the visual.
5. To preview how our mobile report will look on different devices, click the Preview button in the top-right corner of the canvas. You can choose between different device types and orientations from the drop-down menu. You can also interact with the visuals and filters in the preview mode.
6. To publish our mobile report, click the Save button in the top-left corner of the canvas. This will save the mobile layout as part of our report. You can then publish our report to the Power BI service as usual.

By using the Mobile Layout option, you can create a more engaging and user-friendly experience for our mobile audience. You can also ensure that our report is consistent and accessible across different platforms and devices.

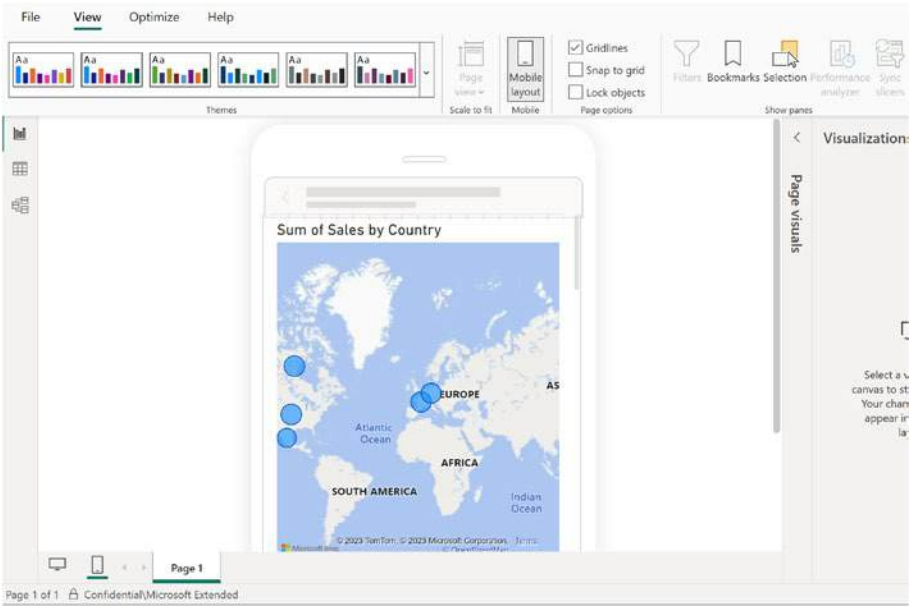


Figure 9-23. *Visualization mobile layout*

To use Power BI Mobile, you first you need to download it at `https://powerbi.microsoft.com/en-sg/mobile/`. Change the url to make it country specific to download the apps accordingly. To use Power BI Mobile, users need to have a Power BI account and sign in with their Microsoft credentials like with the Power BI service. Once you download it, log in, and select the workspace, you will be able to see the Power BI report that you have created, as shown in Figure 9-24.

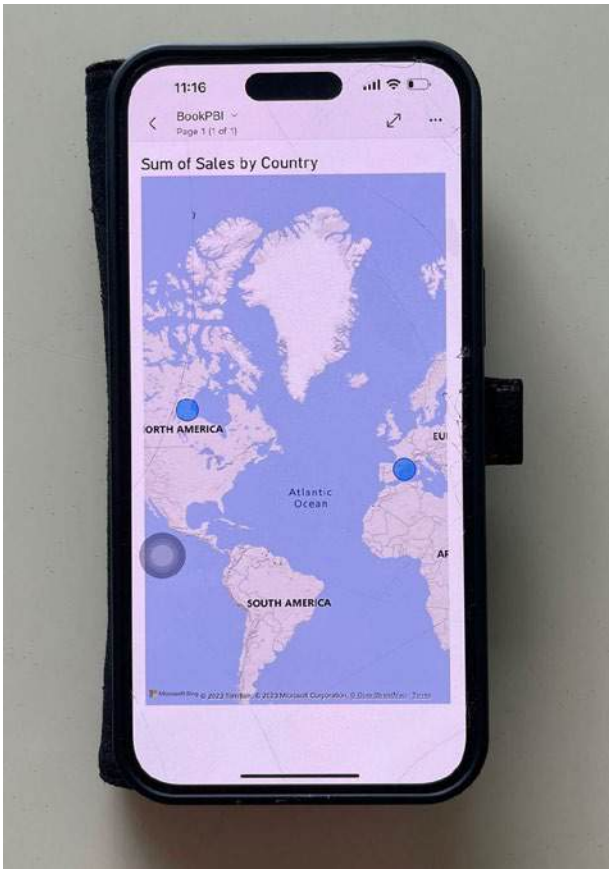


Figure 9-24. *Report displaying on mobile device*

Now that you have completed the three fundamental parts of Power BI (local authoring tool, online cloud service, and mobile version), we now switch our focus to the new GA features of Power BI within the context of Fabric.

Power BI Key New Features (Fabric)

The new features of Power BI focus on the following aspects:

- Seamless integration between Fabric items and Power BI items
- Productivity enhancements for developers

Let's look at some of the new features.

Report Auto-Create

Auto-create is a Power BI feature in the Fabric foundation. Using this capability, you can create a new prebuilt report without doing any Power development. To start using this feature, you need to click Power BI and then click "+ New report," as shown in Figure 9-25.

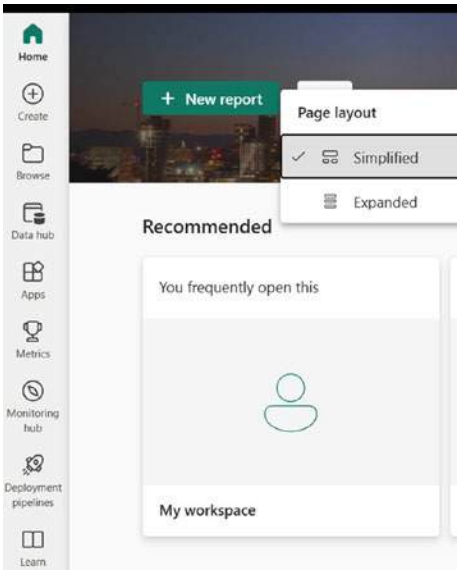


Figure 9-25. *Accessing the auto-create feature*

Now you need to either manually enter the data or pick a semantic dataset for automatic report creation, as shown in Figure 9-26.

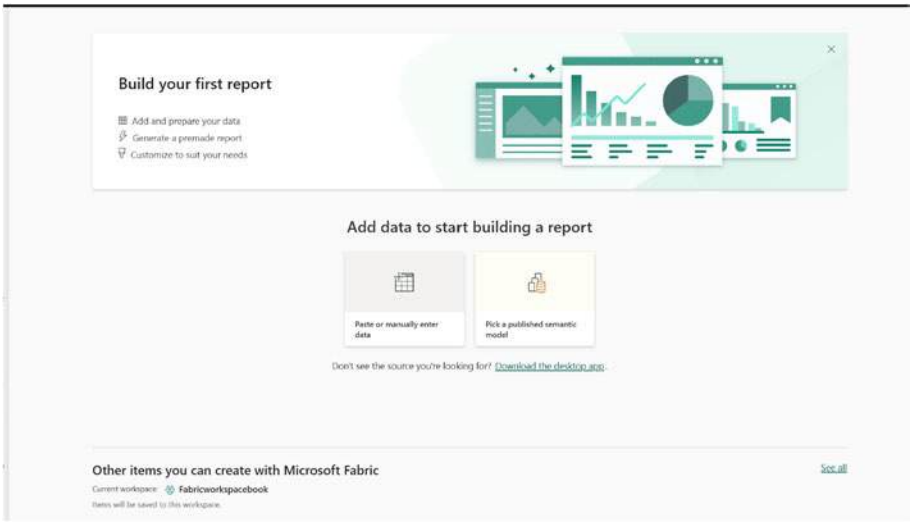


Figure 9-26. *Two ways to add data*

Select the Semantic model that you need here for generating the report and then click the "Auto-create" button. Note that in the previous step when we created the BookPBI report, we leveraged previously created BookPBI Semantic dataset, as shown in Figure 9-27.

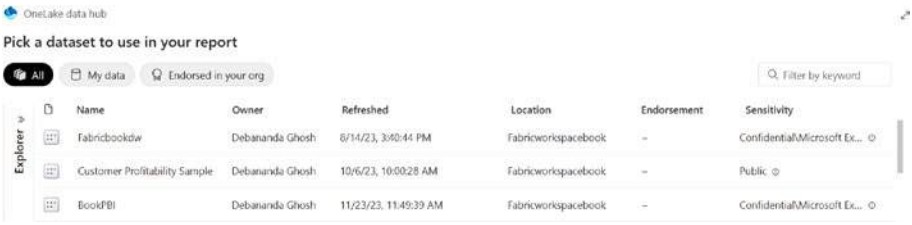


Figure 9-27. *Previously created BookPBI semantic dataset*

As you select the BookPBI semantic model, a report is generated as part of the auto-creation process, as shown in Figure 9-28.

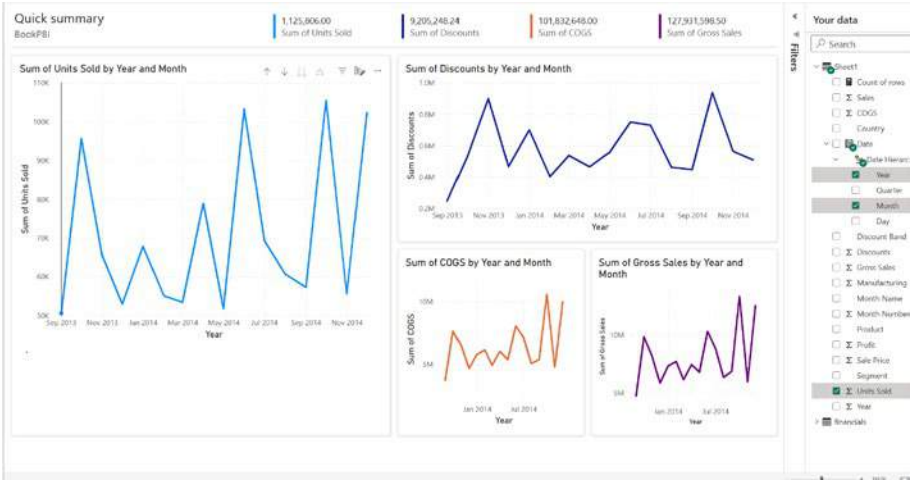


Figure 9-28. *Fabric report generated automatically*

You can also use this feature by going to the corresponding semantic model Fabric item, clicking the three dots (…), and selecting the "Auto-create report" option, as shown in Figure 9-29.

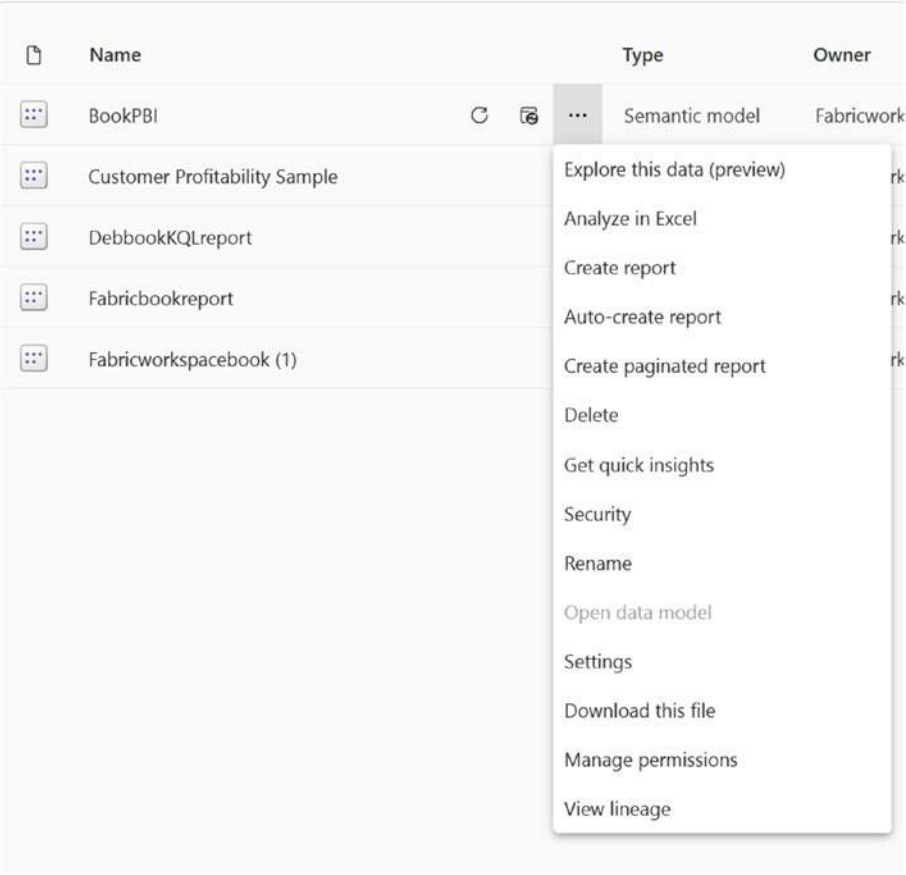


Figure 9-29. *Selecting from the menu*

Quick Insights

The Quick Insight feature provides a one-click feature to get insights from datasets. You need to click the "Get quick insights" feature to generate insights from the report without doing any development. Figure 9-30 shows the automatic analysis of the data and comments such as

"Government and Small Business have noticeably more Discounts." These remarks and charts are the outcome of the automated analytics done using the underlying dataset by Power BI.

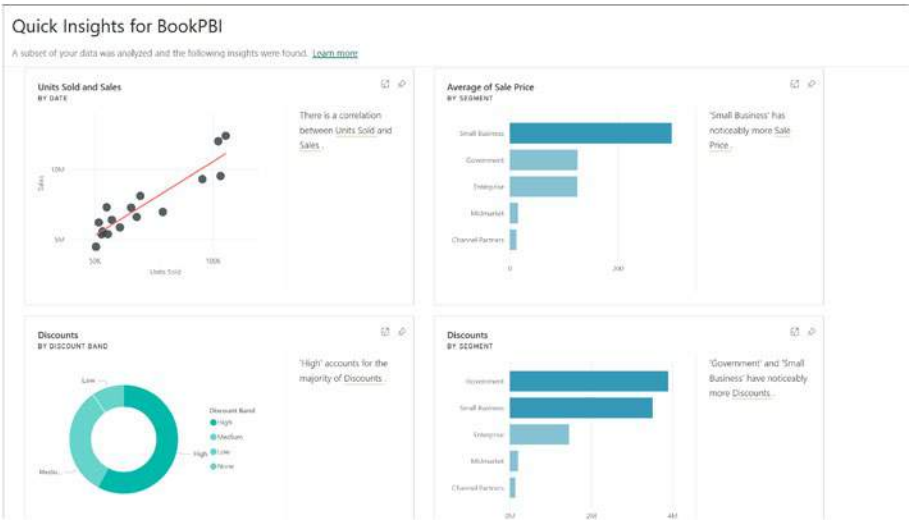


Figure 9-30. *Fabric report quick insights*

Lineage

Lineage is the relationship between the data sources, datasets, data flows, reports, dashboards, and apps that make up your Fabric environment. Lineage helps you to understand the impact of changes, track issues, and optimize performance. You can also use lineage to monitor the refresh status, data protection, and certification of your Fabric items.

To access the Lineage view of a Fabric item, you need to go to the workspace where the item is located and select the item from the list. Then, click the Lineage icon in the top-right corner of the page. Alternatively, you can right-click the item and choose "View lineage" from the context menu. This will open a new tab with a lineage diagram of the item, as shown in Figure 9-31.



Figure 9-31. *Lineage view*

You can see the upstream and downstream dependencies of the item, as well as the details of each dependency, such as the type, name, description, refresh date, and sensitivity label. You can also expand or collapse the nodes, filter by type, search by name, and refresh the lineage view.

Paginated Report

Paginated reports are designed to fit on a page for printing or generating a PDF. We briefly mentioned this capability while explaining the Power BI Report Builder tool. Paginated reports allow you to create pixel-perfect, highly formatted reports with multiple data sources and complex calculations. You can use the Report Builder tool to create and publish paginated reports to Power BI.

To create a paginated report, you need to connect to a data source in Power BI Report Builder and design a report layout. Download Power BI Report Builder at `https://www.microsoft.com/en-us/download/details.aspx?id=58158`. In Fabric, you can start creating a paginated report easily. When you click "Create paginated report" and select the Country and Profit attributes, the page shown in Figure 9-32 appears with the profit details at the country level.

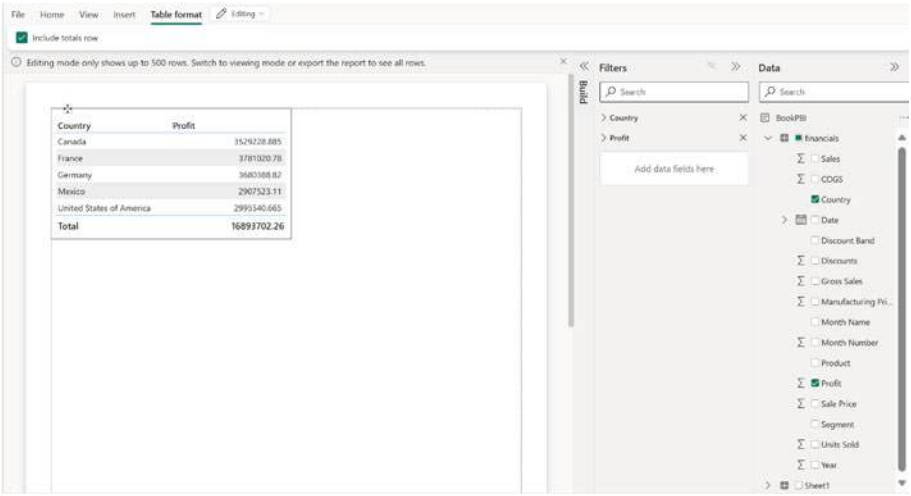


Figure 9-32. *Creating a paginated report*

You can save the report in the workspace or download it in report format or download the report in the RDL file, as shown in Figure 9-33.

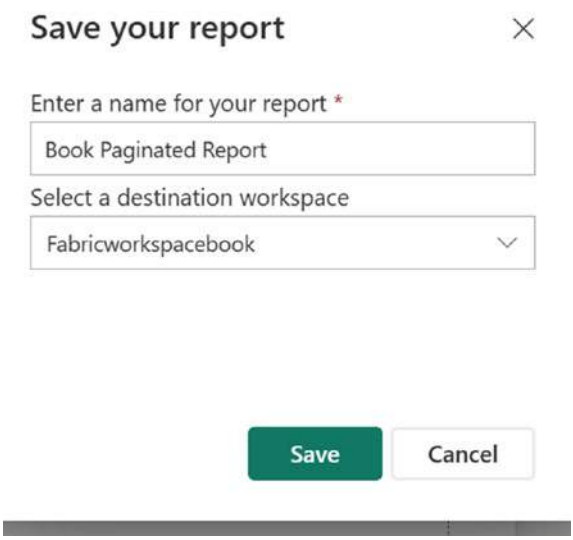


Figure 9-33. *Saving the paginated report*

To design locally using the Power BI Report Builder, you can leverage the RDL file that is downloaded in the previous step. You can use various elements such as tables, charts, maps, images, and text boxes to display your data in an organized and interactive way. You can also apply filters, parameters, expressions, and formatting options to customize your report. Once you have established the connection, you can use the Data Explorer pane to browse and select the tables and fields you want to use in your report. You can also use the Query Designer to write custom queries using DAX. Then, you can drag and drop the fields onto the report design surface and configure the report elements as needed. Figure 9-34 shows the RDL file opened in Power BI Report Builder.

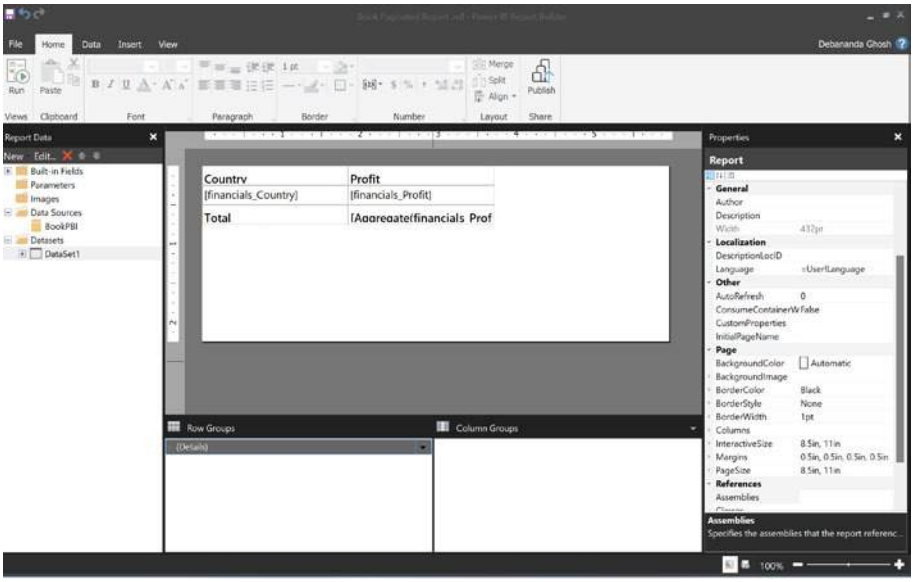


Figure 9-34. *RDL file for paginated report*

Getting Insights

To generate insights from a report in one click, click Get Insights. As shown in Figure 9-35, insights are generated automatically.

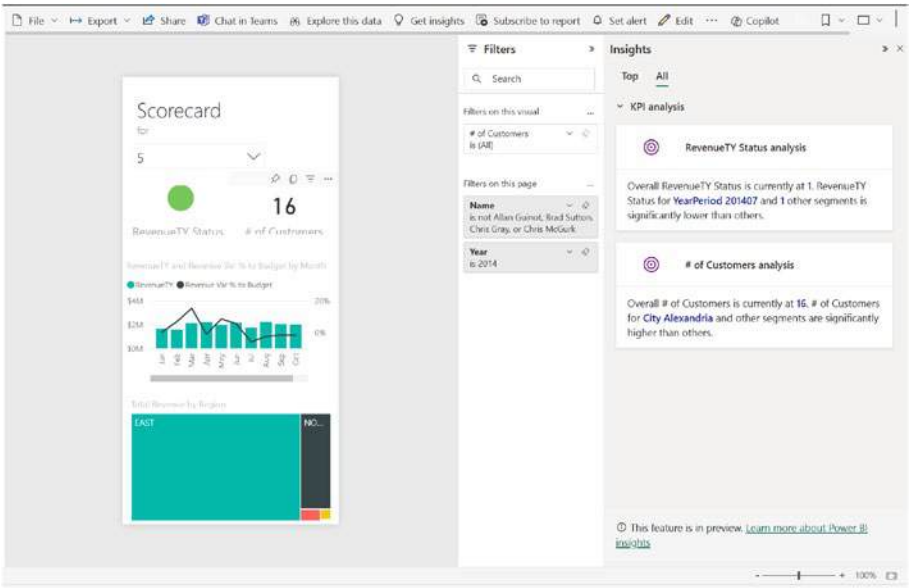


Figure 9-35. *Generating automatic insights*

Power BI Direct Lake Mode

In this section, we will discuss Power BI Direct Lake mode. Previously Power BI had two ways to manage data: physically loading the data using Import mode and directly querying the database using Direct Query mode. Import mode has higher performance, but Direct Query did not need the data to be loaded into Power BI memory, causing better latency and ease of data availability.

Direct Lake is the best of this two world. This is a Power BI feature that allows users to connect to data sources stored in OneLake without importing or loading the data into Power BI. This enables users to access and analyze large volumes of data directly from the lake, without compromising performance or security. Users can reduce the data movement and duplication by querying the data directly from the lake. Figure 9-36 shows the conceptual difference between Direct Lake and Direct Query mode.

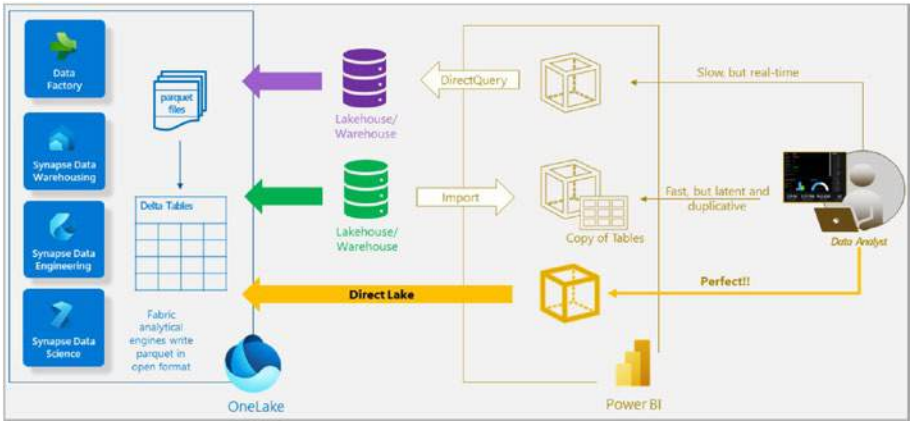


Figure 9-36. *Fabric Direct Lake concept from Microsoft documentation*

To access Direct Lake, you need to go to Lakehouse or Datawarehouse mode and click "New semantic model." In Figure 9-37 we have selected the Fabricworkspacebook lakehouse (from Chapter 3). Once the "New semantic model" appears, you can provide a name (in this case we have provided the name Fabricworkspacebook) and start selecting the table you need in the semantic dataset.

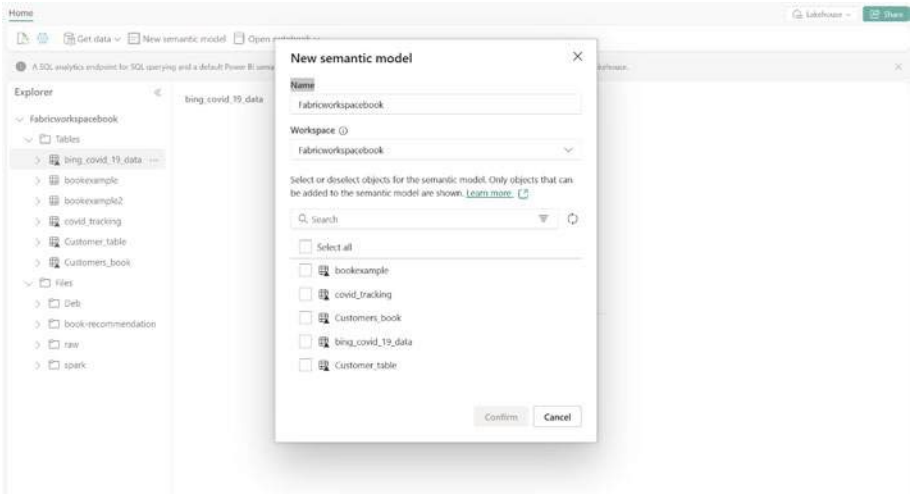


Figure 9-37. *Creating a new semantic model*

As you click the Confirm button, the model will be automatically saved in the workspace with this name (as shown in Figure 9-38).

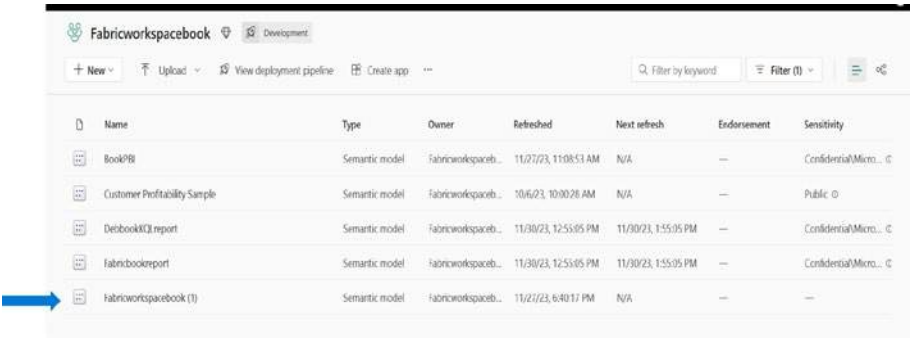


Figure 9-38. *Saved model*

Click this newly created custom semantic model for data modeling purposes. Click "Open data model," as you can see in the Figure 9-39.

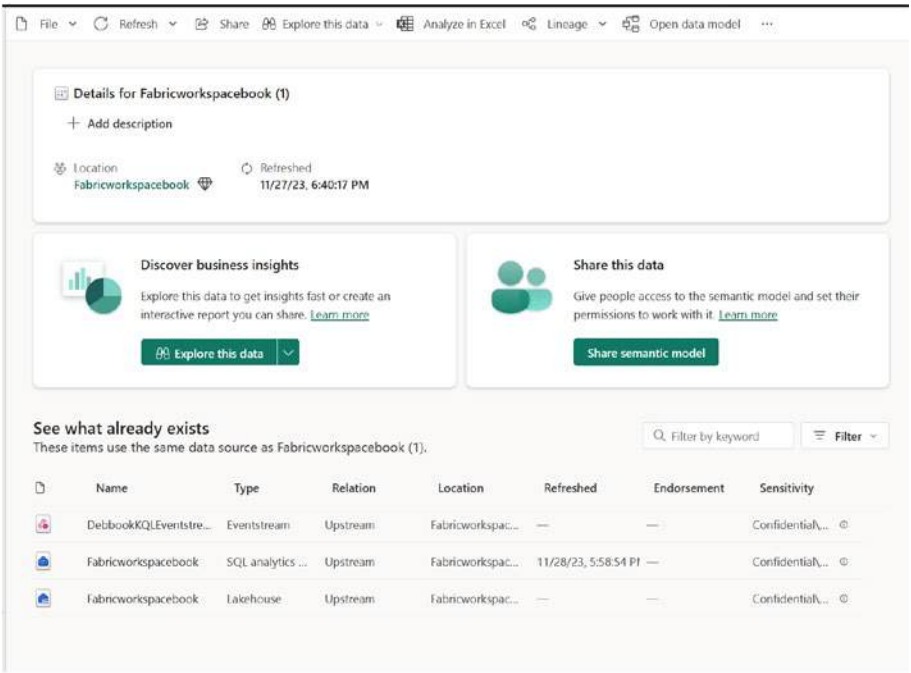


Figure 9-39. *Opening the data model*

Once the semantic model opens, you can start creating relationships by clicking "Manage relationship" or create a new report by clicking "New report," as shown in Figure 9-40. This will be a similar experience to Power BI Desktop–based report authoring.

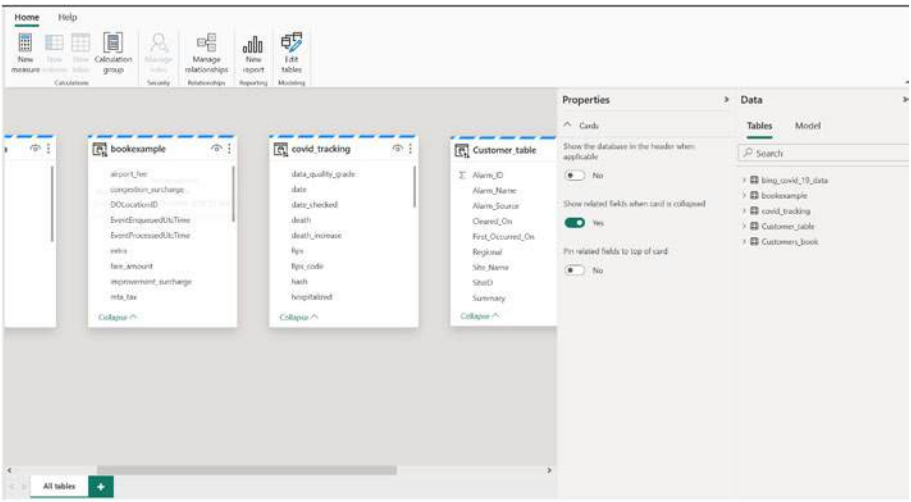


Figure 9-40. *Working with the semantic model*

For existing Power BI report migration to direct lake please continue to monitor reported limitations of Directlake to avoid any migration related blocker - https://learn.microsoft.com/en-us/powerbi/enterprise/directlake-overview#known-issues-and-limitations. Let us now start learning about how to collaborate and share these reports in the next section.

Report Sharing

To share a Power BI report, users need to click the Share button at the top-right corner of the Power BI window and select the recipients and the permissions. Users can also add a message and attach the report as a file or a link. Users can also schedule the report to be sent periodically by clicking the Subscribe option. The receiver will receive an email with a secured report link, as shown in Figure 9-41.

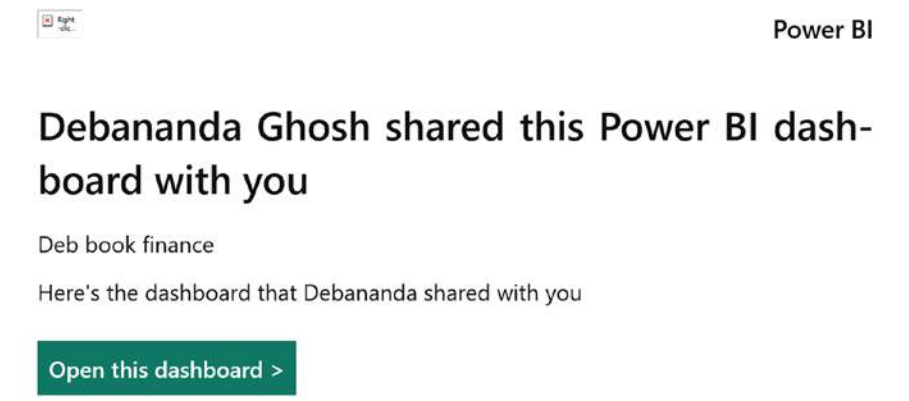


Figure 9-41. *Fabric Power BI dashboard sharing*

Note that you can also share a report in a Teams message. Click Chat in Teams to share it via Microsoft Teams, as shown in Figure 9-42.

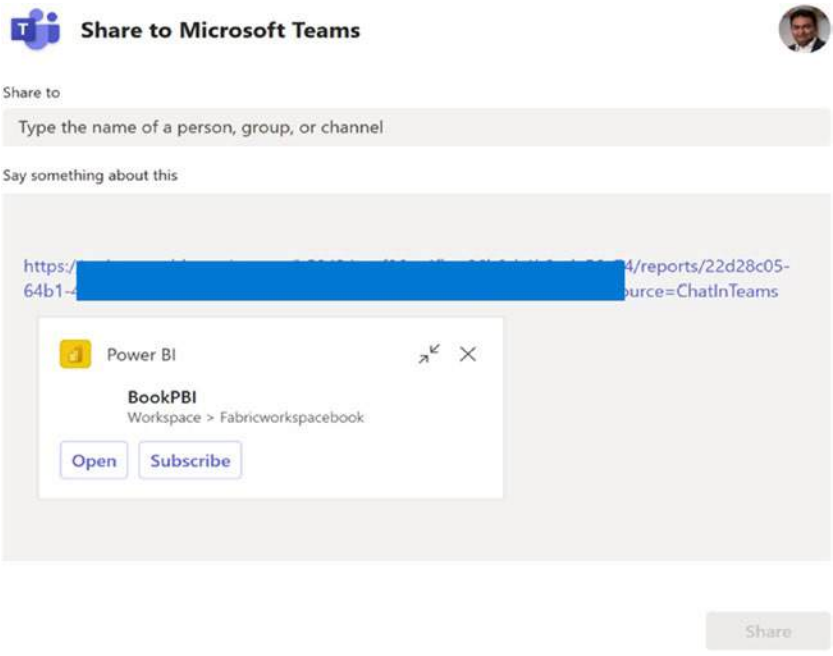


Figure 9-42. *Sharing with Teams group*

To set an alert, click "Set an alert," as shown in Figure 9-43.

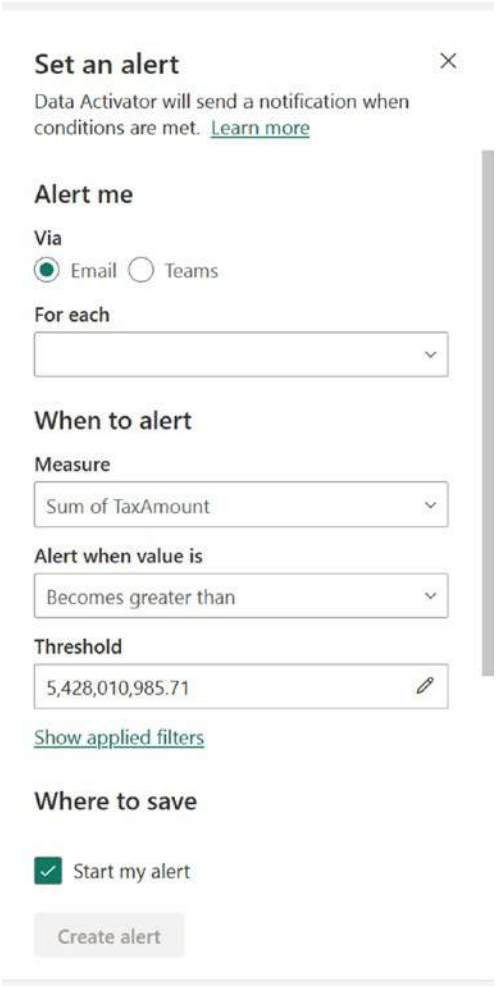


Figure 9-43. *"Set an alert" settings*

Fabric is tightly integrated with Office 365 products. Users can export the data from the report to Excel or CSV format by clicking the Export button at the bottom-right corner of the visual. Users can choose the format, the level of detail, and the destination folder for the exported

file. Users can also export the entire report as a PDF or PowerPoint file by clicking the File menu and selecting the Export to PowerPoint option. There are other tooling integrations like the Export to PDF option available in Fabric as well.

Datamarts

In this section, we will discuss the self-service SQL-compliant database natively provided in Power BI. *Datamarts* enable users to load the data in a fully managed self-service database. The datamart features are as follows:

- Complete UI web-based, no other software required
- No-code database
- Self-service and fully managed
- Integrated with Power BI and Microsoft Office
- SQL support
- Built-in visuals

To create a new datamart database, click New+ and then Datamart, as shown in Figure 9-44.

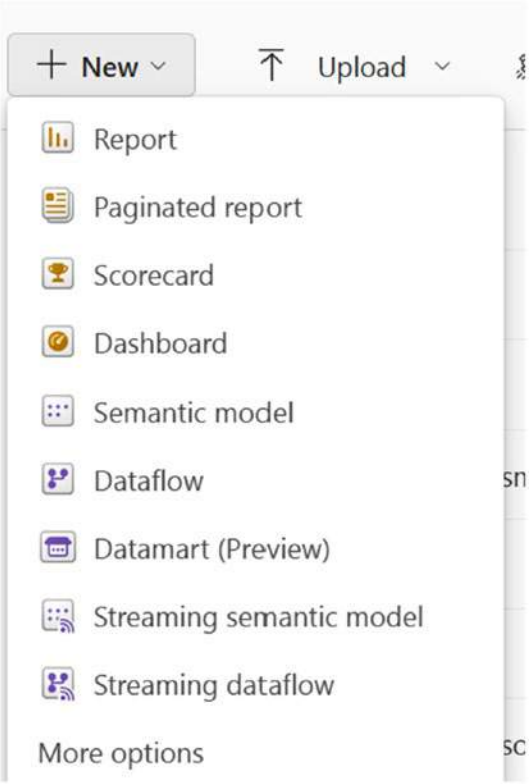


Figure 9-44. *Creating a new datamart*

Then give it a name, as shown in Figure 9-45.



Figure 9-45. *Naming the new datamart*

The canvas shown in Figure 9-46 opens, and you can leverage the self-explanatory canvas to load the data.

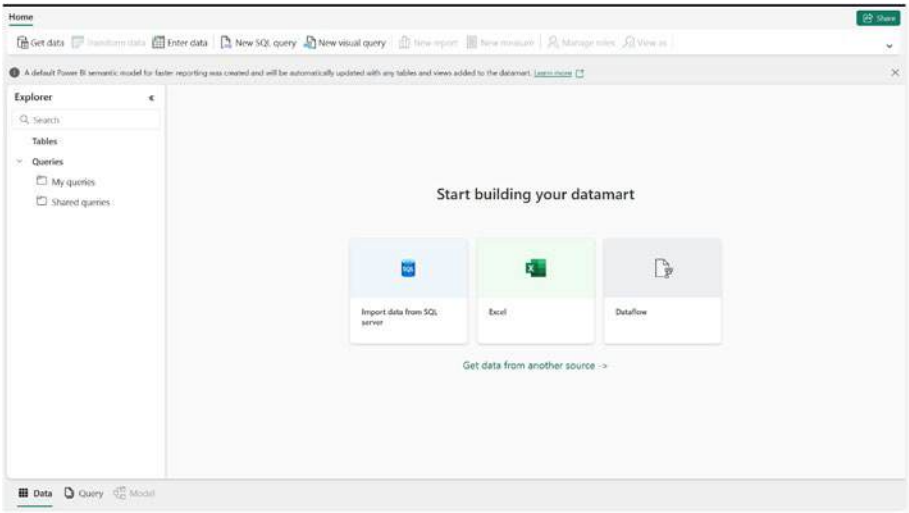


Figure 9-46. *Configuring the datamart*

As you load sample data, you can see data in the several formats, as shown in Figure 9-47.

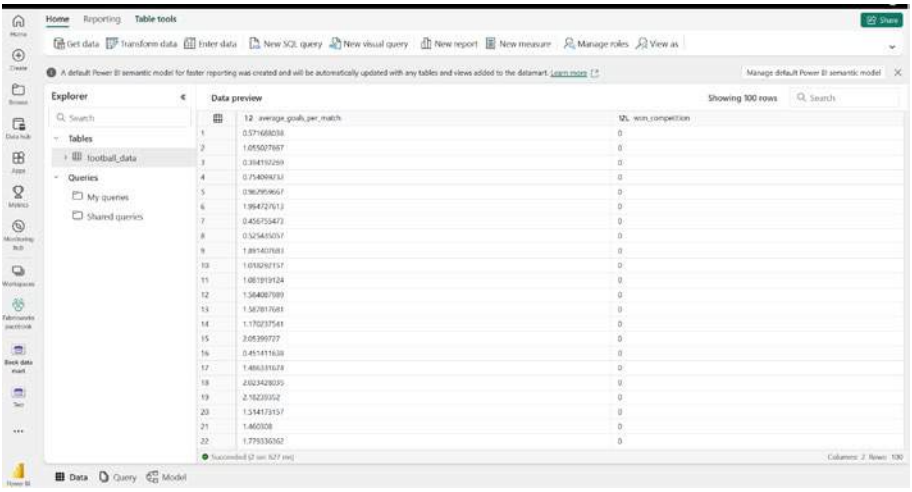


Figure 9-47. *Loading data*

The datamart is saved in the workspace as the Datamart type. The corresponding semantic model is created, as shown in Figure 9-48.



Figure 9-48. *Semantic model for datamart*

You can continue to explore datamarts for learning purposes. Figure 9-49 shows the "Explore Book data mart" capability in Datamart.

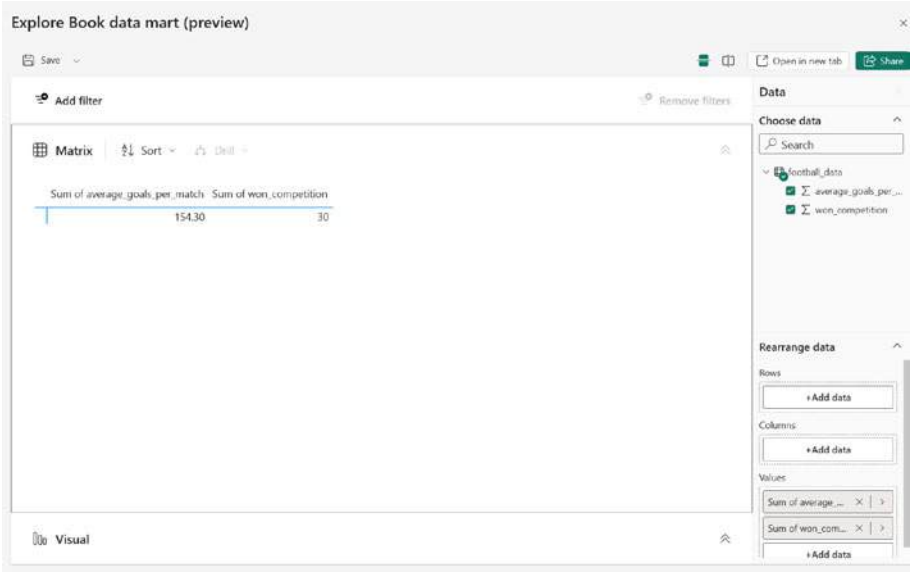


Figure 9-49. *Exploring datamarts*

Finally with the arrival of Fabric new Datawarehouse and Lakehouse features, data mart will be used selectively for self service requirements. As per Microsoft guidelines Datamarts are recommended for customers who need domain oriented, decentralized data ownership and architecture,

such as users who need data as a product or a self-service data platform. In this subsection we covered self service data base part inside traditional Power BI. Let's move on to other exciting AI features within PowerBI.

Fabric Power BI Copilot

Microsoft Copilot is an AI-based assistant that Microsoft is rolling out across its operating systems and apps. Fabric is infused with Copilot capabilities across various computation layer. Power BI Copilot is an interactive assistant that enables users to interact with Fabric solutions within the Power BI environment. Users can launch Copilot from Power BI, select a scenario, and view the corresponding dashboard or report in Power BI. Users can also perform actions such as running a scenario, viewing the results, and comparing scenarios using Fabric Power BI Copilot. Just click Copilot, as you can see in Figure 9-50.

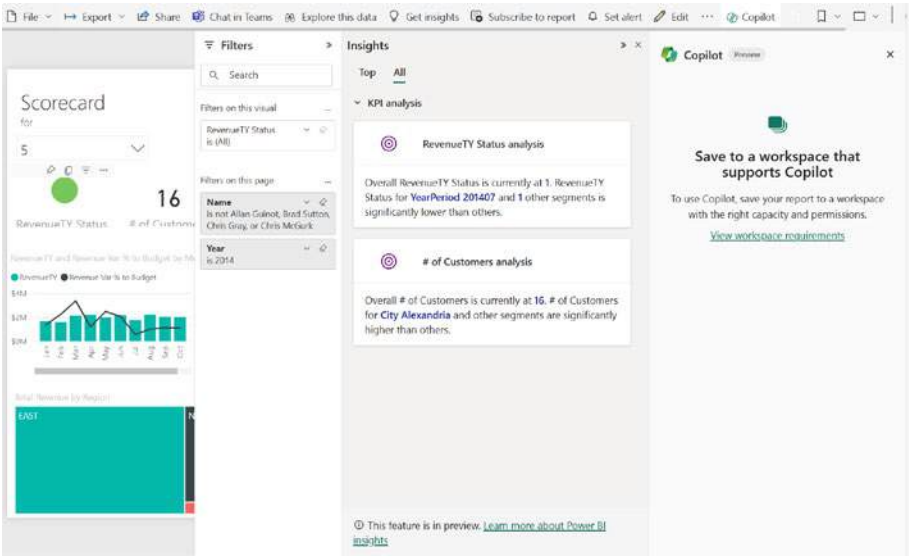


Figure 9-50. *Opening Power BI Copilot*

You can click "Summarize this page" (as prompted by Copilot) and summarize the chart, as shown in Figure 9-51.

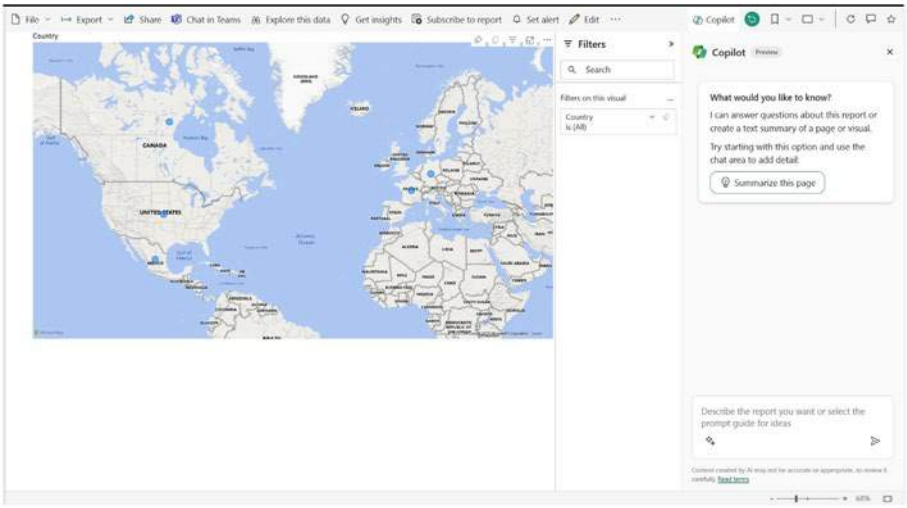


Figure 9-51. *Summarizing a chart*

As you see, Copilot produces the following summary, as shown in the Figure 9-52.

- This page contains a single visual that shows the names of five countries: Canada, France, Germany, Mexico, and United States of America.
- The visual is a categorical text table that lists the countries in alphabetical order. No other information is provided in the visual.
- The purpose of this page is unclear, as it does not provide any context or analysis on the countries. It may be an introduction or a filter for other pages in the report.
- The user may want to see more details or comparisons on the countries, such as population, GDP, or trade statistics.

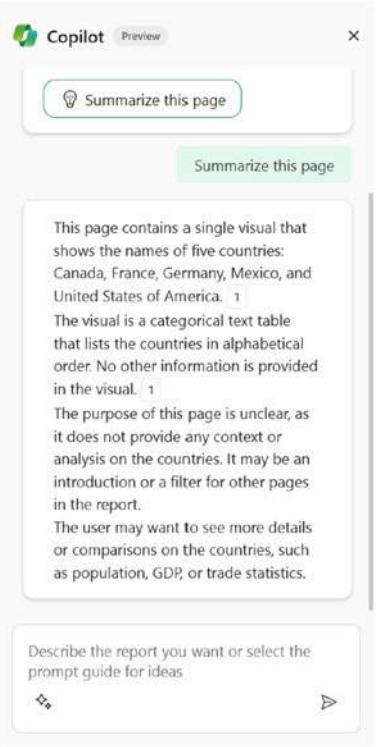


Figure 9-52. *Power BI Copilot summary*

Copilot also mentions the following:

- Canada has a population of about $37.6 million people and a GDP of about $1.7 trillion USD.
- Canada ranks 9th among the countries by total revenue, with about $2.4 billion USD, and 8th by total profit, with about $485 million USD.
- Canada has six product categories, with the highest revenue coming from Books ($1.1 billion USD) and the lowest from Software ($9.4 million USD). The most profitable category is Books ($292 million USD), and the least profitable is Electronics ($-18.7 million USD).

We have now this Open AI–based AI assistant tool capability in Power BI to generate insights. Very exciting!

Summary

You learned how to do Power BI–based data imports and transformations in this chapter. You also learned how to publish the reports to the Web or on mobile devices. We went through the details related to new key Fabric features introduced in Power BI. You also saw how to leverage the Microsoft Copilot capability. Refer to Figure 9-53 for a recap.

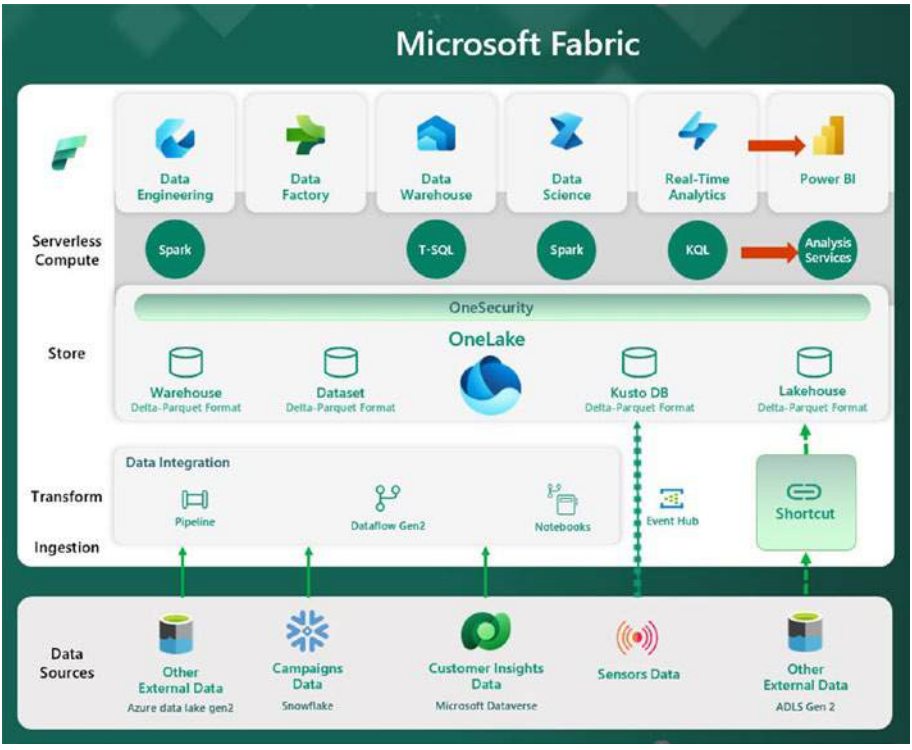


Figure 9-53. *Fabric-based data architecture covered in this chapter*

In the next chapter, we will discuss licensing. Also, we will cover a few key topics that haven't been covered yet.

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Power BI:** https://www.microsoft.com/en-us/power-platform/products/power-bi
- **Power BI blogs:** Power BI Blog—Updates and News | Microsoft Power BI
- **Power BI semantic model:** https://learn.microsoft.com/en-us/fabric/data-warehouse/semantic-models

CHAPTER 10

Microsoft Fabric: Inside and Out

In the previous chapter, you learned about the key features of Fabric for building data architecture platforms. You learned about the data orchestration, data engineering, data science, and visualization capabilities of Microsoft Fabric. You also learned how this data analytics / data integration (DI) platform helps to modernize an organization's footprint.

In this chapter, we will go through a few more exciting topics such as data security and governance. Figure 10-1 shows the Microsoft Fabric spectrum in the DI space; we will focus on trusted DI in this chapter.



Figure 10-1. *Microsoft Fabric is an end-to-end data integration platform*

Finally, we will talk about pricing and the different licensing options you have for Microsoft Fabric.

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D. Ghosh, *Mastering Microsoft Fabric*, https://doi.org/10.1007/979-8-8688-0131-0_10

Fabric Security and Governance

Let's start with Fabric data security and data governance. You need to learn to use the Fabric admin portal before jumping into the Fabric data security part. The Fabric admin portal is a web-based interface for managing and monitoring your Fabric data platform. The Fabric admin portal allows you to perform various tasks, such as the following:

- Create and manage workspaces
- Assign roles and permissions to users and groups, and control access to your workspaces and data assets
- Monitor the health and performance of your Fabric environment and troubleshoot issues using logs and metrics
- Configure settings and preferences for your Fabric account, such as billing, security, and notifications

To access the Fabric admin portal, you need to have an appropriate role, such as Fabric Administrator or Fabric Contributor (refer to the next section). You can sign up for the Fabric admin portal using your Microsoft credentials. The Fabric admin portal is available at `https://fabric.microsoft.com/admin`. It's also available as "Admin portal" under Governance and Insights in the Settings menu, as shown in Figure 10-2.

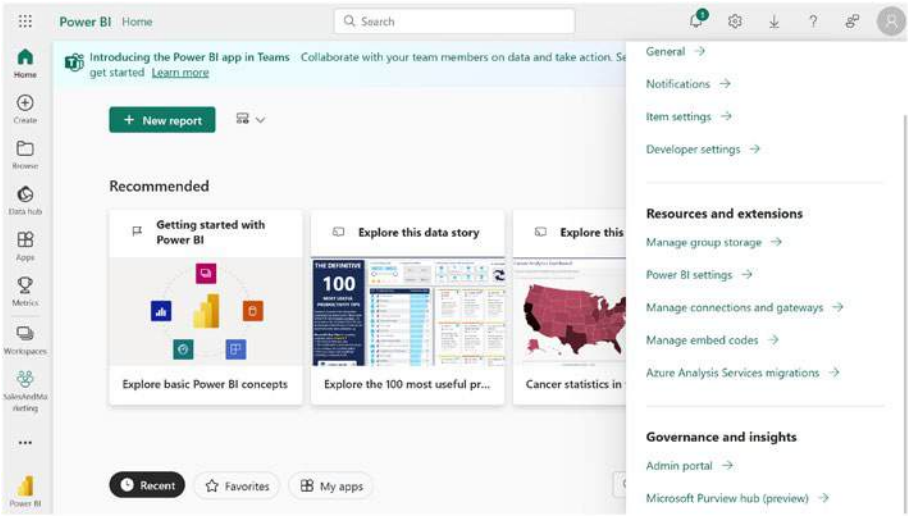


Figure 10-2. *Settings menu*

The Fabric admin portal has a simple and intuitive user interface, with a navigation pane on the left and a main pane on the right. The navigation panel contains the following options:

Fabric Admin Portal	Description
Tenant settings	Access tenant-level settings for the workspace, information protection, exporting and sharing settings, Power BI visuals, and many more.
Usage Metrics	Get Fabric usage metrics.
Premium per user	Not applicable for the Fabric P SKU or F SKU context. This is meant for Premium per user-based licensing.
Audit Logs	Access audit Microsoft Fabric logs.
Domain	Create a domain and organize your data, as covered in Chapter 2.

(*continued*)

Fabric Admin Portal	Description
Capacity Settings	Manage Microsoft Fabric SKUs and all other Power BI SKU features. Schedule capacity refresh.
Embed codes	View embedded codes across organizations.
Organization Visual	View and manage organization visuals.
Azure connections	Connect to Azure storage at the tenant level.
Workspace	View and manage the workspace.
Custom Branding	Customize the PowerBI look and feel with your organization branding.
Information protection	Integrate with Microsoft Purview and Defender.
Featured content	Manage the reports, dashboards, and apps that were promoted to the Featured section on your home page.

Figure 10-3 shows the Fabric admin portal.

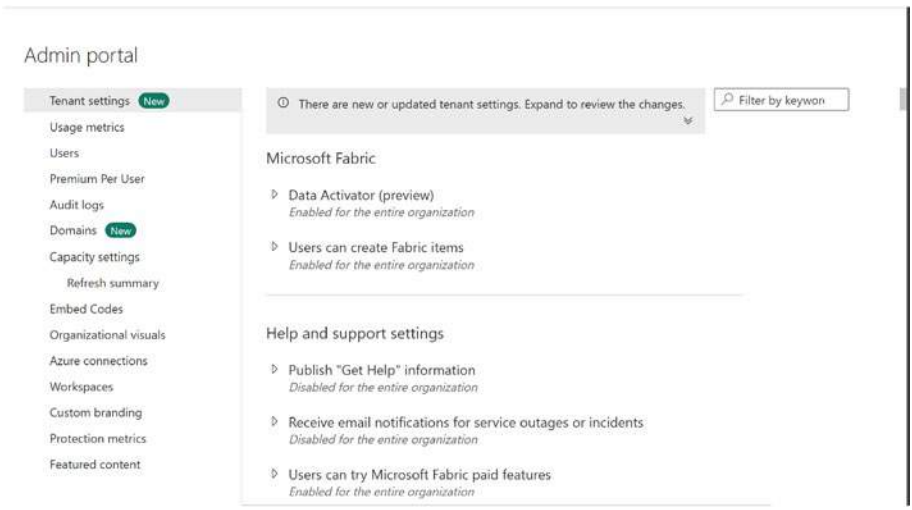


Figure 10-3. *Fabric admin portal*

Now let's zoom into Tenant settings of Microsoft Fabric which will govern critical Fabric security features.

Fabric Tenant Security

Within the Microsoft product portfolio, Entra ID/Previously Active Directory (AD) authentication access is the key authentication mechanism. Active Directory is now known as Microsoft Entra ID (https://www.microsoft.com/en-us/security/business/identity-access/microsoft-entra-id).

The Fabric admin tenant allows you to use the Entra ID (Previous AD ID) authentication mechanism using AD security groups. To apply the AD authentication mechanism for any specific group, you need to select the "Specific security groups" radio button.

Once the specific security groups are selected, click the Edit button to see the list of available groups in Active Directory. You can search for the group name or browse the categories to find the desired group. You can also create a new group by clicking the "Create group" button. Figure 10-4 shows an example of editing the security groups in the Fabric admin portal.

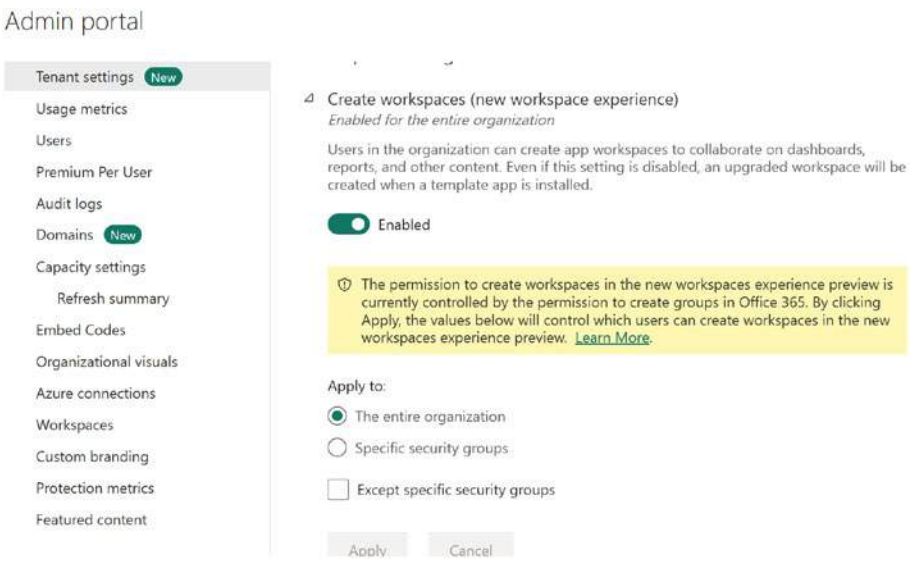


Figure 10-4. *Editing security groups in the Fabric admin portal*

Once you select the security groups, you need to assign them the appropriate roles for accessing the Fabric Admin portal. There are four roles available: Fabric Administrator, Fabric Member, Fabric Contributor, and Fabric Reader. Each role has different capabilities and permissions for managing the Fabric workspaces, data sources, pipelines, and reports. The following table summarizes the roles and their capabilities:

Fabric Workspace Roles	Capabilities
Admin	Users can update and delete the workspace and add and remove other admins from workspace. This role also includes the capabilities in the following rows.
Member	Users can add other members with lower permissions. Allows others to reshare and has the capabilities in the following rows.
Contributor	View, read, write, and delete Fabric artifacts.
Viewer	View and read Fabric artifacts.

By assigning the right security group and Fabric role, you can achieve Tenant live authentication and authorization.

Now you have learned about Fabric Workspace role level security. You will next learn how you can leverage Fabric roles not just in a tenant-level authorization but in the next layer (the Fabric workspace and below layers).

Microsoft Fabric OneLake provides the following layered security:

- Fabric workspace-level security
- Fabric item security
- Fabric compute-specific security

You will learn how to assign role level security in these layers or authorize each layer in following sections.

Fabric Workspace Security

To grant access in Microsoft Fabric at the workspace level, you need to go to the workspace and click the three dots. Then click "Workspace accesses." As covered in Chapter 2, click "Add people" and select the required role for the member. Figure 10-5 shows the role assignment features at the workspace level.

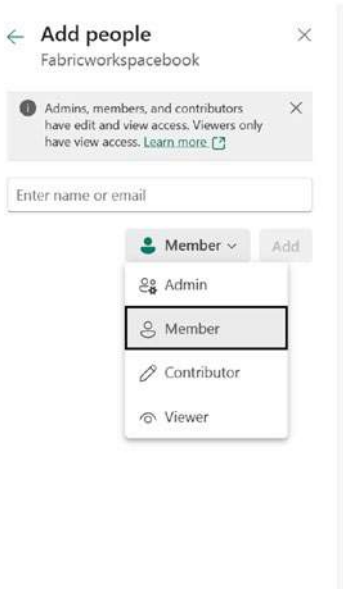


Figure 10-5. *Assigning roles*

Different Fabric roles provide different viewing and writing file capabilities inside the workspace. Refer to the following mapping to understand the capabilities and role assignments:

Fabric Workspace Roles	Capabilities
Admin	View file, write file
Member	View file, write file
Contributor	View file, write file
Viewer	No view file, no write file

Once you have set up workspace-level security, it is time to dive into the next layer of security known as Fabric item security.

Fabric Item Security

Fabric item security uses sharing capability so you can share and grant access at the same time to people across organization. To share an item, you need to follow these steps:

1. Open an item (in this case, the bookdatascience notebook), and click the Share button . The "Create and send link" dialog opens, as shown in Figure 10-6.

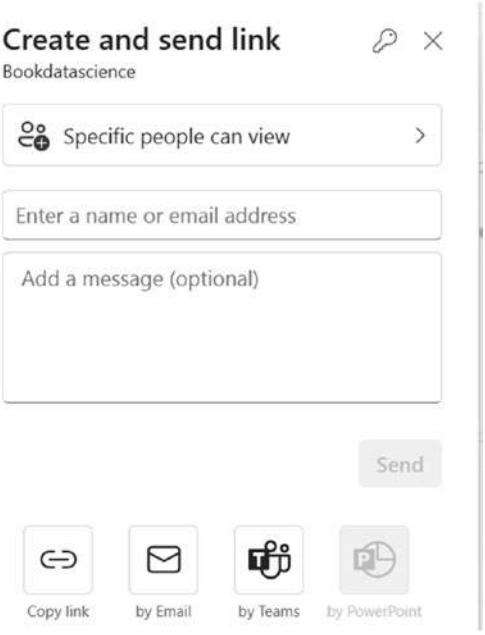


Figure 10-6. *Creating and sending a link*

2. Click “Specific people can view” and then select the “Additional permissions” check box, as shown in Figure 10-7.

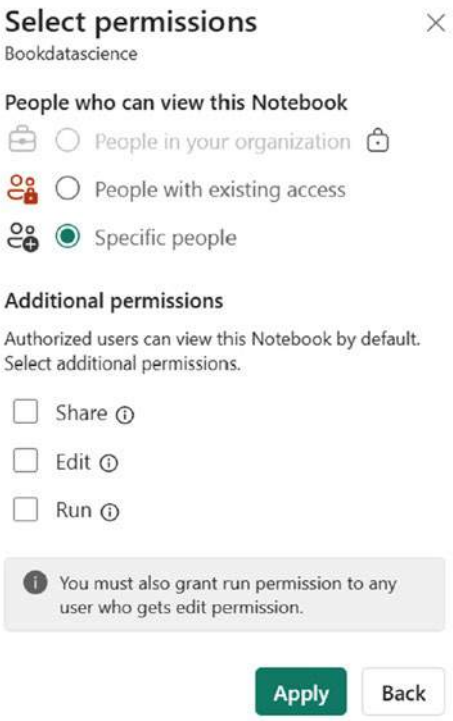


Figure 10-7. *Assigning permissions*

Select the Share, Edit, and Run check boxes accordingly to provide users with additional access beyond Viewing. While this is one way to share and grant access, there are alternate ways as well. For example, follow these steps to achieve Fabric item-level security:

1. Click the three dots (…) beside an item, as shown in Figure 10-8.

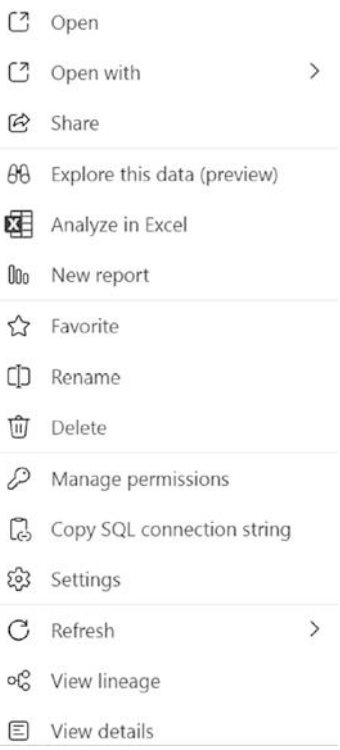


Figure 10-8. *Item management taskbar*

2. Click "Manage permissions."
3. Click "+Add user" and then "Enter a name or email address" in the "Grant people access" taskbar to share and grant item-level access to people, as shown in Figure 10-9.

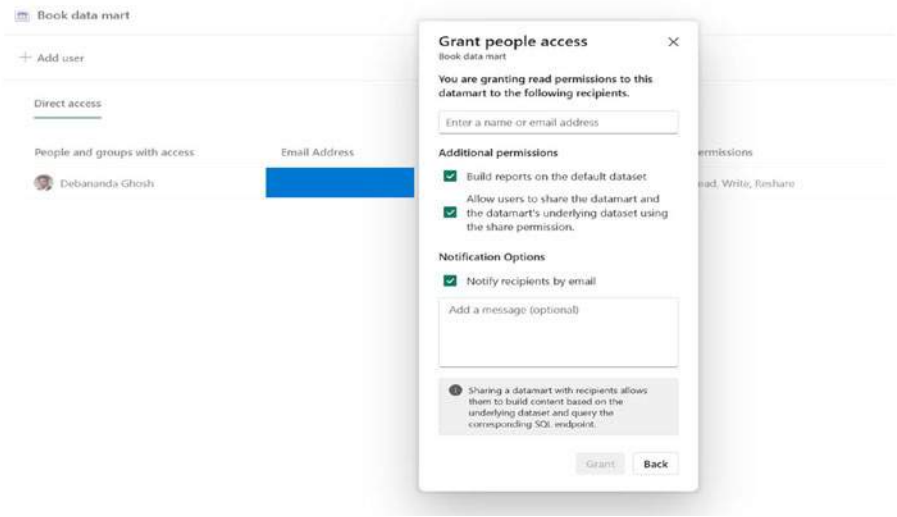


Figure 10-9. *Assigning permissions*

Using these approaches, you can achieve item-level security. Now you will move on to the next level of security, which takes place at the computation layer.

Fabric Computation Security

Fabric provides computing-specific security and is the third layer of its data security. The following computation security features are available today:

- **Object-level security:** Fabric includes specific database object-level access. Object-level security allows you to control granular access for both collaboration and consumption. The Fabric data warehouse supports traditional T-SQL security constructs. GRANTS, REVOKE, and DENY can be used to secure objects with the warehouse.

- **Column-level security:** This provides authorized column-level access.
- **Row-level security:** This provides an authorized viewing of rows in tables.
- **Dynamic data masking**: This is an authorized viewing of sensitive data by using masks.

Storage Encryption

Finally, built-in OneLake storage encryption is a native storage security provided by default with Microsoft Fabric OneLake. Data stored in OneLake is encrypted and decrypted transparently using 256-bit AES encryption, which is FIPS 140-2 compliant.

Fabric Conditional Access

Fabric conditional access is a security feature that allows you to control who can access Fabric, under what conditions, and from which devices. Fabric conditional access leverages Entra ID (Previously Azure Active Directory) to enforce policies based on user identity, location, device state, app sensitivity, and data classification. For example, you will require multifactor authentication (MFA) for users who access sensitive datasets from outside your corporate network, and you will want to block access from devices that are not compliant with your security standards. Fabric conditional access helps you to protect your data from unauthorized or risky access, while enabling productive and secure collaboration across your organization and beyond.

To use Fabric conditional access, you need to have an Azure AD Premium license and assign roles and permissions to your users and groups in Fabric. You also need to configure the conditions and controls for your access policies in the Azure portal. You can create and manage multiple policies for different scenarios and apply them to specific users, groups, and datasets.

To set up Microsoft Fabric conditional access with Entra ID, execute the following steps:

- Sign into the Azure portal (you need global administrator permissions).
- Select Microsoft Entra ID.
- On the Overview page, choose Conditional Access.
- Go to the Conditional Access | Overview page, and select "+Create new policy."
- Provide a name for the policy and configure the details such as user groups, target resources, and other details.

Figure 10-10 shows the "Conditional access" feature in Microsoft Entrap ID.

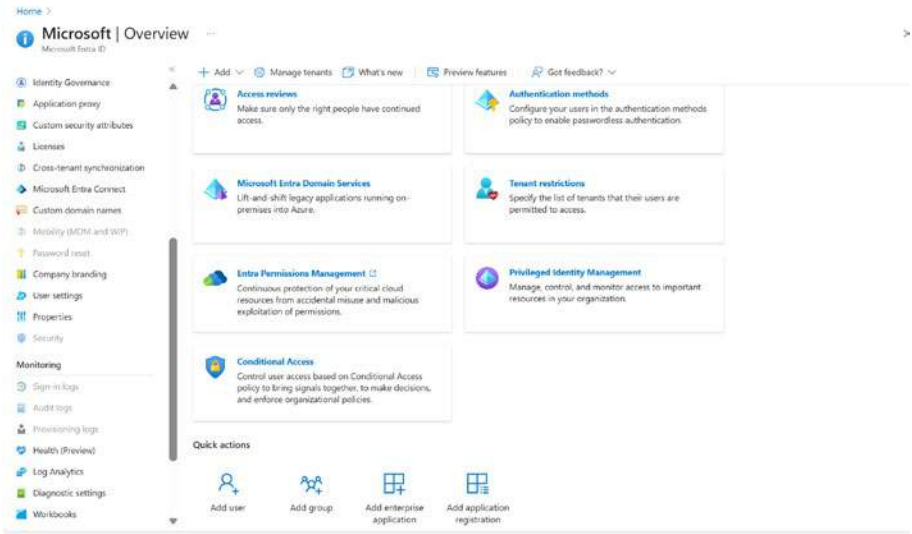


Figure 10-10. *Fabric Entra ID conditional access*

Fabric conditional access supports the following conditions and controls:

- **Sign-in risk:** This gives you the probability that a sign-in attempt is not authorized by the owner of the account. Sign-in risk is calculated by Azure AD based on various signals, such as unfamiliar locations, IP addresses, or devices. You can set a policy to block or challenge sign-ins with high or medium risk.
- **Device state:** This is the compliance and management status of the device used to access Fabric. The device state is determined by Microsoft Intune or another mobile device management (MDM) solution. You can set a policy to allow or deny access from devices that are compliant, managed, hybrid Azure AD joined, or marked as trusted.

- **Location:** This is the physical or network location of the user who tries to access Fabric. The location is based on the IP address of the device. You can set a policy to allow or deny access from specific countries, regions, or IP ranges.
- **Client app:** This is the application used to access Fabric. The client app can be a web browser, a mobile app, a desktop app, or a legacy authentication protocol. You can set a policy to allow or deny access from specific client apps or protocols.
- **Data sensitivity:** This is the level of confidentiality and business impact of the dataset accessed in Fabric. Data sensitivity is defined by the data classification labels that you assign to your datasets in Fabric. You can set a policy to allow or deny access based on the data sensitivity label.
- **Require MFA:** You can require MFA for any access to Fabric datasets with a high sensitivity label from any device.
- **Block access:** You can block access to Fabric datasets from devices that are not compliant with your security standards or managed by your organization.
- **Allow access:** You can allow access to Fabric datasets only from web browsers that support modern authentication and encryption protocols.
- **Deny access:** You can deny access to Fabric datasets from outside your corporate network or from specific countries or regions.

To recap, you have learned about various Microsoft Fabric security aspects. To learn about the future plans for Microsoft Fabric administration and security, you can refer to https://learn.microsoft.com/en-us/fabric/release-plan/admin-governance.

Fabric and Purview: Data Governance

Now let's look at the data governance aspect of Microsoft Fabric. Note that for data governance, Microsoft has a separate tool known as Microsoft Purview (this is a separate service outside the scope of this book). You can learn more about Microsoft Purview at https://www.microsoft.com/en-sg/security/business/microsoft-purview.

In a nutshell, Purview is a unified data governance service that helps you discover, catalog, and classify data, wherever it resides. Purview scans your data sources and creates a comprehensive inventory of your data assets, along with their lineage, properties, and classifications. Purview also helps you to comply with data privacy and security regulations by identifying and tagging sensitive data, such as personal or financial information. Purview integrates with Fabric to provide a holistic view of your data landscape and enable data governance at scale.

In this section, we will discuss the Microsoft Purview hub in Microsoft Fabric. It has a centralized page for Fabric administrators to govern the data estate built on the Fabric platform. This hub is a link between Purview and Fabric to provide Purview-based data protection, auditing, data catalogs, and data loss prevention.

The Purview hub report contains the following pages:

- **Overview report:** This report details Fabric workspace and item distribution and use of endorsement and sensitivity labeling, as shown in Figure 10-11.

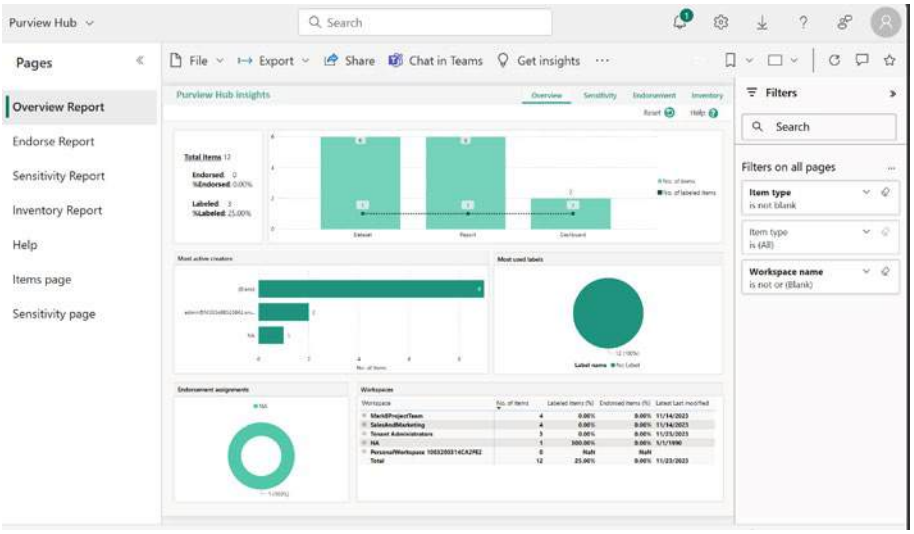


Figure 10-11. *Fabric Overview report*

- **Endorsement report:** This report drills down and analyzes distribution on the endorsement level, as shown in Figure 10-12.

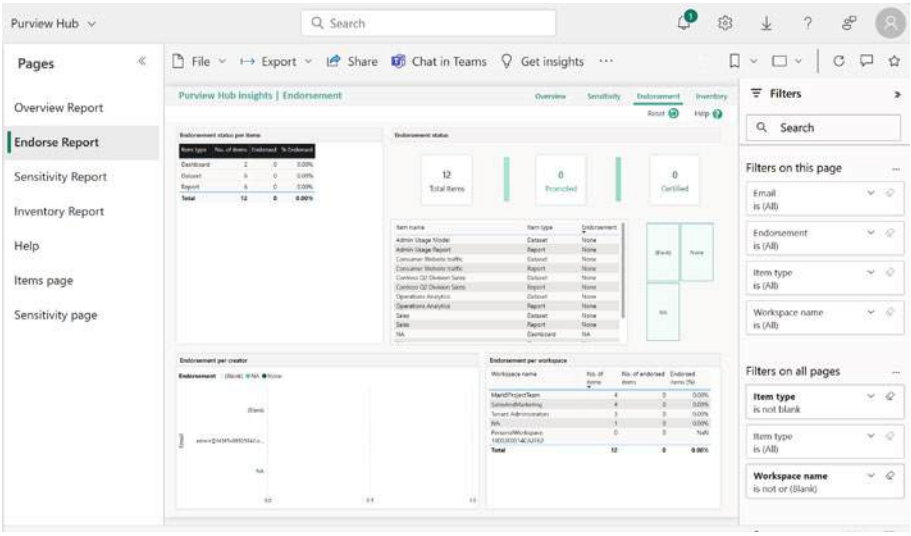


Figure 10-12. *Fabric Endorsement report*

- **Sensitivity report:** This report drills down and analyzes distribution based on sensitivity labeling, as shown in Figure 10-13.

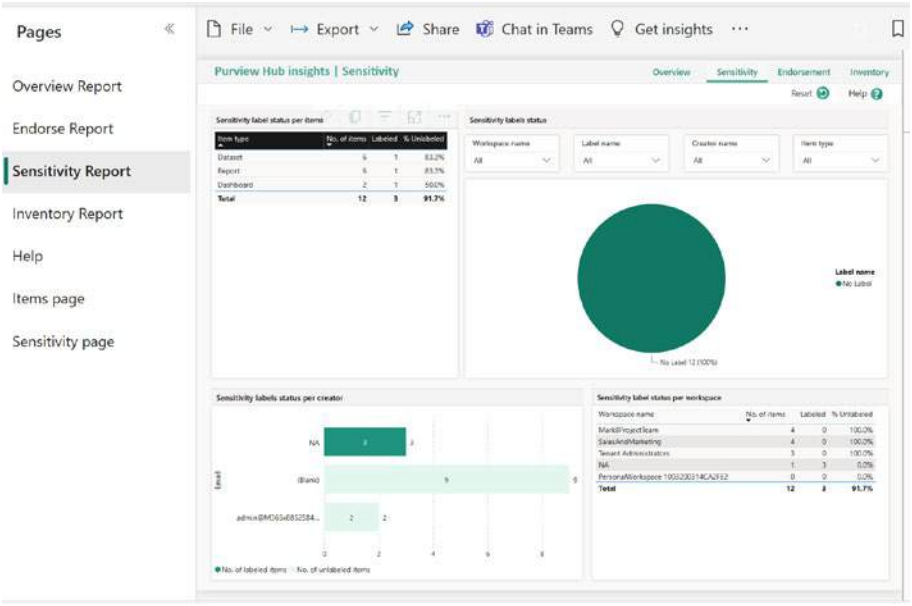


Figure 10-13. *Fabric Sensitivity report*

- **Inventory report:** This report provides details about labeled and endorsed items. You can apply date ranges and filter by workspace, item type, username, email, and modified date, as shown in Figure 10-14.

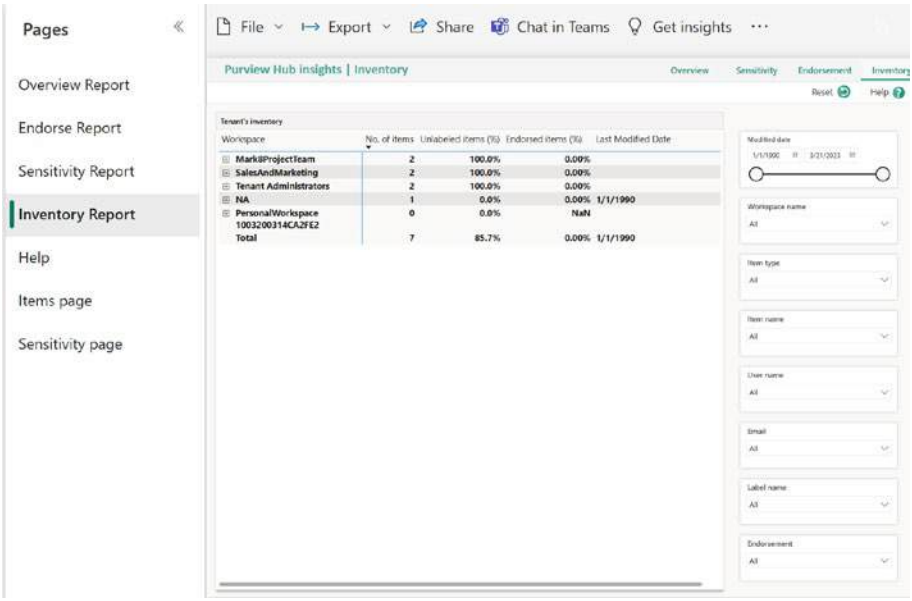


Figure 10-14. *Fabric Inventory report*

- **Items page:** This report details analytics about the distribution of items across organizations.
- **Sensitivity page:** This report details insights about sensitivity labeling across the entire organization.

Note that based on authorization, the Purview hub provides two views.

- **Fabric Admins view:** End-to-end Fabric-based data estate insights
- **Other users view:** User-specific Fabric items and related Microsoft Purview features

By leveraging these reports, you can get end-to-end data insights for your Microsoft Fabric data platform. To learn more about Microsoft Purview, refer to `https://learn.microsoft.com/en-us/fabric/governance/Microsoft-purview-fabric`.

Fabric Pricing and Licensing

We are almost at the end of the book. In this last section of this chapter, we will discuss the Fabric pricing model and licensing options. The following are the ways to adopt Fabric at the time of writing:

- Microsoft Fabric trial (for Fabric test-driving purposes)
- Power BI Premium per capacity (P SKU) (for Fabric enterprise deployment purposes)
- Microsoft Fabric capacity via Azure portal (F SKU) (for Fabric enterprise deployment purposes)

Concisely, Microsoft Fabric is available only through the Power BI Premium per capacity license. There are other ways to use Power BI; however, Fabric is not offered currently through those SKUs. Refer to the following table to learn more:

Power BI Licensing Option	Fabric Availability
Power BI Pro	No
Power BI Premium	No
Power BI Embedded	No
Power BI Premium per capacity	Yes (P SKU)

Fabric can be also provisioned via the Azure portal at `https://ms.portal.azure.com/#home`. This capacity is known as the Fabric capacity (F SKU).

Azure Portal	Fabric Availability
Azure portal	Yes (F SKU)

To provision the Microsoft Fabric F SKU, you need to go to the Azure portal and execute the following steps:

1. Search for *Microsoft Fabric* in the Azure portal search bar, as shown in Figure 10-15. Subsequently select Microsoft Fabric from the list.

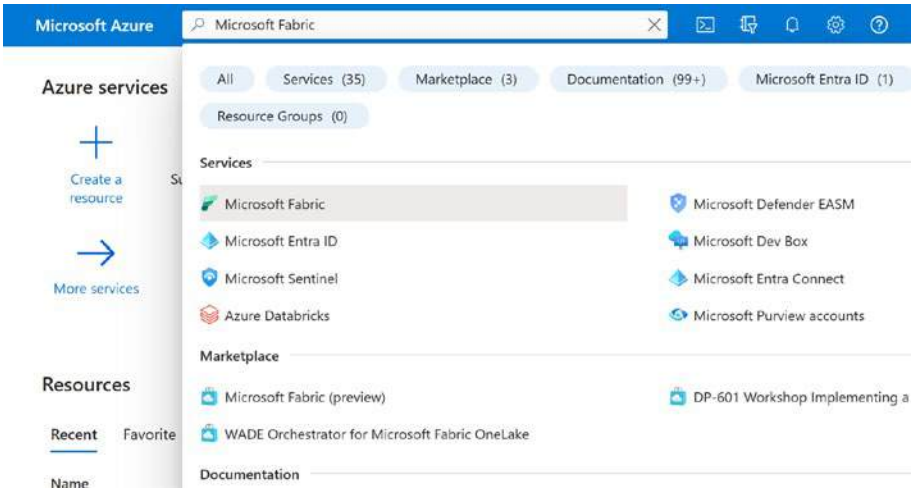


Figure 10-15. *Searching for Microsoft Fabric*

2. Click +Create (as shown in Figure 10-16) to create a Fabric F SKU capacity.

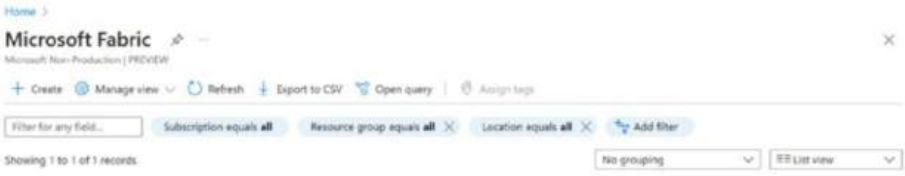


Figure 10-16. *Creating a Fabric capacity*

3. When the "Create Fabric capacity" appears, click Subscription, create a new resource group (or choose an existing one), provide a capacity name, and select Region. Then you need to "click Review + create" to create a Fabric capacity via the Azure portal, as you can see in Figure 10-17.

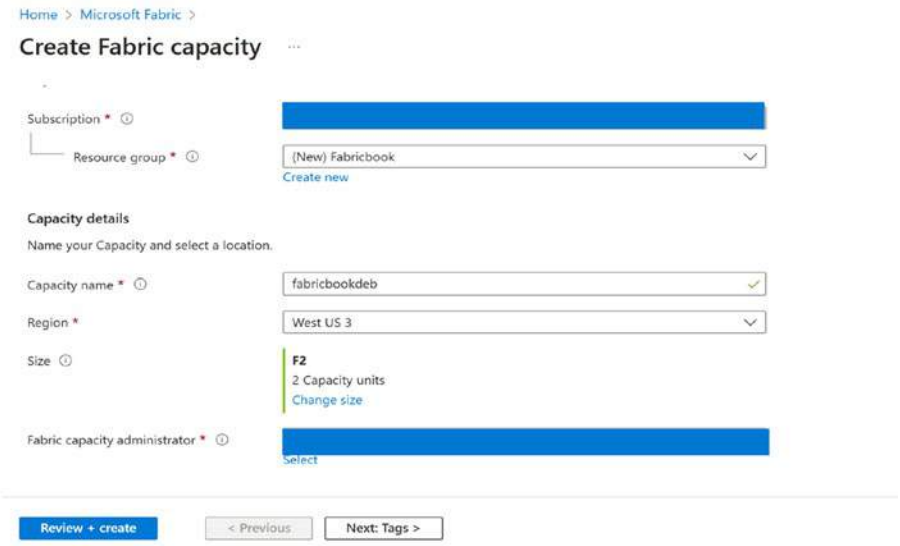


Figure 10-17. *Reviewing the capacity*

4. The Azure portal will validate the details, and once you get the "Validation succeeded" message, you need to click Create to deploy a Microsoft Fabric capacity. Azure will provision the Fabric capacity. Note that here we are choosing F2, which is the smallest capacity for our academic purposes.

Assuming this service is running 24/7, an estimated pay-as-you-go cost will be provided on the canvas (see Figure 10-18).

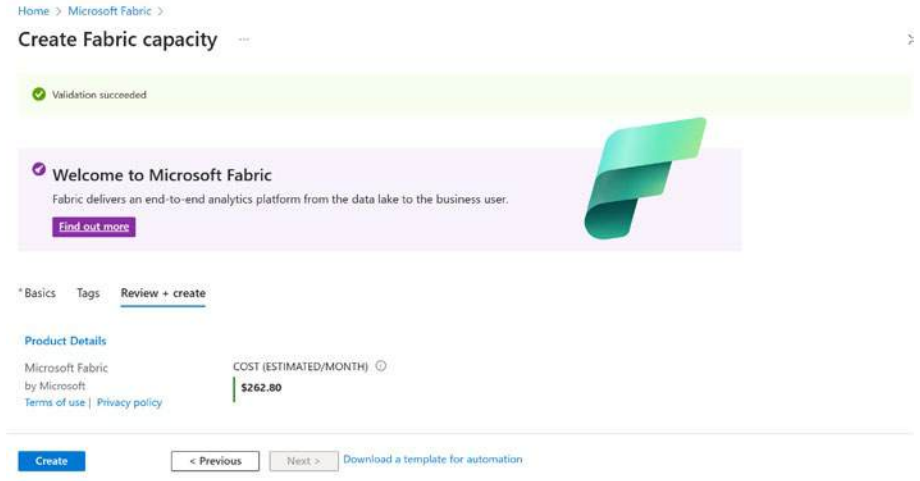


Figure 10-18. *Checking the cost*

5. Once the Fabric F SKU is provisioned in the Azure portal, the Azure portal Fabric capacity canvas looks like Figure 10-19. Note that you need to click Pause whenever you are not using this workspace to lower your costs. Once it's paused, you can click Resume to turn it on.

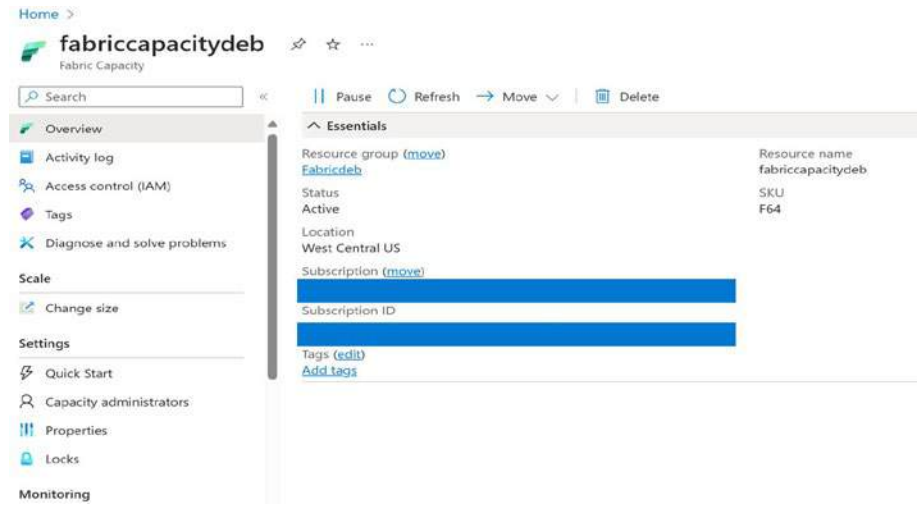


Figure 10-19. *Optimizing the resource costs*

Once you have the capacity up and running, you can follow the capacity assignment steps. Refer to Chapter 2. Follow those steps and select the "Fabric capacity" radio button and the Fabric capacity name "License capacity," as you can see in Figure 10-20.

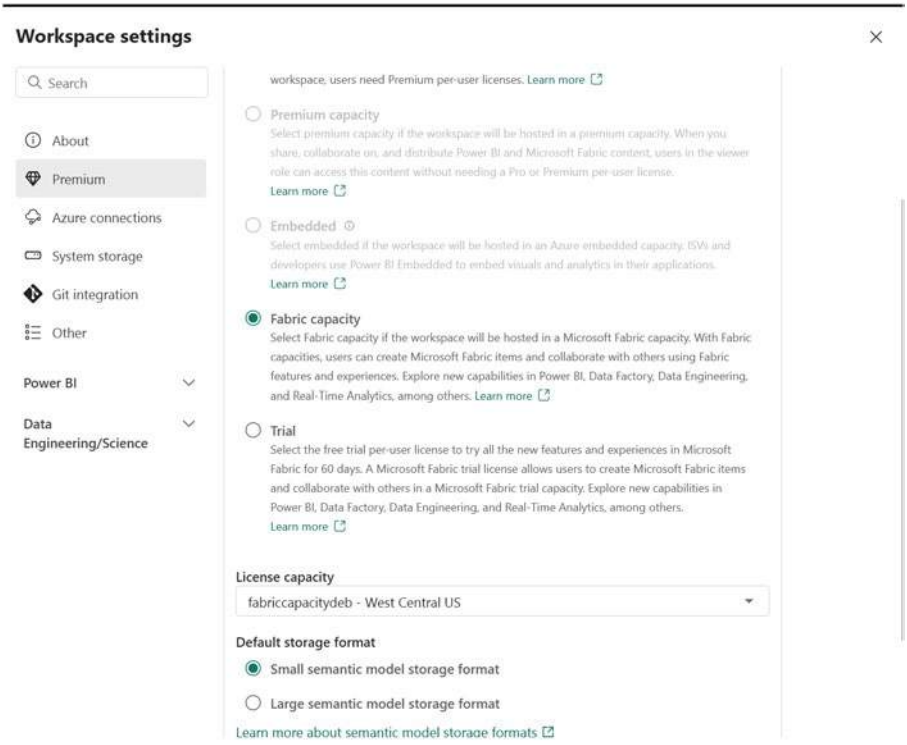


Figure 10-20. *Assigning the Fabric capacity (F SKU) to the Fabric workspace*

We have now this Open AI–based AI assistant tool capability inside Power BI to generate insights. Fabric Pricing has two line items, as shown here:

Fabric Cost Line Item	Cost Dimension (US $)
Computation capacity	Capacity unity/hour
Data storage (OneLake storage, BCDR, caching)	GB/hour

You need to go to the following site to understand the Fabric F SKU pricing: https://azure.microsoft.com/en-us/pricing/details/microsoft-fabric/.

To adopt the Microsoft Power BI Premium per capacity (P SKU), refer to https://powerbi.microsoft.com/en-us/pricing/.

While the previous links highlight the pay-as-you-go model, you may notice that F SKU also has another pricing shown as Reserved at the previous F SKU pricing link. This is known as reserved instance–based pricing for Azure services. The reserved instance is a pricing model that allows you to pre-purchase computer resources for one or three years, at a significant discount compared to pay-as-you-go prices. Reserved instances are valid for Microsoft Fabric, Azure virtual machines, SQL databases, Azure Synapse SQL pools, and other Azure services.

By using a reserved instance, you can optimize your budget and reduce the total cost of ownership of your Power BI solutions. To purchase a reserved instance, you need to specify the region, size, term, and quantity of the resources you want to reserve. You can also choose to pay up front or monthly. Once you purchase a reserved instance, the reservation discount is automatically applied to the matching resources in your subscription.

A reserved instance can be especially beneficial for Power BI Premium capacity deployments, as they use Azure virtual machines to run the dedicated capacity. By reserving the F SKU that you want to use, you can save more compared to pay-as-you-go prices. You can also exchange or cancel your reservations at any time, subject to some limitations and fees. The following steps show you how to buy reserved instances of the Microsoft Fabric F SKU from the Azure portal:

1. Go to the Azure portal and search for *Reservations,* as shown in Figure 10-21. Then click "Purchase now."

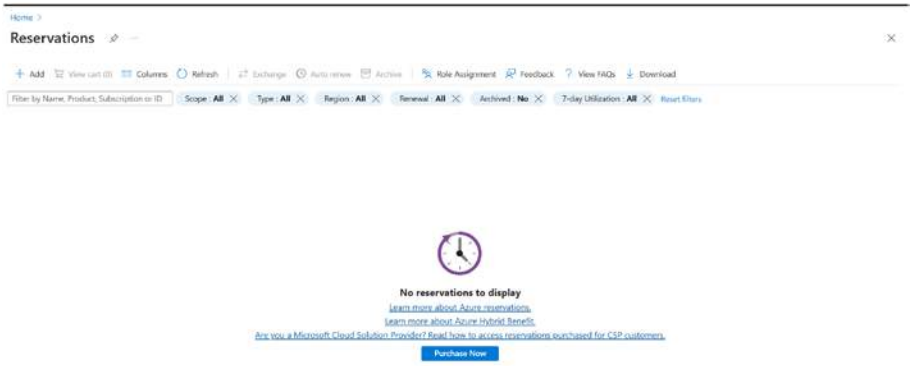


Figure 10-21. *Searching for reservations*

2. Select Products and then Microsoft Fabric, as shown in Figure 10-22.

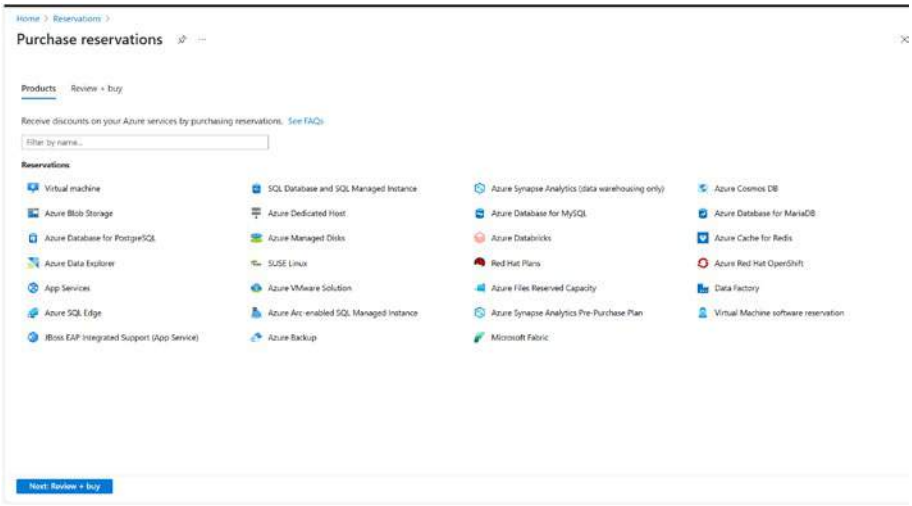


Figure 10-22. *Reserved instances*

3. Set the subscription and scope, as you can see in Figure 10-23. Then click "Next: Review + buy" to go ahead with the reservation-based procurement.

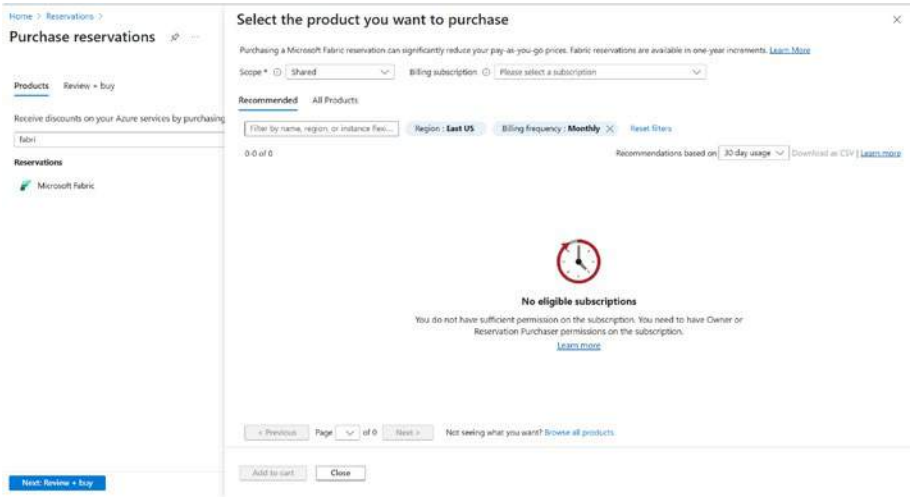


Figure 10-23. *Reserving your services*

Summary

In this book, you learned about the cloud analytics evolution and how the field moved from IaaS to SaaS. We then introduced Microsoft Fabric. After that, we did a deep dive into the ETL, data engineering, and data science capabilities of Fabric. You learned how to build real-time analytics in Microsoft Fabric. Also, we mapped your learning to the corresponding data architecture at the end of some chapters, like in Figure 10-24.

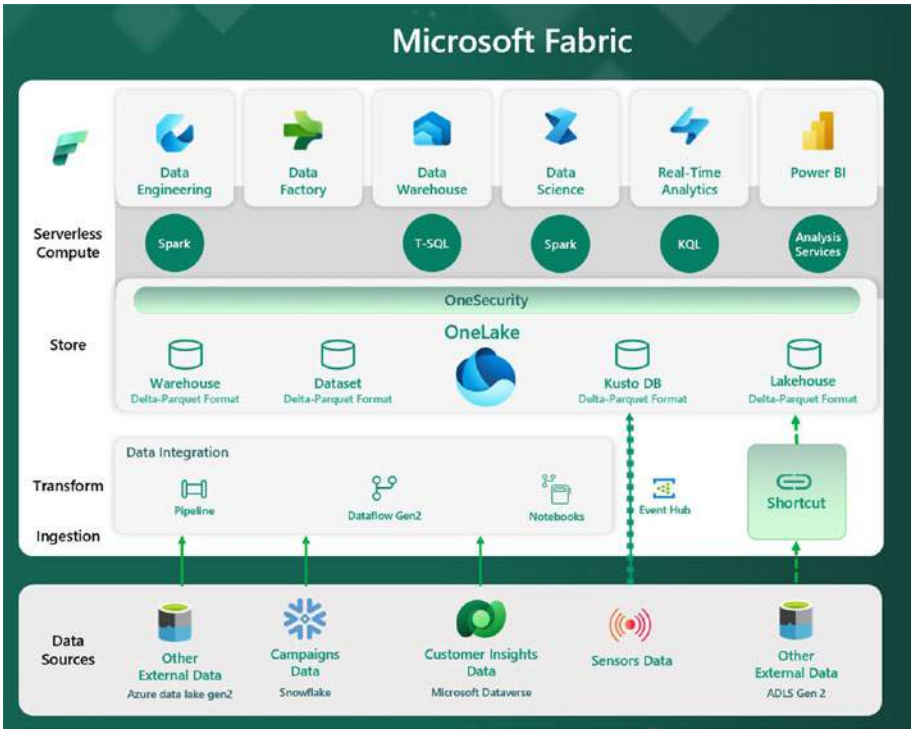


Figure 10-24. *Microsoft Fabric end-to-end data architecture*

You also learned how to deploy and adopt Microsoft Fabric via the Power platform and the Azure portal. You learned about the features of each key pillar and how to leverage the Copilot capability in the Fabric workspace. Note that we tried to cover as many GA features as possible while writing this book. By now you should be able to work and deploy data platforms into production using the Microsoft Fabric capabilities.

Note that these features will further evolve, and hence we recommend you refer to `https://www.microsoft.com/en-us/microsoft-fabric` for information about new capabilities.

Also, for future release plans, visit `https://learn.microsoft.com/en-us/fabric/release-plan/`.

Happy Microsoft Fabric learning! Cheers!

Further Reading

Learn more about the topics covered in this chapter at the following locations:

- **Fabric workspace access:** https://learn.microsoft.com/en-us/fabric/get-started/give-access-workspaces
- **Fabric row-level security:** https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security
- **Fabric workspace roles:** https://learn.microsoft.com/en-us/fabric/get-started/roles-workspaces
- **Fabric reserved instance:** https://learn.microsoft.com/en-us/azure/cost-management-billing/reservations/fabric-capacity

Index

A, B

Activator anatomy
  core parts, 246
  event data and visualizations, 248
  landing page, 246
  object trigger visualization, 249
  package option, 247
  primary workflow, 250
  reflex configuration, 248, 249
  Reflex object, 246
Application programming interface (API), 126, 127, 219, 230
Azure Data Factory (ADF), 204–206

C

Capacity units (CUs)
  Fabric capacity (F SKU), 28, 29
  platform resources
    access management, 39
    creation, 36
    demo capacity, 37
    portal capacity, 34, 35
    workspace allocation, 37, 38
  Power BI licenses, 25
  Power BI Premium per capacity
    admin portal, 29, 30
    configuration, 29
    landing page, 33, 34
    Premium capacity, 32
    workspace assignment, 30–33
  Power BI Premium series (P SKU), 26, 27
  technology landscape, 24
  trial license, 25, 26
Cloud analytics evolution
  business users/technical users, 2
  data management capabilities, 2
  differences, 3
  IaaS to SaaS, 4
  open-source tools, 4
  scalability, 3
  unified analytics platforms, 4
Cloudera Data Platform (CDP), 4

D

Data Analysis Expressions (DAX), 271
Data engineering
  architecture, 61, 92
  batch job monitoring
    capacity/node size mapping, 85

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Data engineering (*cont.*)
directed acyclic graph (DAG), 80, 81
environment, 85, 86
file creation, 87
monitoring hub, 81, 82
pool customization, 83–85
starter pool, 82, 83
status/tasks, 80
explorer, 70–72
features, 49, 60
ingestion/pipeline, 61–70
Lakehouse (*see* Lakehouse)
notebook
data exploration activity, 74
drag-and-drop feature, 76
preparation/ transformation, 72
sample option, 74, 75
schedule tab, 77, 78
source code, 75
Spark code, 73, 74
variables feature, 77
source code, 89
Spark job
definition, 78, 79
monitoring, 80–87
notebook, 72–78
SQL endpoint, 87–90
visualization, 90–92
Data integration (DI)
architecture, 349
features, 319
platform, 319
Data warehouses (DWs)
architecture, 165
canvas, 132
capabilities, 133, 134
cardinality selection, 159
components, 134
computation isolation, 154
copying data tab, 137, 138
creation, 134
cross-database/virtual queries, 146, 147
data model/relationship, 157–159
development, 141
explorer, 135
fabricdwpipeline creation, 136
features, 131
ingestion/pipeline, 136–141
integration
auto-creation, 161
Excel pivot page, 160
Power BI dataset, 159
report creation, 161
learning process, 132
monitor hub, 162–164
multiple layers/query optimization, 154–156
multiple tables, 139
OneLake explorer, 141
pipeline monitoring, 163
pipeline status, 139
Power BI Dataset, 159–162
professional developer, 133
Server management studio, 148–152

SQL editor, 148, 149
statement types and lock mapping, 157
tables, 140
transaction capabilities, 156, 157
visual query transformation
column profiling features, 143
GUI capabilities, 142
Power Query editor, 144
prepopulated table, 142
tables, 145
workspace canvas, 144, 145
workload management, 154
Distributed Query Processing (DQP), 152

E

Exploratory data analysis (EDA), 97, 108–110
Extraction, transformation, and loading (ETL), 7, 168
Extract/transform/load (ETL), 168

F

Fabrics
activator, 13
admin portal, 324
anatomy, 9
assigning permissions, 328
computation security, 330
conditional access, 331–335
conversational language integration, 5
data analytics capabilities, 5
data engineering capabilities, 10
factory, 12
foundation, 6
governance
authorization, 339, 340
definition, 335
endorsement report, 337
inventory report, 339
overview report, 336
Purview hub reports, 335
sensitivity report, 338
high-level concepts, 1
item security, 327–330
Power BI, 13
pricing model/licensing options
assigning process, 345
capacity, 341
capacity (F SKU), 340
cost checking, 343
line items, 345
Power BI, 340
reservations, 346, 347
resource costs, 344
reviewing option, 342
search bar, 341
steps, 340
subscription/scope, 348
real-time analytics, 12
roles/capabilities, 324
science capabilities, 11

Fabrics (*cont.*)
security/governance, 320–323
settings menu, 321
storage encryption, 331
tenant security, 323–325
warehouse, 11
workspace, 5
workspace accesses, 325, 326
workspace landing page, 10
Flow Gen2
append/replace capability, 182
authentication, 174, 175
capabilities, 170
code-free transformation, 179
connecting source, 173
connectors, 172
credentials, 181
data source, 173
designer canvas, 171
destination, 180
export template, 187, 188
features, 170, 174
filtering rows, 183
gateway, 174, 175
import mechanisms, 172
item published, 184
naming table, 182
orchestration, 185
Power Query reference, 188
Power Query ribbon, 177, 178
preview, 175, 176
replacing values, 178
scheduling window, 185–187
searching option, 184
search pane, 179
steps, 170, 185
tabs, 171
transformation, 178

G, H

Graphical user interface (GUI), 141, 179

I, J

Infrastructure-as-a-service (IaaS), 3–5, 19
Ingestion/pipeline
activity details, 70
canvas, 64
copy data details, 67, 68
copy tool steps, 62
creation, 63
data destination, 65–67
datasets, 65
data warehouses (DWs), 136–141
destinations, 61
FabricBookPipeline, 69
icon details, 69
name creation, 63
preview window, 62
science development
book-recommendation folder, 102
code statement, 100

CSV file, 102
file/external path, 100
Pandas dataframe, 101
Parquet file, 101
table name, 68
Integrated development environment (IDE), 230
Integration
business benefits, 169
capabilities, 167
Copilot experience
attribute creation, 191
capabilities, 189
column option, 191
factory context, 189, 190
Data Factory, 168
e-commerce websites, 167
factory mounting, 204–206
features, 169
Flow Gen2, 170–188
pipeline (Preview)
capabilities, 192
channel selection, 201
creation, 193
execution, 203
execution failure, 200
experience, 197
factory condition execution, 200
features, 192
file creation, 197
group chat selection, 202
integrating teams, 202
name option, 193, 194
notebook selection, 199
running option, 203
searching tool, 198
steps, 198
tabs, 194, 195
team chat, 204
template gallery, 195, 196

K

Kusto (KQL) database
API/SDK, 230
benefits, 223
BI visualization, 223
code-free tooling, 219–221
create button, 224
creation, 213, 214
development/integration features, 240
empty database, 214
execution/outcome, 222
gallery, 218
home page, 216
ingestion options, 216–219
menu option, 225
naming table, 218
Python plugin, 228, 229
query language, 223–228
retention and caching policy, 231, 232
streaming analytics platform, 213–216
Kusto Query Language (KQL), 11

L

Lakehouse
  advantages/disadvantages, 51
  bin compaction, 53
  Datawarehouse mode, 303
  Data warehouses (DWs), 131
  delta file, 52–54
  history, 51
  integration, 180
  pinning workspace, 57
  SQL endpoint, 87–90
  structure, 89
  V-ordered Parquet file, 52
  workspace creation, 54–60
Large language modeling/model (LLM), 97, 124–127
Logical data modeling (LDM), 157

M, N

Massively parallel processing (MPP), 2
Mobile device management (MDM), 333
Monitoring/alert generation
  activator (*see* Activator anatomy)
  activator offering, 245
  challenges, 244
  data source
    alert and power automate, 253
    data instructions, 252
    detecting/generating alert, 254
    event stream, 255–259
    Power BI, 252–255
    reflex object, 250–252
  define/detect actionable patterns, 260, 261
    act feature, 261
    detecting/acting, 261
    trigger creation, 260
  event stream
    canvas, 255, 256
    definition, 255
    destination, 256
    home page/search, 255
    integration, 257
    item, 258
    key column/properties/object, 259
    linking object, 257
    object assignment, 258
    workspace filter, 255
  high level process, 243
  trigger generation, 244
  trigger workflow, 262
Multifactor authentication (MFA), 331

O

OneLake workspace
  configuration details, 45
  definition, 39
  explorer experience, 41
  ingestion, 100
  integration, 232

openness, 42
security/serverless computation concept, 40
shortcut feature, 43
storage encryption, 331
On-premises data gateway (OPDG), 174

P, Q

Platform as a service (PaaS), 3, 19
Point-of-sale (POS), 167
Power BI platform
admin portal, 29, 30
auto-create report
automatic report creation, 295
BookPBI semantic dataset, 295
feature, 294
generation, 295
menu selection, 297
business/technical users, 13
capabilities, 266
concepts, 266
Copilot
chart summarization, 313, 314
product categories, 315
scenarios, 313
summarization, 314, 315
datamarts
configuration, 311
database creation, 309, 310
exploring option, 312
features, 309
loading data, 311
naming option, 310
semantic model, 312
desktop/report builder
canvas, 271
desktop, 269
development, 268
Excel data preview, 273
filter/formatting options, 282
geospatial plugin, 280, 281
import option, 276
loading/transforming dataset, 275
managing relationships, 278
none phrase, 274
ribbon icons, 272
saving file, 283
steps, 269
transformation, 274
views, 271
visualization, 275–277
worksheets, 279
direct lake mode
data model, 304
documentation, 302, 303
import mode, 302
saved model, 304
semantic model, 304, 306
Fabric capacity, 28
features, 294

Power BI platform (*cont.*)
fundamentals, 267, 268
insights, 302
integration, 159
landing page, 33
licenses, 25
mobile apps
learning process, 290
lineage view, 299, 300
quick insight feature, 297, 298
report display, 292, 293
visualization mobile layout, 291, 292
work offline/online, 290
paginated reports
builder tool, 299
creation, 300
format/download report, 300
RDL file, 301
Premium per capacity license, 29
Premium series, 26, 27
pricing model/licensing options, 340
services
dashboard creation, 288, 289
dashboards/flow (Gen1), 284
features/capabilities, 283, 284
message notification, 286
online cloud workspace, 286
.pbix file, 285
published report, 287
reports/datasets, 284
steps, 287
sharing report
alert settings, 308
dashboard, 306, 307
teams group, 307
visualization, 265
workspace
assignment, 31
landing page, 33, 34
Premium capacity, 31
trial license, 31
Python language, 228, 229

R

Real-time analytics (RTA), 1, 243
analytics, 209
analytics canvas, 211
canvas, 212
capabilities, 210
challenges, 210
components, 211
conceptual illustration, 209, 210
data retention/catching policies, 231, 232
definition, 209
event stream, 232–239
intuitive tab, 213
KQL (*see* Kusto (KQL) database)
Kusto (KQL) database, 213–216

learning process, 211
monitoring and alert (*see* Monitoring/alert generation)
Report Definition Language (RDL), 269

S

SaaSification analytics
auto-visualization, 8
capabilities, 7
code-free data wrangling, 8
code-free real-time alert configuration, 8
developers/data analysts, 6
machine learning user interface, 8
OneLake/intuitive experience, 7
virtual data warehouse, 8
visual editor/Dataflow Gen2, 7
world-class lakehouse, 8
Science development
canvas, 99
code and no-code capabilities, 98
dataset exploration, 104, 105
definition, 95
EDA/visualization, 108–110
home page, 98
ingestion, 100
large language modeling, 124–127
learning process, 97
machine learning experimentation
auto-logging capabilities, 121
book recommendation, 119
comparison, 120
experiment comparison, 121
item selection, 119
model experimentation, 119
model performance comparison, 124
registered model, 123
source code, 118
machine learning model, 116
model development, 116–118
notebook code, 110, 111
process overview, 96
recommendation systems, 117
requirements, 95
steps, 95
user interface (UI), 98
VS Code, 112–115
workspace, 96, 97
Wrangler tool
code-free, 105
drop column operation, 107
features, 104
loading data, 108
operations, 106
source code, 107
user interface, 106
Software-as-a-service (SaaS), 1, 3, 4, 39, 168, 211
configuration details, 45

Software-as-a-service (SaaS) (*cont.*)
data hub, 41
data mesh capabilities, 45–47
multicloud virtualization, 42–45
OneLake, 39–42
security/serverless computation concept, 40
See also SaaSification analytics
Software development Kit (SDK), 230
SQL Server Management Studio (SSMS), 141, 149–153
SQL Server Reporting Service (SSRS), 159
Streaming pipeline
apps development, 236
capabilities, 232
creation, 234
destination sources, 235
empty canvas, 234
event stream pipeline, 235, 236
feature, 233
source code, 237–239
sources, 235

T

Tenant/workspace/capacities
components, 21
CU (*see* Capacity units (CUs))
hierarchy, 20
icons, 24
items, 21–23
organization/subscription/license concept, 18
power platform admin settings, 19
security groups, 323–325
user interface (UI), 21
workspace creation, 21
Transact-SQL (T-SQL), 141

U

User-defined functions (UDFs), 228

V, W, X, Y, Z

Visualization
dataset, 91, 92
data warehouses (DWs), 159
geospatial chart, 91
results, 90
science development, 108–110
Visual Studio Code (VS Code)
detailed instructions, 112
kernel selection, 114
notebook/integration, 113
Spark setup, 113
web pages, 114
workspace integration, 115